

GE Healthcare

Signa HD, HDx & HDxt 1.5T Block Diagrams & Supplemental Schematics (Release 12.x, 14.x, 15.x, 16.x & HD23.x)

Technical Publication



5133310
Rev 15

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366. Select “Install Support Services for FOA and MIS”.
- Contact your local service coordinator for more information on this process.

© 2004 – 2011, General Electric Company
GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3200 N. Grandview Boulevard
Waukesha, Wisconsin 53188
USA

www.gehealthcare.com



LANGUAGE POLICY

ПРЕДУПРЕЖДЕНИЕ Това упътване за работа е налично само на английски език.
(BG)

- Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод.
- Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа.
- Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.

警告 本维修手册仅提供英文版本。
(ZH-CN)

- 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。

警告 本服務手冊僅提供英文版本。
(ZH-HK)

- 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。
- 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。
- 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他危險。

警告 本維修手冊僅有英文版。
(ZH-TW)

- 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。
- 請勿試圖維修本設備，除非您已查閱並瞭解本維修手冊。
- 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。

UPOZORENJE Ovaj servisni priručnik dostupan je na engleskom jeziku.
(HR)

- Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod.
- Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik.
- Zanimarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

VÝSTRAHA Tento provozní návod existuje pouze v anglickém jazyce.
(CS)

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL Denne servicemanual findes kun på engelsk.
(DA)

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
(NL)

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING This service manual is available in English only.
(EN)

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS See teenindusjuhend on saadaval ainult inglise keeles.
(ET)

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS Tämä huolto-ohje on saatavilla vain englanniksi.
(FI)

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION Ce manuel d'installation et de maintenance est disponible uniquement en anglais.

(FR)

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

ADVASEL Diese Serviceanleitung existiert nur in englischer Sprache.

(DE)

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

(EL)

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

(HU)

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

Þessi þjónustuhandbók er aðeins áanleg á ensku.

(IS)

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.
(IT)

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告 このサービスマニュアルには英語版しかありません。
(JA)

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고 본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.
(KO)

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.
(LV)

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprikojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS Šis eksploatavimo vadovas yra tik anglų kalba.
(LT)

- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas.
- Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo.
- Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL Denne servicehåndboken finnes bare på engelsk.
(NO)

- Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
(PL)

- Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go.
- Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.

ATENÇÃO Este manual de assistência técnica encontra-se disponível unicamente em inglês.
(PT-BR)

- Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.

ATENÇÃO Este manual de assistência técnica só se encontra disponível em inglês.
(PT-PT)

- Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE Acest manual de service este disponibil doar în limba engleză.
(RO)

- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolului de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО! Данное руководство по техническому обслуживанию представлено только на английском языке.

(RU)

- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE

Ovo servisno uputstvo je dostupno samo na engleskom jeziku.

(SR)

- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge.
- Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.

UPOZORNENIE

Tento návod na obsluhu je k dispozícii len v angličtine.

(SK)

- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

ATENCION

Este manual de servicio sólo existe en inglés.

(ES)

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING
(SV)

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

OPOZORILO
(SL)

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

DİKKAT
(TR)

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

This page left blank intentionally.

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1. . .	Sep 29, 2004	Initial Release
2 . . .	Apr 14, 2005	Overview Tab – Sec 3: Added new acronyms and descriptions. System Tab – Added XW8200 host computer, removed CAN Link for ACGD, added HFD–S CAN Link, and other minor corrections. Operator Workspace Tab – Added XW8200 host computer and other minor corrections. Patient Handling Tab – Made minor corrections. Gradient Tab – Removed ACGD cabinet pages and inserted HFD–S Cabinet page. MGD/RRF/RF Tab – Corrected minor errors and added informational callouts. Added MRC II Tool & Cable and MRC II Tool – 8 Channel Connector pages.
3 . . .	Oct 15, 2005	System Tab – Added J116 and J119 to TAC Cabinet and relocated Terminator. Patient Handling Tab – Added test points and LED's to Magnet Enclosure and PHPS. Added MNS connection information to SSM. MGD/RRF/RF Tab – Added LED info to Exciter and Receiver boards. Corrected RRF Receive Chain connection labels. Added MNS connection info to Driver Module and SSM. Added MC Preamp Bias Filter pinouts. Added LED and Pinout info for 16 Ca'hannel Switch. Corrected MNS connection info on ASC Chassis. Revised and added interconnects for forward production and upgrade MNS options.
4 . . .	Mar 31, 2006	Major update to manual with the addition of HDx (Release 14.x) product. Changed title page and banners on all pages. Overview Tab – Sec 3: Added new acronyms and descriptions. System Tab – Added HP9300 computer info, 32 Channel option info, and the new RFS cabinet power distribution using two power strips. Revised all sheet references. OW Tab – Added HP9300 computer info. MGD/RRF/RF Tab – Added HDx (14.x) info as needed which includes VRE modules: ICN's, 16/24 Port Ethernet Switch, Infiniband board and module. Added HDx A, B, and C port connection information for the HDx LPCA. Added MCR III Tool
5 . . .	Jun 30, 2006	Overview Tab – Added definitions to Section 3. Added TOC's to System, Operator Workspace and Patient Handling Sections. System Tab – Added ASC Module to System/RFS cabinet depictions. Patient Handling Tab – Updated J20 Pin Signal table. Added SSM block diagram. MGD/RRF/RF Tab – Added 32 Channel information and other minor corrections and updates.
6 . . .	Oct 20, 2006	Added Environmental Sensor to System/RFS cabinet in System. Operator Workspace, and MGD/RRF/RF Sections (MRIhc19337). MGD/RRF/RF Tab – Added Port 2 to Port 2 connection for 4 ICN configuration. Added 4 ICN configuration without Infiniband switch. Updated Multi–Coil Adapter by adding MC Bias 17–24 for 14.x systems (MRIhc18216). Updated several pages with additional MC signal information (MRIhc18217).
7 . . .	Dec 20, 2006	SYSTEM TAB – Added 32 Channel configuration without Infiniband Switch. GRADIENT Tab – Added Pang block configuration info (MRIhc20731). OW TAB – Added Audio IN/OUT signals for PC Host for 14.x system. MGD/RRF/RF Tab – Revised ASC Chassis to reflect functionality using new Detector Board, 5251054.
8	Jun 6, 2007	SYSTEM Tab – Added note for designating components required for 16 channel systems. MGD/RRF/RF Tab – Added notes to designate parts not included with newer 8 Channel systems and for parts that are required for 16 Channel systems.
9 . . .	Oct 26, 2007	SYSTEM Tab – Added CAM Chassis designation to MGD Chassis where applicable. Added CAN LINK block diagram for RFS cabinet with CAM Chassis. MGD/RF/RRF Tab – Added board slot locations for CAM Chassis.
10	Sep 11, 2008	Changed title of manual and all page headers to <i>Signa HD, HDx & HDxt 1.5T (12.x, 14.x & 15.x) Block Diagrams & Supplemental Schematics</i> . HDi and Signa Vibrant are information is also included with this name change. Updates for HDxT (15.x) in all sections. Added information on Sun ICN's. Added depopulated 16–channel switch for 8–channel installations.
11	Dec 17, 2008	SYSTEM Tab – Added new environmental sensor diagram (ECO 2055310). PATIENT HANDLING Tab – Revised E–Stop circuit voltage from 24V to 12V in Teal PDU (PQR 13207104).
12	Jun 30, 2010	SYSTEM Tab – Added new Z400 computer page (ECO2056587) and added 5337894–3 and –4 ICN's (ECO 2101939). MGD/RRF/RF Tab – Added new ICN's (ECO 2101939).
13	Jan 20, 2011	Update entire manual headers and title page to include 16.x software version (DOC0853690). Added XW8400 Host computer to SYSTEM & OW tabs. Removed terminator from all SCSI towers. Removed J11, SCSI tower power cord, from power strip for Z400 host computers (MRIhc54665).
14	Aug 4, 2011	Changes made to reflect HD16.0v02 update, ICN end–of–life, and IRF2b & RRF–DIF3 hardware upgrade configurations.
15	Nov 28, 2011	MGD/RRF/RF Tab: Added sheets 2C, 6B, & 7C. Existing sheet 7C became 7D. Updated sheets 3D, 7A, 13, 1'4A & 14C. Changes made to reflect HD23 upgrade configurations (SPR HCSDM00101291).

LIST OF EFFECTIVE PAGES

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page	15	2–1	15	i	15	PDU		2–1 to 2–12	15
GE Logo Page	–	3–1 to 3–6	15	1–1 to 1–10	15	Tab MR406	–		
Language Policy	15	SYSTEM		PATIENT HANDLING		i	15	COOLING SYSTEM	
A	15	Tab MR364	–	Tab MR405	–	MGD/RRF/RF		Tab 2308173	–
i	15	i	15	i	15	Tab MR451	–	i	15
OVERVIEW		1–1 to 1–20	15	1–1 to 1–8	15	i to ii	15	1–1 to 1–2	15
Tab MR367	–	OPERATOR		GRADIENT		1–1 to 1–50	15		
i	15	WORKSPACE		Tab MR418	–				
1–1	15	Tab 2158195	–	1–1 to 1–3	15				

* This revision number/letter corresponds to the indicated document's revision control system.

CONTENTS

OVERVIEW

Purpose
Standard Conventions
Abbreviations



SYSTEM

System Block Diagrams



OPERATOR WORKSPACE

Operator Workspace Block Diagrams



PATIENT HANDLING

Patient Handling Block Diagrams



GRADIENT

Gradient Block Diagrams (HFD & ACGD)



PDU

PDU Block Diagrams

Note:

Phoenix PDU found in Vendor Manuals only. (SEE Service Methods CDROM)



MGD/RRF/RF (1.5T)

MGD/RRF/RF Block Diagrams (1.5T)
RF Supplemental Schematics (1.5T)



COOLING SYSTEM

Water Chillers Block Diagrams

Note:

For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDROM)



OVERVIEW

TABLE OF CONTENTS

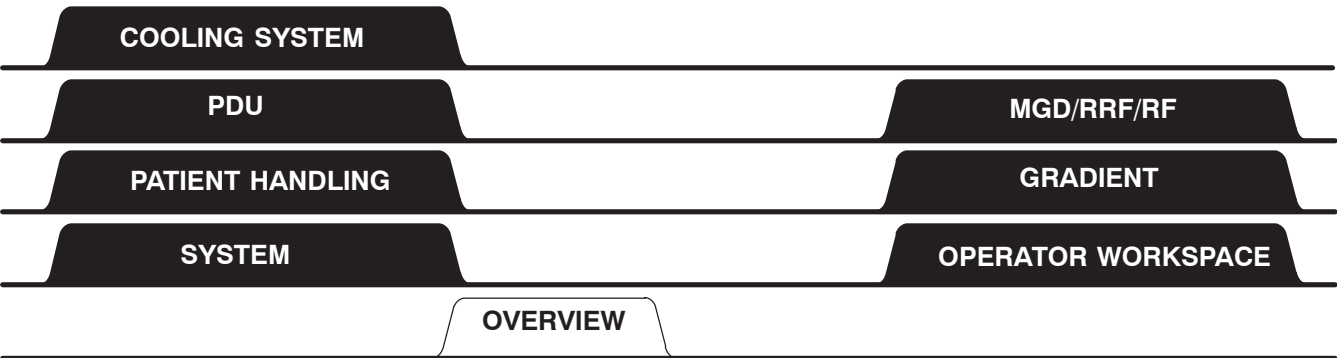
<u>SECTION</u>	<u>PAGE</u>
1 – PURPOSE	1–1
2 – STANDARD CONVENTIONS	2–1
3 – ABBREVIATIONS	3–1

SECTION 1 – PURPOSE

The purpose of this document is to provide an overview of the hardware which makes up the MR Signa System, and a quick means to troubleshoot major signals within the system.

It is recommended that you page through this manual to become familiar with its content and organization, beginning with System Tab, OVERALL SYSTEM. This section shows how the major subsystems are grouped by cabinet, and the main communication lines between them.

The remaining block diagrams are organized into sections which correspond to major subsystems. Some diagrams show hardware from another subsystem in order to complete an entire signal flow route. For such exceptions, the board/module title of the circuitry shown will generally help identify the applicable subsystem. See Illustration 1–1 for tab organization of this manual.



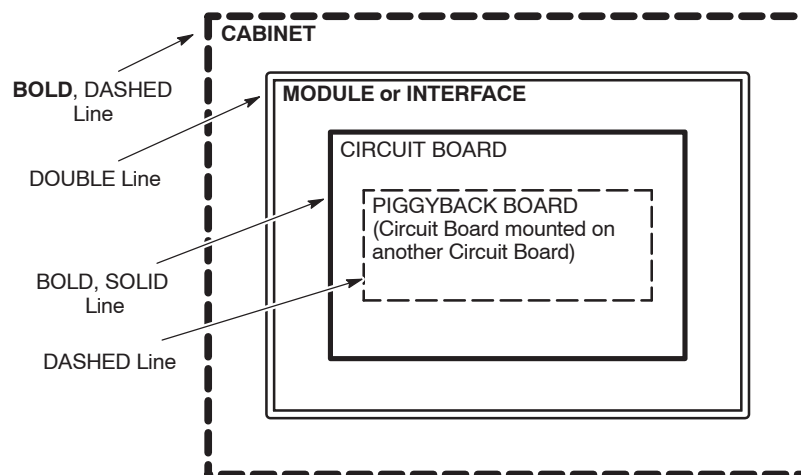
TAB ORGANIZATION FOR THIS DIRECTION
ILLUSTRATION 1

These block diagrams, along with provided diagnostics, are intended to be the primary tools for troubleshooting the MR System from a system or subsystem level on down to the specific module or board level. Since some of the RF subsystem block diagrams are presently insufficient for troubleshooting to the FRU (Field Replaceable Unit) level, schematics have been provided to supplement those block diagrams. When using the diagnostics, the block diagrams can provide a better understanding of what circuitry the diagnostics are testing, and how to further isolate a problem if required.

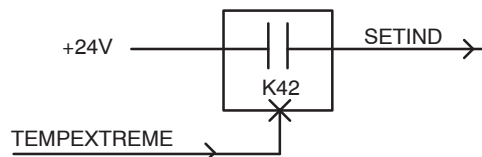
SECTION 2 – STANDARD CONVENTIONS

The following standard conventions are used within the block diagrams:

- The type of line used to enclose a major box on a block diagram indicates the following:



- The component designator for cabinets, modules, circuit boards, etc. is enclosed in “()” after or below their name on the block diagram.
- Logic levels: A “0” indicates a logic “low” level (typ. <0.8VDC), a “1” indicates a logic “high” level (typ. > 2.0VDC).
- An “*” is used to identify an “active low” signal (e.g. “INT*” signal when at a logic “0” indicates that an interrupt is present; a logic “1” level indicates that no interrupt is present).
- A “#” indicates the number of wires in a cable (e.g. $\frac{9}{\text{---}}$ indicates 9 wires).
- Signal flow is generally from left to right on a block diagram. Arrows are used to identify exceptions to this rule and to aid in identifying input/output signal lines on the diagram.
- The “x” symbol (e.g. $\text{---}x$) shown on boxes in a block diagram are used to indicate that the function within the box is enabled by the “active” signal on this input line.



For this example, when signal TEMPEXTREME goes active (“1”), relay K42 will close (become enabled) and the signal SETIND now becomes active to set a temperature indicator.

SECTION 3 – ABBREVIATIONS

Note

This list has been compiled from a wide variety of sources. Not all of these abbreviations will apply to your system. For an electronic copy of over 1000 acronyms, see *Direction 2124201–2*.

A

AC = Alternating Current
ACGD = Advanced Concept Gradient Driver
ACK = Acknowledge
ACQ = Acquisition
A/D = Analog-to-Digital
ADC = Analog to Digital Converter
ADD or ADDR = Address
ADJ = Adjust
ADV = Advance
AGC = Automatic Gain Control
AGP = Applications Gateway Processor (Board)
AI = Applications Interface, Artificial Intelligence
ALGN = Align
ALU = Arithmetic Logic Unit
AM = Amplitude Modulation
AMP = ampere, amplifier
A/N = Alpha-numeric
ANSI = American National Standards Institute
ANT = Antenna
AP = Array Processor
APM = Analog Power Monitor
APS = Auto Pre-Scan or Acquisition Processing System (Board)
APSGT = Auto Pre-Scan Gain Control
ARCH = Archive/Remove Process
ASC = Automated Support Center or Amplifier Support Controller
ASCII = American Standard Code for Information Interchange
ASM or ASSY = Assembly
ASM = Analog Service Module
ATTEN = Attenuator
AUTO = Automatic
AUX = Auxiliary

B

B = magnetic field
BATT = Battery
BB = Buffered Data or Broadband
BC = Body Coil

BD = Board
BDS = Buffered Data Strobe
BLK = Black
BNC = BNC Connector
BRM = Body Resonance Module
BRT = Brightness
BUF = Buffer
B/W = BandWidth

C

C = centigrade, Celsius
CAB = Cabinet
CAL = Calibration
CALC = Calculate
CAN = Controller Area Network
CATTEN = Course Attenuation
CB = Circuit Breaker
CCC = CAN Core Controller card
CCW = Counter Clockwise
CD-ROM = Compact Disk Read-Only Memory
CE = Chip Enable
CF = Center Frequency
CFA = Center Frequency Adjust
CHAN or CHNL = Channel
CHK = Check
CKT = Circuit
CLR = Clear
CMD = Command
CMMR = Common Mode Rejection Ratio
CMOS = Complimentary Symmetry Metal Oxide Semiconductor
CMP = Compare or Compressor
CNT = Contrast
COAX = Coaxial
COF = Cut Off Frequency
CONFIG = Configuration
CONT = Continuous
CONV = Converter
CPD = Communication Pin Driver

CPU = Central Processing Unit
 CRT = Cathode-Ray Tube
 CTL = Clock Time Control
 CTRL = Control
 CV = Configuration Variable
 CW = Clockwise or Controller Wave
 CYL = Cylinder

D

D/A = Digital-to-Analog
 DAC = Digital to Analog Converter
 dB = Decibel
 dBm = Decibels above or below one Milliwatt
 DC = Direct Current
 DCB = Driver Control Board
 DCKSTP = Dock Stop
 DD = Dynamic Disable
 DDS = Direct Digital Synthesizer
 DDSB = Dynamic Disable Switch Box
 DEC = Decimal
 DEG = Degree(s)
 DEMUX = Demultiplexer
 DEV = device
 DIA = Diagnostic Imaging Accessories
 DIAG = Diagnostic
 DIFF = Differential
 DIG = Digit
 Dig B0 = Digital B0 Adjustment
 DIP = Dual In-line Package
 DIR = Direction
 DIS = Disable
 DM = Driver Module
 DMA = Direct Memory Access
 DMAC = Direct Memory Access Controller
 DMM = Digital Multi-Meter
 DPR or DPRAM = Dual Port Random Access Memory
 DQA = Daily Quality Assurance
 DRAM = Dynamic Random Access Memory
 DRC = Dynamic Ram Controller
 DRF = Digital Receive Filter (Board)
 DRVR = Driver
 DSP = Digital Signal Processor
 DVM = Digital Volt Meter
 DWG = Drawing

E

ECC = Extended Cursor Control or Error Checking and Correction
 ECG = Electro-Cardiogram
 EEPROM = Electrically Erasable Programmable Read Only Memory

EFB = Envelop Feedback
 EPI = Echo Planer Imaging
 e.g. = for example
 EM or EMERG = Emergency
 EMI = Electro-Magnetic Interference
 EN or ENA = Enable
 ENV = Envelope
 EOS = End Of Sequence
 EOT = End Of Travel
 EOW = End Of Waveform
 EPO = Emergency Power Off
 EPROM = Erasable Programmable Read Only Memory
 ERU = Emergency Rundown Unit
 EXER = Exerciser
 EXCITE = Expanding Applications with Multi-channel Imaging Technology

F

f = farad
 F = Fahrenheit, Fuse (designator)
 FBD = Functional Block Diagram
 FBK = Feedback
 FC = footcandle
 Fc = Center Frequency
 FDBK = Feedback
 FDO = Future Delivery Order
 FE = Field Engineer
 FET = Field-Effect Transistor
 FF or FIFO = First In – First Out
 FFD = Full Field Distortion
 FFT = Fast Fourier Transform
 FID = Free Induction Decay
 FIDAT = FIFO In Data
 FIFO or FF = First In – First Out
 FILT = Filter
 FM = Frequency Modulated
 FMI = Field Modification Instruction
 FOV = Field Of View
 FREQ = Frequency
 FRNT = Front
 FRU = Field Replaceable Unit
 ft = foot
 FT = Feet
 FTP = File Transfer Protocol

G

G = gauss
 G/A = Gradient Amplifier
 GEN = Generator
 GND = Ground
 GOC AA = Global Operator Console Audio Assembly
 GRAD = Gradient
 GRN = Green
 GRY = Grey
 GP = Gradient Processor
 GSW = Gradient Switch

H

H = henry
 H = Hexadecimal
 HART = Hardware Tracker
 HC = Head Coil
 HD = Head or High Definition
 He = Helium
 He-ALARM = Helium Alarm
 HEX = Hexadecimal
 He-WARNING = Helium Warning
 HFA = High Fidelity Amplifier
 HFD = High Fidelity Driver
 HFD-S = High Fidelity Driver – Special
 HORIZ or HORZ = Horizontal
 HPC = Hardware and Protocol Control
 HPFS = Host Platform Software
 hr = hour
 HSN = High Speed Network
 HSS (D) = High Speed Serial Data Link
 HTR = Heater
 HV = High Voltage
 HV PSU = High Voltage Power Supply Unit
 HVAC = Heating, Ventilation, & Air Conditioning
 Hz = hertz

I

I = Current or Phase or Real
 IAMP = Amplifier Current (Gradient)
 ICD = Installation Certification Document
 ICL = Coil Overcurrent (Gradient)
 ICN = Image Compute Node
 ID = inner diameter
 ID = Identity, Identifier
 i.e. = that is to say
 IEC = International Electrical Code
 IEEE = Institute of Electrical & Electronics Engineers
 I/F = Interface
 in. = inch
 INFO = Information

INS, INST, or INSTR = Instruction
 INT or INTR = Interrupt
 I/O = Input/Output
 IP = Image Processor or Plate Current (RF amp) or
 Instruction Processor or Internet Protocol
 IPA = Intermediate Power Amplifier
 IPC = Interprocess Communication
 IQ = Image Quality
 IR = Image Reject
 IRF = Interface and Remote Functions (Board)
 IRF I/O = Interface and Remote Functions
 Input/Output (Board)
 IRQ = Interrupt Request
 ISA = Interrupt Start Address
 ISI = Inter-Sequence Interrupt
 ISO = Isolator or Isolation
 IT-MGD = Image Transfer-MDG (Board)
 IT-Host = Image Transfer-Host (Board)
 IVI = Interactive Vascular Imaging

J

J = Jack Connector (designator)
 JP = Jumper

K

K = Kelvin (unit of temperature) or Kilo (Thousand)
 KB = one thousand bytes of memory
 KCal = kiloCalorie
 kg = kilogram
 kHz = kilohertz
 KVA = kilo Volt Amperes
 kW = Kilowatt

L

L or LFT = Left
 LAN = Local Area Network
 lb = pound
 LBK = Loopback
 LCA = Logic Cell Arrays
 LED = Light Emitting Diode
 LFC = Load From Cold
 LHe = Liquid Helium
 LIN = Linear
 LM or LMK = Landmark
 LMP = Lamp
 LNK = Link
 LPCA = Low Profile Carriage Assembly
 LO = Local Oscillator
 LOEP = List of Effective Pages
 LONG = Longitudinal
 LVLE = Low Voltage, Low Energy

M

m = meter
 M = mega – one million
 mA = millamps
 MAG = Magnet
 MB = megabyte
 MC = Multi-Coil
 MCD = Multi-Coil switch Driver
 MCR = Multicoil Receive (Tool or Diagnostic)
 MDS = Multidrop Serial interface
 MEM = Memory
 Mfg. = Manufacturing
 MFU = Multi-Format Unit
 MGD = Multi-Generational Data acquisition (Chassis)
 MGMNT = Management
 MHz = megahertz
 mm = millimeter
 MOD = Modifier
 MON = Monitor
 MOSFET = Metal Oxide Semiconductor Field Effect Transistor
 MOV = Metal Oxide Varistor
 MPS = Manual Pre-Scan
 MR = Magnetic Resonance
 MSG = Message
 MTR = Motor
 MUX = Multiplexer
 mV = millivolts
 MW = megawatt
 mW = milliwatt

N

N = Negative
 N₂ = Nitrogen
 n.c. = normally closed
 NEC = National Electrical Code
 Ni-Cad = Nickel Cadmium
 n.o. = normally open
 NO. = Number
 NOPROC = No Processing
 NOREC = No Reconstruction
 nS = Nanosecond

O

O₂ = oxygen
 OC = Operator's Console
 OD = outside diameter
 OM = Oxygen Monitor
 OPI = Oblique Plane Imaging
 OPTO = Optical
 ORG = Orange

OT = Overtemp
 OV = Over voltage
 OVR = Over
 OVRD or OL = Overload
 oz = ounce
 OW = Operator Workspace

P

P = Positive
 PA = Patient or Power Amplifier
 PAC = Physiological Acquisition Controller
 PAL = Programmable Array Logic or Patient Alignment Lights
 PAN = Primary Archive Node
 PAT = Patient
 PCI = Programmable Communications Interface or Peripheral Component Interconnect
 PCT = Pulse Controller Task
 PDI = Product Delivery Instruction
 PDU = Power Distribution Unit
 PEN = Penetration
 PHPS = Patient Handling Power Supply
 PIN = Positive-Intrinsic-Negative
 PIO = Programmed Input/Output
 PLL = Phase Lock Loop
 PIXEL = Picture Element
 PM = Phase Modulation, or Planned Maintenance
 PMC = PCI Mezzanine Connector
 P/N = Positive/Negative
 PNL = Panel
 POS = Position
 POT = Potentiometer
 PP = Penetration Panel
 PPC = Power PC
 P-P = Peak to Peak
 ppm = parts per million
 PREAMP = Preamplifier
 PROC = Process, Processor
 PROG = Program, -ed, -able
 PROM = Programmable Read Only Memory or Program
 PS = Power Supply
 PSD = Pulse Sequence Database
 PSG = Pulse Sequence Generator
 PSU = Power Supply Unit
 PSS = Power Supply System Board
 PT = Patient Transport or Patient Table
 PUR = Purple
 PVC = Polyvinylchloride

PW = Pulse Width

PWA = Printed Wiring Assembly

PWB = printed wiring board

PWR = Power

Q

Q = Quadrature, Quality Factor or Imaginary

Qty = Quantity

QUAD = Quadrature or Quadrant

R

R or RT = Right

RAM = Random Access Memory

RCV = Receive

RD = Read

RDY = Ready

RECON = Reconstruction

REF = Reference or Reflected

REG = Register or Regulator

RET or RTN = Return

REV = Revision or Reverse

RF = Radio Frequency

RFAMP = Radio Frequency Amplifier

RFA = RF Amplifier

RFI = Radio Frequency Interference or RF Interface
(Module)

RFS = RF System Cabinet

RGB = Red Green Blue

RLA or RLY = Relay

rms = root mean square

ROI = Region of Interest

ROM = Read Only Memory

RPC = Remote Control Panel

RRF = Remote RF (Chassis)

RRF-DIF = Remote RF-Digital Interface board

RSR = Register

RST = Reset

R/W = Read/Write

RX = Receive

RXD = Receive Data

S

S = Switch (designator)

SACK = Slave Acknowledge

SAG = Sagittal

SAR = Specific Absorption Rate

S/C = Superconducting

SCA = Scan Control Assembly

SCP = Scan Control Processor (Board)

SCR = Silicon Controlled Rectifier

SCSI = Small Computer System Interface

SEC = second

SEG = Segment

SEL = Select

SEPI = Spin Echo Planer Imaging

SGA = Switchable Gradient Amplifier

SGI = Silicon Graphics Inc.

SIG = Signal

SII = "S-2" (type of magnet)

SIII = "S-3" (type of magnet)

SIMMS = Single In-line Memory Module

SIP = Single In-line Package

SL = Slice

SMC = Status Monitoring and Control

SMPTE = Society of Motion Picture and Television
Engineers

SN = Service Note

SNR = Signal-to-Noise Ratio

SPI = Serial Peripheral Interface

SPT = System Performance Test

SPU = Signal Processing Unit

SRAM = Static Random Access Memory

SRF = Solid State RF Amplifier SGA or Scalable RF
(Cabinet)SRF/TRF = Sequencer Related Function / Trigger and
Rotation Function (Board)

SRI = Scan Room Interface

SRV = Service

SSM = System Support Module

SSP = Sequence Synchronous Protocol

SST = Small Sample Test

STAB = Stability

STAT = Status

STD = Standard

STD DEV = Standard Deviation

STIF Bd = SRF/TRF InterFace Board

SUB-D = Subminiature D (connector)

SW = Switch or Software

S/W = Software

SYNC = Synchronous

SYS = System

SYSID = System Identification

T

T = tesla
TAC = Twin Accessory Cabinet
TBD = To Be Determined
TBL = Table
TEMP = Temperature
TFU = Thyristor Firing Unit
TG = Transmit Gain
TGWC = TwinSpeed Gradient Water Cooler
TNS = Transient Noise Suppressor
TP = Test Point (designator)
TR or T/R = Transmit–Receive
TRMAP = Transmit/Receive Map
TRM = Twin Coil Resonator Module
TS = Terminal Strip
TSCC = TwinSpeed Cooling Cabinet
TST = Test
TTL = Transistor to Transistor Logic
TX = Transmit
TXD = Transmit Data

U

UART = Universal Asynchronous Receiver–Transmitter
uf = microfarad
UFI = Ultra Fast Imaging
UNBLK = Unblank
uP = Microprocessor
UPLMT = Up Limit
UPM = Universal Power Monitor
USART = Universal Synchronous/Asynchronous
Receiver–Transmitter
uS or usec = microsecond
UTNS = Universal Transient Noise Suppressor (Board)
UV = Under voltage

V

VAC = Volts Alternating Current
VAR = Variable
VBUS = Bus Voltage
VCR = Video Cassette Recorder
VCNTRL = Control Voltage for 8645 Techrons
VDC = Volts Direct Current
VERT = Vertical
Vp–p = Volts Peak–to–Peak
VRE = Volume Recon Engine
VRMS = Volts, Root Mean Square
VTR = Video Tape Recorder

W

WARP = Waveform And Rotation Processor
WC = Water Chiller
WR = Write
WHT = White
WIM = Workspace Interface Module
WND = Window
WORM = Write Once Read Many
WRT = Write

X

XCVR = Transceiver
XFMR = Transformer
XMT or XMIT = Transmit

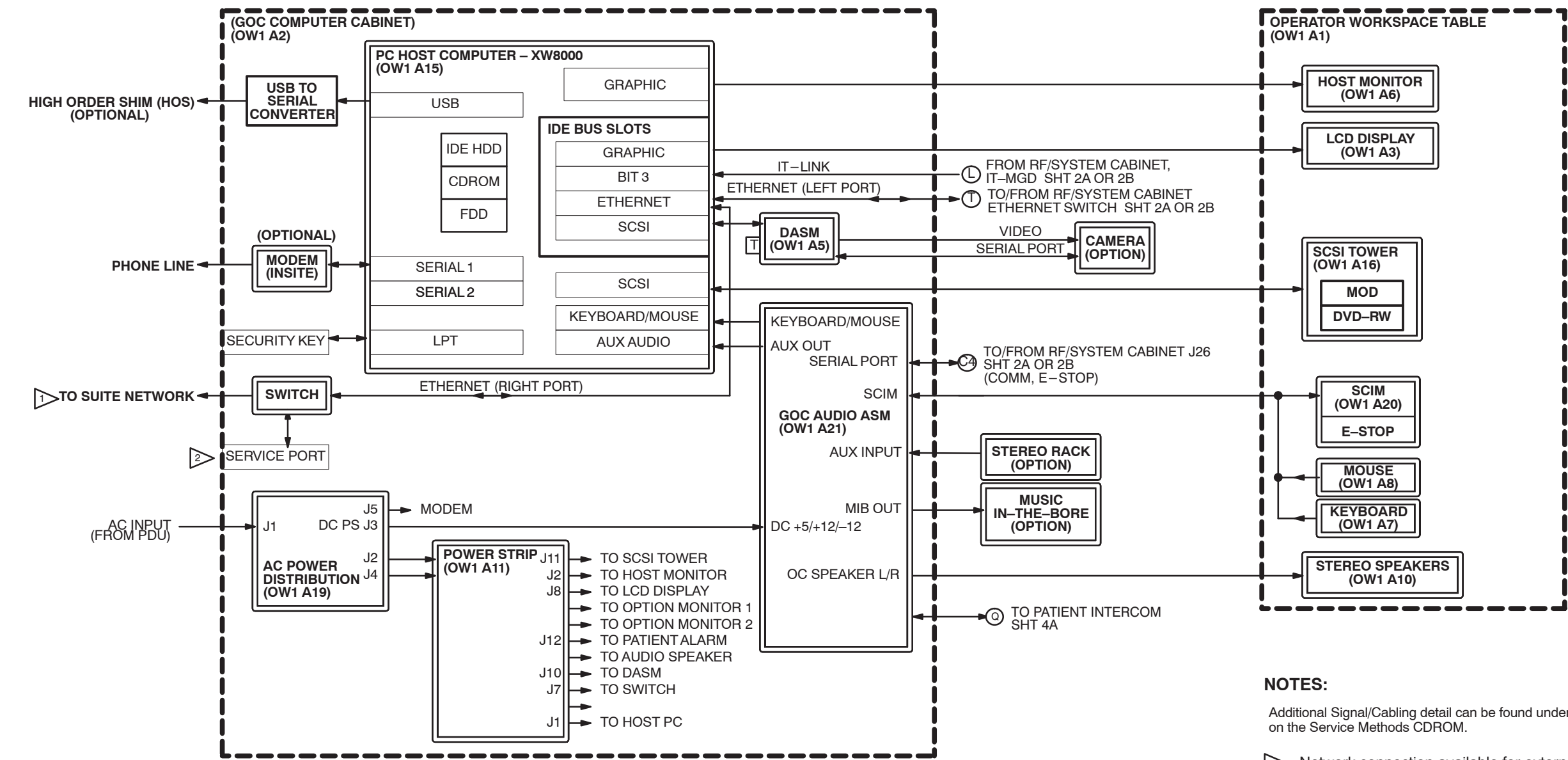
Y

YEL = Yellow

SYSTEM

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET w/8000 & TABLE (Forward Production or Pre–EXCITE to HD Upgrades)	1–1
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET w/8000 & TABLE (EXCITE to HD Upgrades)	1–2
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET w/8200 & TABLE (Forward Production or Pre–EXCITE to HDx Upgrades)	1–3
HDx & HDxt OVERALL SYSTEM – OPERATOR WORKSPACE CABINETw/9300 & TABLE (Forward Production, EXCITE, or Pre–EXCITE Upgrades)	1–4
HDx & HDxt OVERALL SYSTEM – OPERATOR WORKSPACE CABINETw/9300 & TABLE (EXCITE HD to HDx Upgrades)	1–5
HDx & HDxt OVERALL SYSTEM – OPERATOR WORKSPACE CABINETw/Z400 & TABLE (EXCITE HD to HDx Upgrades)	1–6
HD OVERALL SYSTEM – 4 or 8 Channel	1–8
HD OVERALL SYSTEM – 16 Channel	1–9
HDx & HDxt OVERALL SYSTEM – 8 Channel	1–10
HDx & HDxt OVERALL SYSTEM – 8 Channel, IRF2b/RRF–DIF3 H/W Configuration	1–11
HDx & HDxt OVERALL SYSTEM – 16 Channel	1–12
HDx & HDxt OVERALL SYSTEM – 16 Channel, IRF2b/RRF–DIF3 H/W Configuration	1–13
HDx & HDxt OVERALL SYSTEM – 32 Channel with Infiniband Switch	1–14
HDx & HDxt OVERALL SYSTEM – 32 Channel without Infiniband Switch	1–15
HDx & HDxt OVERALL SYSTEM – 32 Channel	1–16
HD, HDx & HDxt OVERALL SYSTEM (Forward Production & Upgrades)	1–17
HD, HDx & HDxt OVERALL SYSTEM (Forward Production & Upgrades)	1–18
HD, HDx & HDxt OVERALL SYSTEM – 4, 8, or 16 Channel (HD Forward Production or HDx Upgrades)	1–19
HDx & HDxt OVERALL SYSTEM – 8 or 16 Channel (HDx Forward Production)	1–20
HDx & HDxt OVERALL SYSTEM – 32 Channel	1–21
HD & HDx & HDxt OVERALL SYSTEM – CAN Link Block Diagrams	1–22



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection

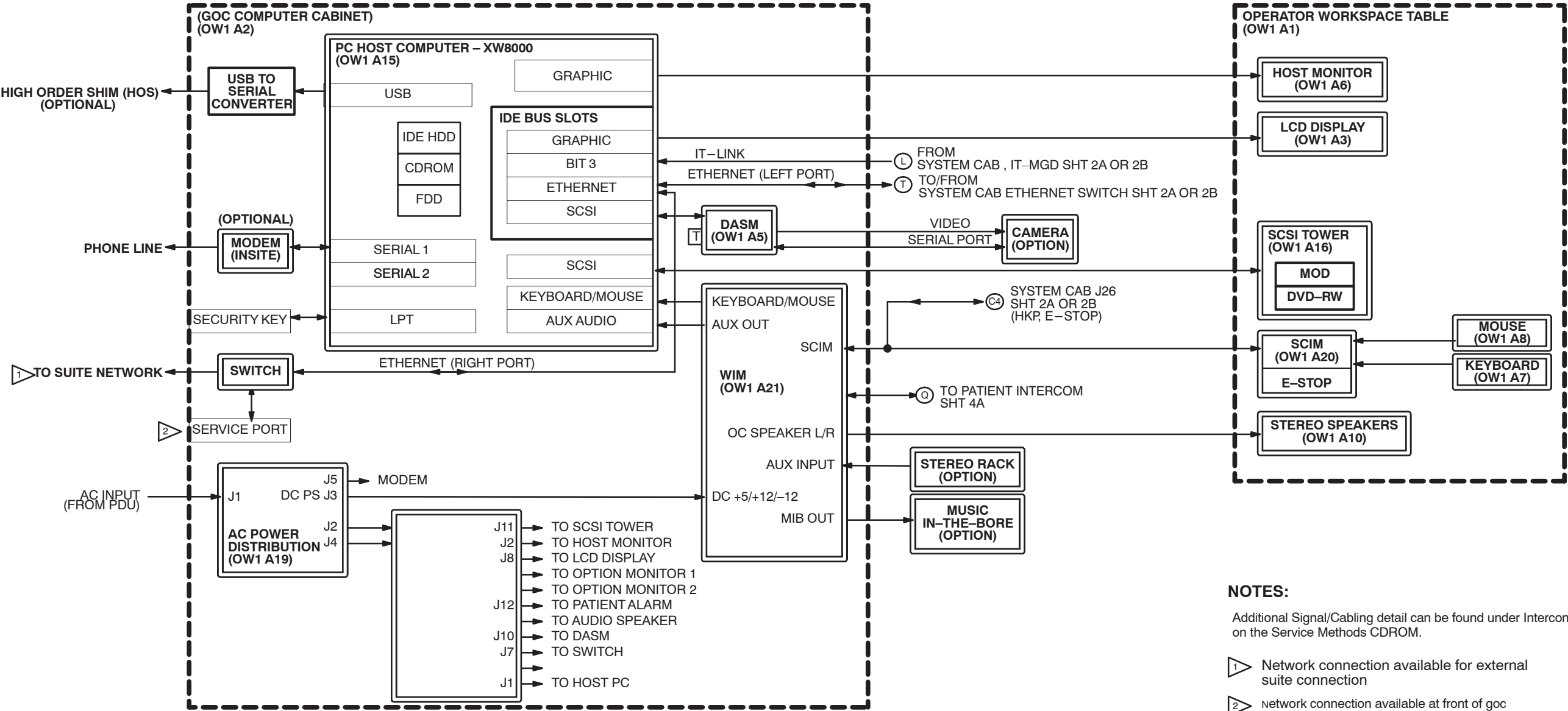
2 network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

1) PATIENT HANDLING

2) MGD/RRF/RF



NOTES:

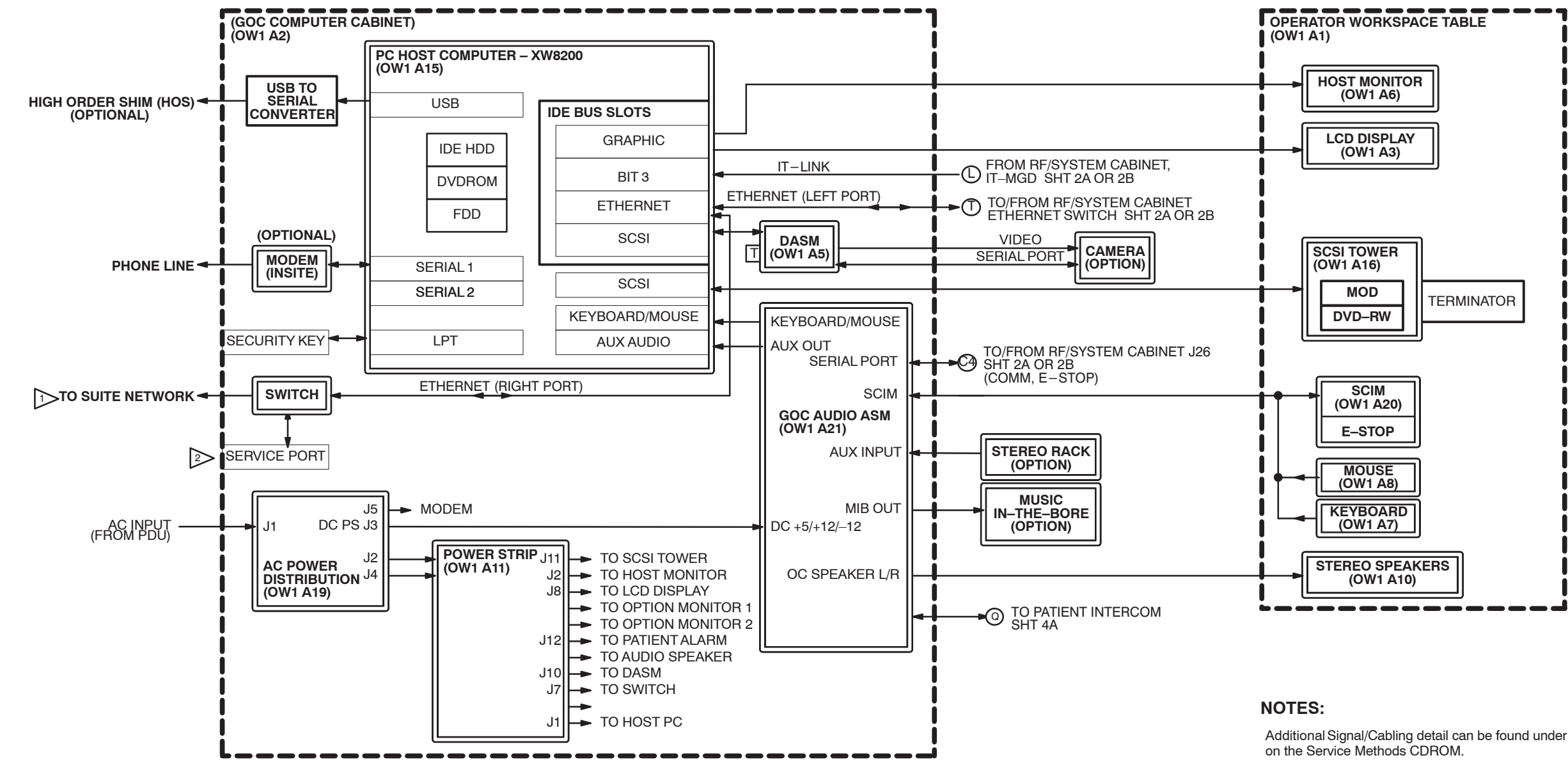
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

- 1 Network connection available for external suite connection
- 2 network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

EXCITE HD (12.x) OVERALL SYSTEM
EXCITE to HD Upgrades
(OPERATOR WORKSPACE CABINET w/8000 & TABLE)



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection

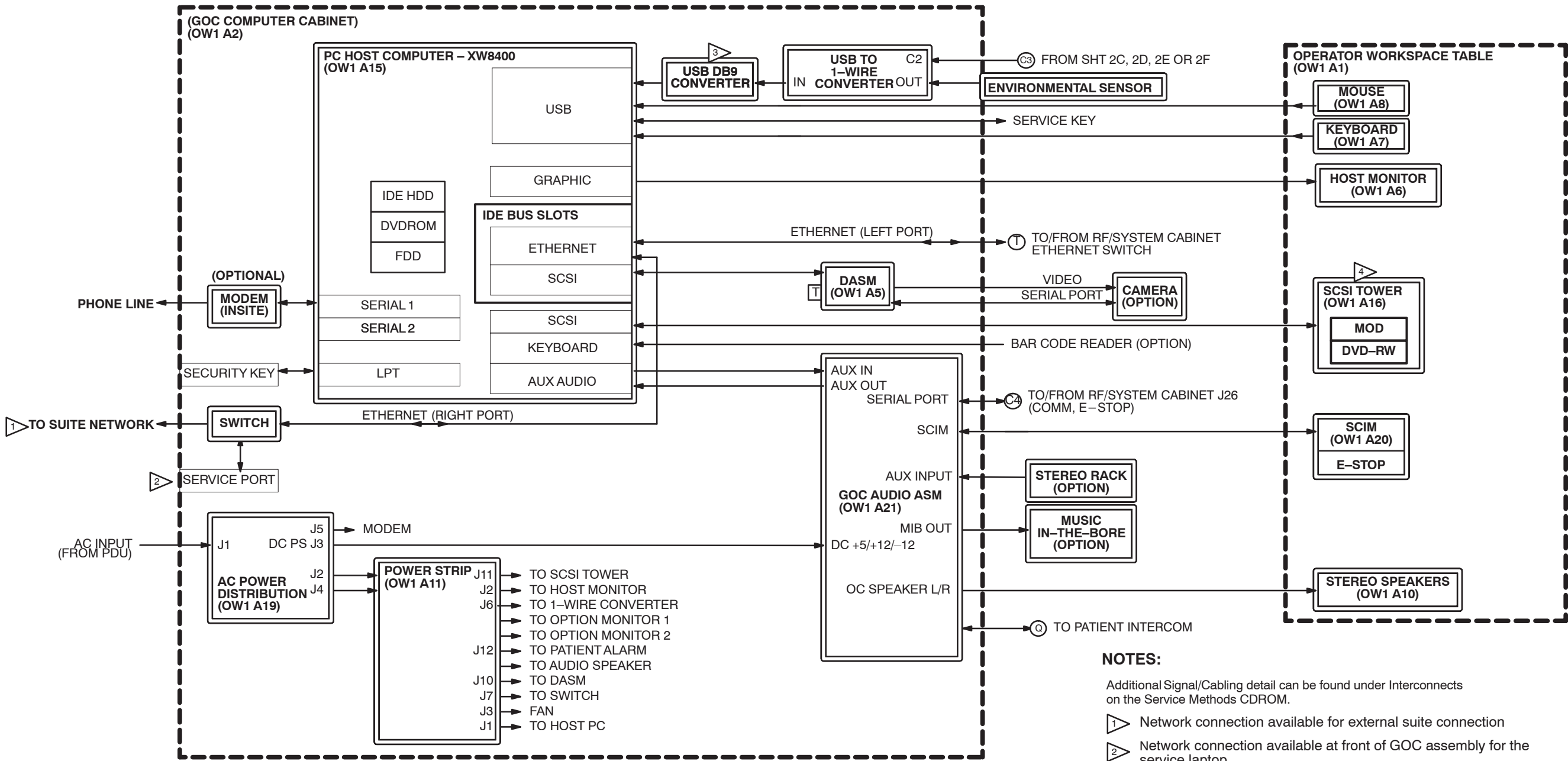
2 network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

1) PATIENT HANDLING

2) MGD/RRF/RF



Refer to the following functional block diagram for additional information:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

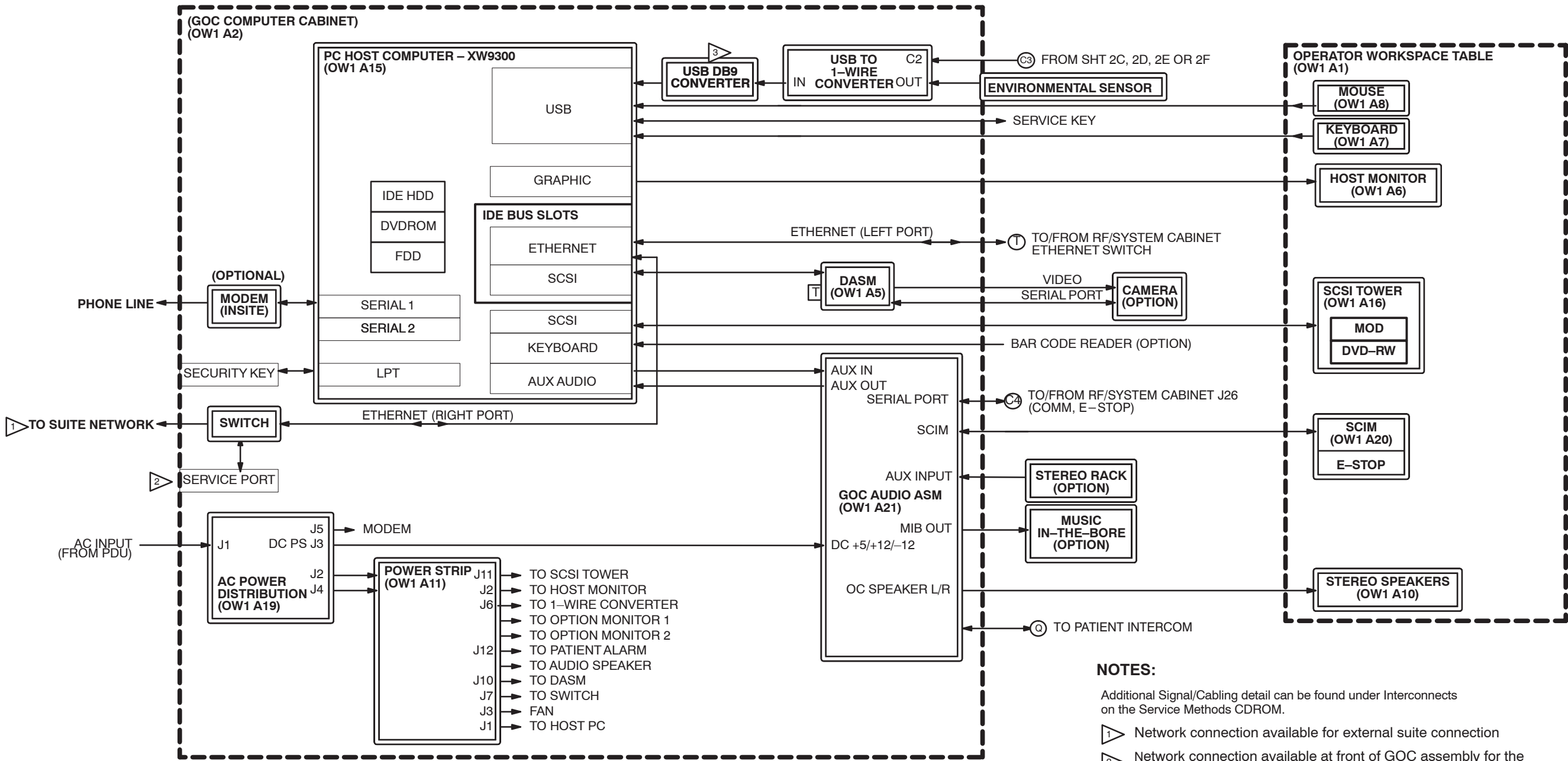
1) Network connection available for external suite connection

2) Network connection available at front of GOC assembly for the service laptop

3) Beginning August 2008, design changed to 1-wire to USB converter. Environmental Sensors Block Diagram utilizing 1-wire 5147136-2

4) 8400 Host Computer is also available without SCSI Tower.

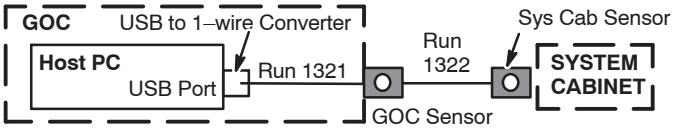
HDxt (15.x, 16.x) OVERALL SYSTEM
Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/8400 & TABLE)



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

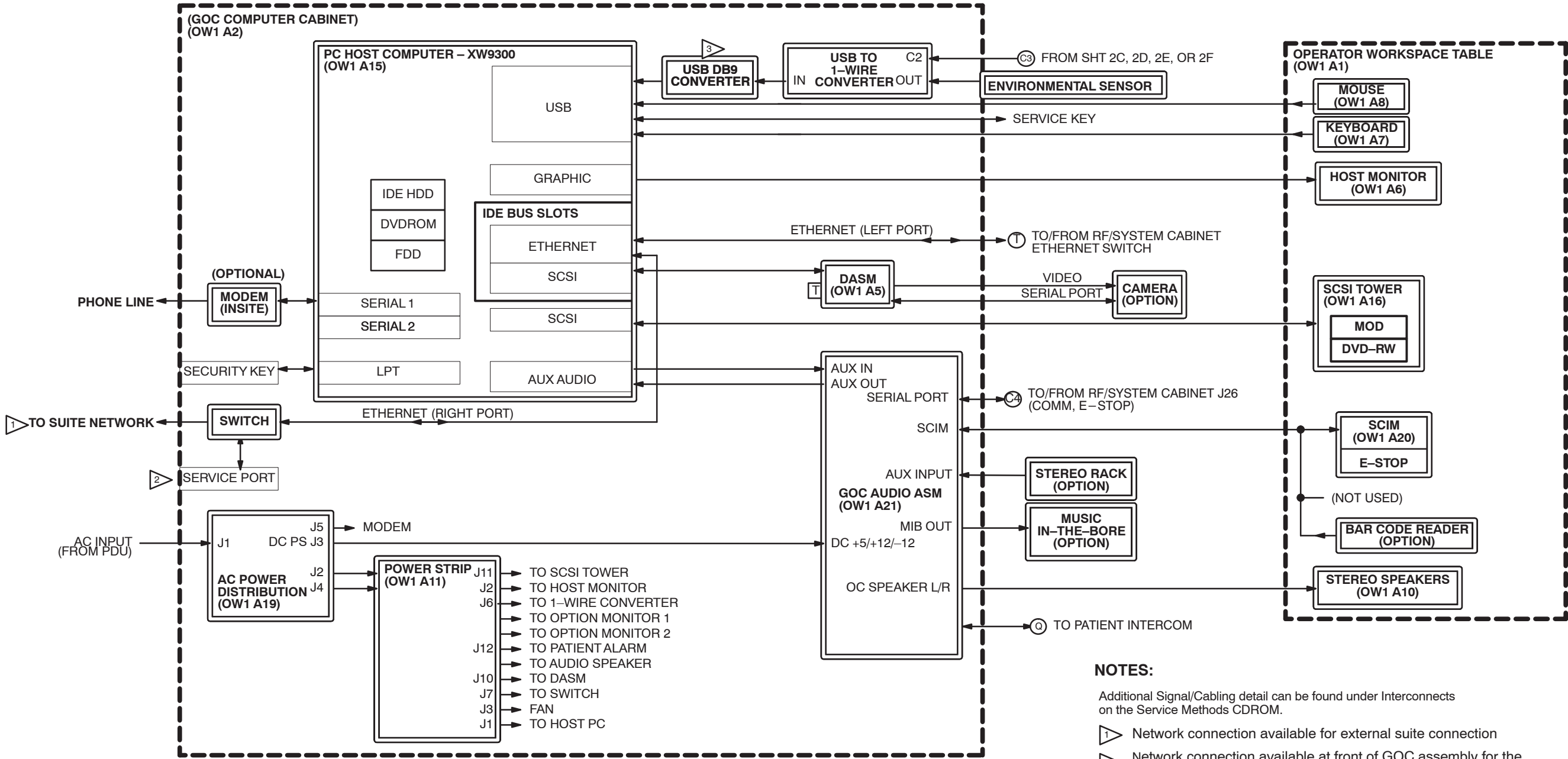
- 1 Network connection available for external suite connection
- 2 Network connection available at front of GOC assembly for the service laptop
- 3 Beginning August 2008, design changed to 1-wire to USB converter. Environmental Sensors Block Diagram utilizing 1-wire 5147136-2



Refer to the following functional block diagram for additional information:

- NOTE:**
- 1) PATIENT HANDLING
 - 2) MGD/RRF/RF

HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/9300 & TABLE)



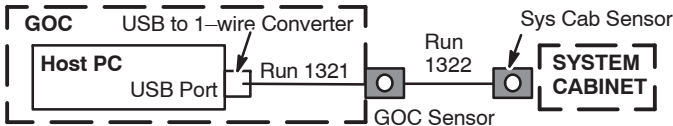
NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection

2 Network connection available at front of GOC assembly for the service laptop

3 Beginning August 2008, design changed to 1–wire to USB converter. Environmental Sensors Block Diagram utilizing 1–wire 5147136–2



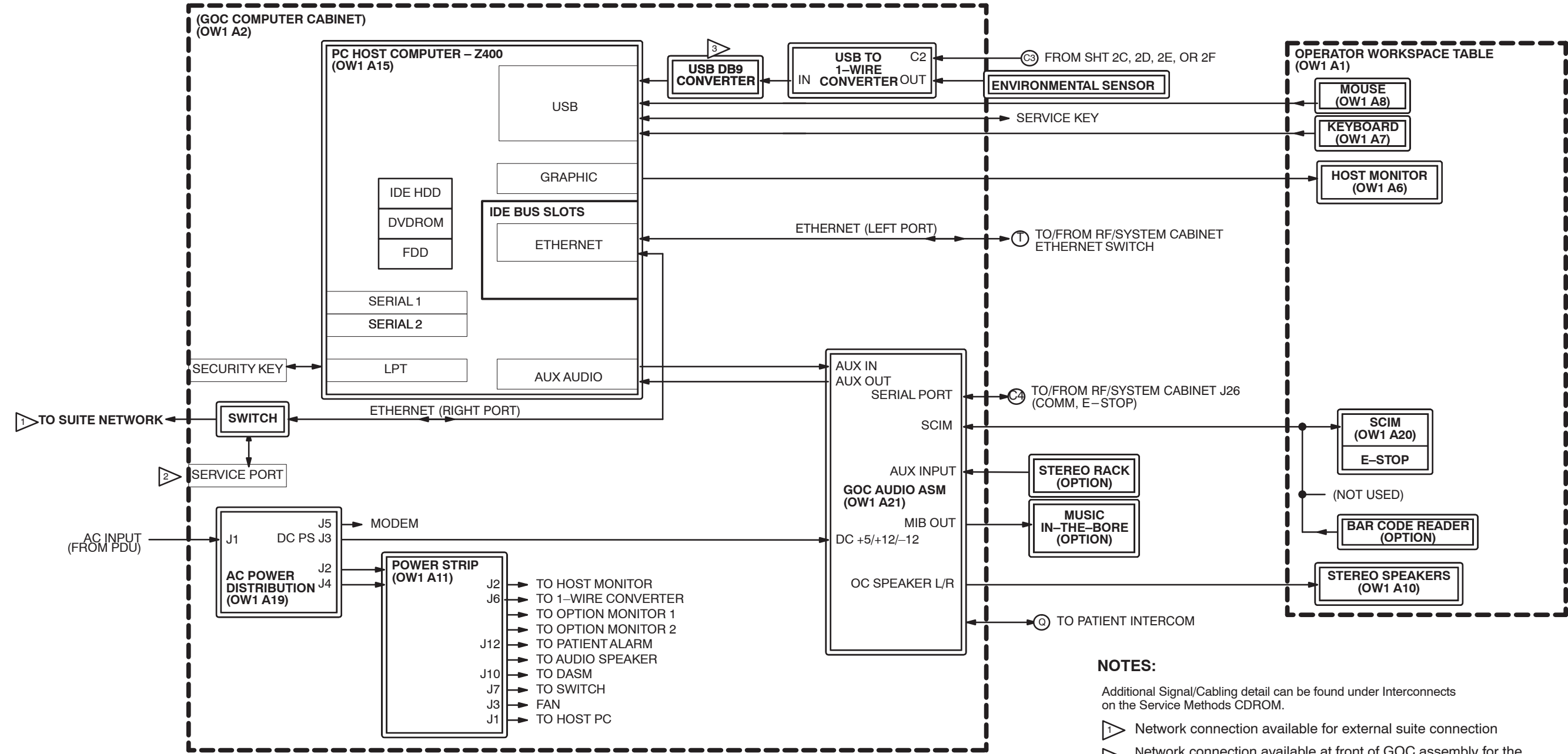
Refer to the following functional block diagram for additional information:

NOTE:

- 1) PATIENT HANDLING
- 2) MGD/RRF/RF

HDx (14.x) & HDxt (15.x) OVERALL SYSTEM
EXCITE HD to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/9300 & TABLE)

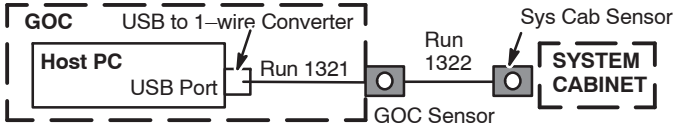
SHEET 1F OF 5



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

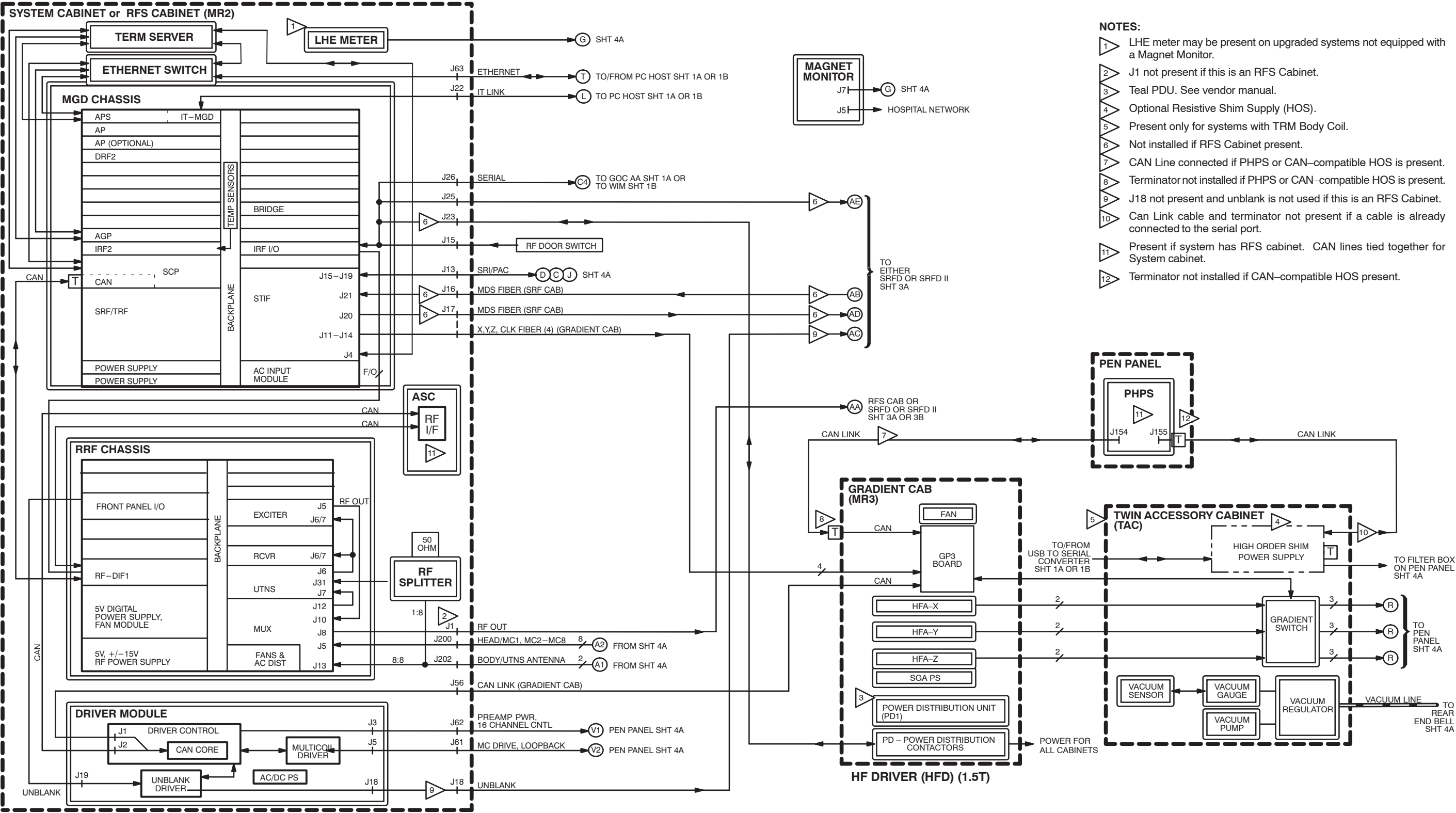
- 1 Network connection available for external suite connection
- 2 Network connection available at front of GOC assembly for the service laptop
- 3 Beginning August 2008, design changed to 1-wire to USB converter. Environmental Sensors Block Diagram utilizing 1-wire 5147136-2



Refer to the following functional block diagram for additional information:

- NOTE:**
- 1) PATIENT HANDLING
 - 2) MGD/RRF/RF

HDxt (15.x, 16.x) OVERALL SYSTEM
EXCITE HD to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/Z400 & TABLE)



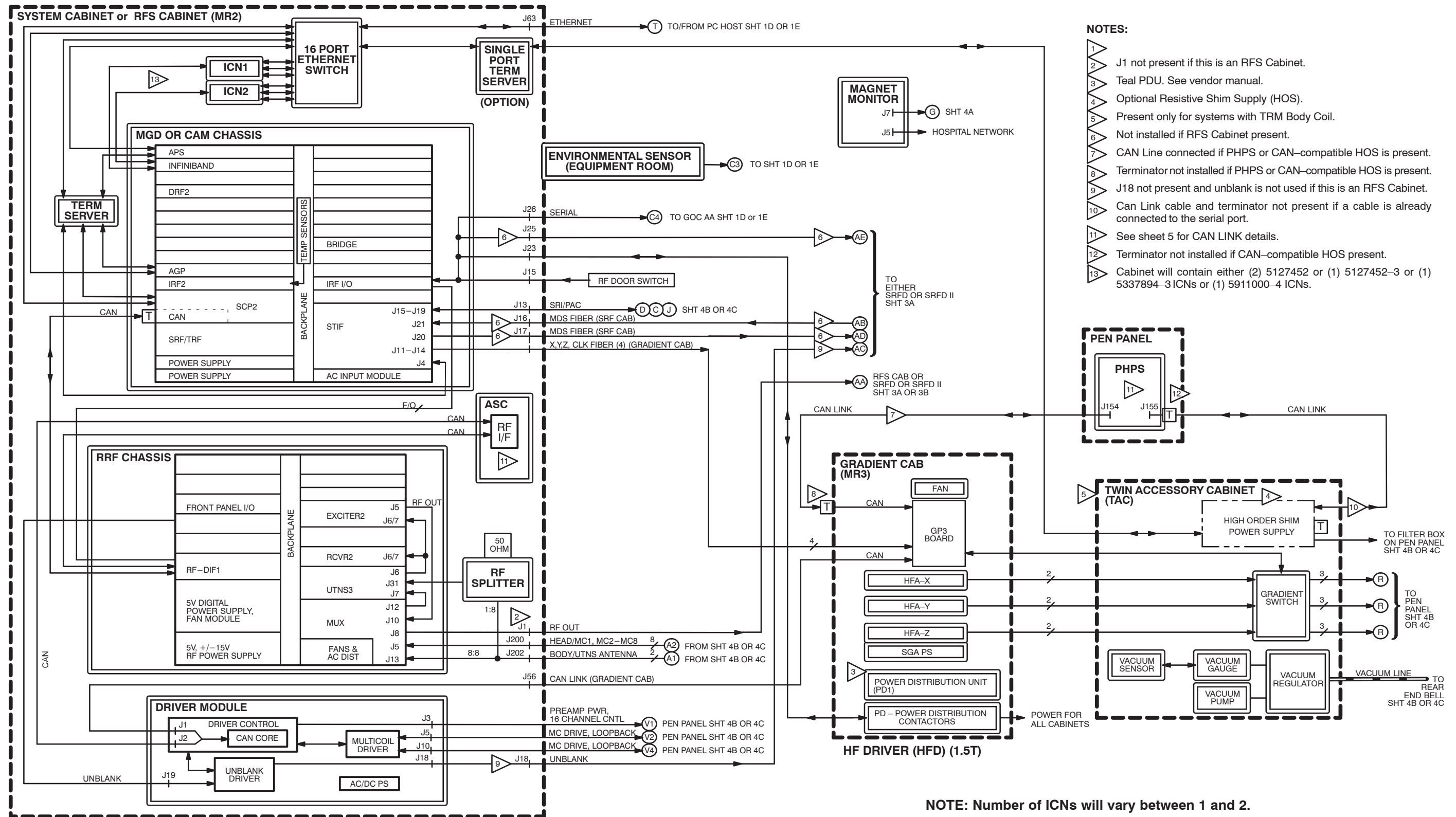
EXCITE HD (12.x) OVERALL SYSTEM – 4 Channel or 8 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 2A OF 5



- 1 J1 not present if this is an RFS Cabinet.
- 2 Teal PDU. See vendor manual.
- 3 Optional Resistive Shim Supply (HOS).
- 4 Present only for systems with TRM Body Coil.
- 5 Not installed if RFS Cabinet present.
- 6 CAN Line connected if PHPS or CAN-compatible HOS is present.
- 7 Terminator not installed if PHPS or CAN-compatible HOS is present.
- 8 J18 not present and unblank is not used if this is an RFS Cabinet.
- 9 Can Link cable and terminator not present if a cable is already connected to the serial port.
- 10 Present if system has RFS cabinet. CAN lines tied together for System cabinet.
- 11 Terminator not installed if CAN-compatible HOS present.

EXCITE HD (12.x) OVERALL SYSTEM – 16 Channels
(ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2B OF 5



HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 8 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2C OF 5

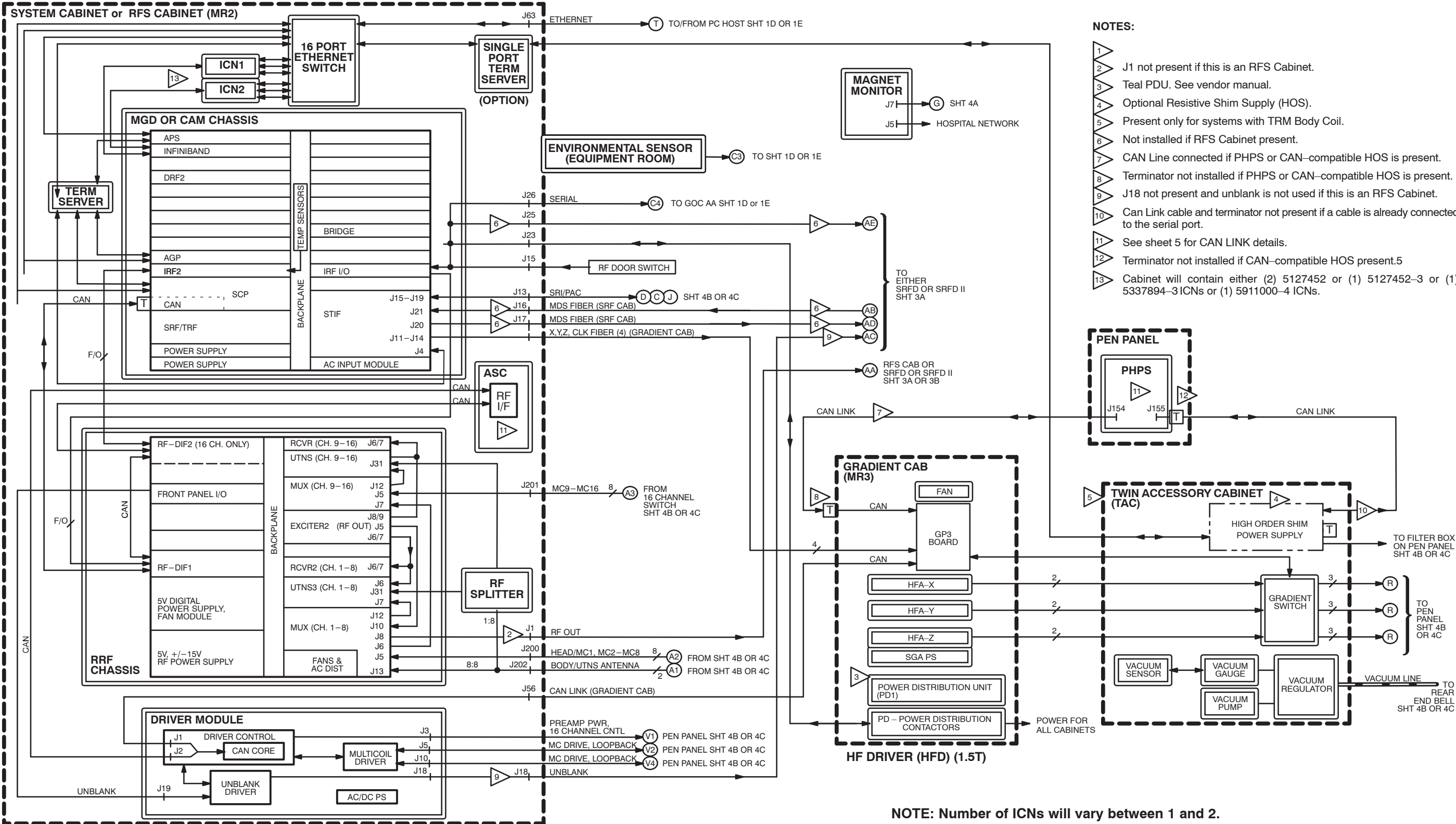
NOTE: Number of ICNs will vary between 1 and 2.



- | | |
|----|---|
| 1 | J1 not present if this is an RFS Cabinet. |
| 2 | Teal PDU. See vendor manual. |
| 3 | Optional Resistive Shim Supply (HOS). |
| 4 | Present only for systems with TRM Body Coil. |
| 5 | Not installed if RFS Cabinet present. |
| 6 | CAN Line connected if PHPS or CAN-compatible HOS is present. |
| 7 | Terminator not installed if PHPS or CAN-compatible HOS is present. |
| 8 | J18 not present and unblank is not used if this is an RFS Cabinet. |
| 9 | Can Link cable and terminator not present if a cable is already connected to the serial port. |
| 10 | See sheet 5 for CAN LINK details. |
| 11 | Terminator not installed if CAN-compatible HOS present. |

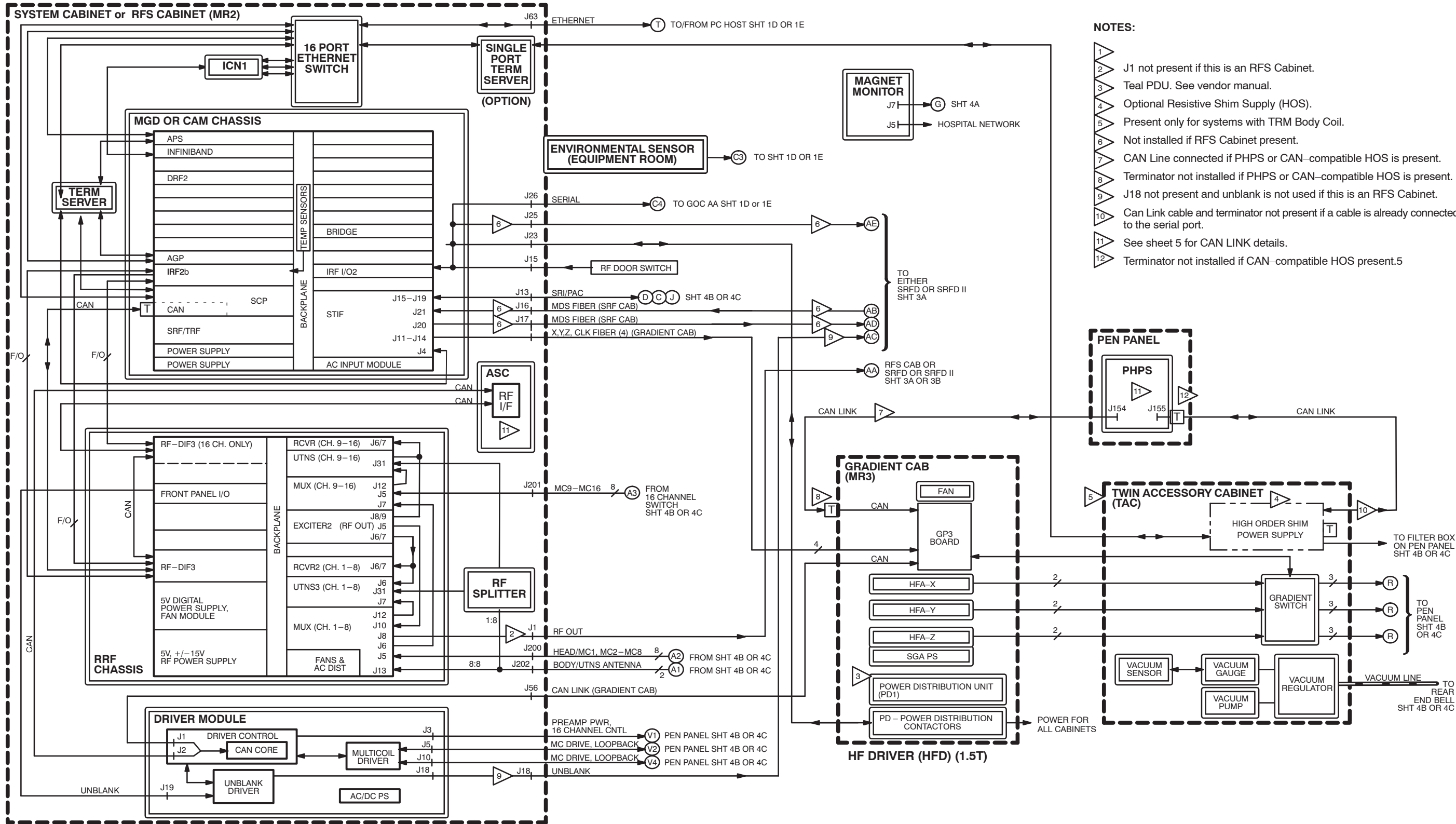
HDx & HDxt OVERALL SYSTEM – 8 Channel, IRF2b/RRF-DIF3 Hardware Configuration
(ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2D OF 5



HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 16 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 2E OF 5

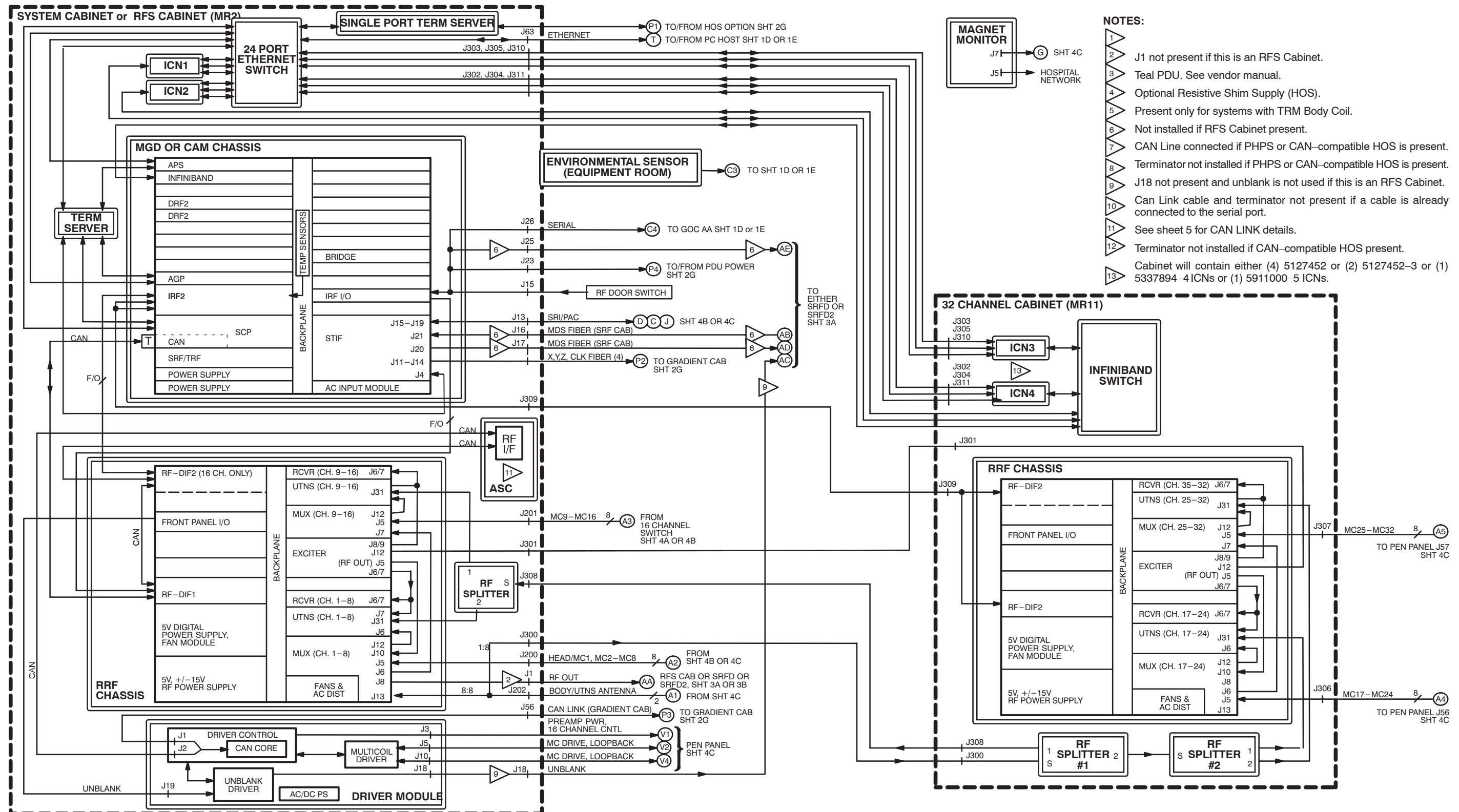
NOTE: Number of ICNs will vary between 1 and 2.



NOTES:

- 1 J1 not present if this is an RFS Cabinet.
- 2 Teal PDU. See vendor manual.
- 3 Optional Resistive Shim Supply (HOS).
- 4 Present only for systems with TRM Body Coil.
- 5 Not installed if RFS Cabinet present.
- 6 CAN Line connected if PHPS or CAN-compatible HOS is present.
- 7 Terminator not installed if PHPS or CAN-compatible HOS is present.
- 8 J18 not present and unblank is not used if this is an RFS Cabinet.
- 9 Can Link cable and terminator not present if a cable is already connected to the serial port.
- 10 See sheet 5 for CAN LINK details.
- 11 Terminator not installed if CAN-compatible HOS present.
- 12

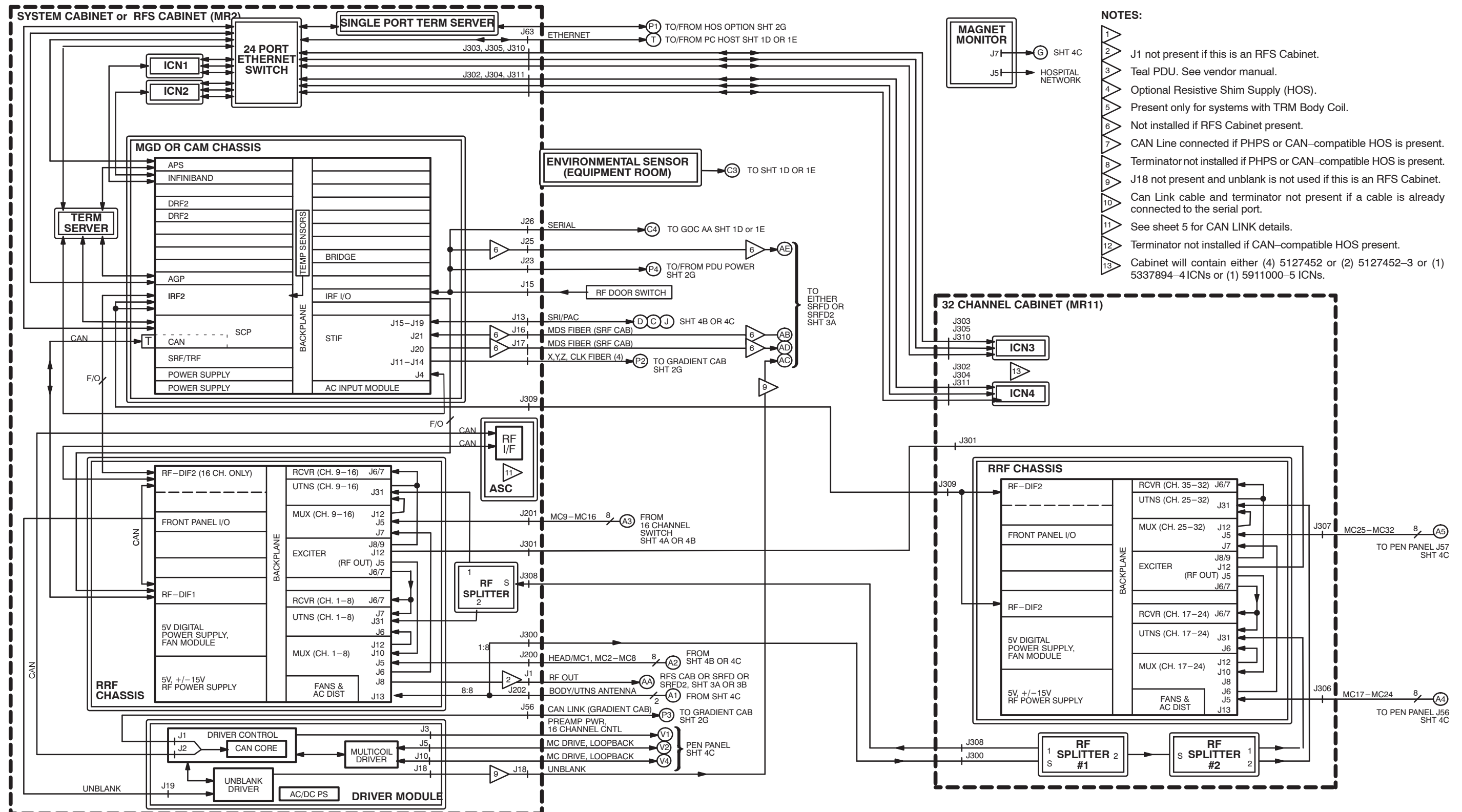
HDxt - 16 Channel, IRF2B/RRF-DIF3 Hardware Configuration
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 2F OF 5



HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 32 Channel with Infinband Switch
(ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2G OF 5

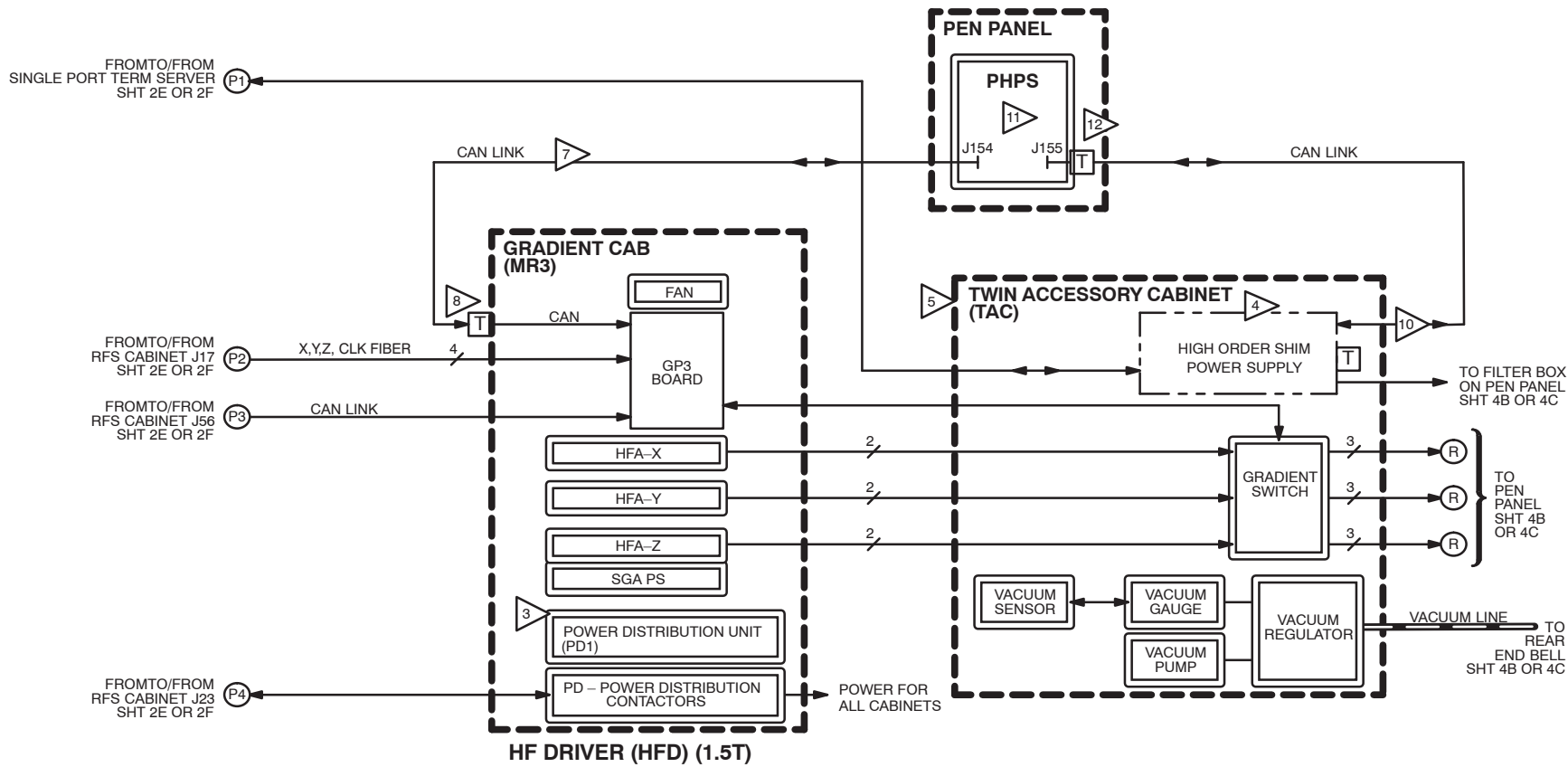
NOTE: Number of ICNs will vary between 1, 2 or 4.



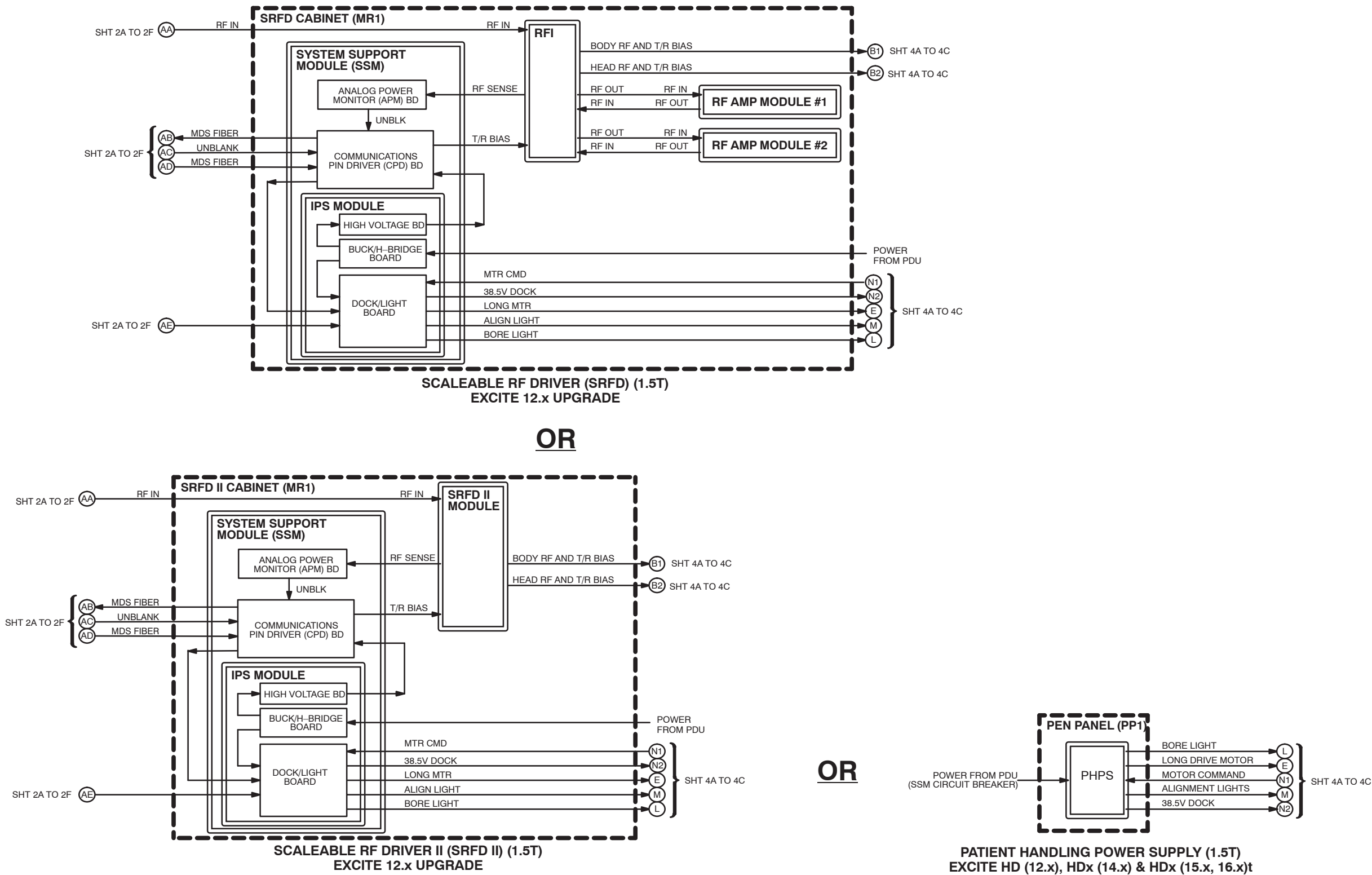
HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 32 Channel without Infinband Switch (ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 2H OF 5

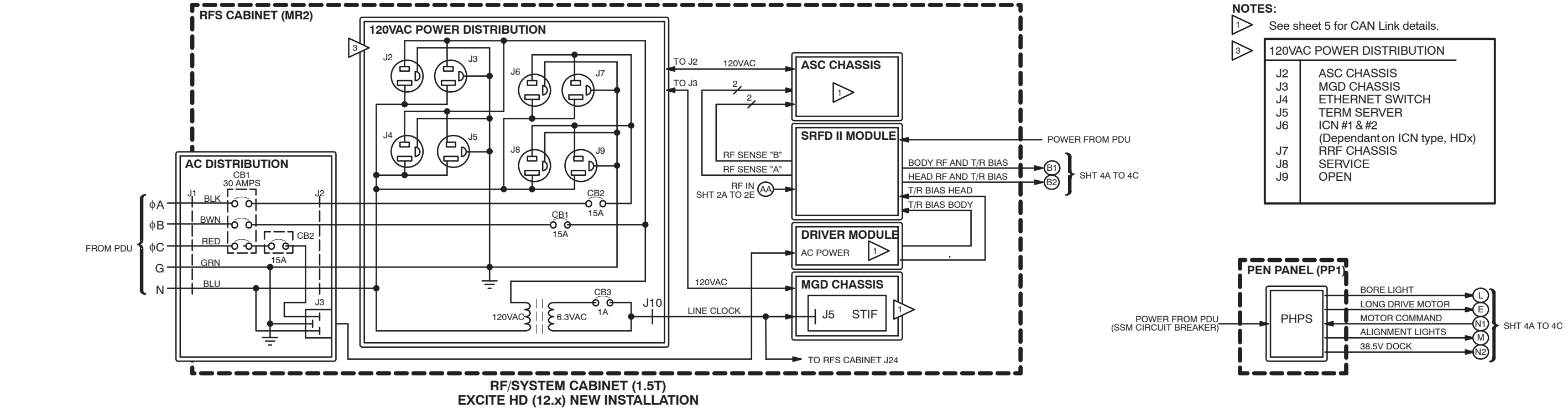
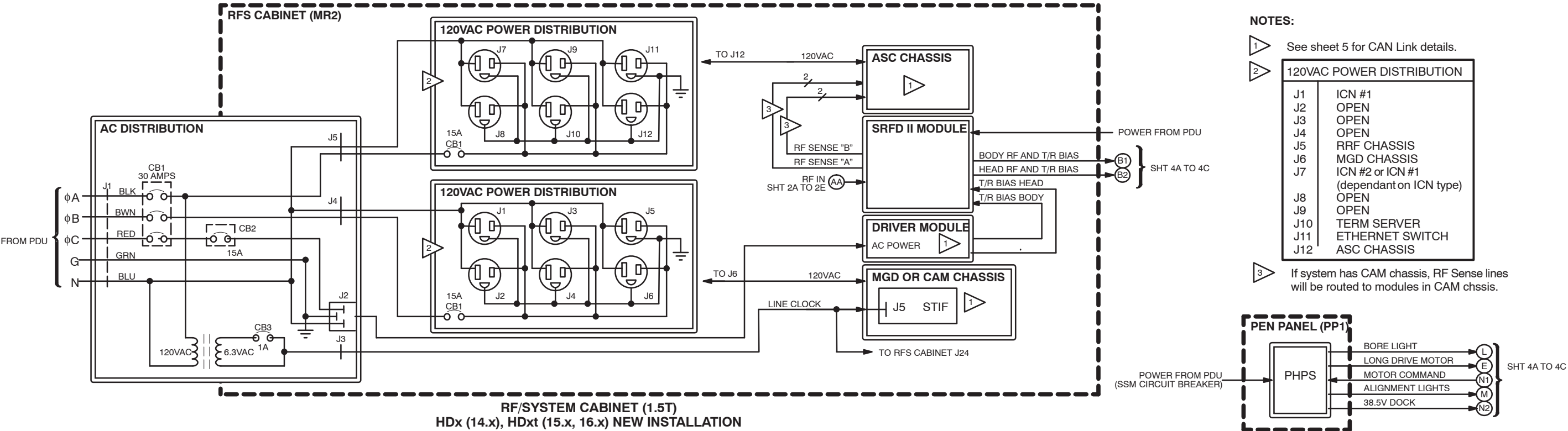
NOTE: Number of ICNs will vary between 1, 2 or 4.



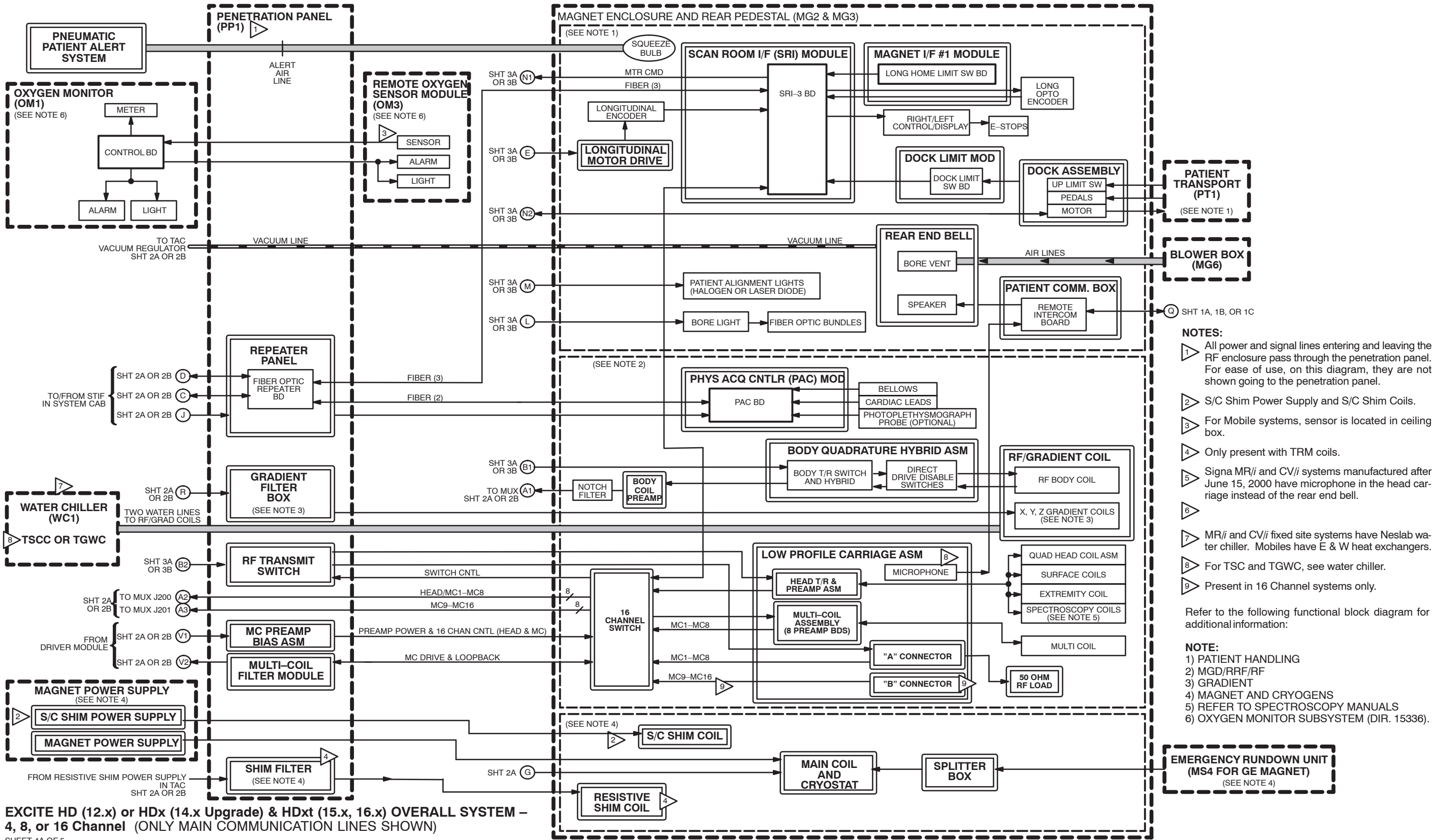
- NOTES:**
- 4 Teal PDU. See vendor manual.
 - 5 Optional Resistive Shim Supply (HOS).
 - 10 Can Link cable and terminator not present if a cable is already connected to the serial port.
 - 11 PHPS present if system has RFS cabinet.
 - 12 Terminator not installed if CAN-compatible HOS present.

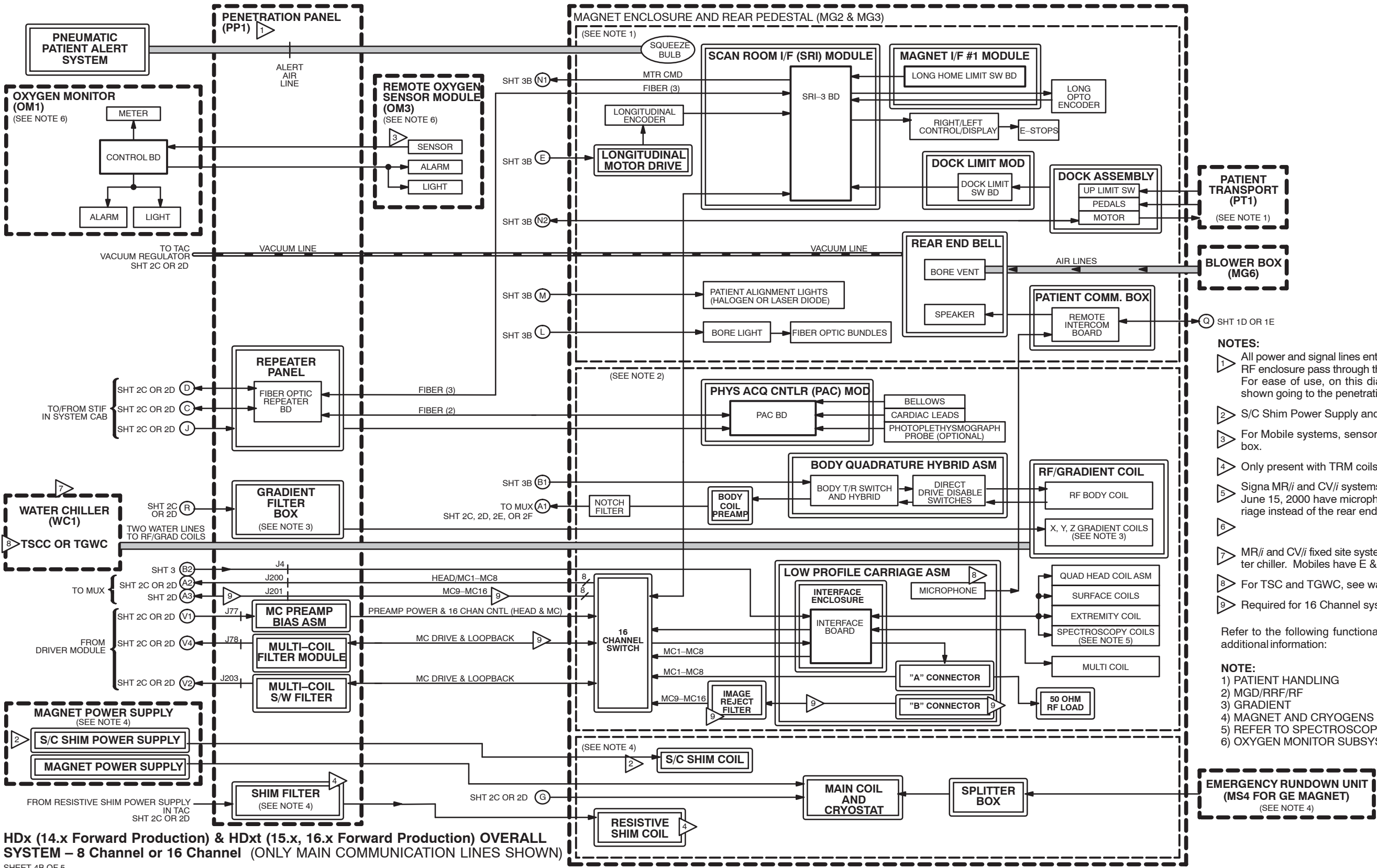


EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) OVERALL SYSTEM – New Installation & Up-grade
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 3A OF 5



EXCITE HD (12.x) and HDx (14.x) & HDxt (15.x, 16.x) OVERALL SYSTEM – New Installation & Upgrade
(ONLY MAIN COMMUNICATION LINES SHOWN)
SHEET 3B OF 5





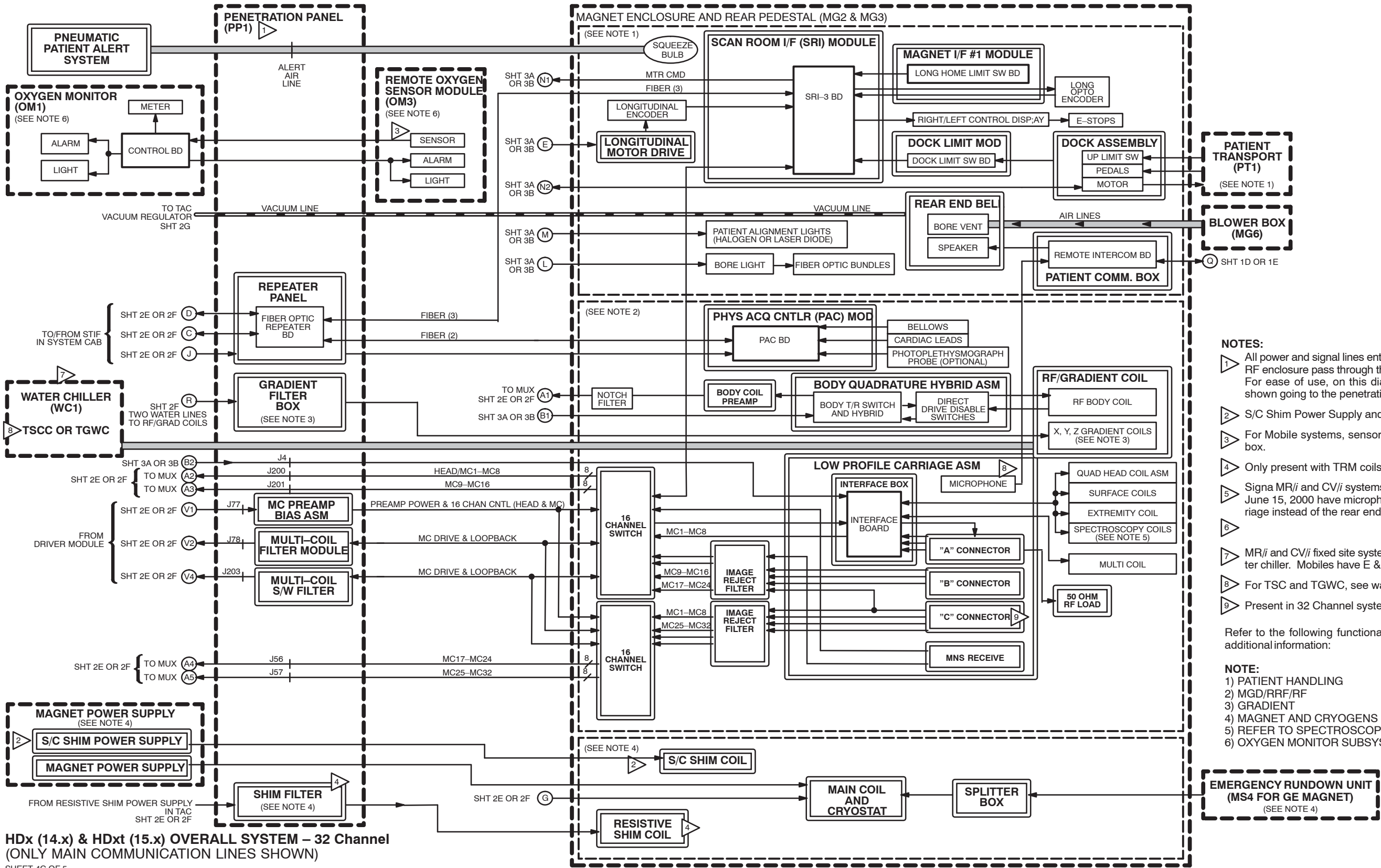
- NOTES:**
- 1 All power and signal lines entering and leaving the RF enclosure pass through the penetration panel. For ease of use, on this diagram, they are not shown going to the penetration panel.
 - 2 S/C Shim Power Supply and S/C Shim Coils.
 - 3 For Mobile systems, sensor is located in ceiling box.
 - 4 Only present with TRM coils.
 - 5 Signa MR/i and CV/i systems manufactured after June 15, 2000 have microphone in the head carriage instead of the rear end bell.
 - 6
 - 7 MR/i and CV/i fixed site systems have Neslab water chiller. Mobiles have E & W heat exchangers.
 - 8 For TSC and TGWC, see water chiller.
 - 9 Required for 16 Channel systems only.

Refer to the following functional block diagram for additional information:

NOTE:

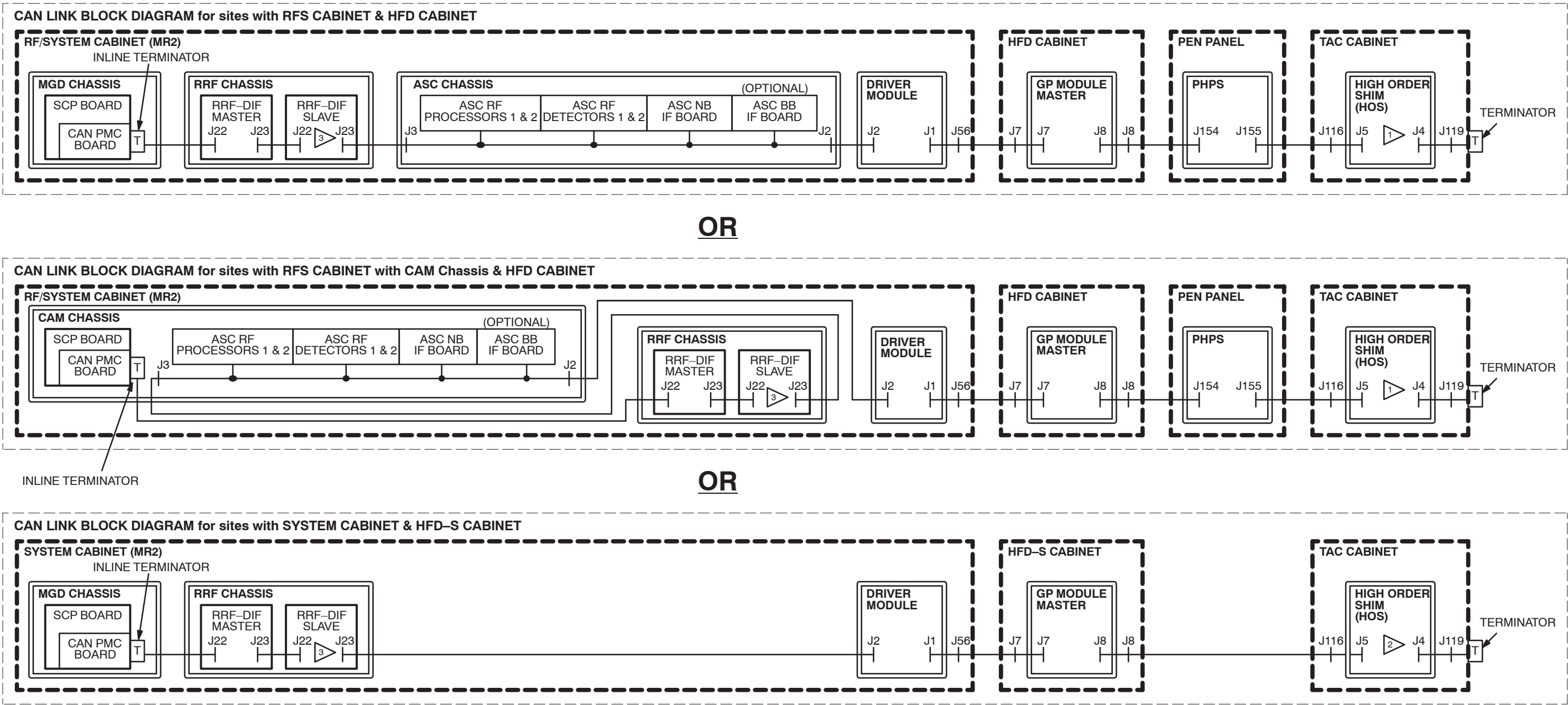
- 1) PATIENT HANDLING
- 2) MGD/RRF/RF
- 3) GRADIENT
- 4) MAGNET AND CRYOGENS
- 5) REFER TO SPECTROSCOPY MANUALS
- 6) OXYGEN MONITOR SUBSYSTEM (DIR. 15336).

HDx (14.x Forward Production) & HDxt (15.x, 16.x Forward Production) OVERALL SYSTEM – 8 Channel or 16 Channel (ONLY MAIN COMMUNICATION LINES SHOWN)



HDx (14.x) & HDxt (15.x) OVERALL SYSTEM – 32 Channel
(ONLY MAIN COMMUNICATION LINES SHOWN)

SHEET 4C OF 5



EXCITE HD (12.x) and HDx (14.x) & HDxt (15.x, 16.x) OVERALL SYSTEM
CAN Link Block Diagrams
(ONLY MAIN COMMUNICATION LINES SHOWN)

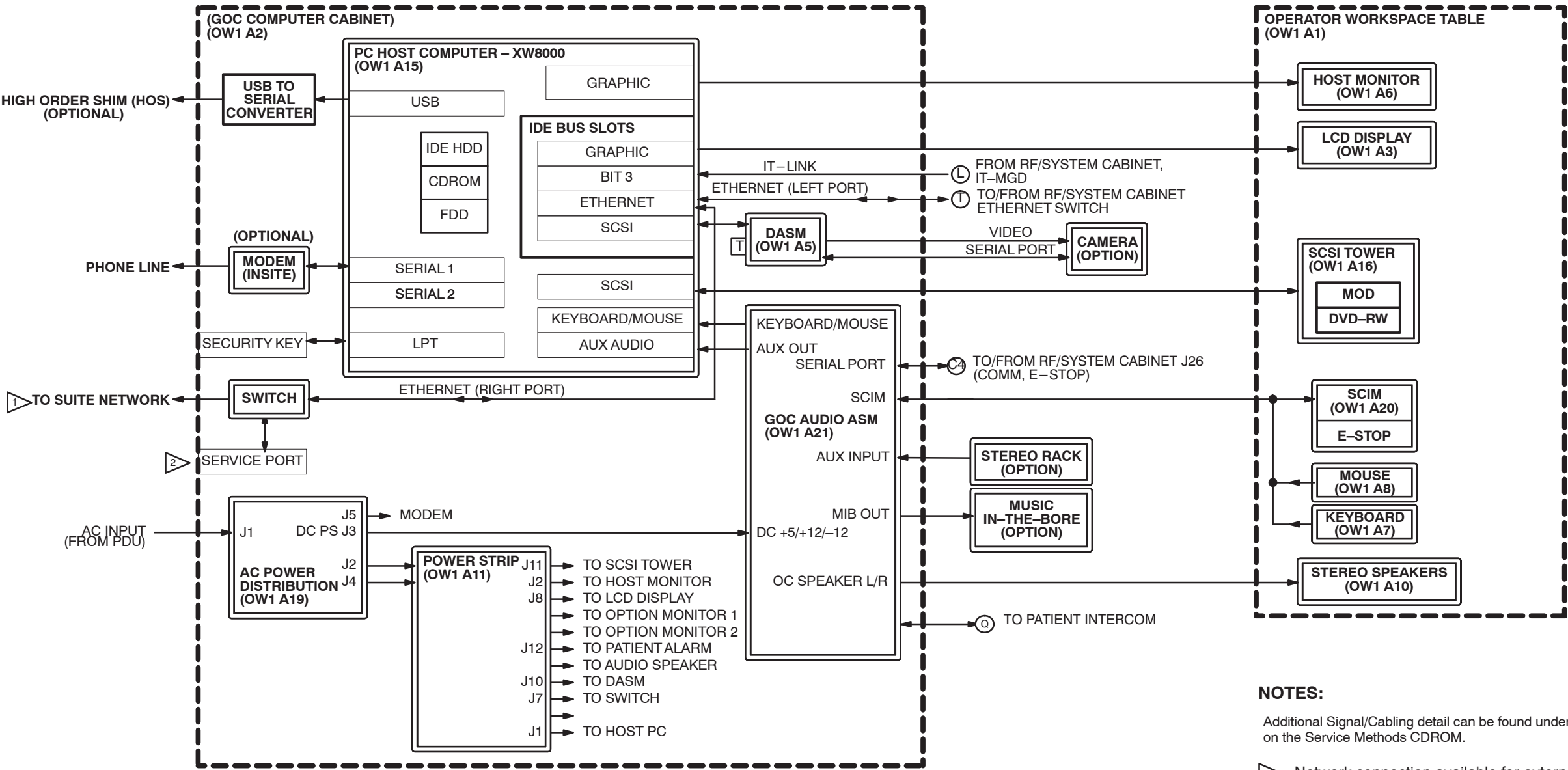
SHEET 5 OF 5

- NOTES:**
- 1 HOS is optional. Terminator must be connected to J155 on PHPS if HOS not present.
 - 2 HOS is optional. Terminator must be connected to J8 on HFD Cabinet I/F if HOS not present.
 - 3 Only present in systems equipped with 16 Channels. If system is 8 Channels or less, then RF-DIF Master J23 connects to next unit in link as shown.

OPERATOR WORKSPACE

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
1 – OPERATOR WORKSPACE	
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET w/8000 & GOC AA (Forward Production or Pre–EXCITE to HD Upgrades)	1–1
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET w/8000 & WIM (EXCITE to HD Upgrades)	1–2
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET w/8200 & GOC AA (Forward Production or Pre–EXCITE to HD Upgrades)	1–3
HDx & HDxt OVERALL SYSTEM – OPERATOR WORKSPACE CABINETw/9300 & GOC AA (Forward Production, EXCITE, or Pre–EXCITE to HDx Upgrades)	1–4
HDx & HDxt OVERALL SYSTEM – OPERATOR WORKSPACE CABINETw/9300 & GOC AA (EXCITE HD to HDx Upgrades)	1–5
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET/TABLE w/WIM (Forward Production & Upgrades)	1–6
HD OVERALL SYSTEM – OPERATOR WORKSPACE CABINET/TABLE w/GOC AA (Forward Production & Upgrades)	1–7
HDx & HDxt OVERALL SYSTEM – OPERATOR WORKSPACE CABINET/TABLE w/GOC AA (Forward Production & Upgrades)	1–8



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

- 1) Network connection available for external suite connection
- 2) network connection available at front of goc assembly for the service laptop

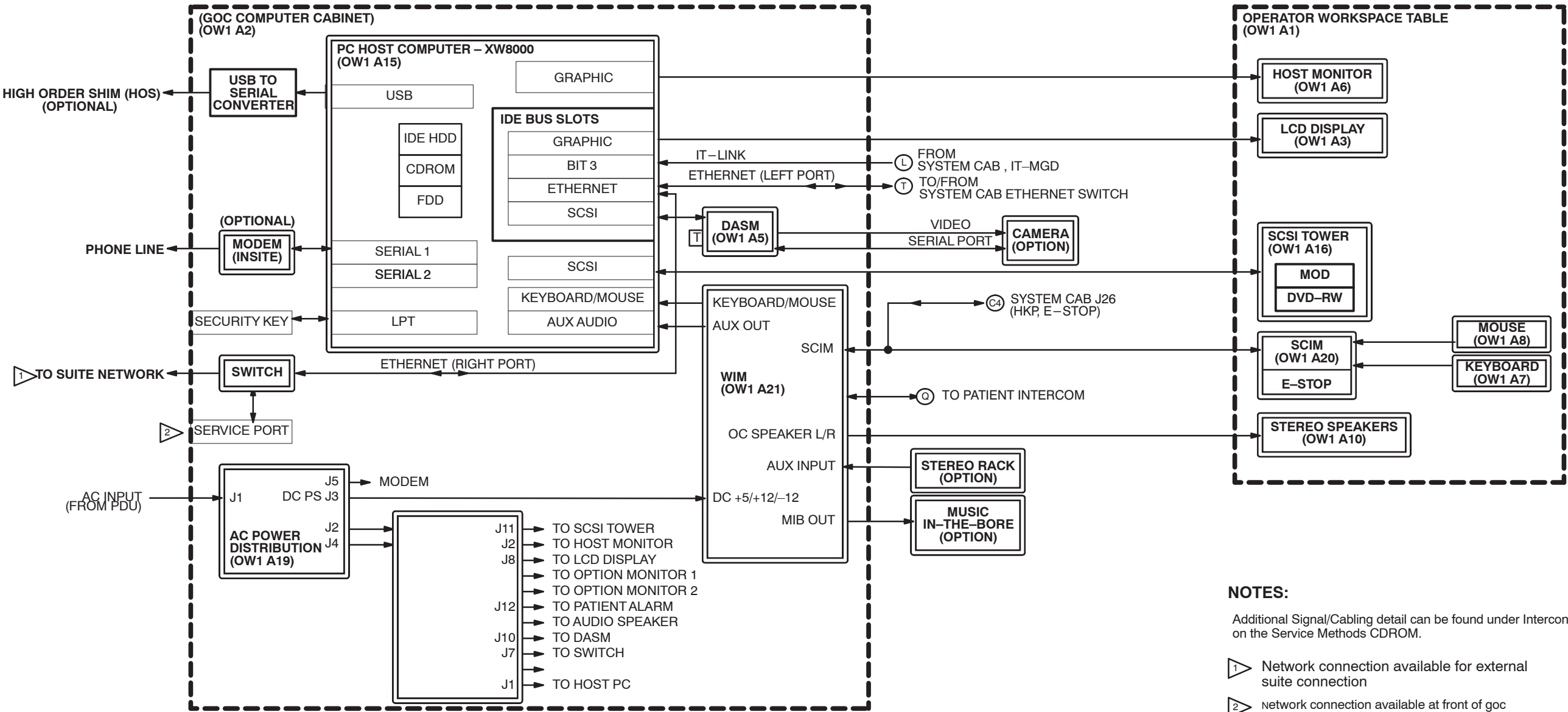
Refer to the following functional block diagram for additional information:

NOTE:

- 1) PATIENT HANDLING
- 2) MGD/RRF/RF

EXCITE HD (12.x) OVERALL SYSTEM
Forward Production or Pre-EXCITE to HD Upgrades
(OPERATOR WORKSPACE CABINET w/8000 & TABLE)

SHEET 1A OF 5



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection

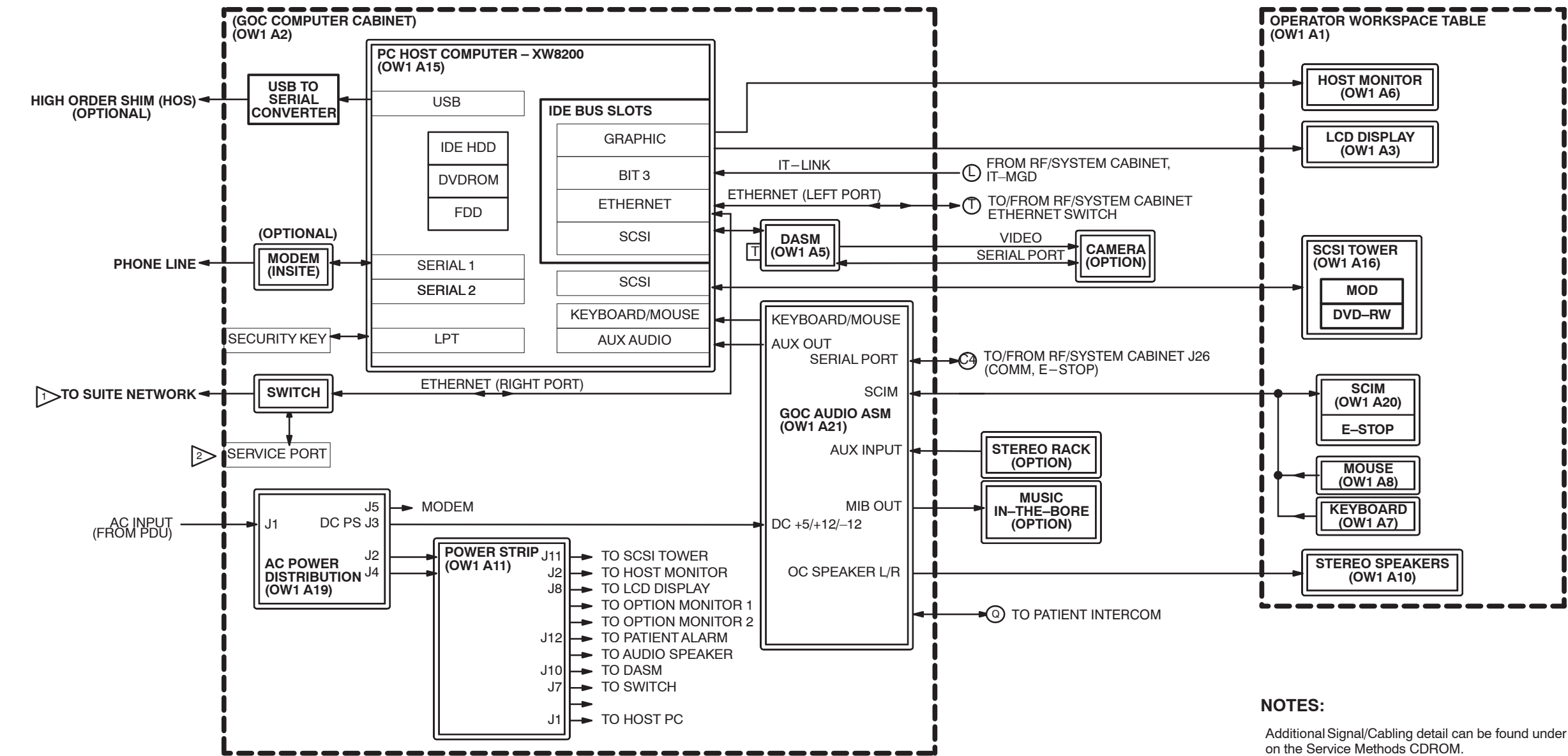
2 network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

1) PATIENT HANDLING

2) MGD/RRF/RF



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

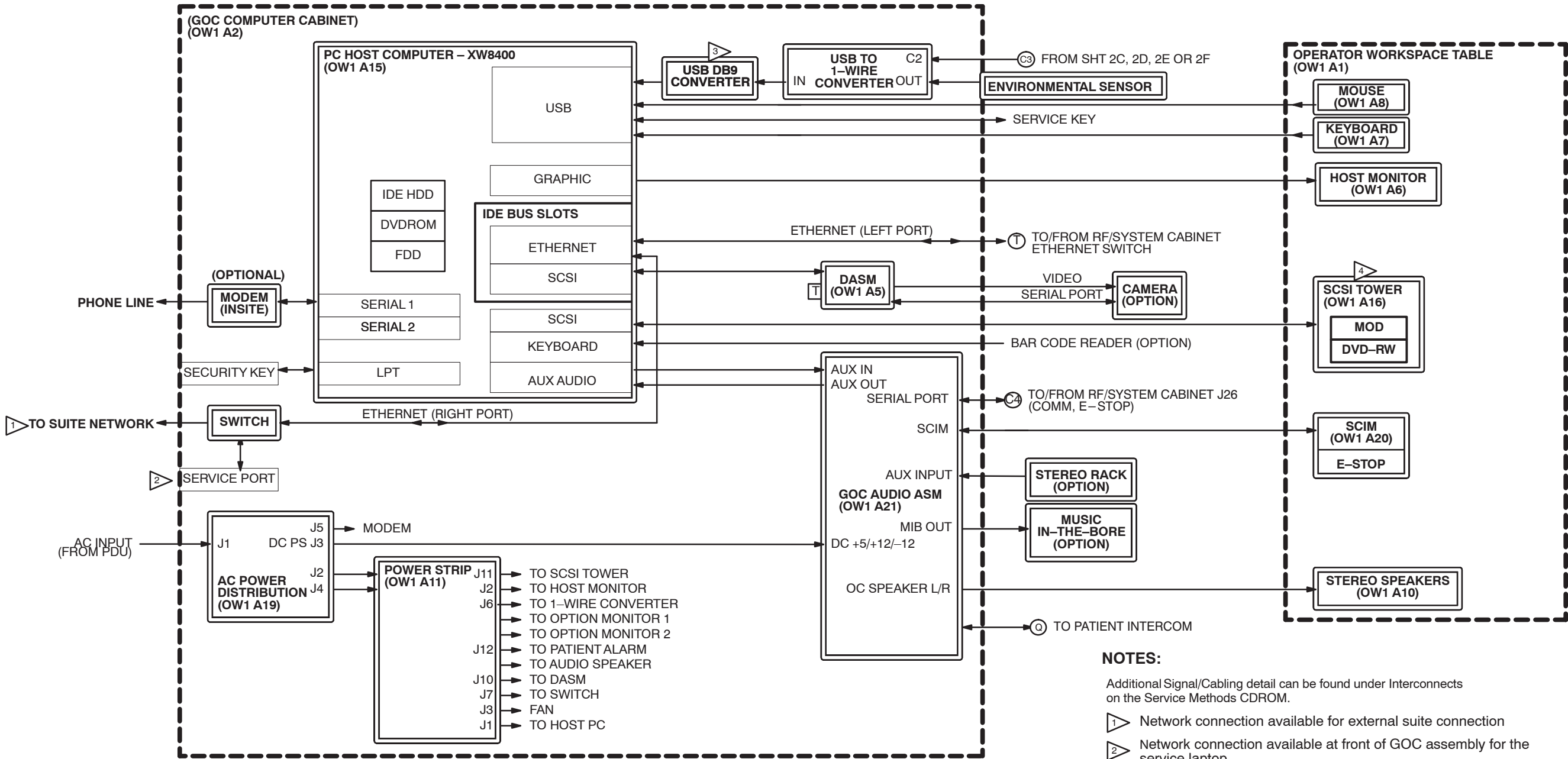
- 1) Network connection available for external suite connection
- 2) Network connection available at front of goc assembly for the service laptop

Refer to the following functional block diagram for additional information:

NOTE:

- 1) PATIENT HANDLING
- 2) MGD/RRF/RF

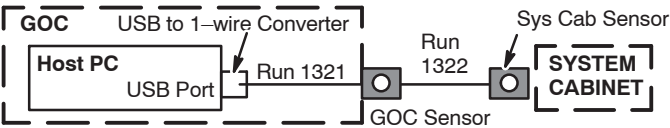
EXCITE HD (12.x) OVERALL SYSTEM
Forward Production or Pre-EXCITE to HD Upgrades
(OPERATOR WORKSPACE CABINET w/8200 & TABLE)



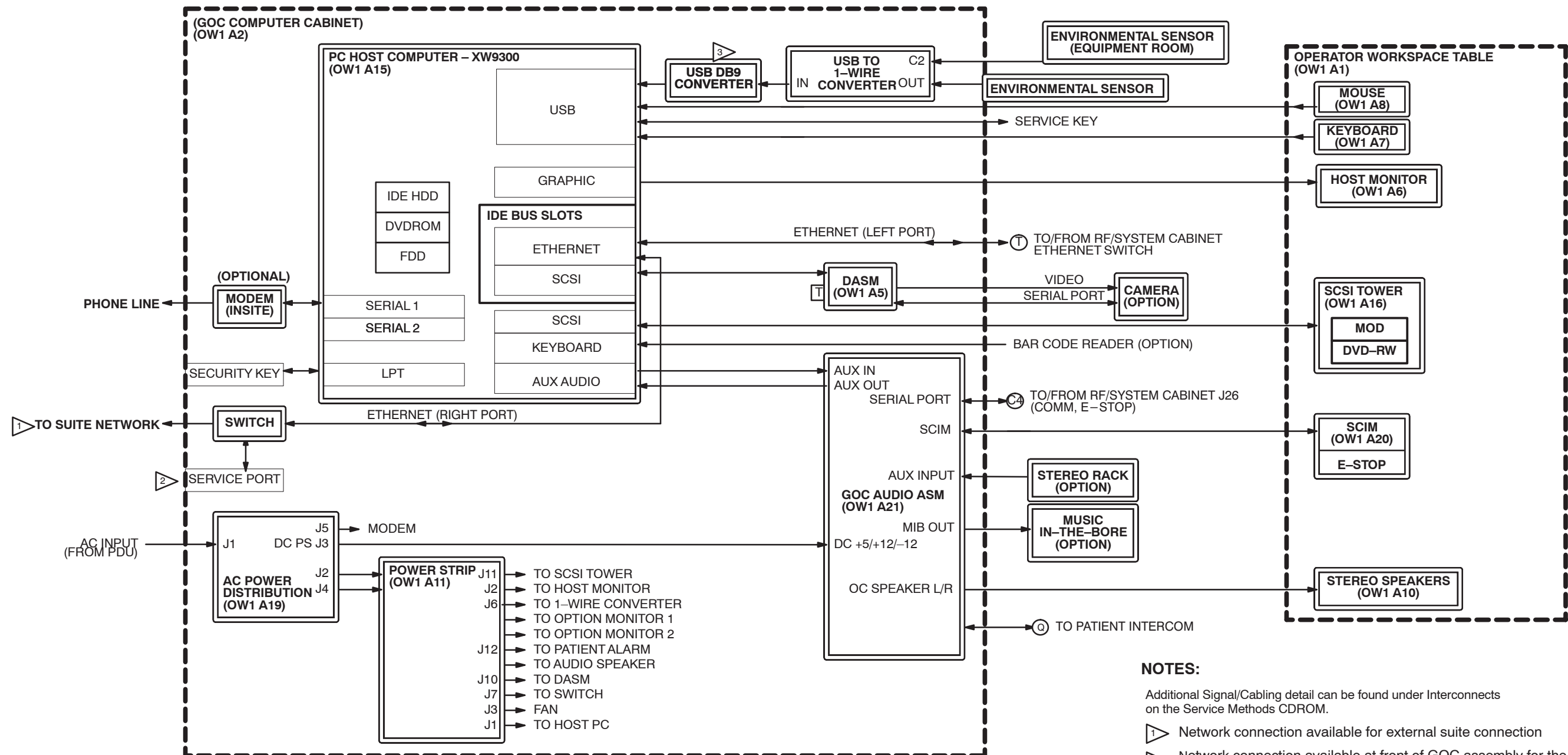
Refer to the following functional block diagram for additional information:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

- NOTES:**
- Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.
- 1 Network connection available for external suite connection
 - 2 Network connection available at front of GOC assembly for the service laptop
 - 3 Beginning August 2008, design changed to 1-wire to USB converter. Environmental Sensors Block Diagram utilizing 1-wire 5147136-2
 - 4 8400 Host Computer is also available without SCSI Tower.



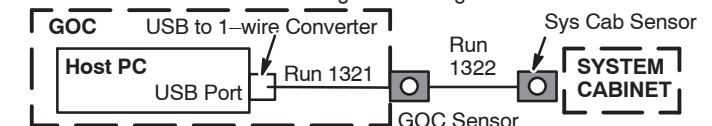
HDx (14.x) & HDxt (15.x, 16.x) OVERALL SYSTEM
Forward Production, EXCITE, or Pre-EXCITE to HDxt Upgrades
(OPERATOR WORKSPACE CABINET w/8400 & TABLE)
SHEET 1D OF 5



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

- 1 Network connection available for external suite connection
- 2 Network connection available at front of GOC assembly for the service laptop
- 3 Beginning August 2008, design changed to 1-wire to USB converter.



Refer to the following functional block diagram for additional information:

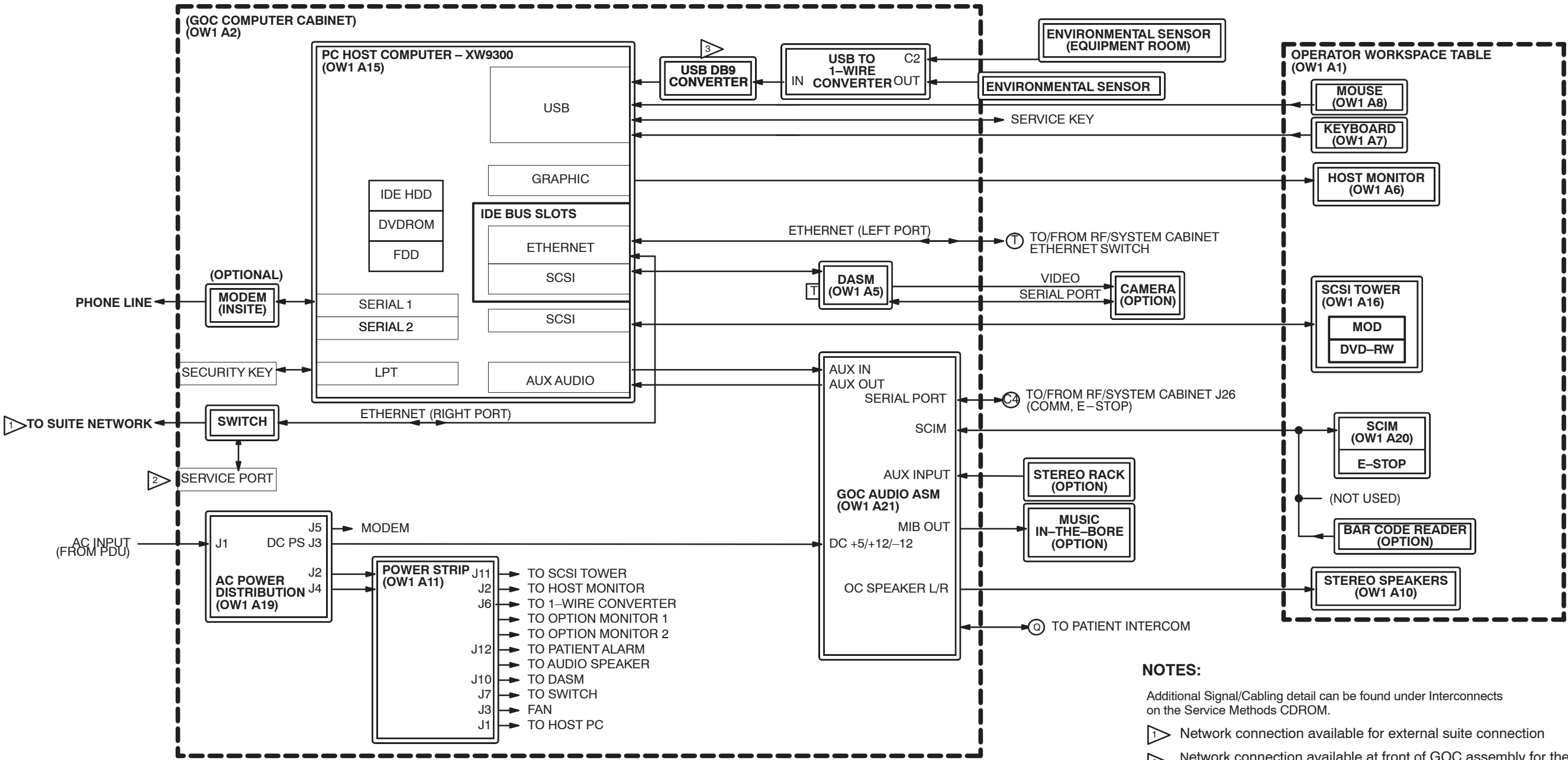
NOTE:

- NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF

HDx (14.x) & HDxt (15.x) OVERALL SYSTEM

Forward Production, EXCITE, or Pre-EXCITE to HDx Upgrades
(OPERATOR WORKSPACE CABINET w/9300 & TABLE)

SHEET 1E OF 5



NOTES:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

1 Network connection available for external suite connection

2 Network connection available at front of GOC assembly for the service laptop

3 Beginning August 2008, design changed to 1-wire to USB converter. Environmental Sensors Block Diagram utilizing 1-wire 5147136-2

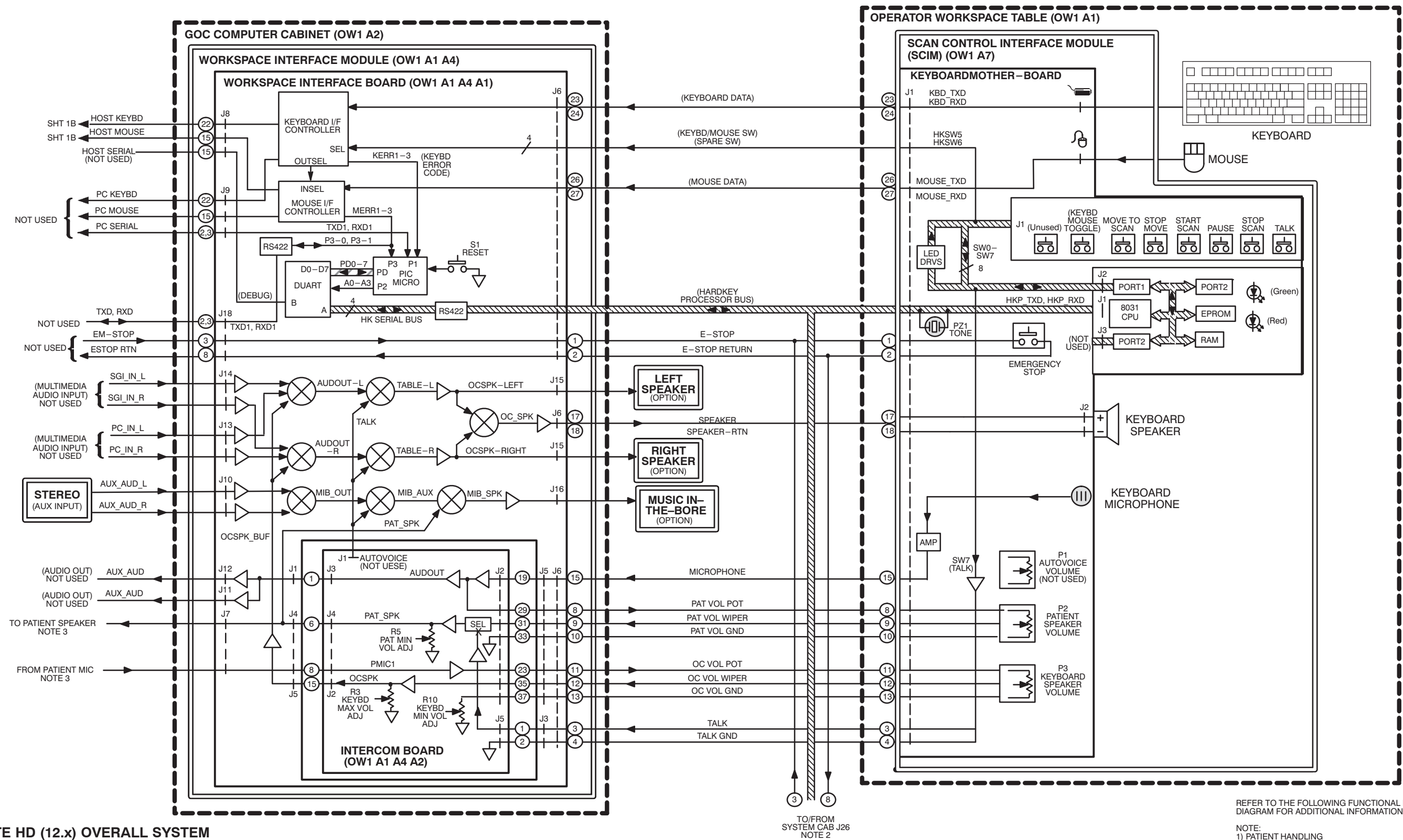
Refer to the following functional block diagram for additional information:

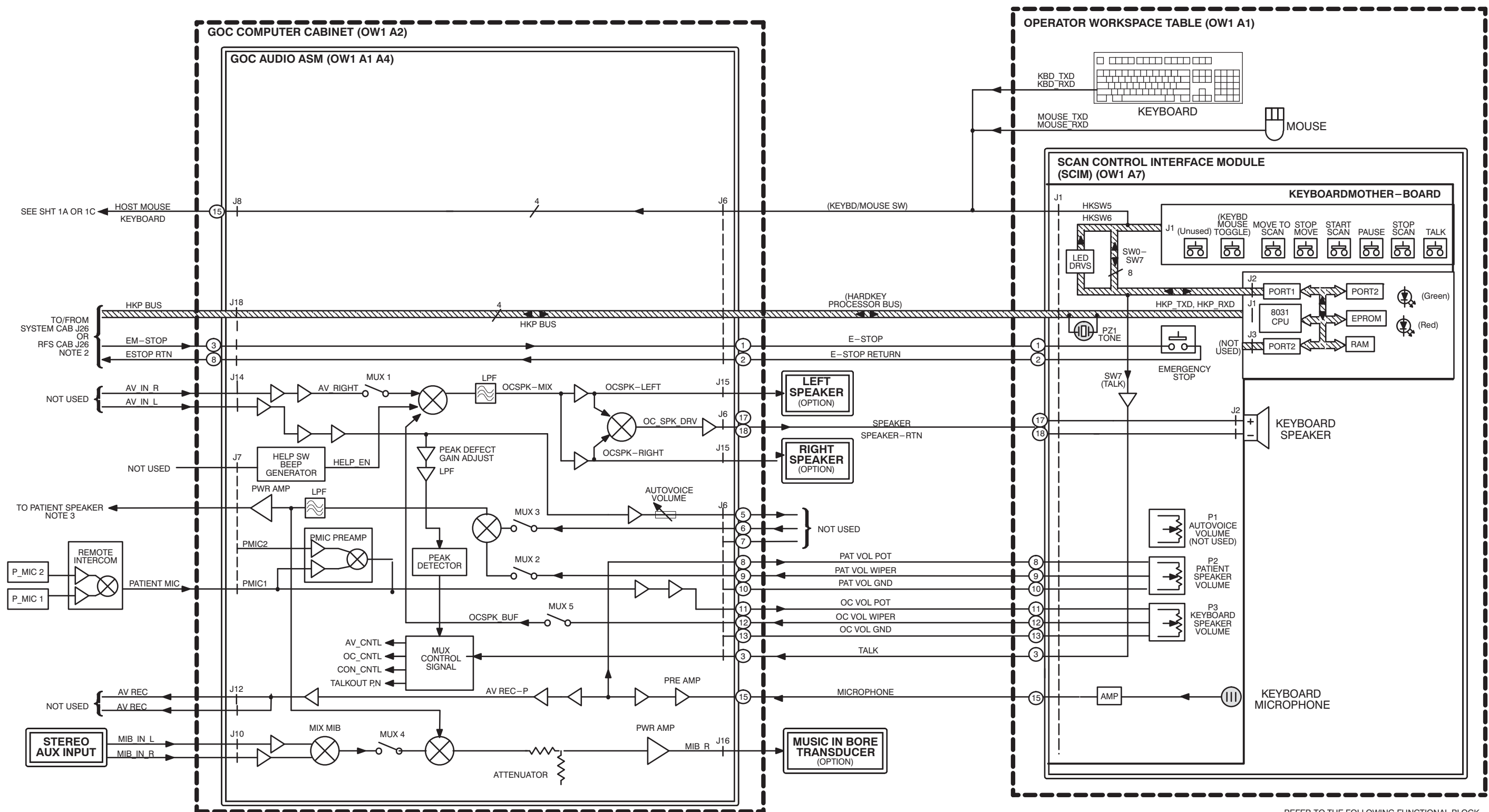
NOTE:

1) PATIENT HANDLING
2) MGD/RRF/RF



SHEET 1G OF 5



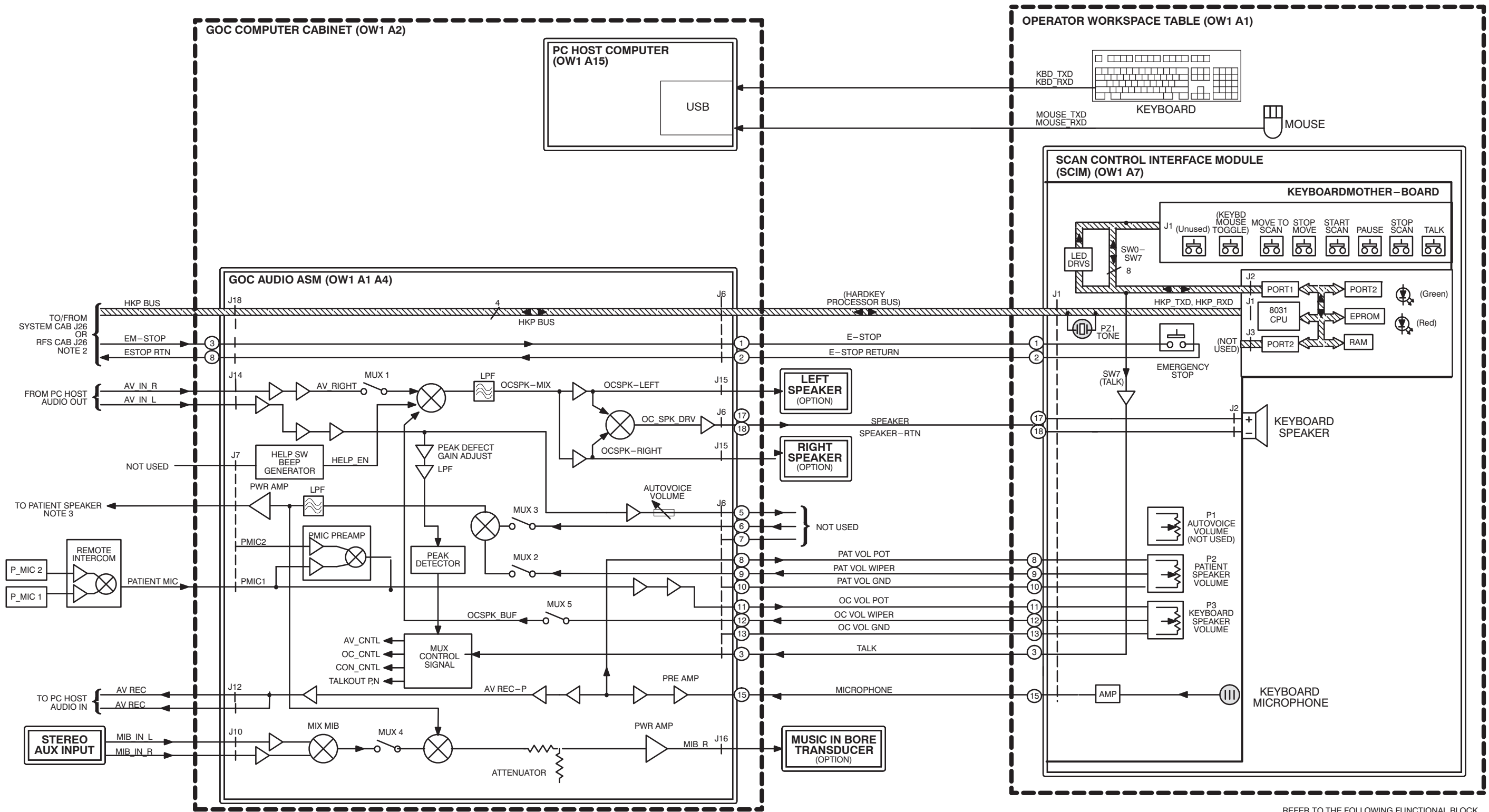


REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:

NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF
3) OVERALL SYSTEM

EXCITE HD (12.x) OVERALL SYSTEM
(OPERATOR WORKSPACE CABINET/TABLE with GOC AUDIO ASM)

SHEET 2B OF 2



HDx (14.x) & HDxt (15.x, 16.x) OVERALL SYSTEM
(OPERATOR WORKSPACE CABINET/TABLE with GOC AUDIO ASM)
SHEET 2C OF 2

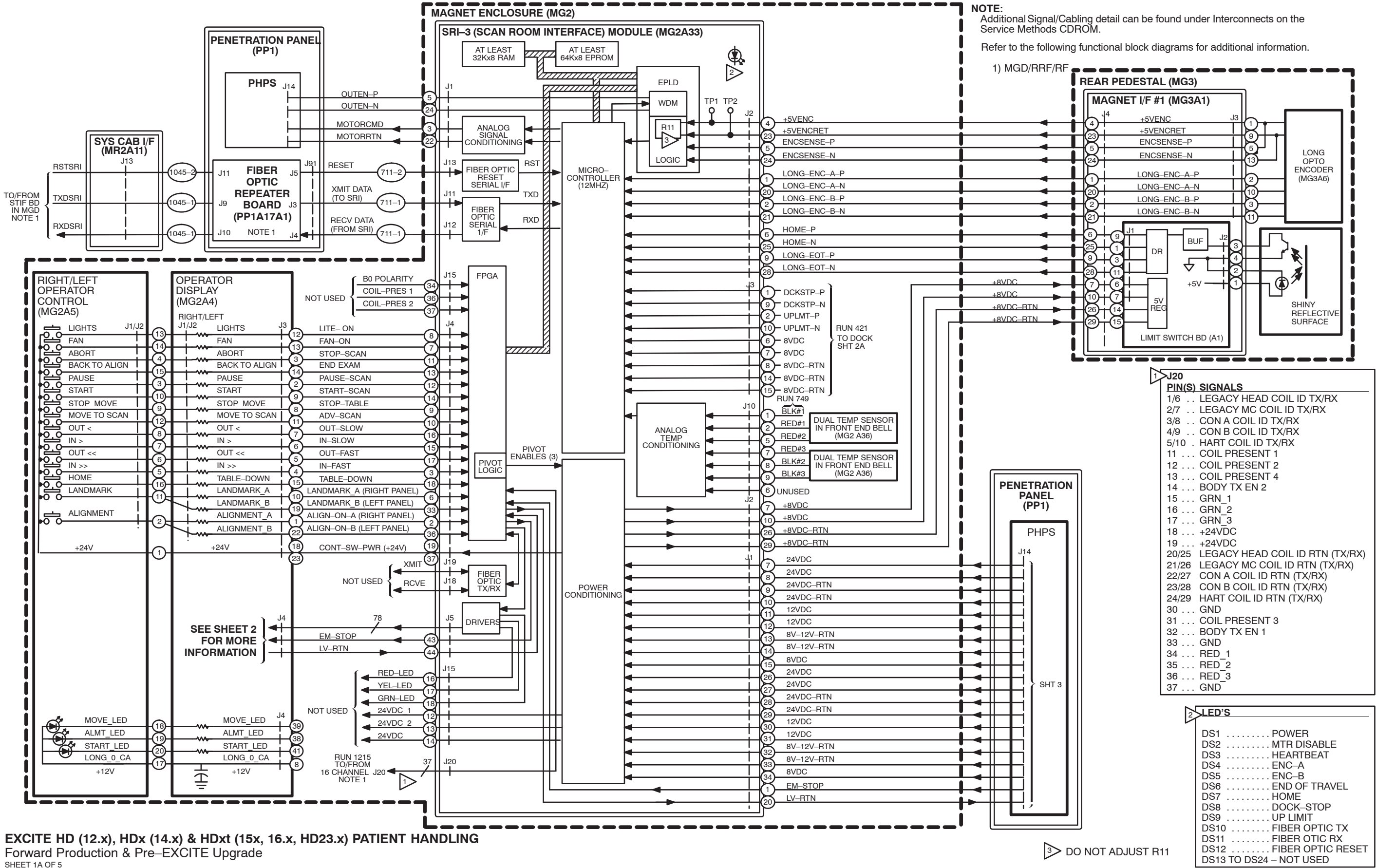
REFER TO THE FOLLOWING FUNCTIONAL BLOCK
DIAGRAM FOR ADDITIONAL INFORMATION:

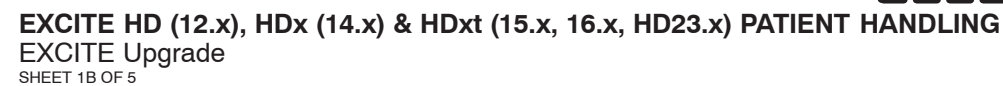
NOTE:
1) PATIENT HANDLING
2) MGD/RRF/RF
3) OVERALL SYSTEM

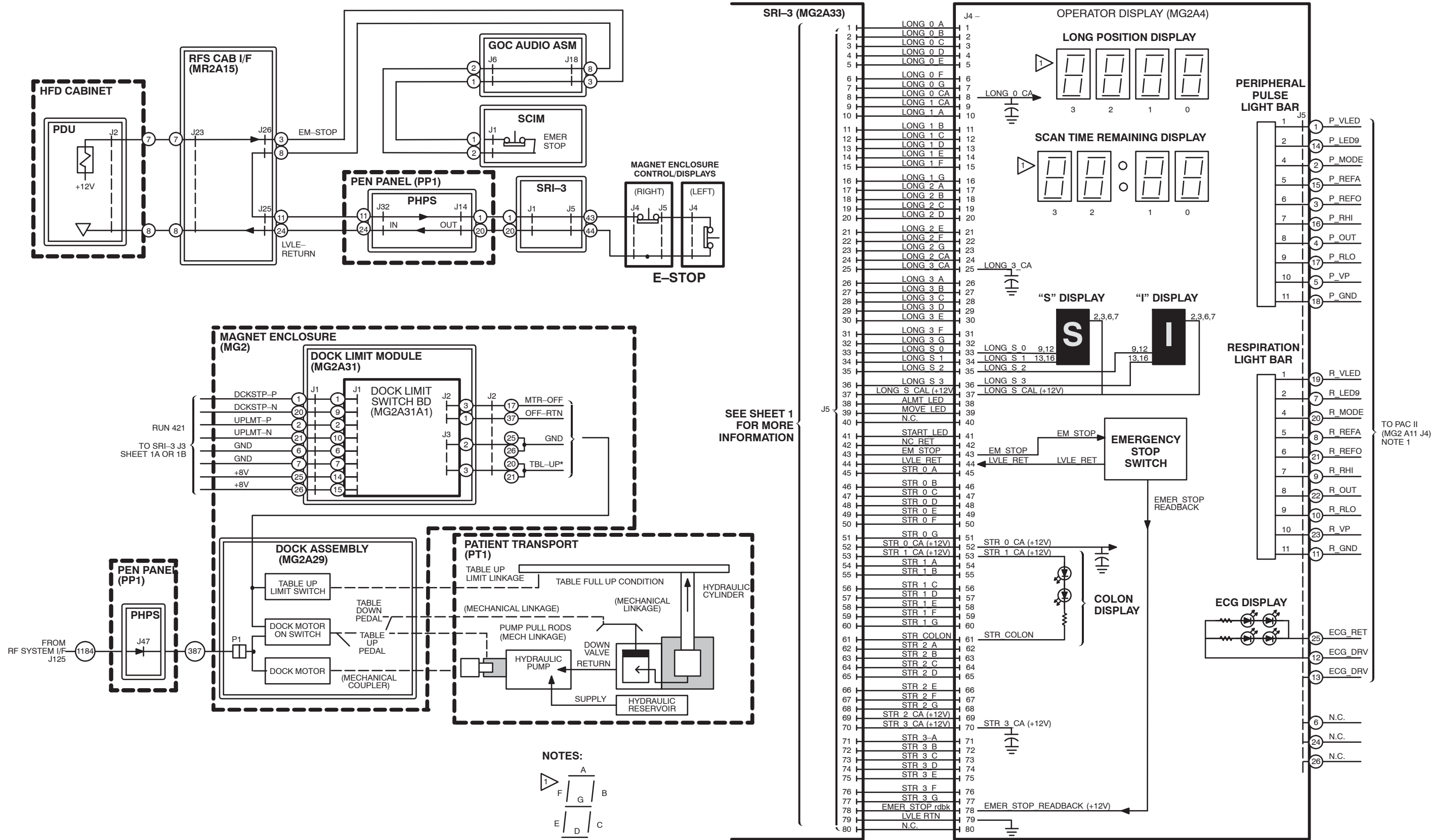
PATIENT HANDLING

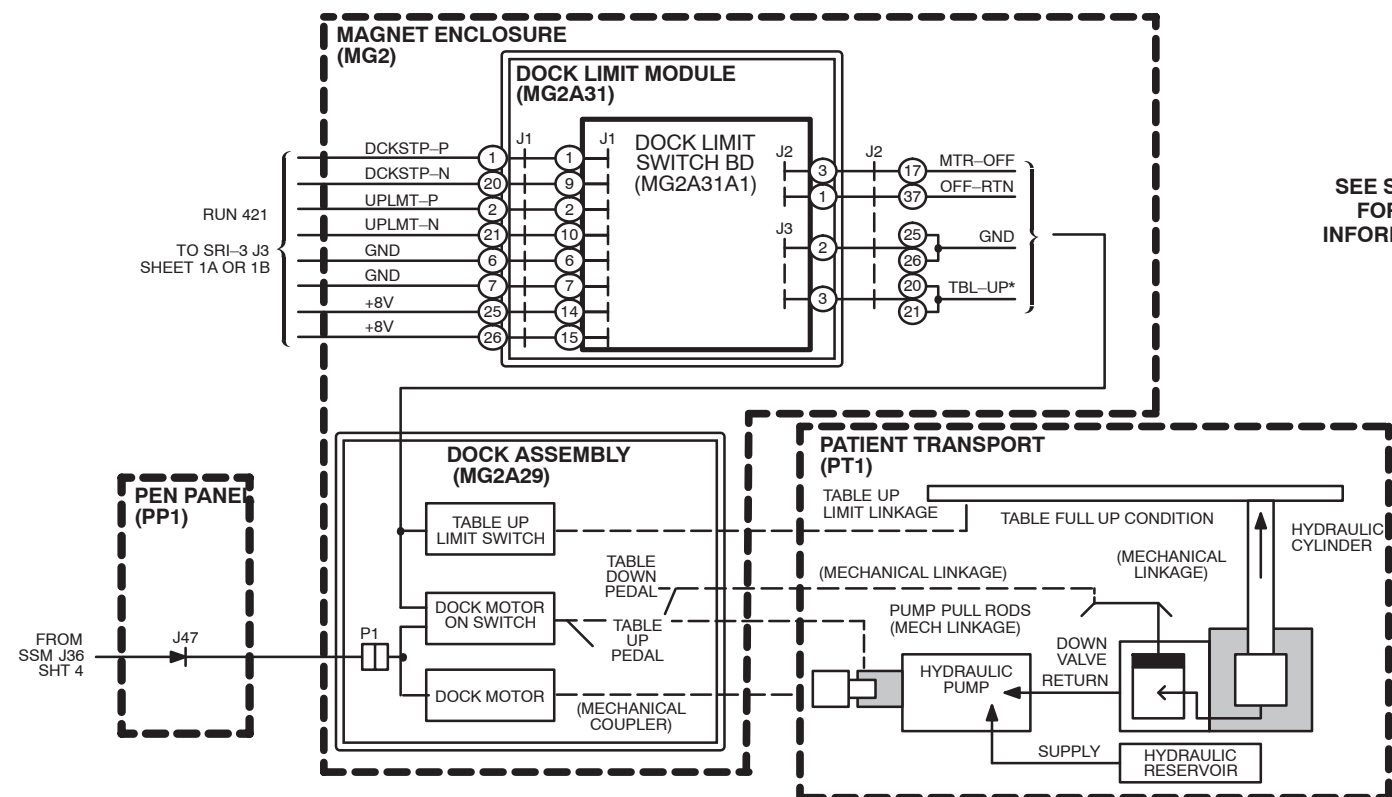
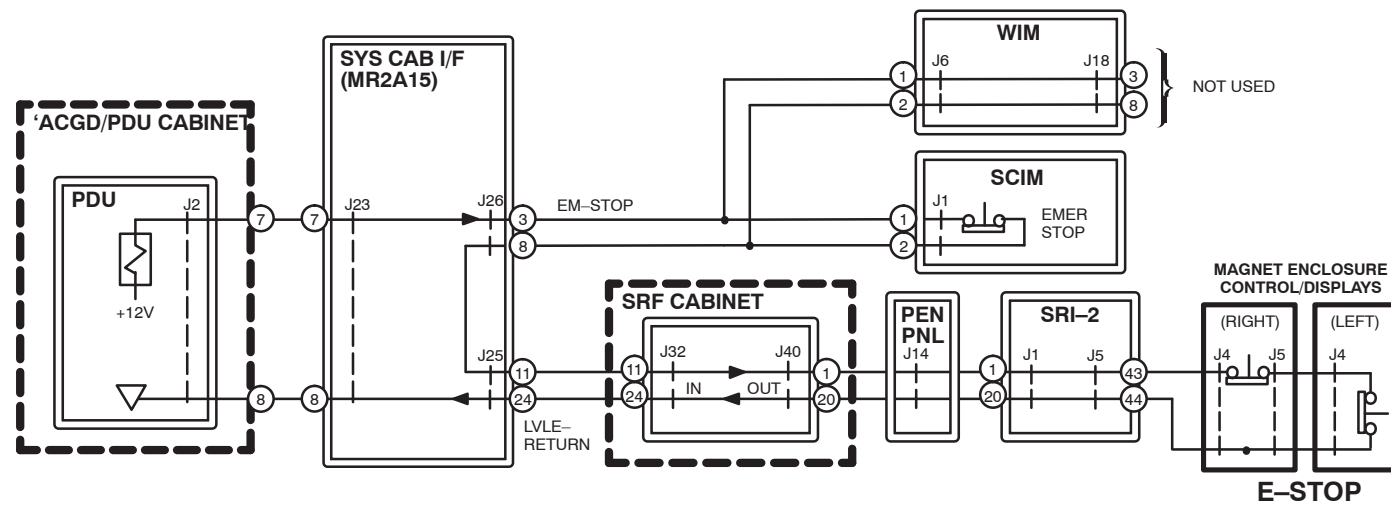
TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
1 – PATIENT HANDLING	
HD, HDx & HDxt PATIENT HANDLING (Forward Production or Pre-EXCITE Upgrades)	1-1
HD, HDx & HDxt PATIENT HANDLING (EXCITE Upgrades)	1-2
HD, HDx & HDxt PATIENT HANDLING (Forward Production or Pre-EXCITE Upgrades)	1-3
HD, HDx & HDxt PATIENT HANDLING (EXCITE Upgrades)	1-4
HD, HDx & HDxt PATIENT HANDLING POWER SUPPLY (PHPS)	1-5
HD PATIENT HANDLING – SSM	1-6
HD, HDx & HDxt PATIENT HANDLING – SSM (EXCITE Upgrades)	1-7
HD, HDx & HDxt PATIENT HANDLING – SSM	1-8

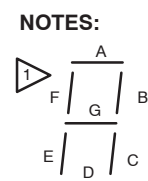






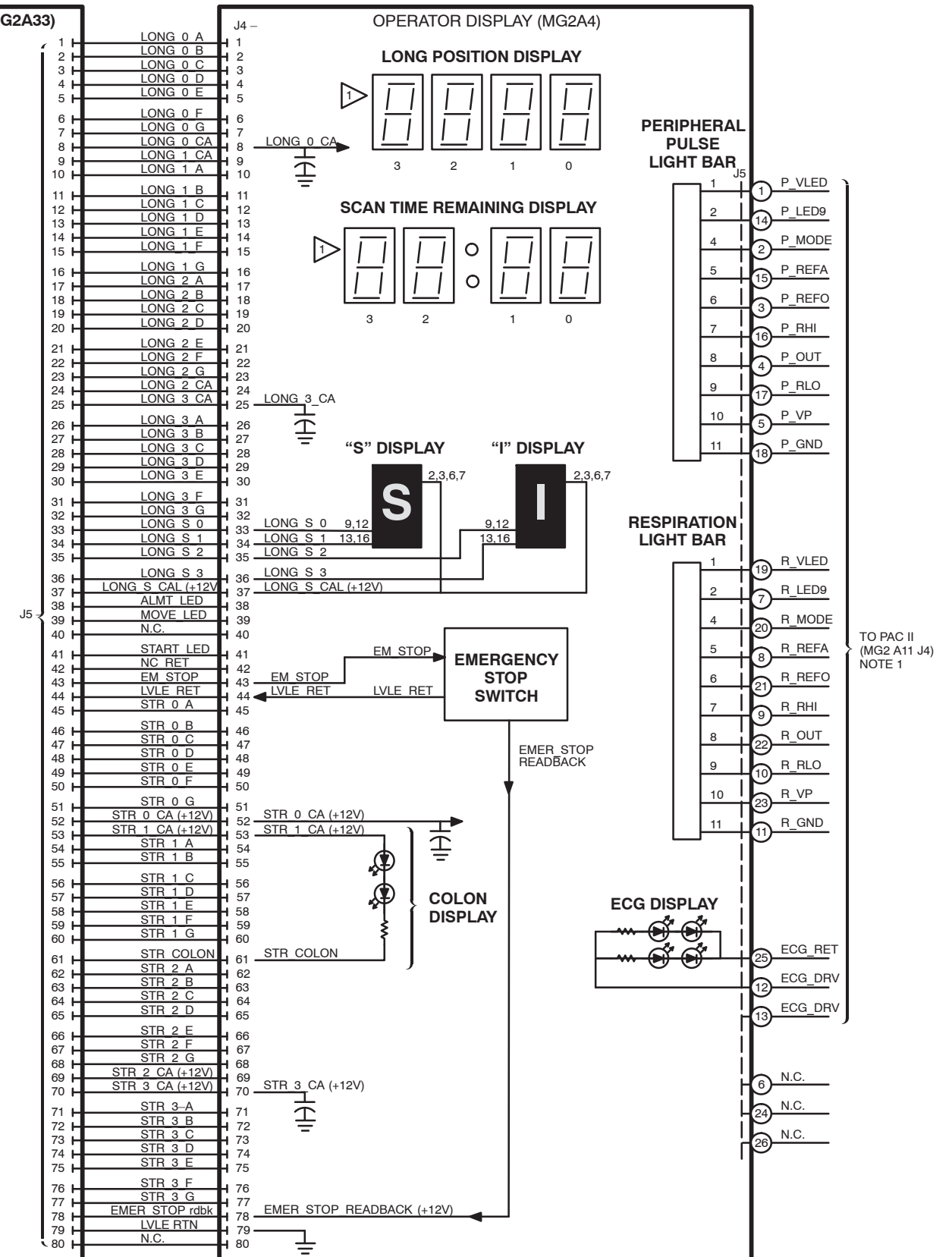


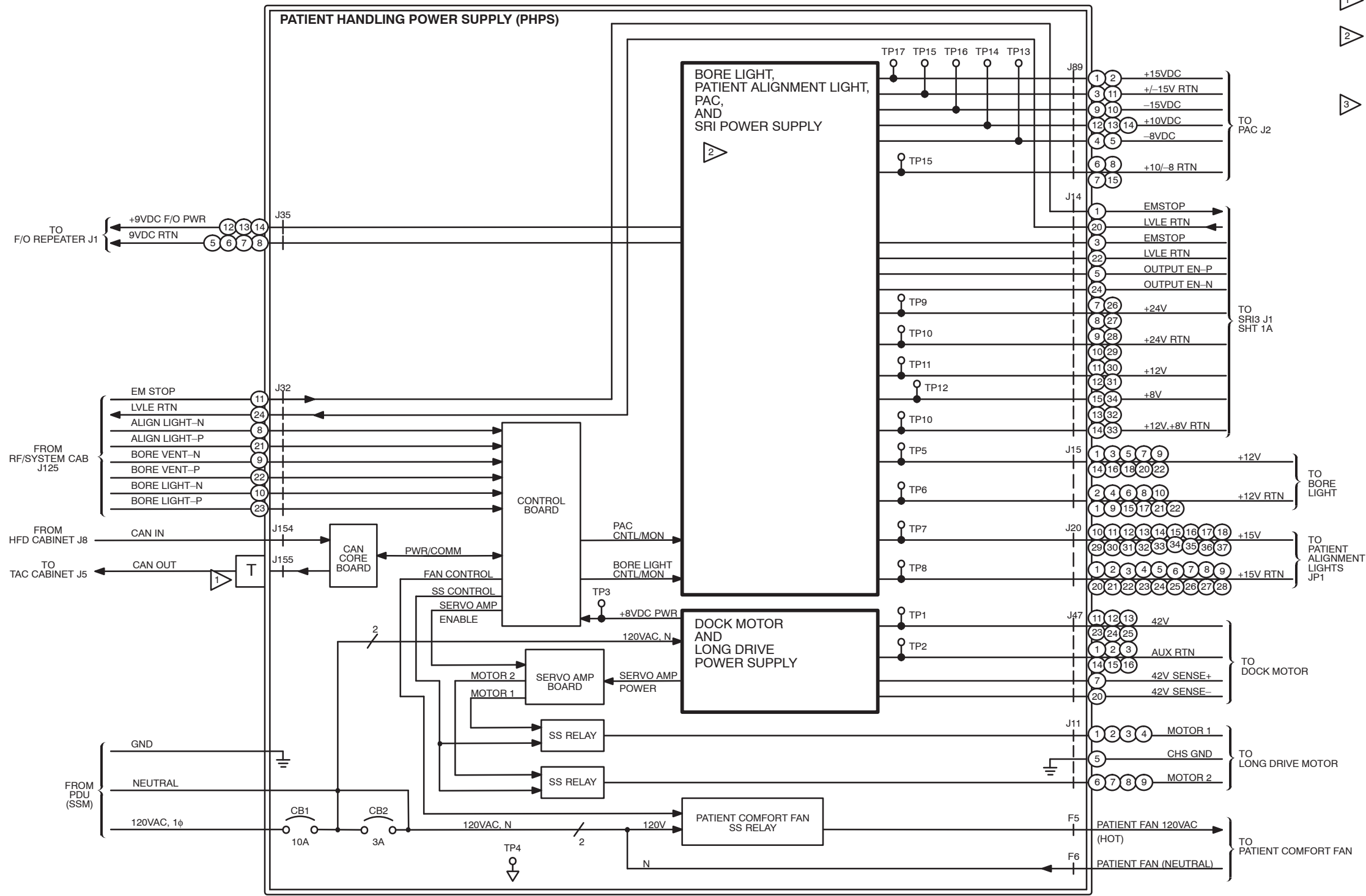
**SEE SHEET 1
FOR MORE
INFORMATION**



Refer to the following functional block diagrams for additional information. :

1) MGD/RRF/RF

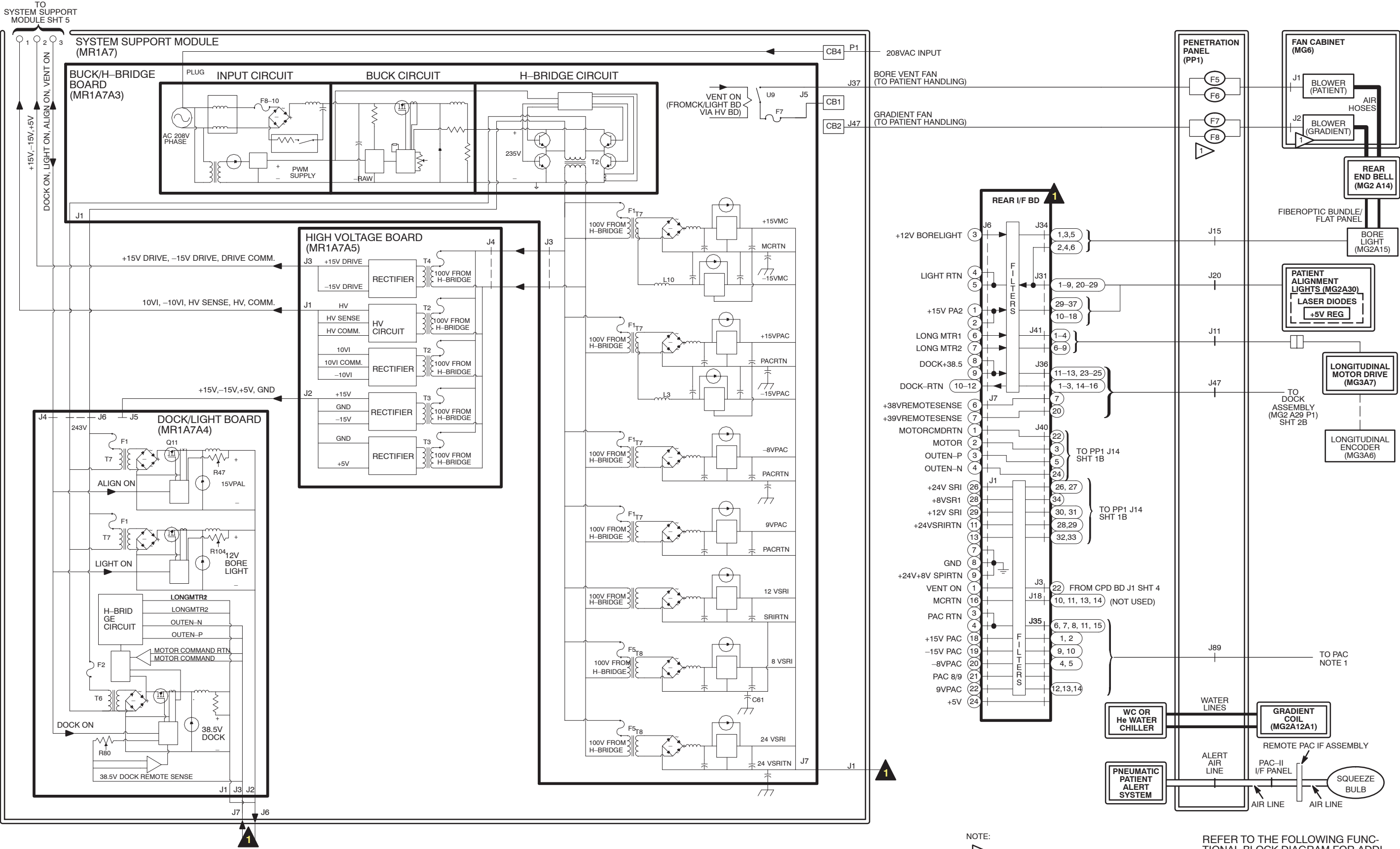




NOTES:

- 1 OPTIONAL: Used only if High Order Shim (HOS) installed.
PP1 PHPS J155 Terminator removed when HOS is installed.
- 2 Systems equipped with Laser Patient Alignment lights use one type of power supply while those equipped with Halogen patient Alignment lights must use another. See the Renewal Parts manual.

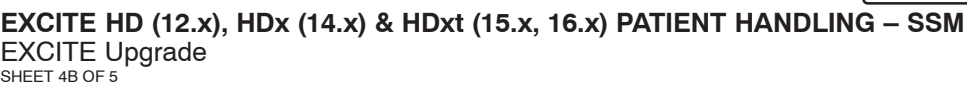
TEST POINT	FUNCTION
TP1	TABLE VOLTS (42VDC)
TP2	TAABLE COM
TP3	CONTROL BD +8V
TP4	+8V COM
TP5	BORE +12V
TP6	BORE COM
TP7	PAL +15V
TP8	PAL COM
TP9	SRI +24V
TP10	SRI COM
TP11	SRI +12V
TP12	SRI +8V
TP13	PAC -8V
TP14	PAC +10V
TP15	PAC COM
TP16	PAC -15V
TP17	PAC +15V



EXCITE HD (12.x) PATIENT HANDLING – PATIENT HANDLING – SSM
SHEET 4A OF 5

NOTE:
F7/F8 AND GRADIENT BLOWER PRESENT
ON SYSTEMS WITH BRM BODY COIL.

REFER TO THE FOLLOWING FUNC-
TIONAL BLOCK DIAGRAM FOR ADDI-
TIONAL INFORMATION:
NOTE:
1) MGD/RRF/RF





NOTE:
1) GRADIENT
2) MGD/RRF/RF

2 Present only on Forward Production or Pre-EXCITE upgrade systems.

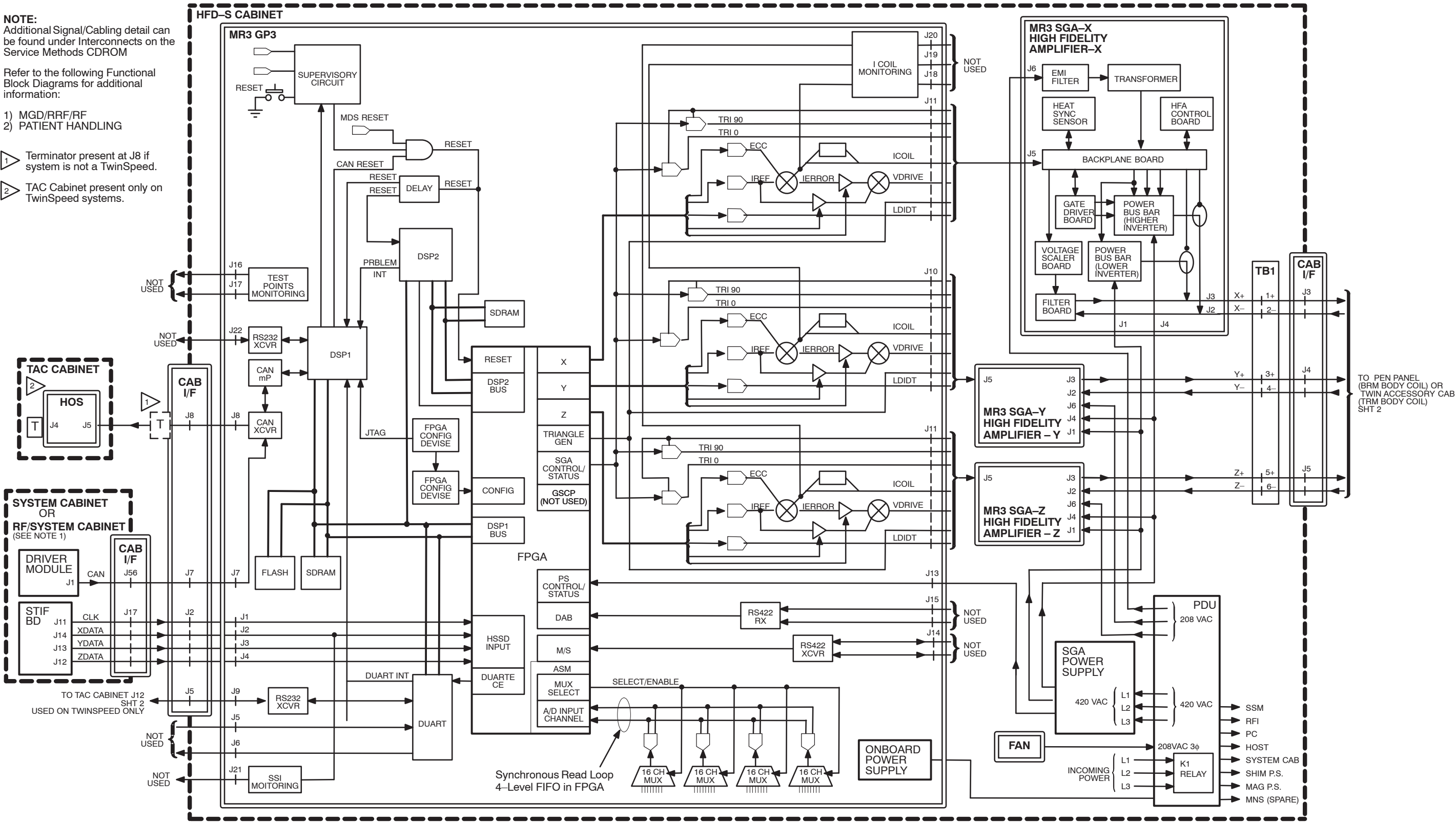


NOTE:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM

Refer to the following Functional Block Diagrams for additional information:

- 1) MGD/RRF/RF
- 2) PATIENT HANDLING

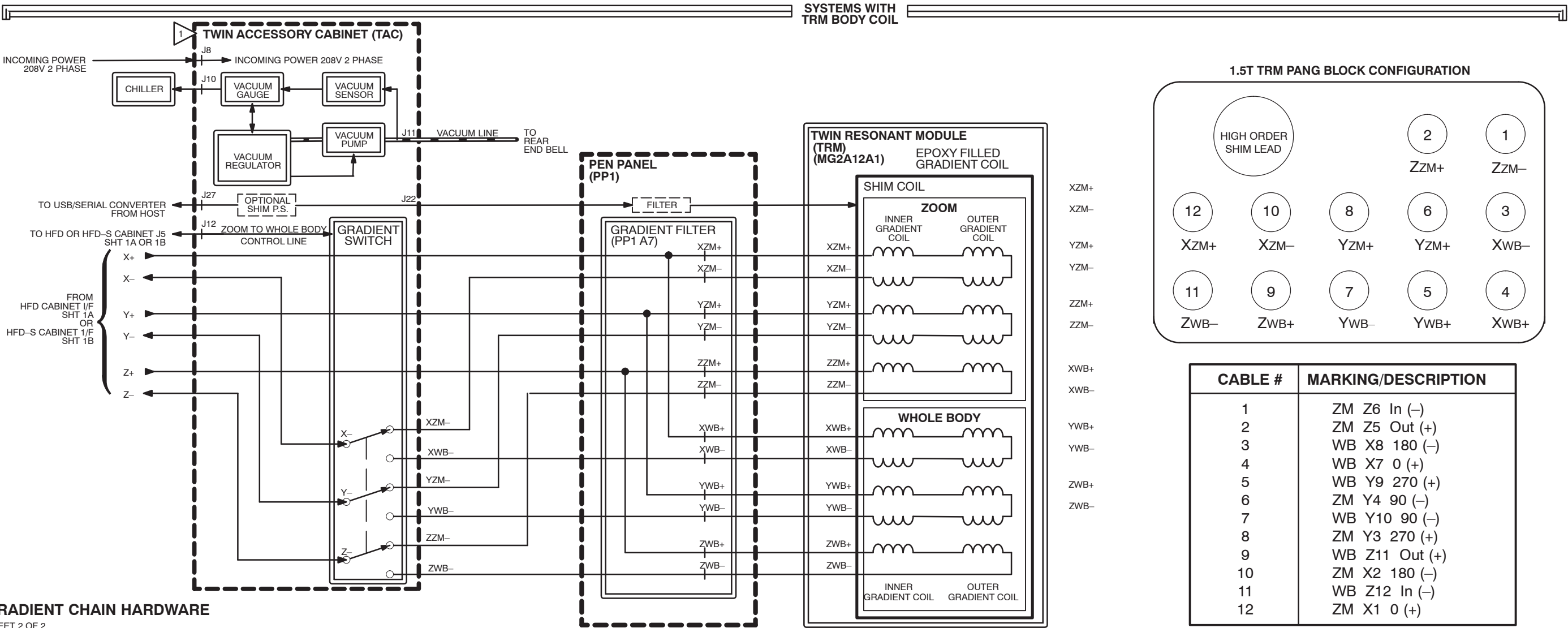
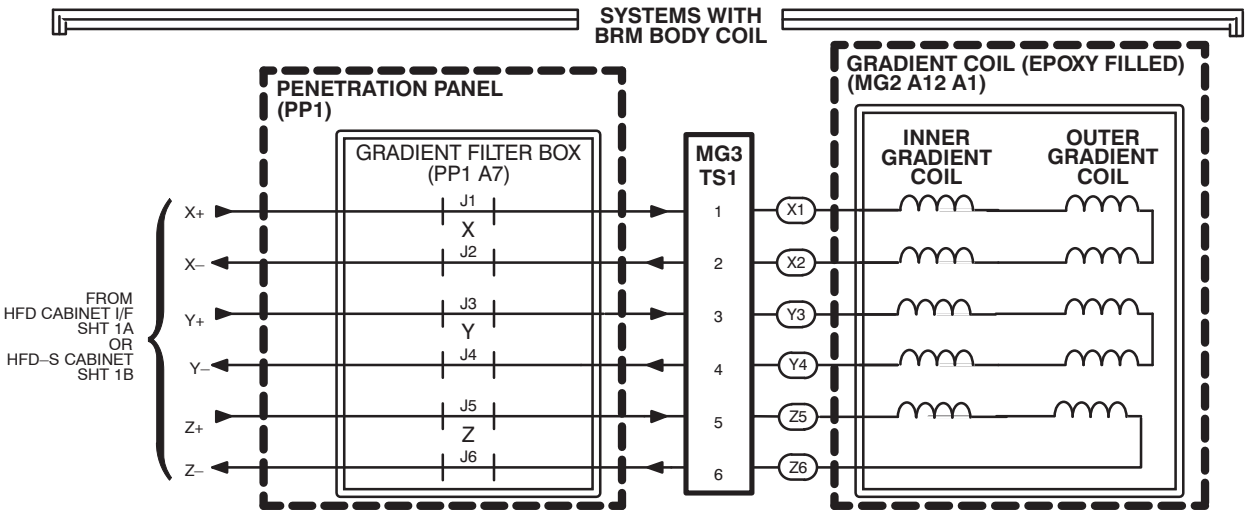
- 1 Terminator present at J8 if system is not a TwinSpeed.
- 2 TAC Cabinet present only on TwinSpeed systems.



NOTE:
Refer to the following Functional Block Diagrams for additional information:

1)OVERALL SYSTEM

1 Twin Accessory Cabinet is present only on TwinSpeed systems (with TRM body coil).



GRADIENT CHAIN HARDWARE

SHEET 2 OF 2

PDU

TABLE OF CONTENTS

Note:

PDU found in Vendor Manuals only. (SEE SERVICE METHODS CDROM)

MGD/RRF/RF (1.5T)

TABLE OF CONTENTS

<u>SECTION</u>	<u>PAGE</u>
1 – MGD/RRF/RF SUBSYSTEM (1.5T)	
HD MGD CHASSIS – HOST, RECEIVE, AND RECON FUNCTIONS (Forward Production & Upgrades)	1–1
HDx, HDxt MGD CHASSIS – HOST, RECEIVE, AND RECON FUNCTIONS (Forward Production & Upgrades)	1–2
HDx HDxt MGD CHASSIS – VOLUME RECON ENGINE (Forward Production & Upgrades)	1–3
HD, HDx & HDxt MGD CHASSIS – SEQUENCE SEGMENT	1–4
HD, HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (4 and 8 Channel Systems)	1–7
HD, HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (16 Channel – Forward Production & Upgrades)	1–9
HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (32 Channel – Forward Production & Upgrades)	1–11
HD, HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (4 and 8 Channel Systems)	1–12
HD, HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (Channels 1–8 of 16 Channel – Forward Production & Upgrades)	1–13
HD, HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (Channels 1–8 of 32 Channel – Forward Production & Upgrades)	1–14
HD, HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (Channels 9–16 of 16 or 32 Channel – Forward Production & Upgrades)	1–15
HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (Channels 17–24 of 32 Channel – Forward Production & Upgrades)	1–16
HDx & HDxt RRF CHASSIS INTERCONNECT BLOCK DIAGRAM (Channels 25–32 of 32 Channel – Forward Production & Upgrades)	1–17
HD, HDx & HDxt RRF RECEIVE CHAIN CONNECTIONS	1–18
HD DRIVER MODULE (Upgrade with SSM)	1–19
HD DRIVER MODULE (Forward Production, RFS Cabinet, no SSM)	1–20
HDx & HDxt DRIVER MODULE (Upgrade with SSM)	1–21
HDx & HDxt DRIVER MODULE (Forward Production, RFS Cabinet, no SSM)	1–22
HD, HDx & HDxt SRFD POWER CABINET – SYSTEM SUPPORT MODULE (Upgrades Only)	1–23
HD, HDx & HDxt SRFD CABINET – RFI AND SOLID STATE AMPS (Upgrades Only)	1–25
HD & HDx SRFD II POWER CABINET – SYSTEM SUPPORT MODULE (Upgrades Only)	1–26
HD, HDx & HDxt SRFD II RF AMPLIFIER	1–28
HD, HDx & HDxt MAGNET ROOM & RELATED COMPONENTS (12.x – Forward Production & Upgrades) (14.x, 15.x– Upgrades)	1–29

TABLE OF CONTENTS (Continued)

<u>SECTION</u>	<u>PAGE</u>
1 – MGD/RRF/RF SUBSYSTEM (1.5T) (Continued)	
HDx & HDxt MAGNET ROOM & RELATED COMPONENTS (14.x, 15.x – Forward Production)	1–30
HD & HDx MULTI-COIL MODULES	1–31
HDx & HDxt 8 CHANNEL & 16 CHANNEL MULTI-COIL	1–32
HDx & HDxt 32 CHANNEL MULTI-COIL	1–33
HDx & HDxt LPCA MULTI-COIL INTERFACE ASM	1–34
HD & HDx MULTI-COIL ADAPTER (J10)	1–35
HD, HDx & HDxt PHYSIOLOGICAL ACQUISITION CONTROLLER II	1–36
HD, HDx & HDxt 16 CHANNEL SWITCH (1.5T)	1–37
HDx & HDxt 8 DEPOPULATED 8 CHANNEL SWITCH (1.5T)	1–38
HD 8 CHANNEL MULTICOIL “A” AND “B” CONNECTOR	1–39
HDx & HDxt MULTICOIL “A+” CONNECTOR (Forward Production)	1–40
HDx & HDxt MULTICOIL “B8” CONNECTOR (Forward Production)	1–41
HDx & HDxt MULTICOIL “B16” AND “C16” CONNECTOR (Forward Production)	1–42
HD, HDx & HDxt ASC CHASSIS	1–43
HD, HDx & HDxt MNS RF Amplifier (Optional Forward Production MNS)	1–44
HD, HDx & HDxt MNS Interconnects (Optional Forward Production MNS)	1–45
HD, HDx & HDxt MNS Interconnects (Optional MNS Upgrade Only)	1–46
HD MCR II TOOL AND CABLE	1–47
HDx & HDxt MCR III TOOL AND CABLE	1–48
HD MCR II TOOL – 8 CHANNEL CONNECTOR	1–49
HDx & HDxt MCR III TOOL – 8 CHANNEL CONNECTOR	1–50

NOTE:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

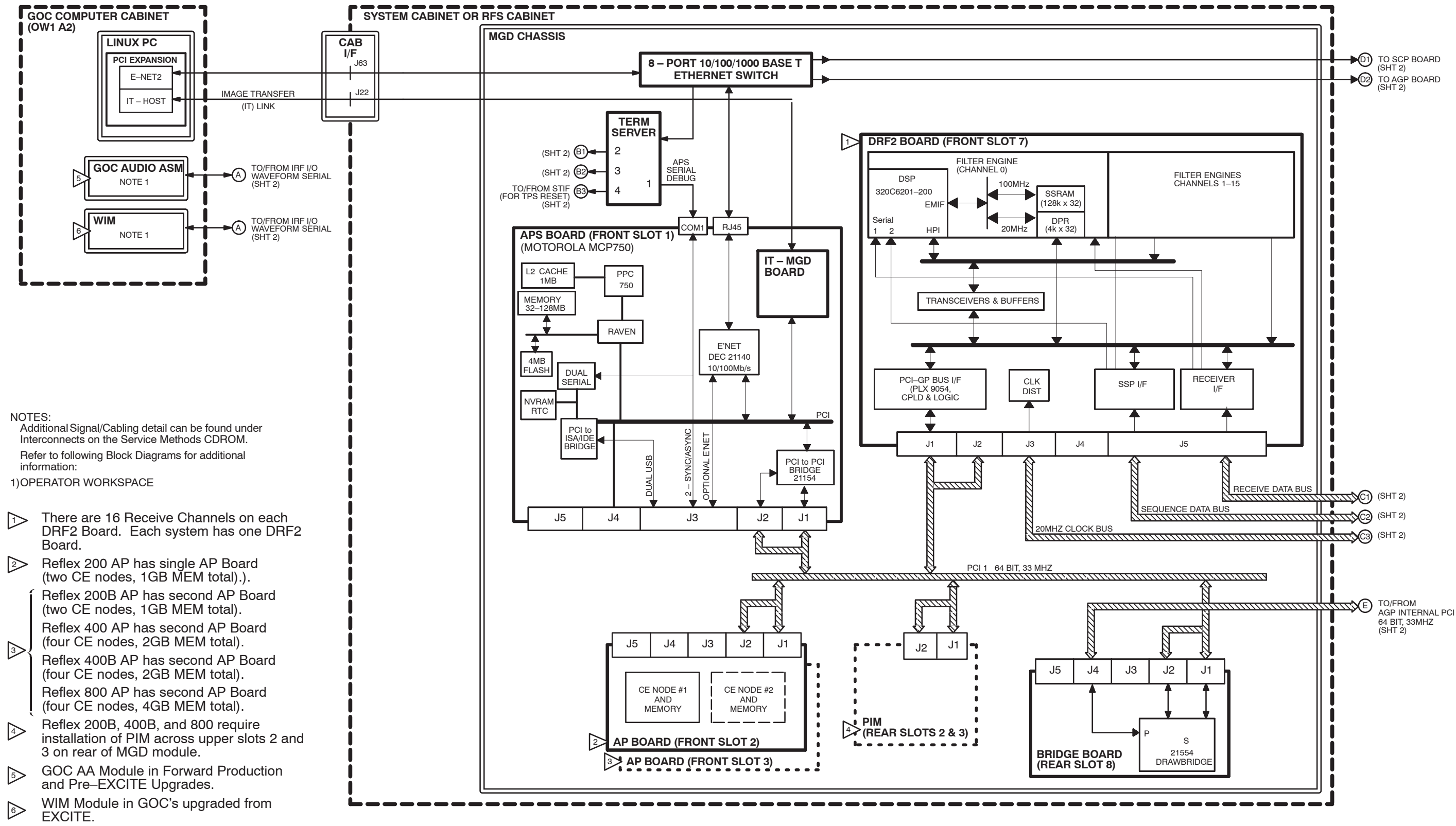
2 – MGD/RRF/RF SUBSYSTEM SUPPLEMENTAL SCHEMATICS (1.5T)

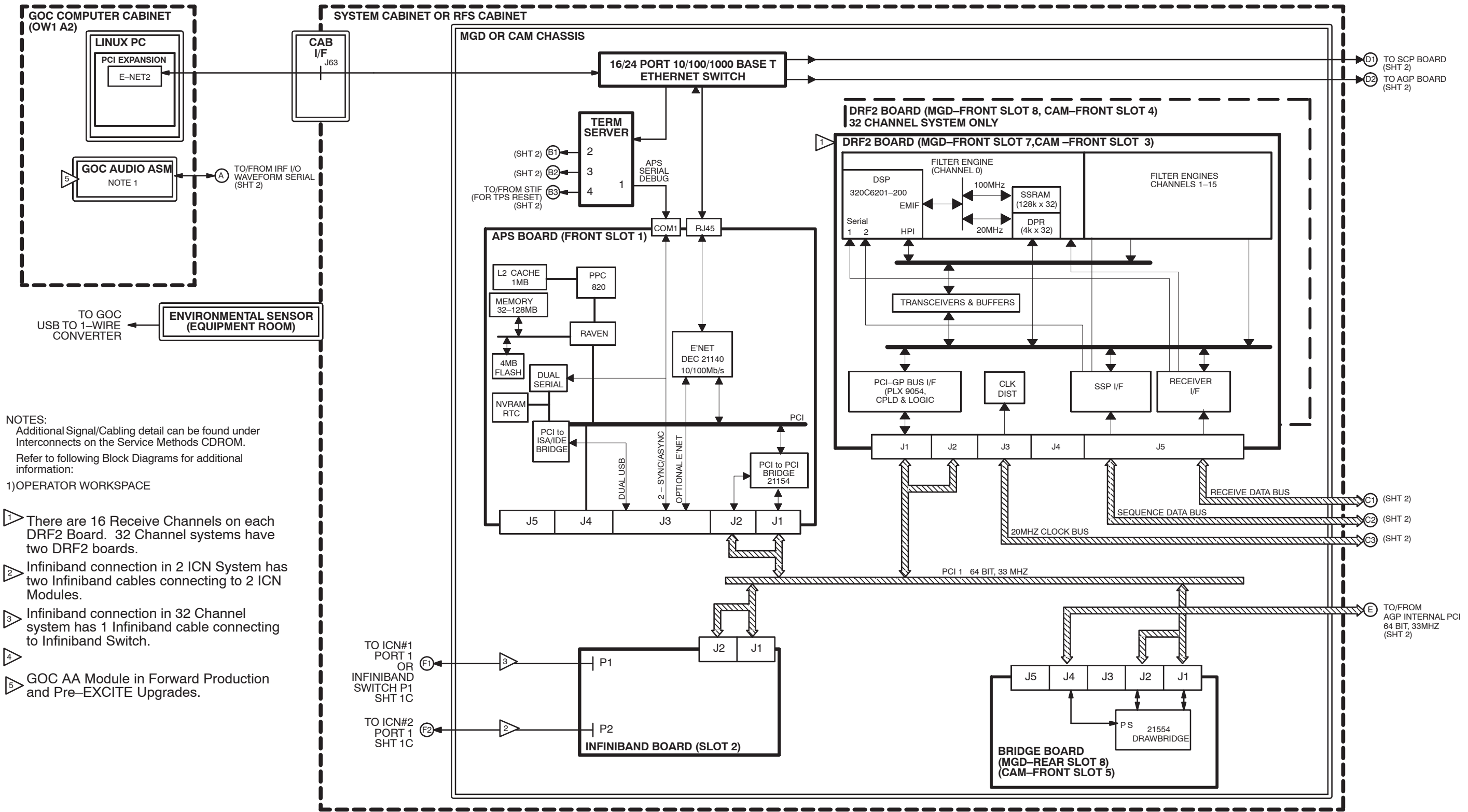
HEAD & BODY PREAMP SCHEMATIC (46–264442S)	2–1
HV DYNAMIC DISABLE FILTER MODULE SCHEMATIC	2–2
HEAD T/R PREAMP FILTER BOARD (2295844, 3 SHEETS)	2–3
T/R BOARD SCHEMATIC (2133404S)	2–6
MULTICOIL MODULE (2295846)	2–7
MOTHER BOARD (2295860, 5 SHEETS)	2–8

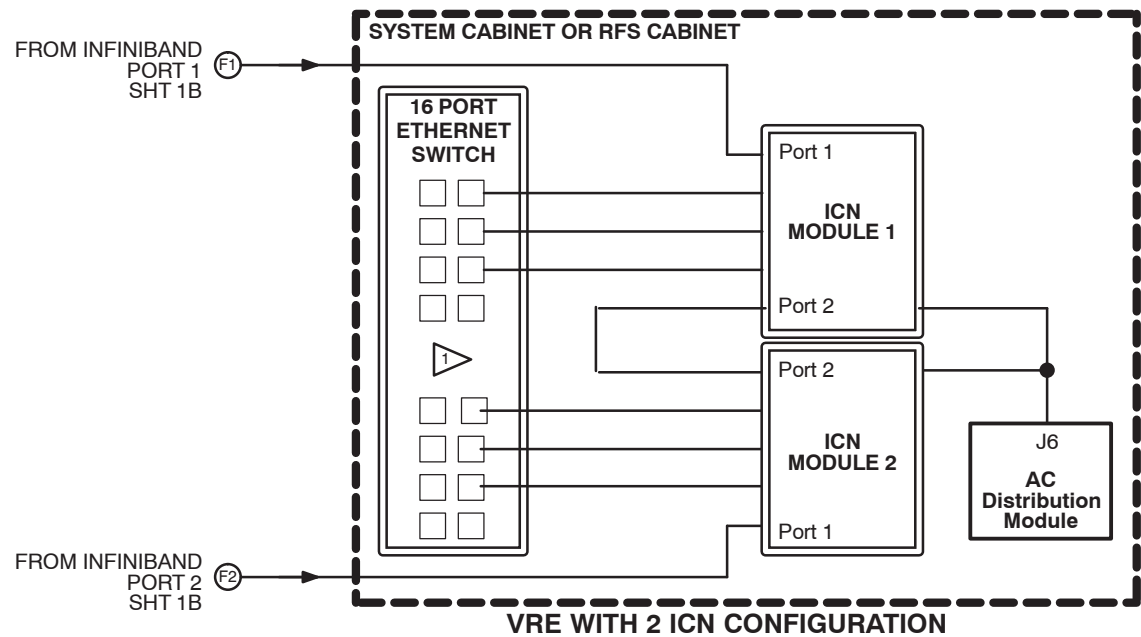
NOTE:

Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.

SECTION 1 – MGD/RRF/RF SUBSYSTEM (1.5T)

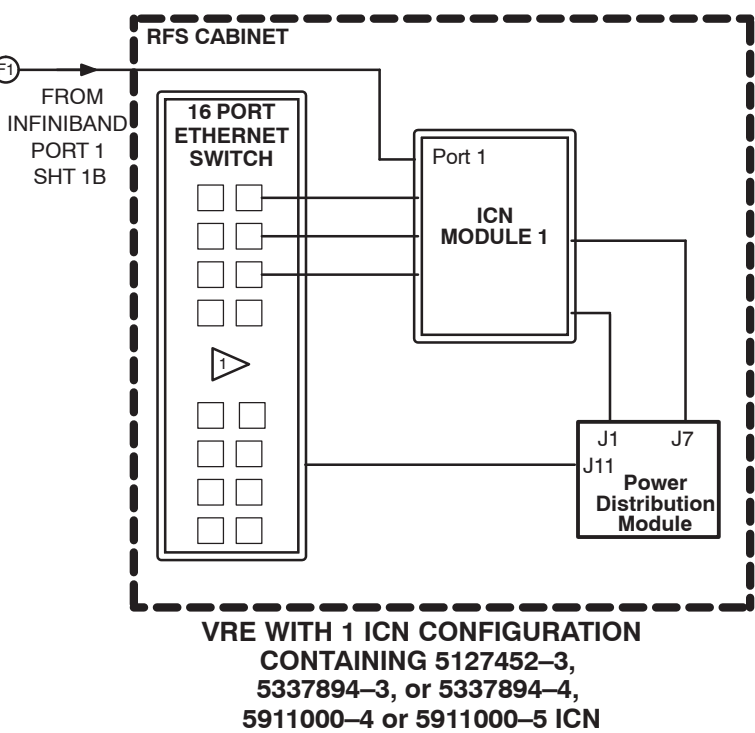
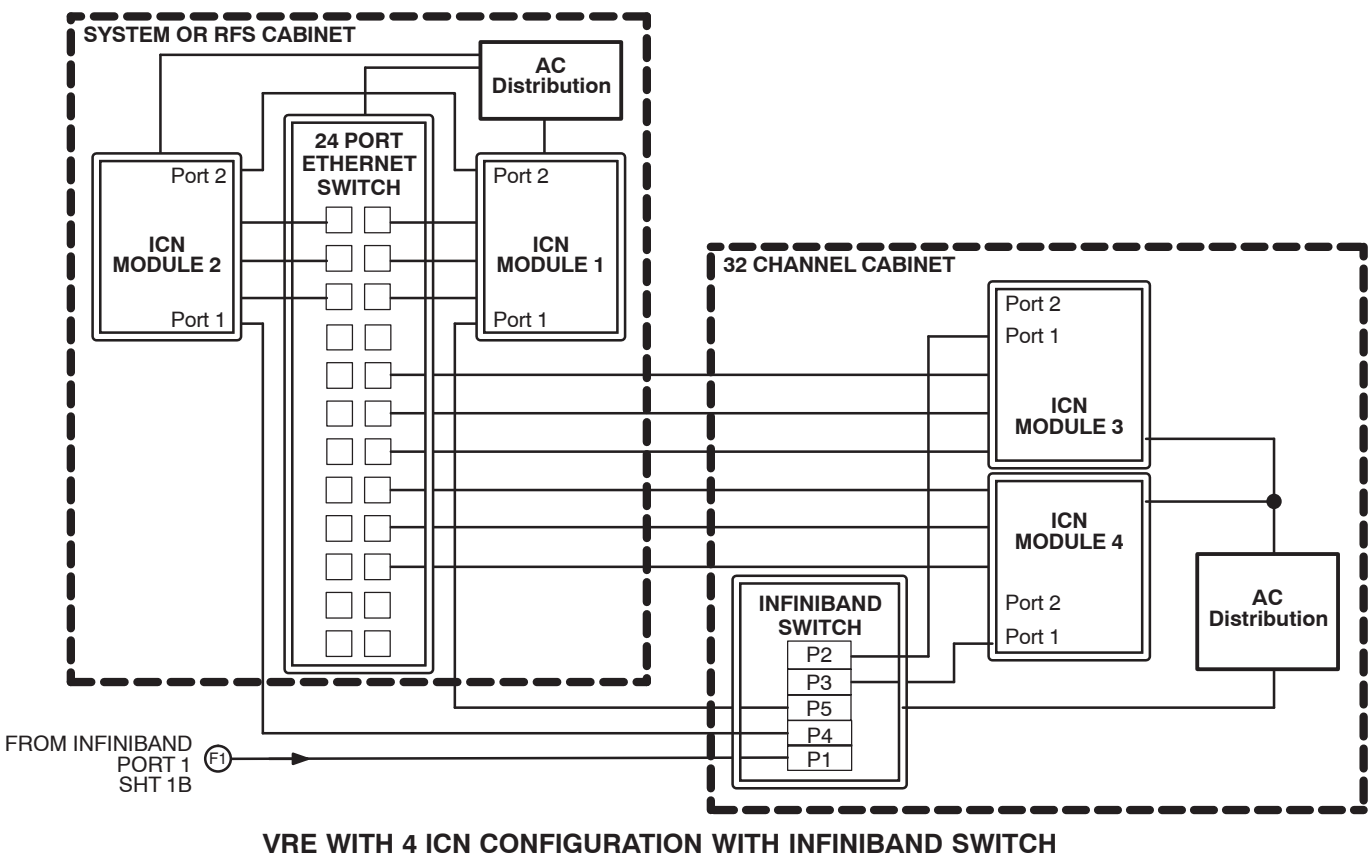




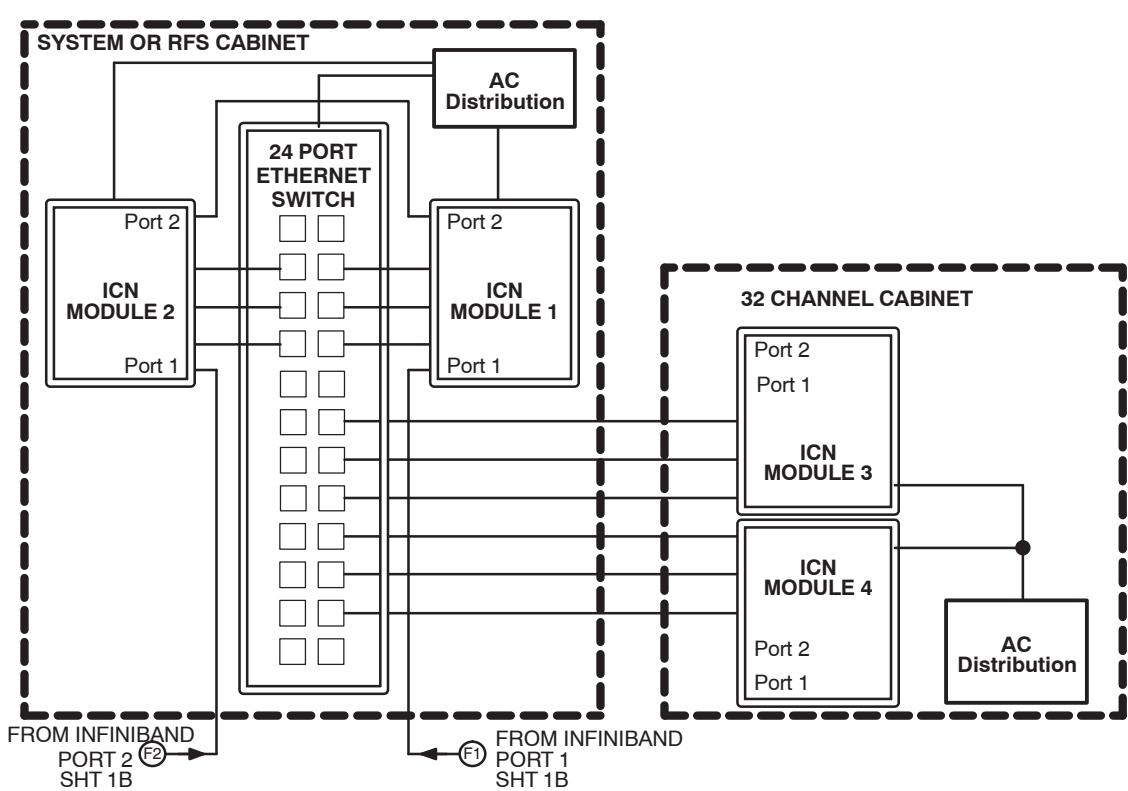
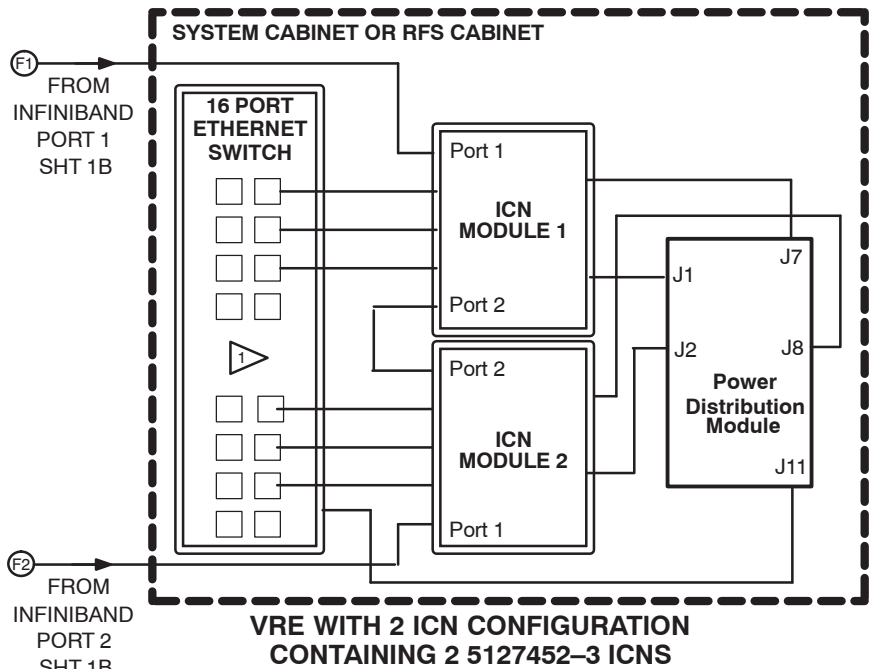


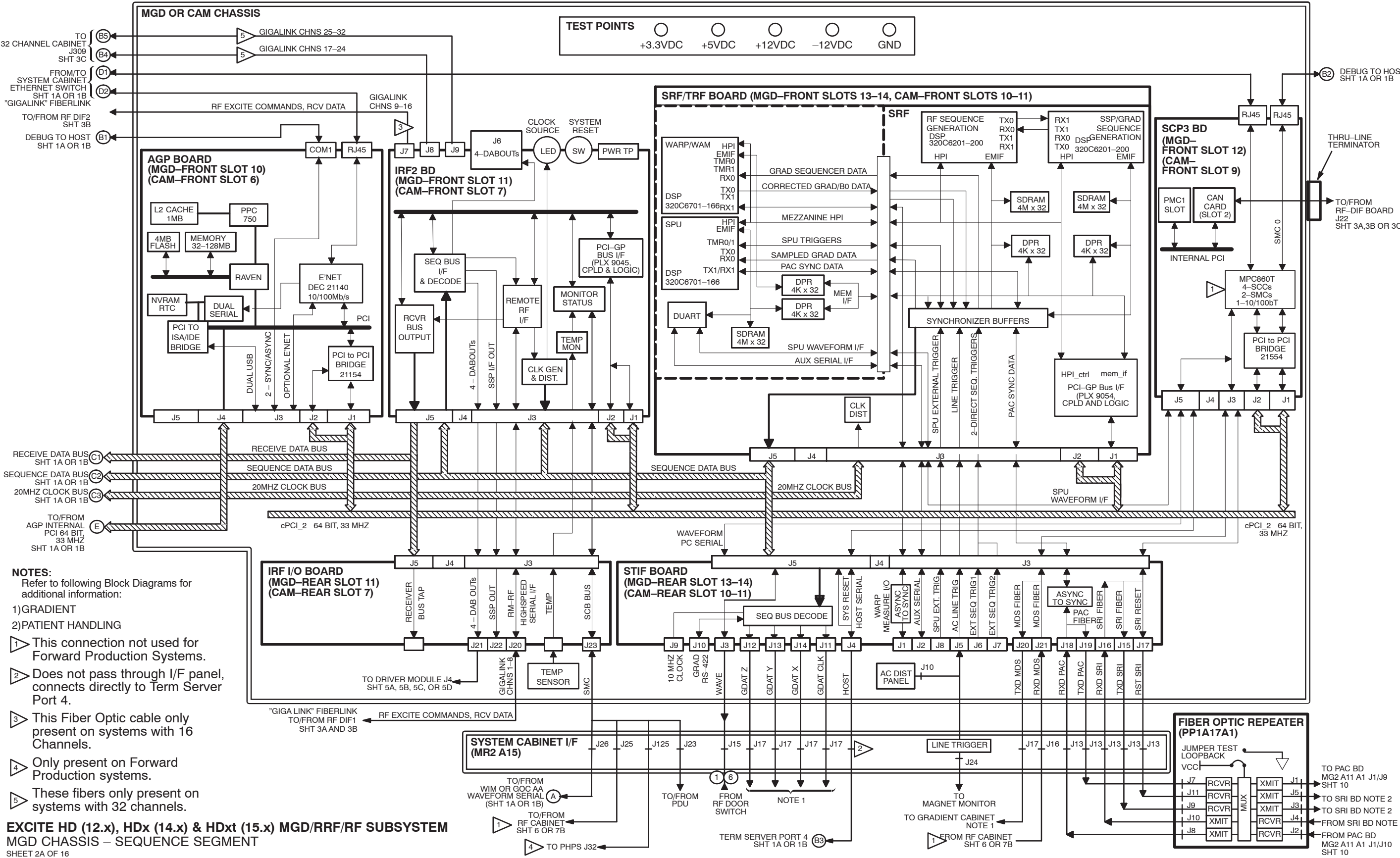
NOTES:
Additional Signal/Cabling detail can be found under Interconnects on the Service Methods CDROM.
Refer to following Block Diagrams for additional information:
1) OPERATOR WORKSPACE

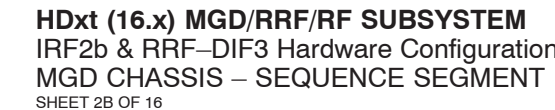
Other connections to 16/24 port Ethernet Switch on sheet 1B.



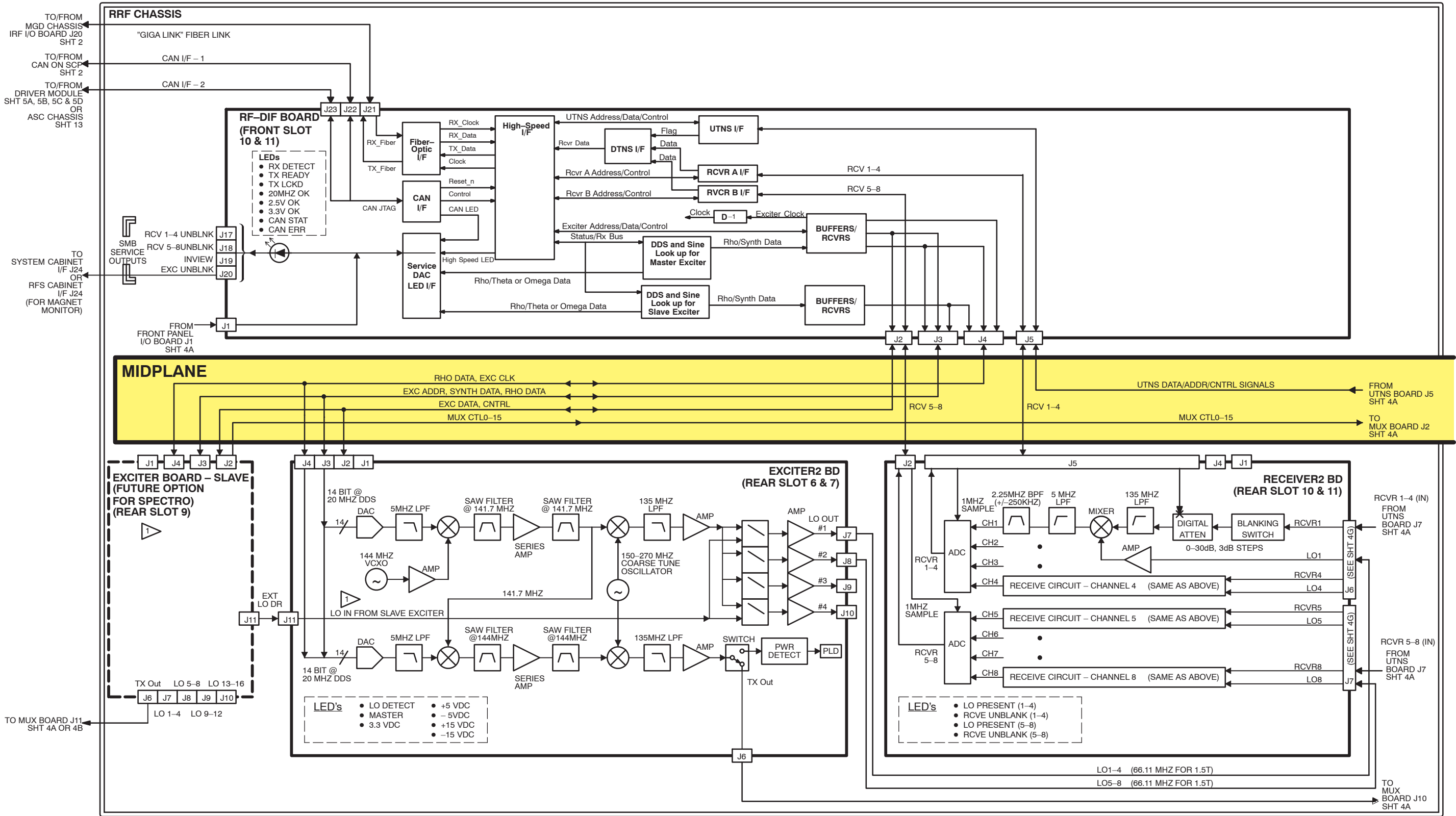
HDx (14.x) & HDxt (15.x, 16.x or Later) MGD/RRF/RF SUBSYSTEM
(Forward Production & Upgrades)
VOLUME RECON ENGINE
SHEET 1C OF 16







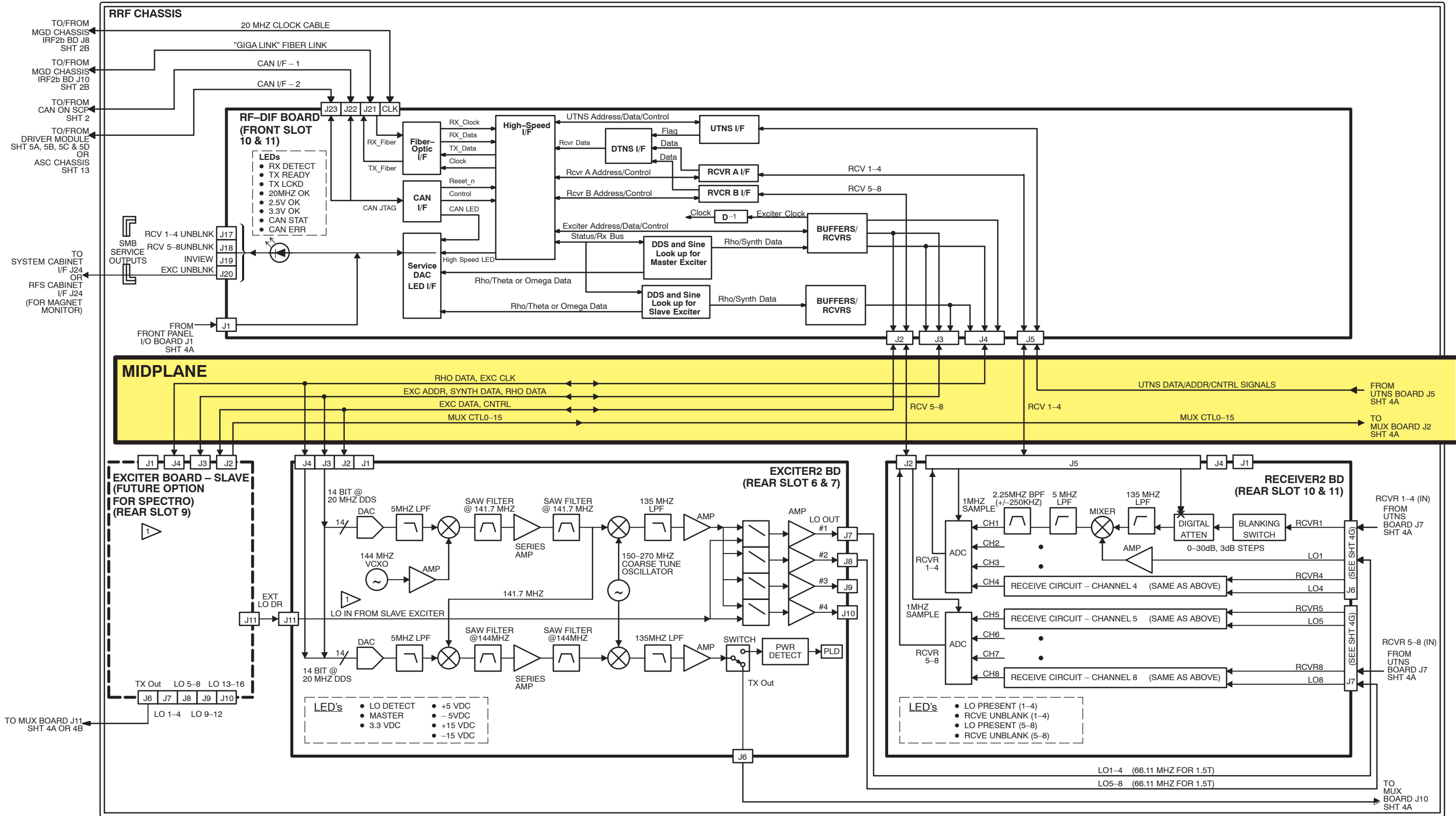




EXCITE HD (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(4 and 8 Channel Systems)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 4
SHEET 3A OF 16

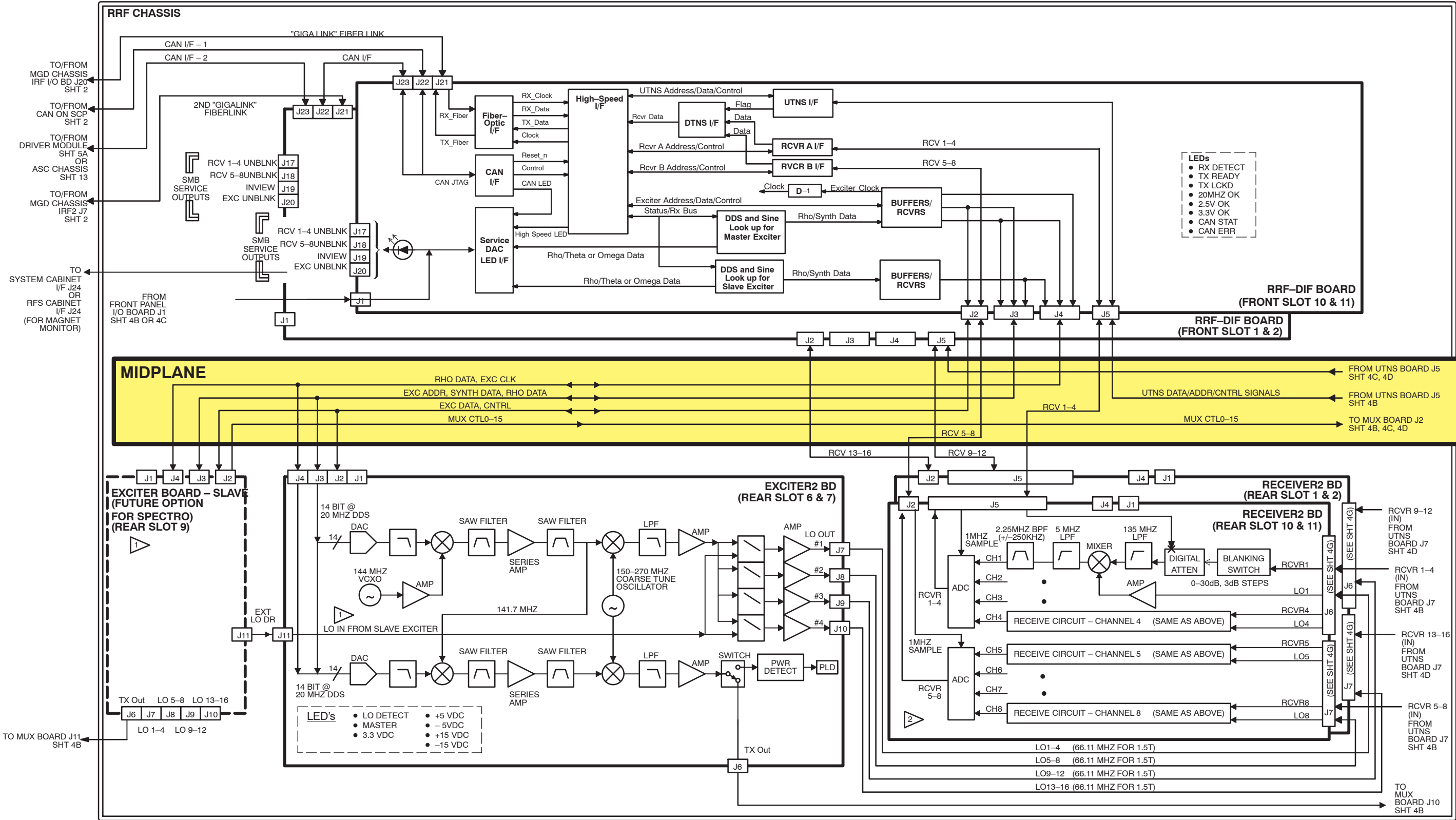
NOTE:

▶ Slave Exciter is a future option for Spectro.



\HDxt (16.x or Later) MGD/RRF/RF SUBSYSTEM
IRF2b and RRF-DIF3 Hardware Configuration (8 Channel Systems)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 2 OF 4
SHEET 3B OF 16

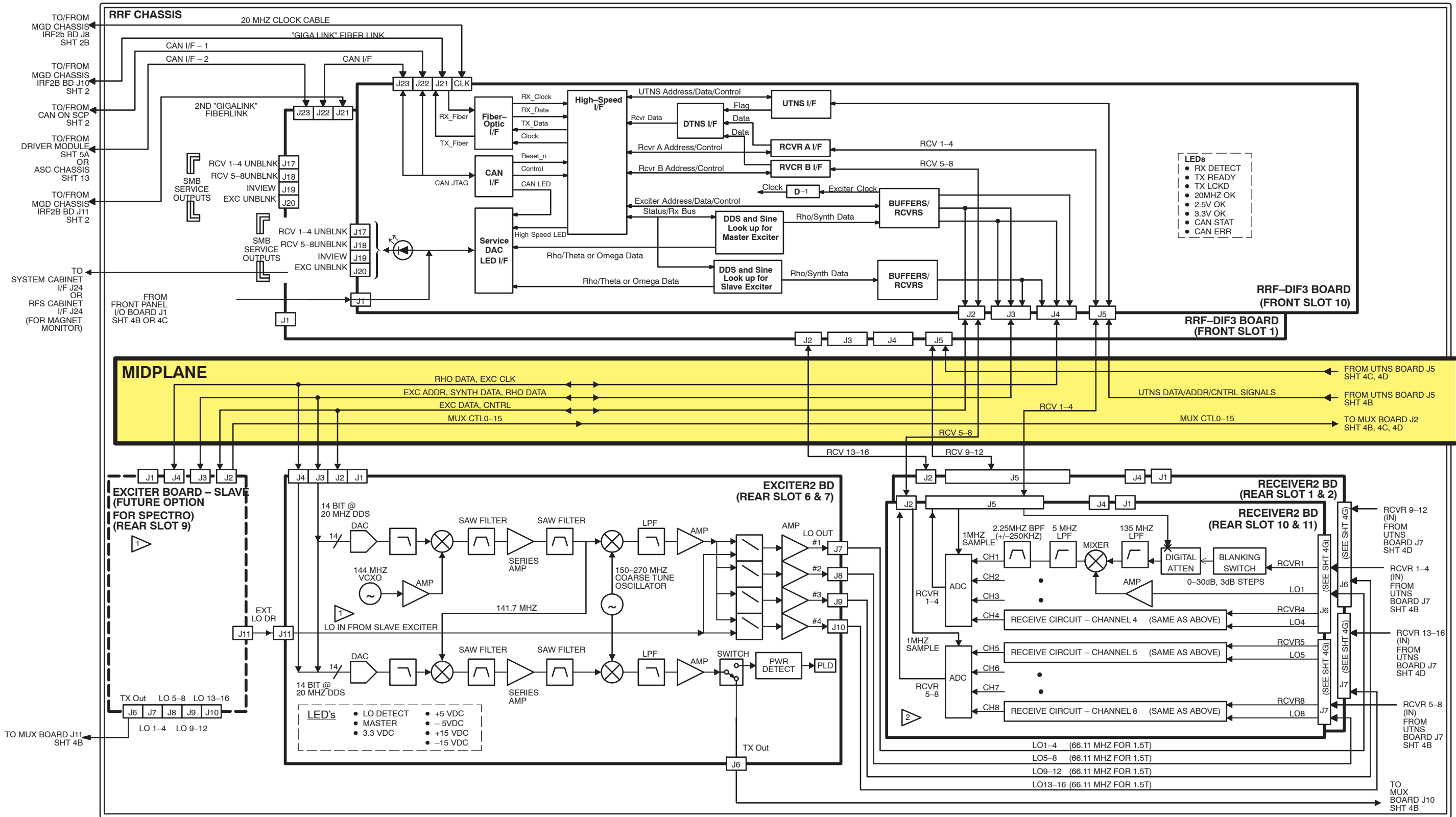
NOTE:
▶ Slave Exciter is a future option for Spectro.



EXCITE HDx (12.x), HDx (14.x) & HDxt (15.x) MGD/RRF/RF SUBSYSTEM
(16 Channel – Forward Production & Upgrades)
2nd RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 3 OF 4
SHEET 3C OF 16

NOTE:
1 Slave Exciter is a future option for Spectro.

2 LED's
• LO PRESENT (1-4)
• RCVE UNBLANK (1-4)
• LO PRESENT (5-8)
• RCVE UNBLANK (5-8)

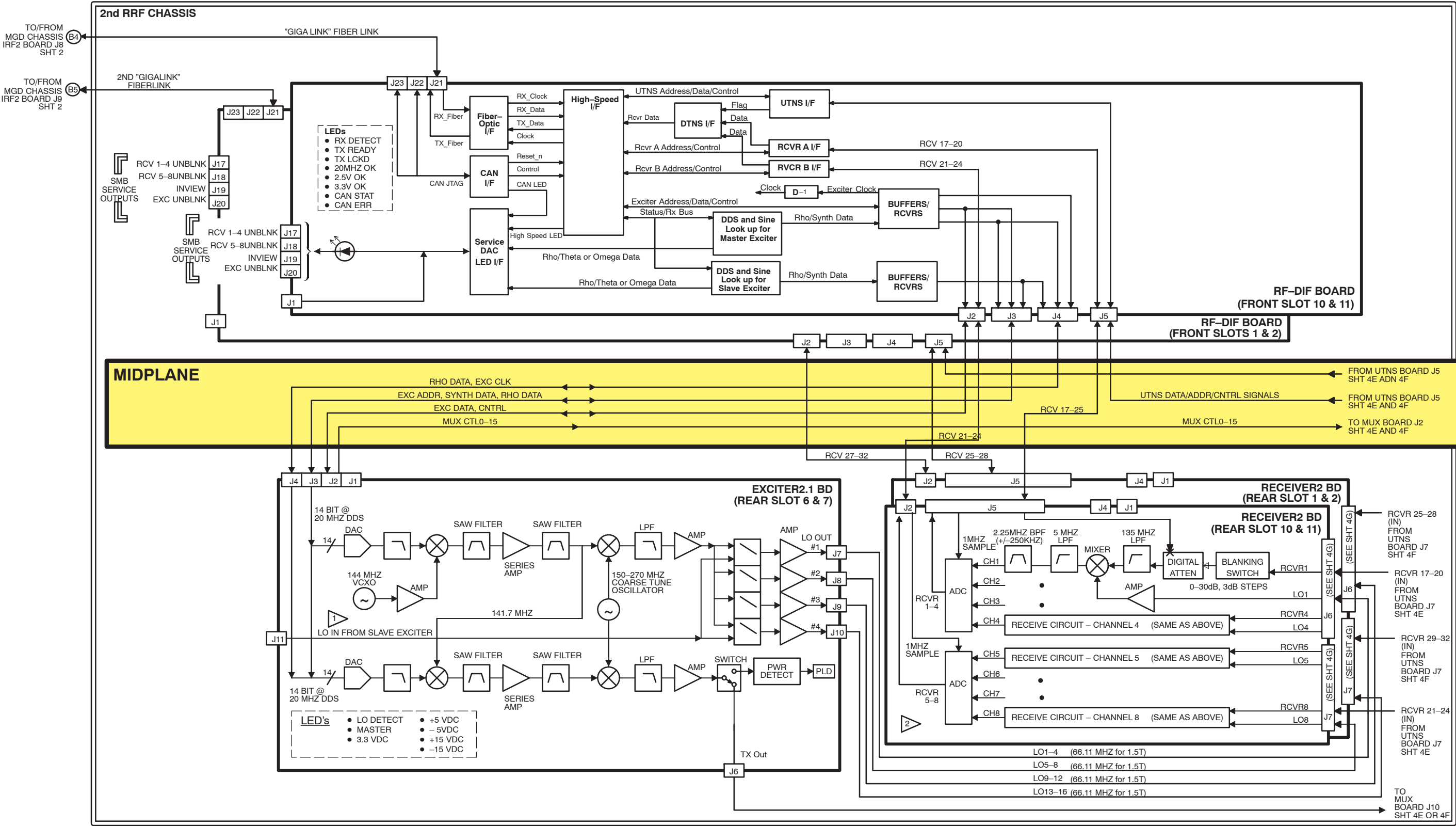


HDxt (16.x or Later) MGD/RRF/RF SUBSYSTEM
IRF2b & RRF-DIF3 Hardware Configuration (16 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 2B
SHEET 3D OF 16

NOTE:
1 Slave Exciter is a future option for Spectro.

2 LED's

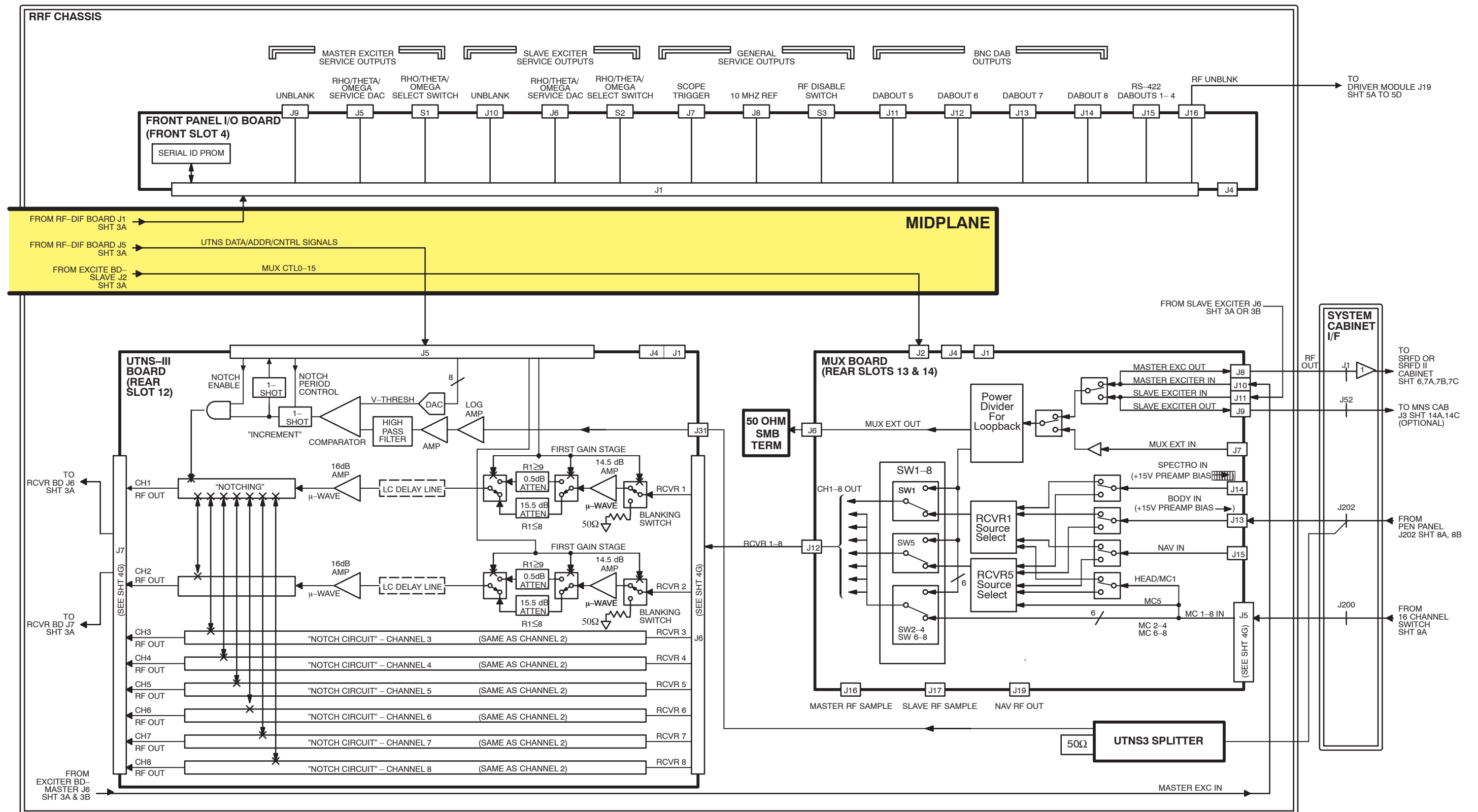
- LO PRESENT (1-4)
- RCVE UNBLANK (1-4)
- LO PRESENT (5-8)
- RCVE UNBLANK (5-8)



HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
(32 Channel – Forward Production & Upgrades)
2nd RRF CHASSIS INTERCONNECT BLOCK DIAGRAM 1 OF 4C
SHEET 3E OF 16

NOTE: LED's

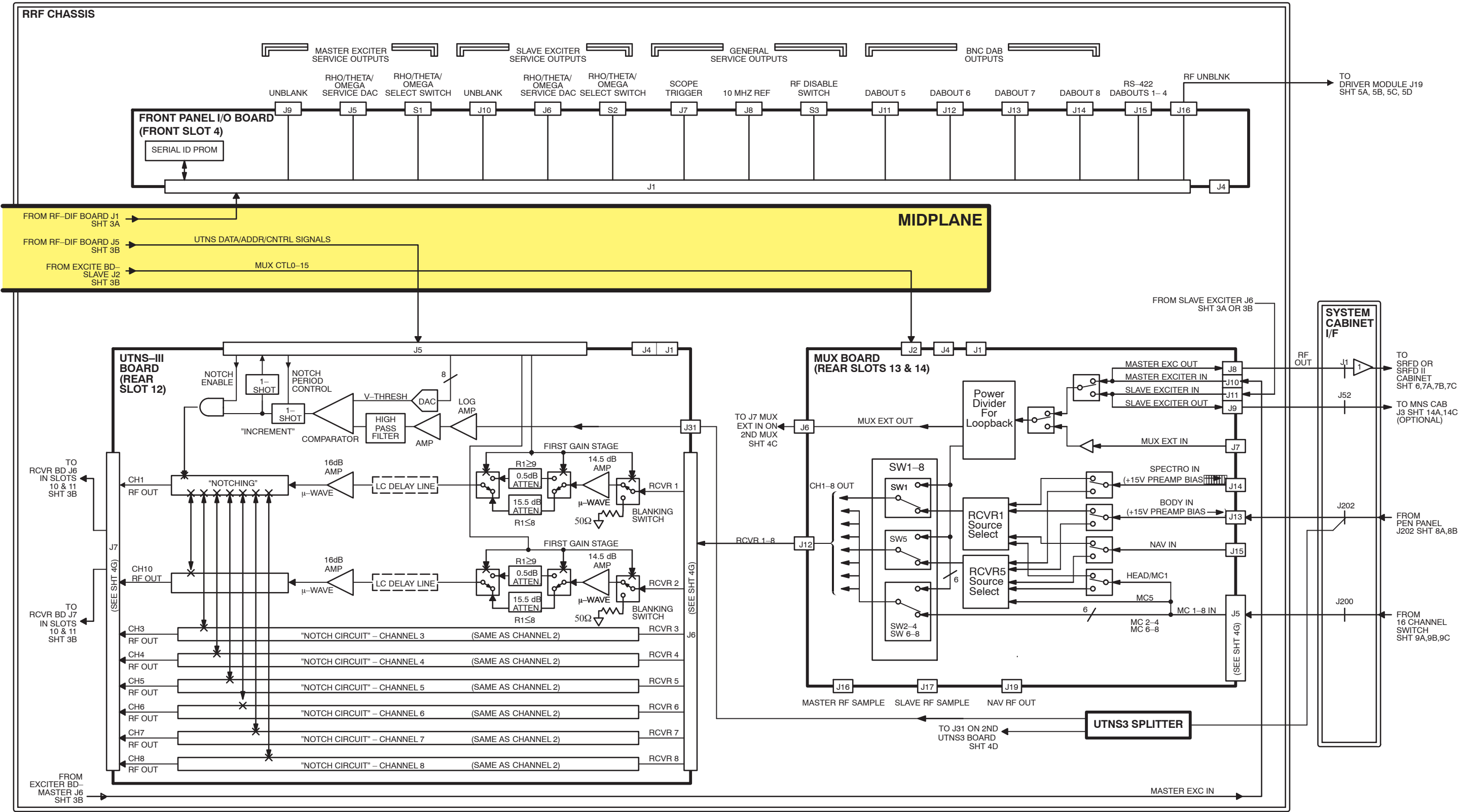
- LO PRESENT (1-4)
- RCVE UNBLANK (1-4)
- LO PRESENT (5-8)
- RCVE UNBLANK (5-8)



EXCITE HD (12.x), HDx (14.x) & HDxt (1.5.x, 16.x) MGD/RRF/RF SUBSYSTEM
(4 and 8 Channel Systems)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM
 SHEET 4A OF 16

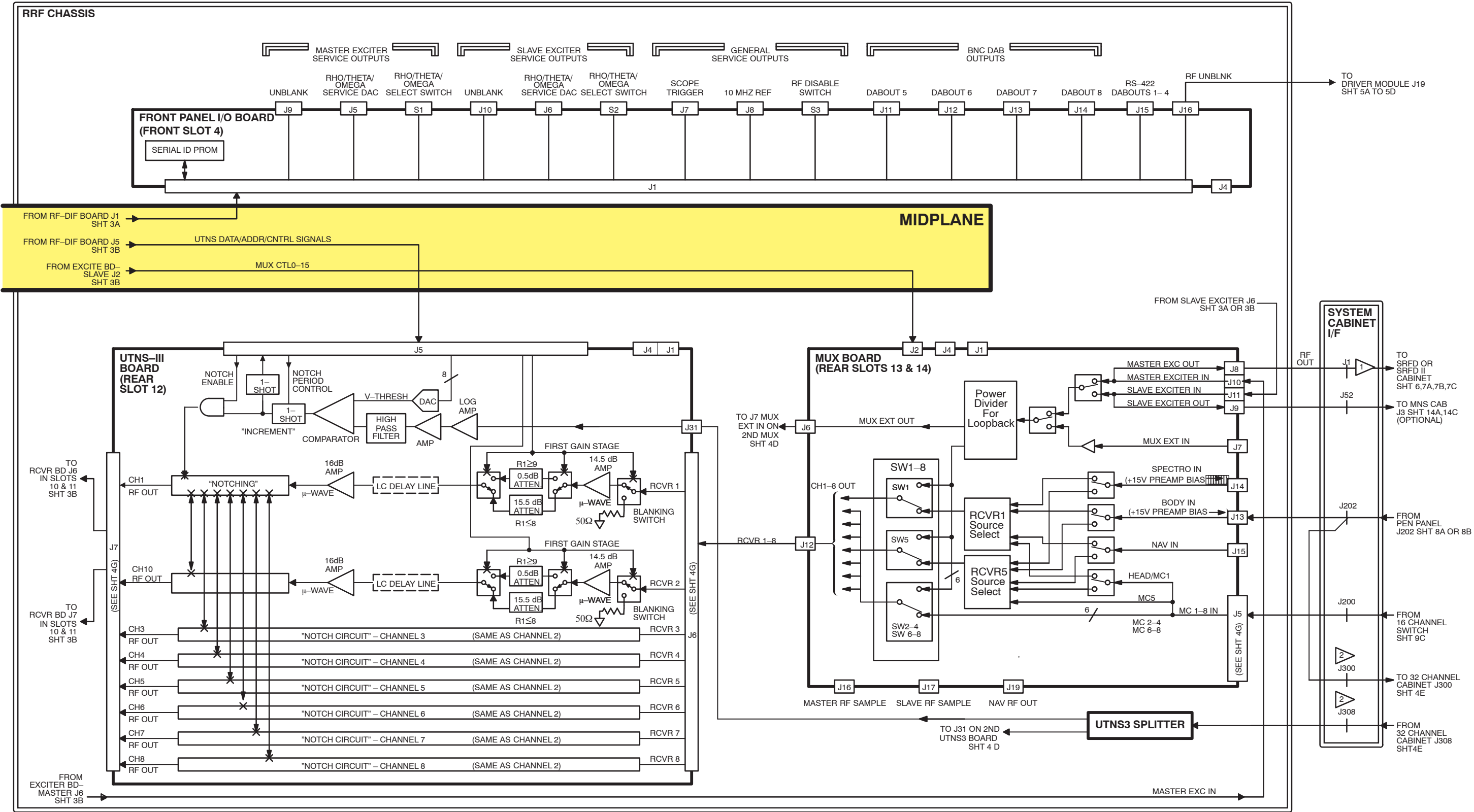
NOTE:

1 J1 not present in RF System Cabinet.



EXCITE HD (12.x), HDx (14.x) & HDxt (1.5.x, 16.x) MGD/RRF/RF SUBSYSTEM
(Channels 1-8 of 16 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM
SHEET 4B OF 16

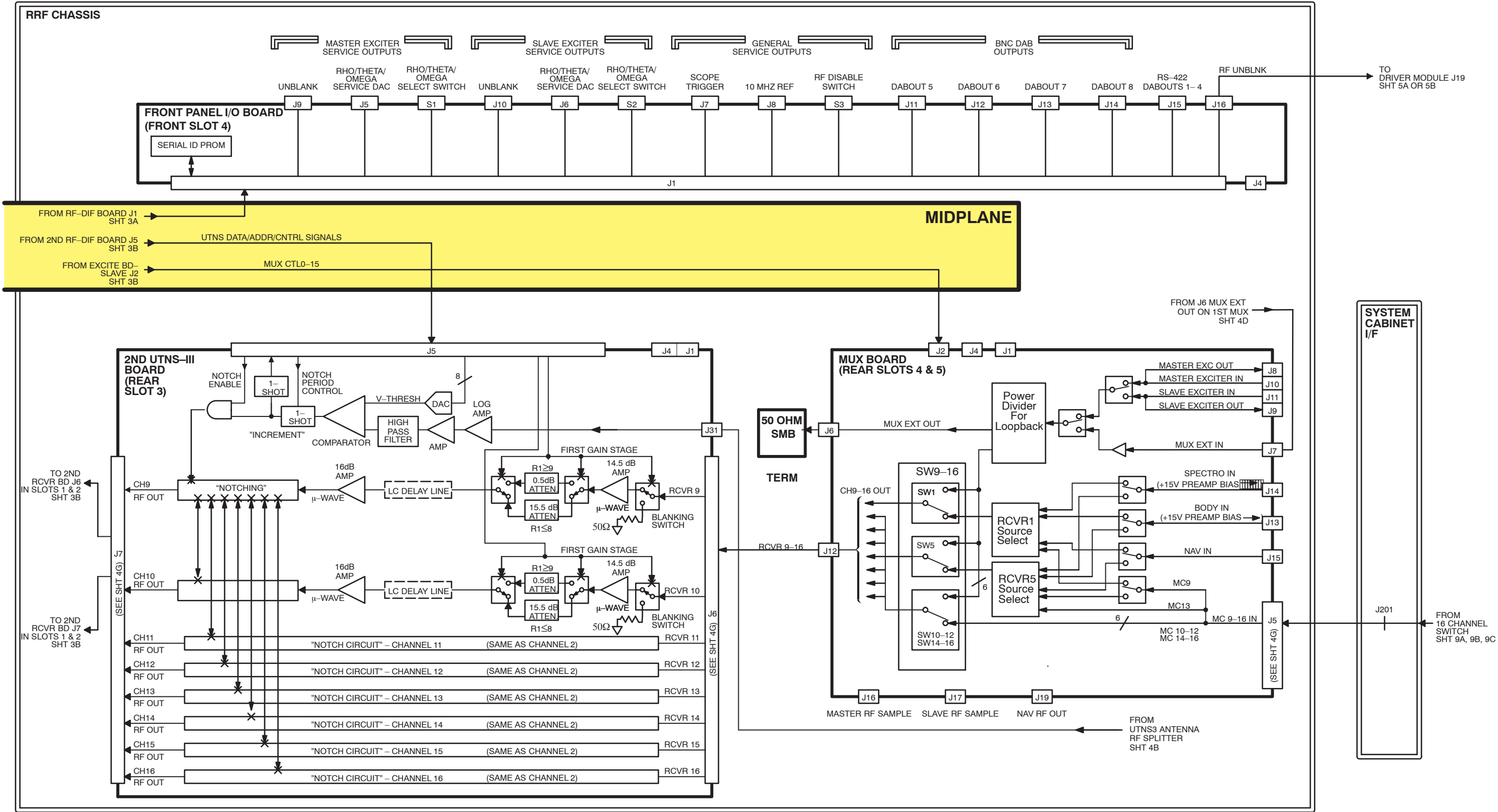
NOTE:
J1 not present in RF System Cabinet.



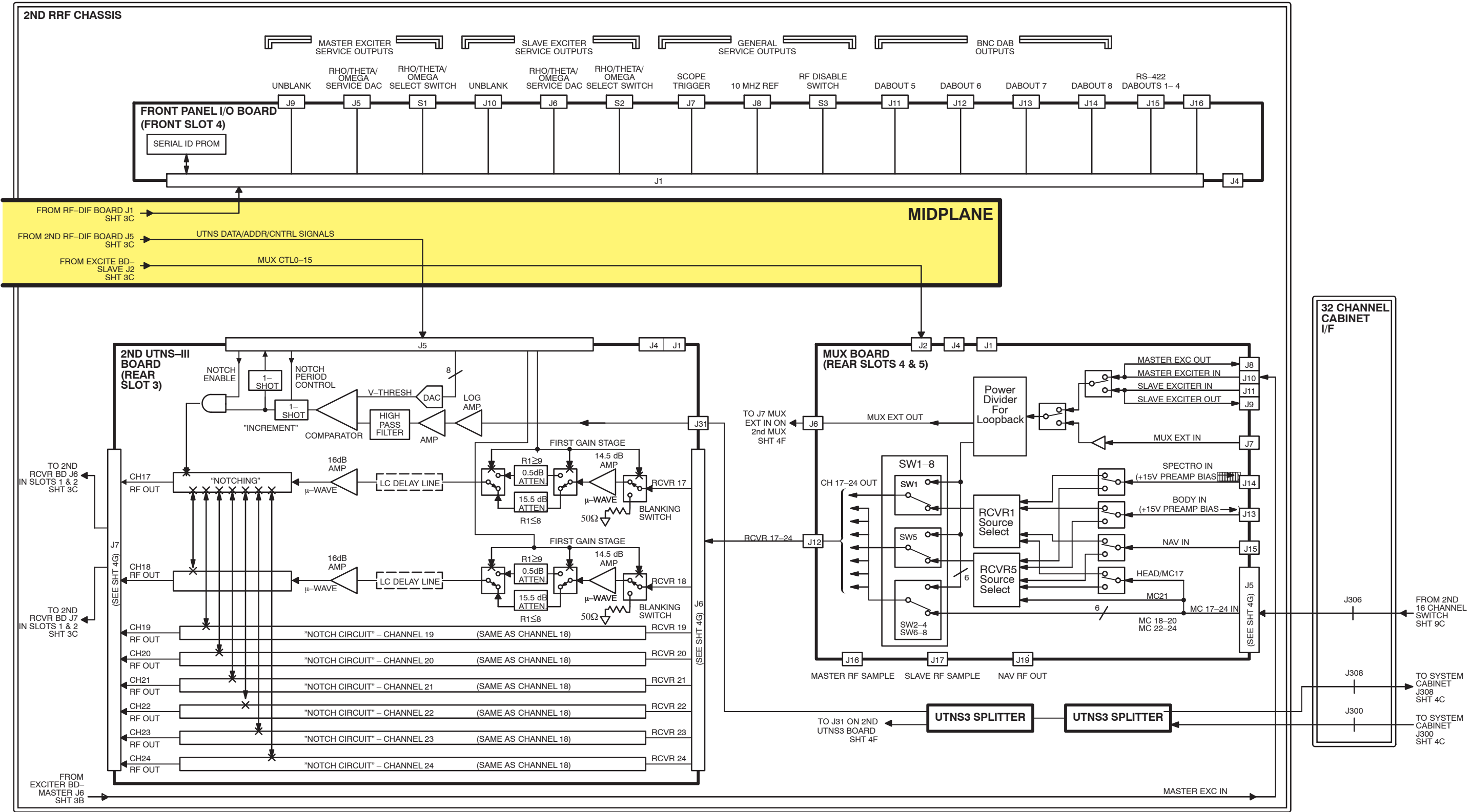
EXCITE HD (12.x), HDx (14.x) & HDxt (1.5.x, 16.x or Later) MGD/RRF/RF SUBSYSTEM
(Channels 1-8 of 32 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM
SHEET 4C OF 16

NOTE:

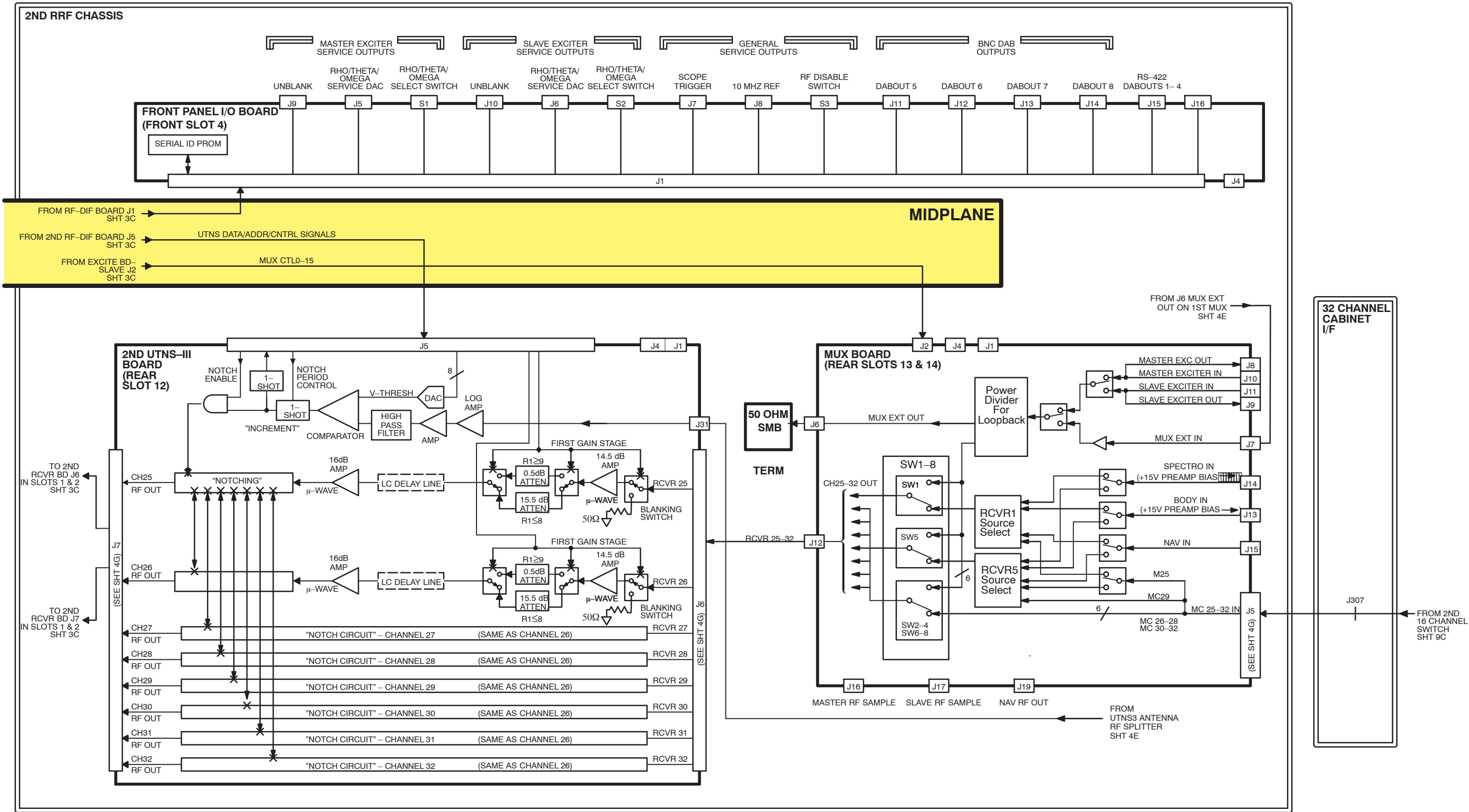
- 1 J1 not present in RF System Cabinet.
- 2 32 Channel systems only.



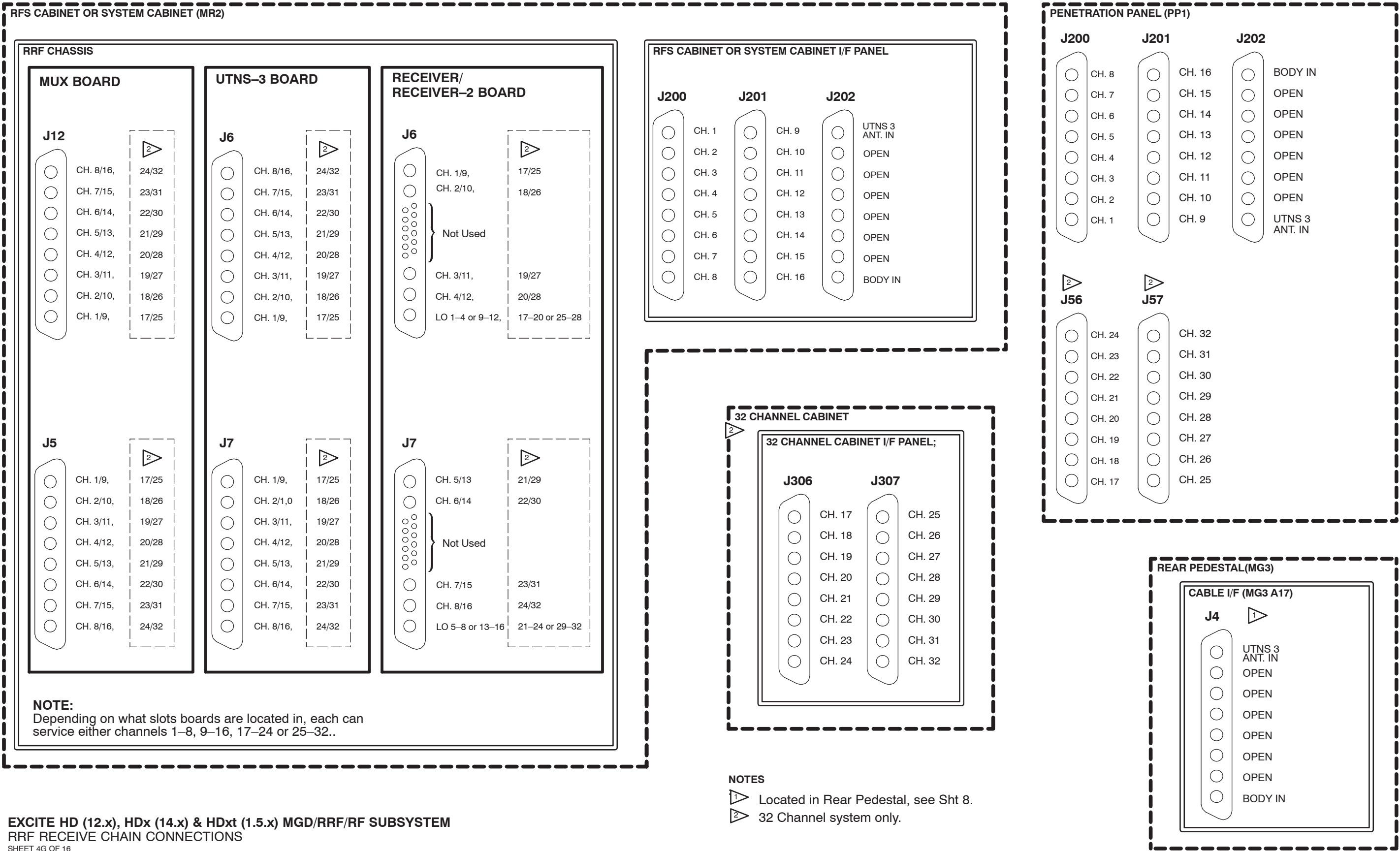
EXCITE HD (12.x), HDx (14.x) & HDxt (1.5.x, 16.x or Later) MGD/RRF/RF SUBSYSTEM
(Channels 9-16 of 16 or 32 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM
SHEET 4D OF 16



HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
(Channels 17-24 of 32 Channel – Forward Production & Upgrades)
2nd RRF CHASSIS INTERCONNECT BLOCK DIAGRAM
SHEET 4E OF 16



HDx (14.x) & HDxt (1.5.x) MGD/RRF/RF SUBSYSTEM
(Channels 25-32 of 32 Channel - Forward Production & Upgrades)
RRF CHASSIS INTERCONNECT BLOCK DIAGRAM
SHEET 4F OF 16



RFS CABINET OR SYSTEM CABINET I/F PANEL

J200

CH. 1

CH. 2

CH. 3

CH. 4

CH. 5

CH. 6

CH. 7

CH. 8

J201

CH. 9

CH. 10

CH. 11

CH. 12

CH. 13

CH. 14

CH. 15

CH. 16

J202

UTNS 3 ANT. IN

OPEN

OPEN

OPEN

OPEN

OPEN

OPEN

BODY IN

32 CHANNEL CABINET

32 CHANNEL CABINET I/F PANEL;

J306

CH. 17

CH. 18

CH. 19

CH. 20

CH. 21

CH. 22

CH. 23

CH. 24

J307

CH. 25

CH. 26

CH. 27

CH. 28

CH. 29

CH. 30

CH. 31

CH. 32

PENETRATION PANEL (PP1)

J200

CH. 8

CH. 7

CH. 6

CH. 5

CH. 4

CH. 3

CH. 2

CH. 1

J201

CH. 16

CH. 15

CH. 14

CH. 13

CH. 12

CH. 11

CH. 10

CH. 9

J202

BODY IN

OPEN

OPEN

OPEN

OPEN

OPEN

OPEN

UTNS 3 ANT. IN

J56

CH. 24

CH. 23

CH. 22

CH. 21

CH. 20

CH. 19

CH. 18

CH. 17

J57

CH. 32

CH. 31

CH. 30

CH. 29

CH. 28

CH. 27

CH. 26

CH. 25

REAR PEDESTAL(MG3)

CABLE I/F (MG3 A17)

J4

UTNS 3 ANT. IN

OPEN

OPEN

OPEN

OPEN

OPEN

OPEN

BODY IN

1

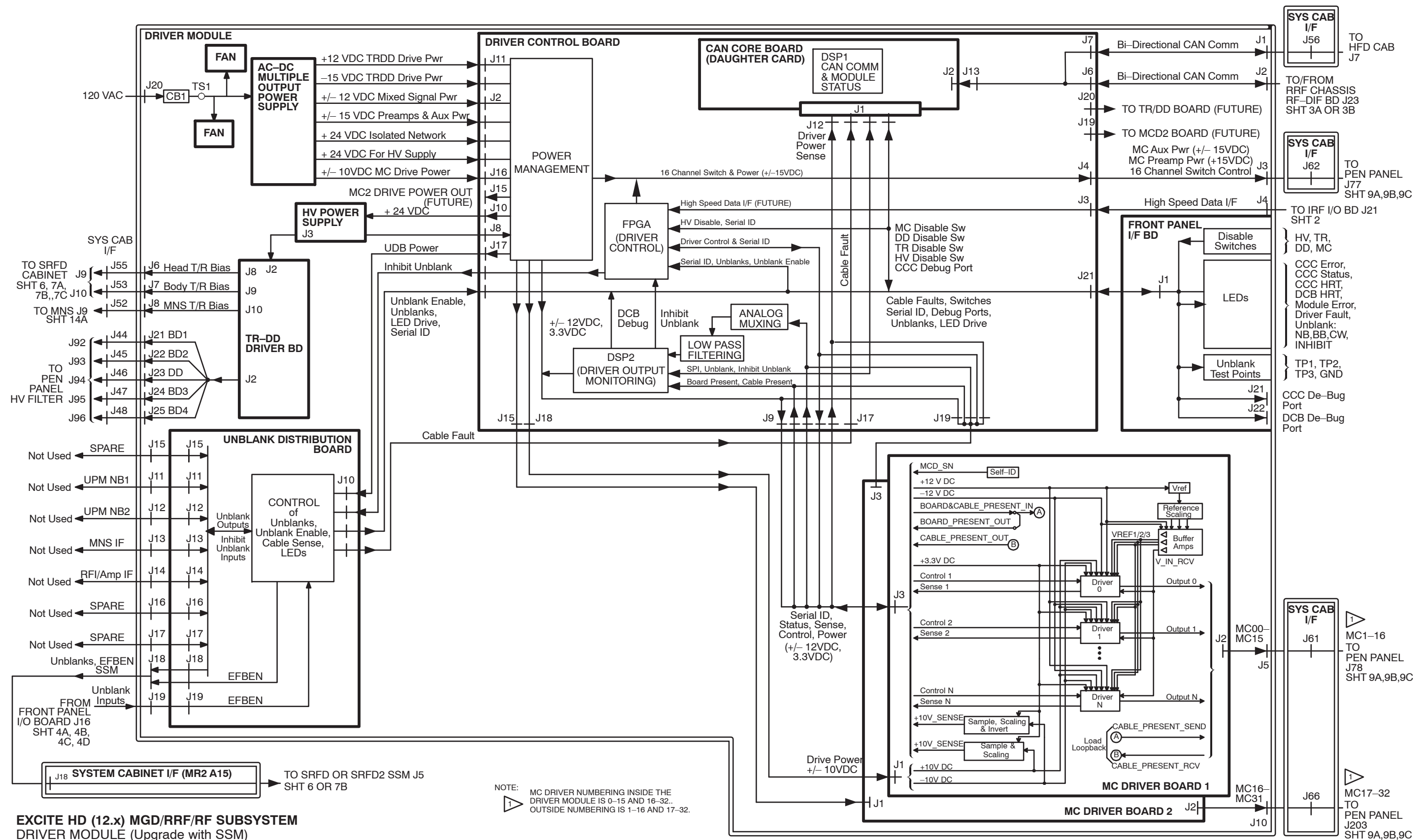
NOTES

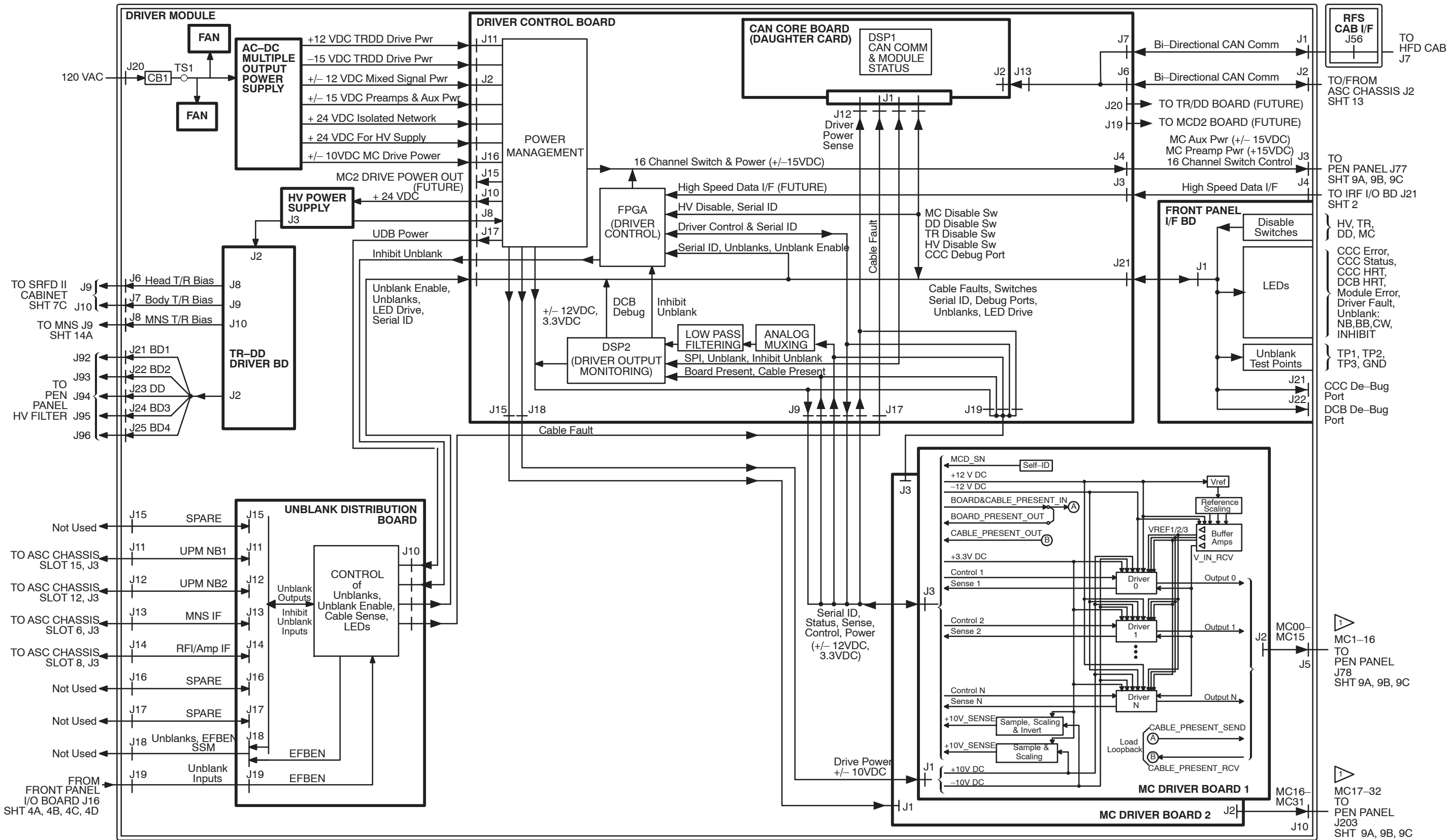
1

 Located in Rear Pedestal, see Sht 8.

2

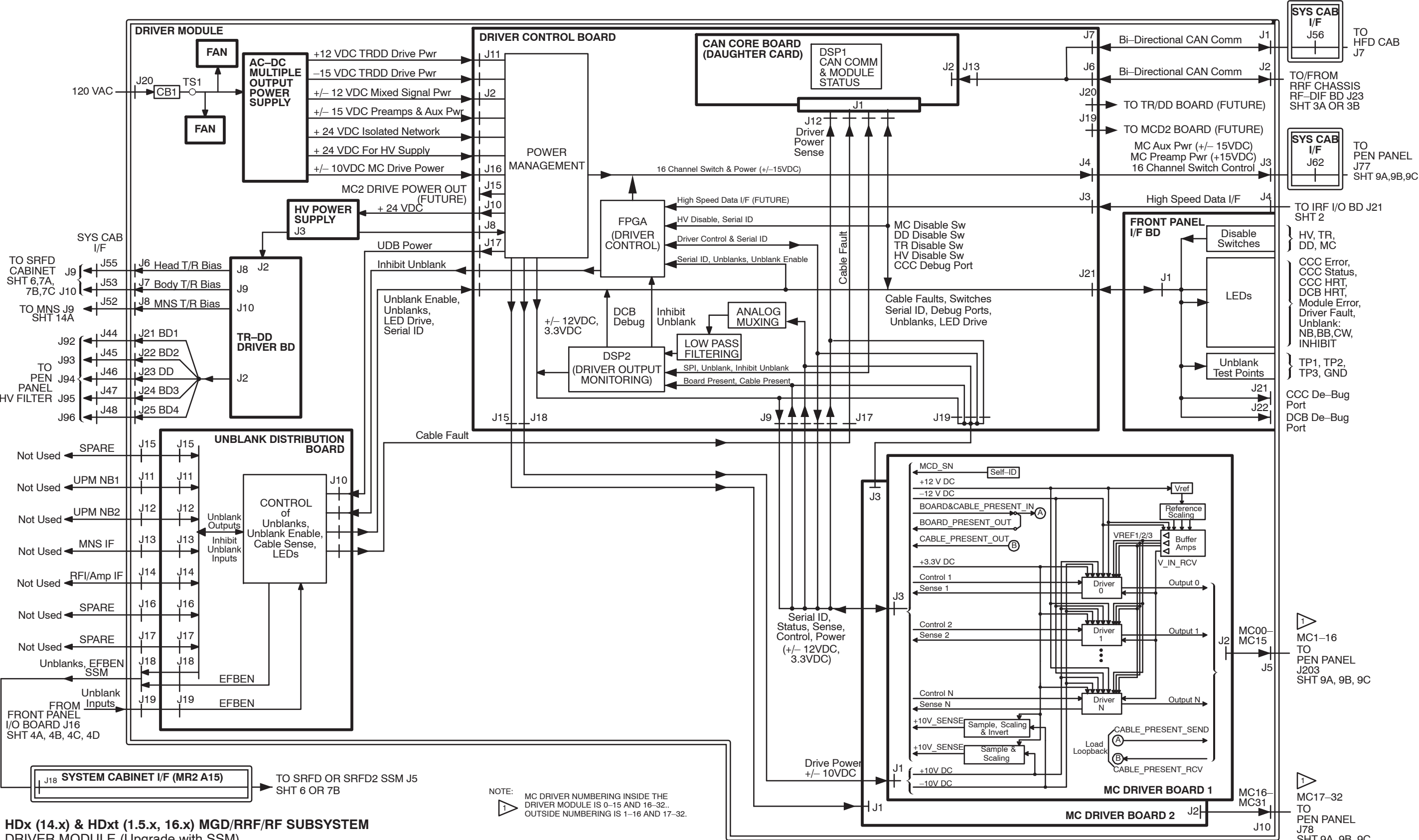
 32 Channel system only.



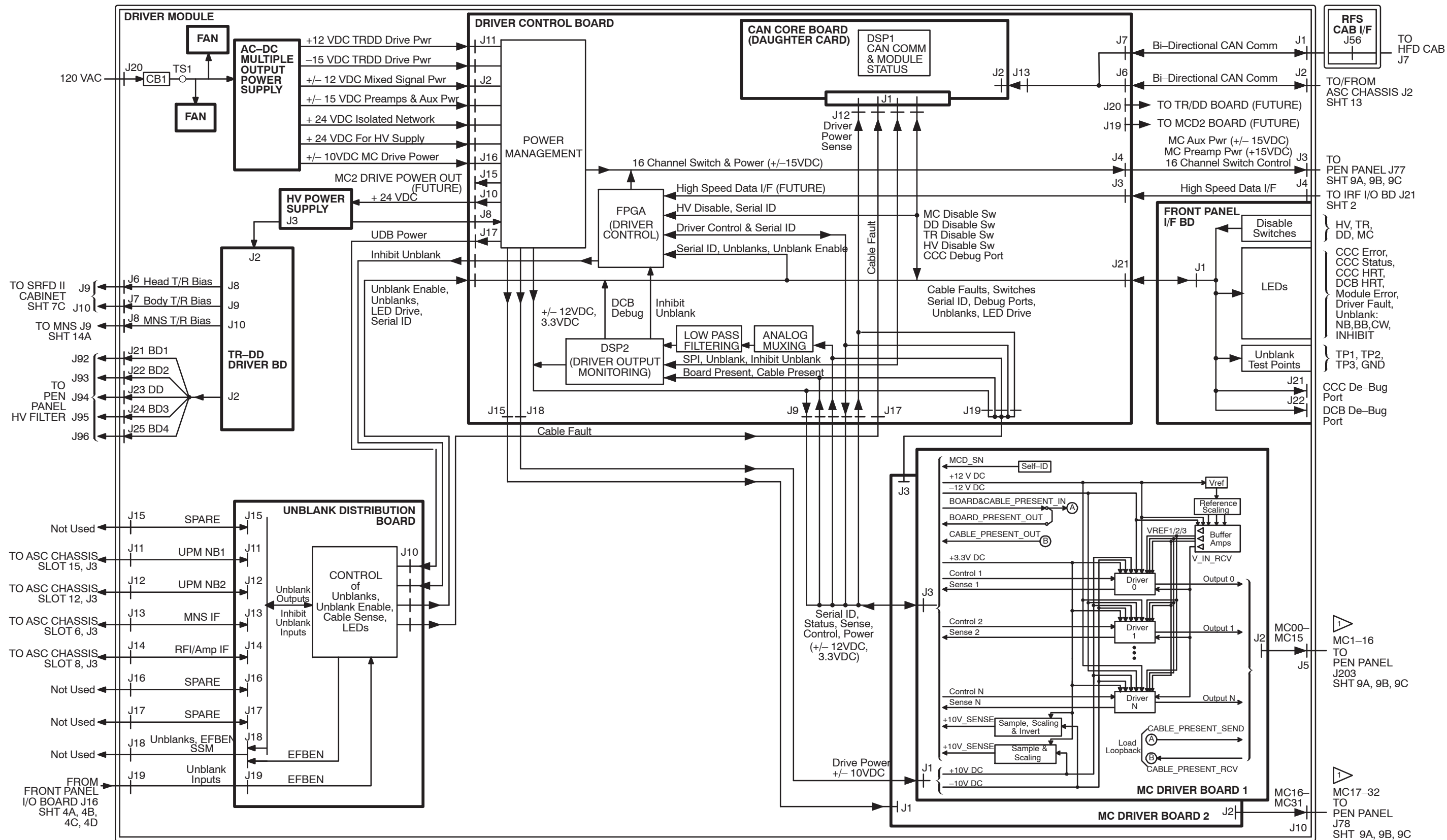


EXCITE HD (12.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Forward Production, RFS Cabinet, no SSM)
SHEET 5B OF 16

NOTE:
MC DRIVER NUMBERING INSIDE THE DRIVER MODULE IS 0-15 AND 16-32. OUTSIDE NUMBERING IS 1-16 AND 17-32.

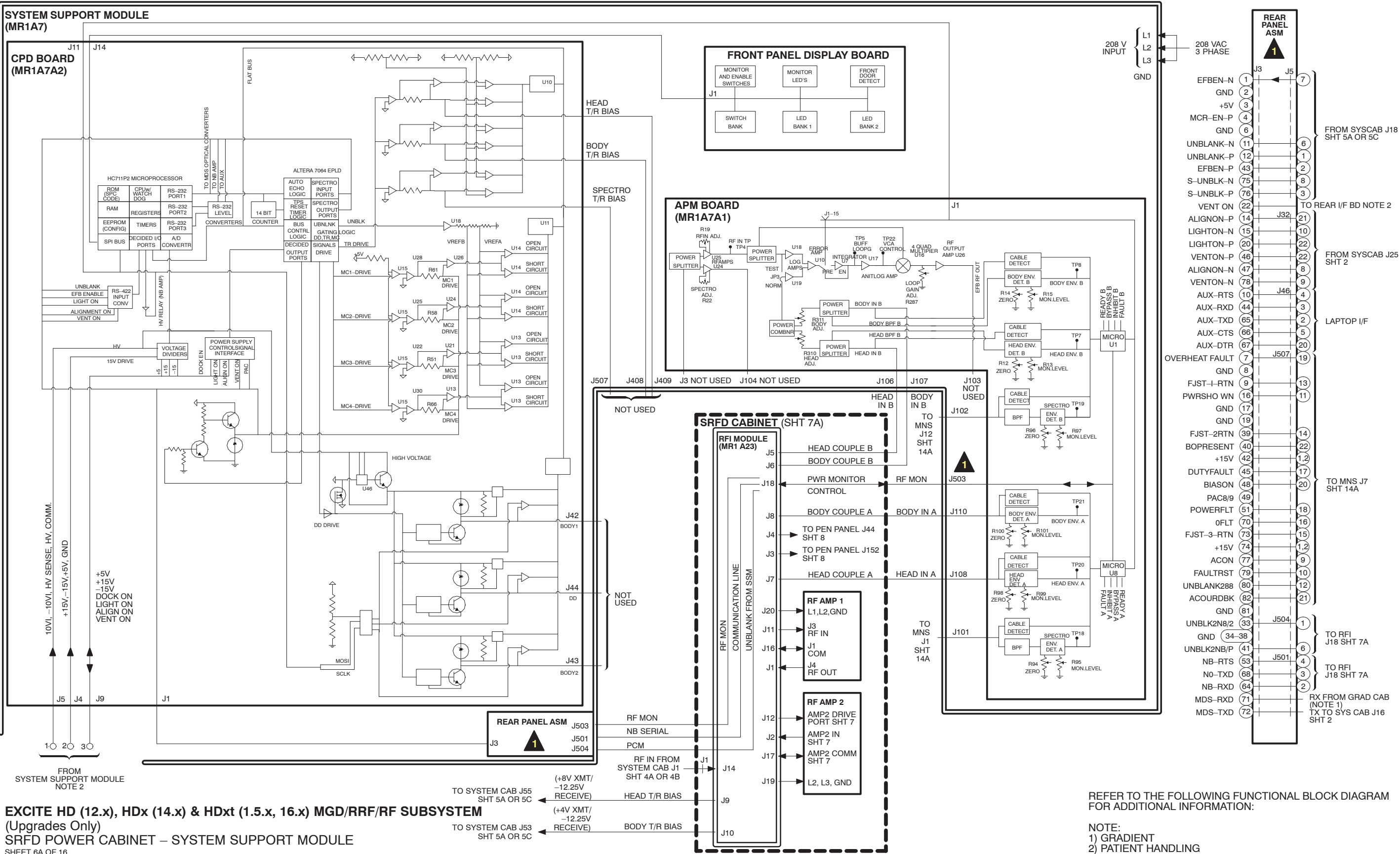


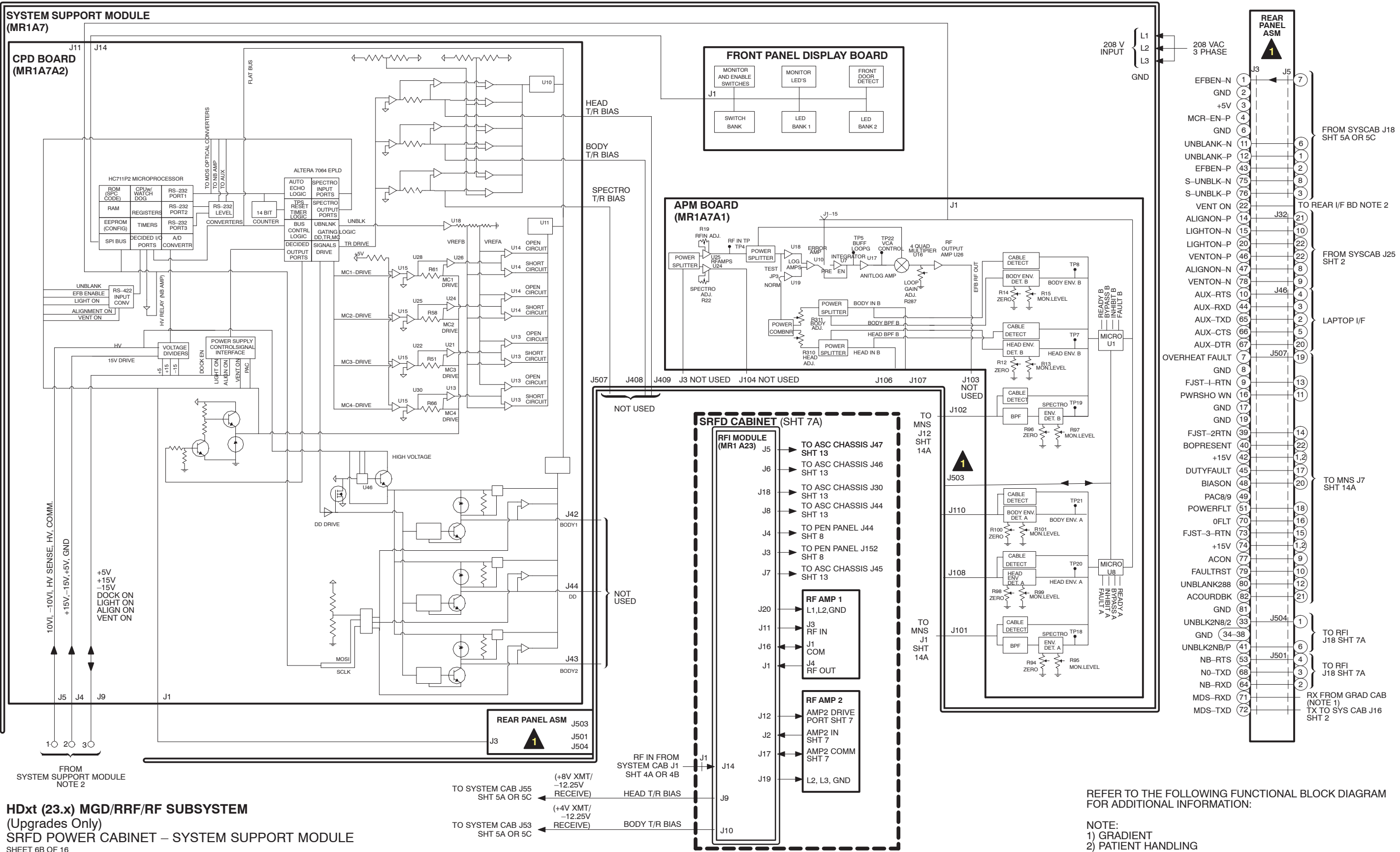
HDx (14.x) & HDxt (1.5.x, 16.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Upgrade with SSM)
SHEET 5C OF 16

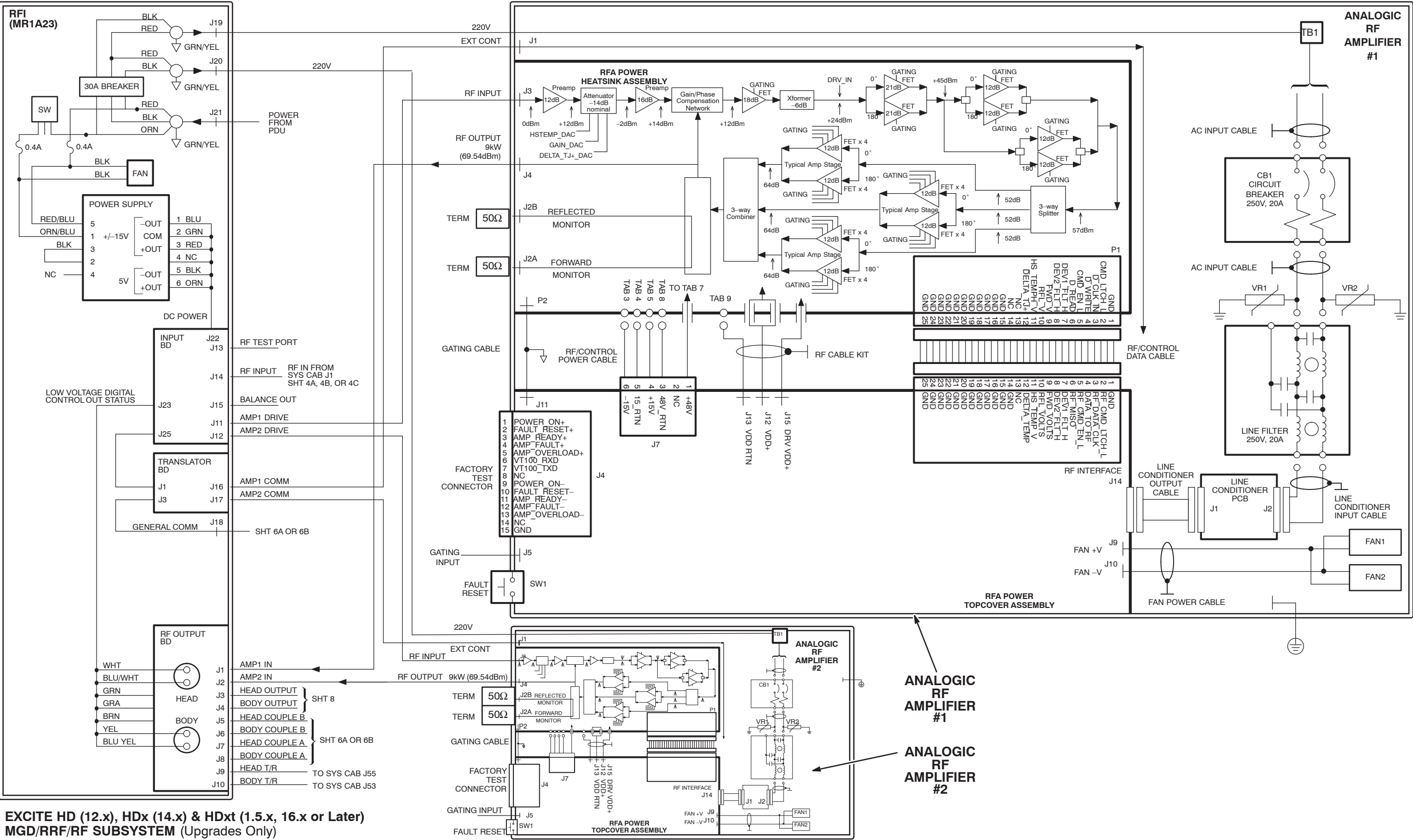


HDx (14.x) & HDxt (1.5.x, 16.x) MGD/RRF/RF SUBSYSTEM
DRIVER MODULE (Forward Production, RFS Cabinet, no SSM)
SHEET 5D OF 16

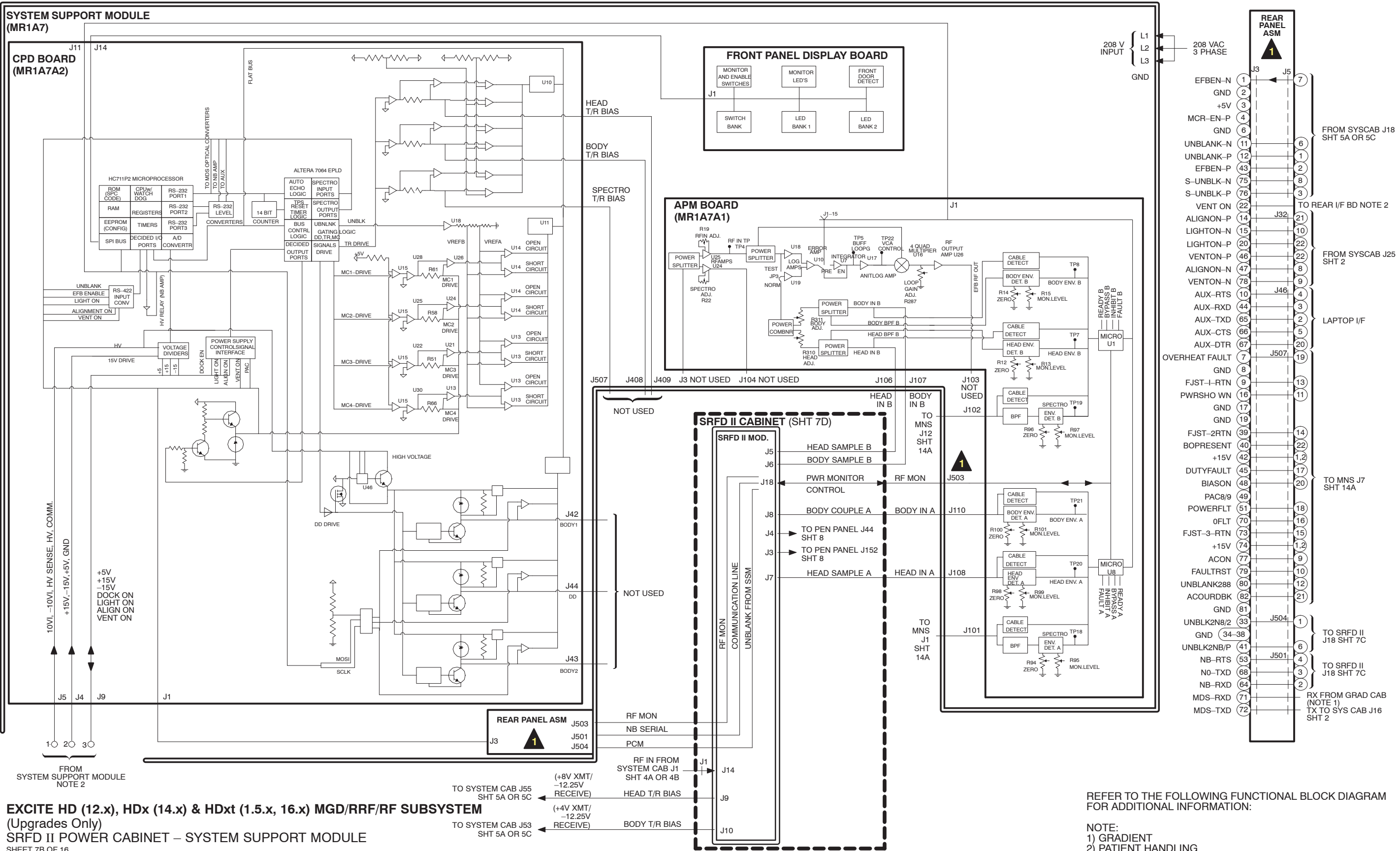
NOTE:
MC DRIVER NUMBERING INSIDE THE
DRIVER MODULE IS 0-15 AND 16-32.
OUTSIDE NUMBERING IS 1-16 AND 17-32.

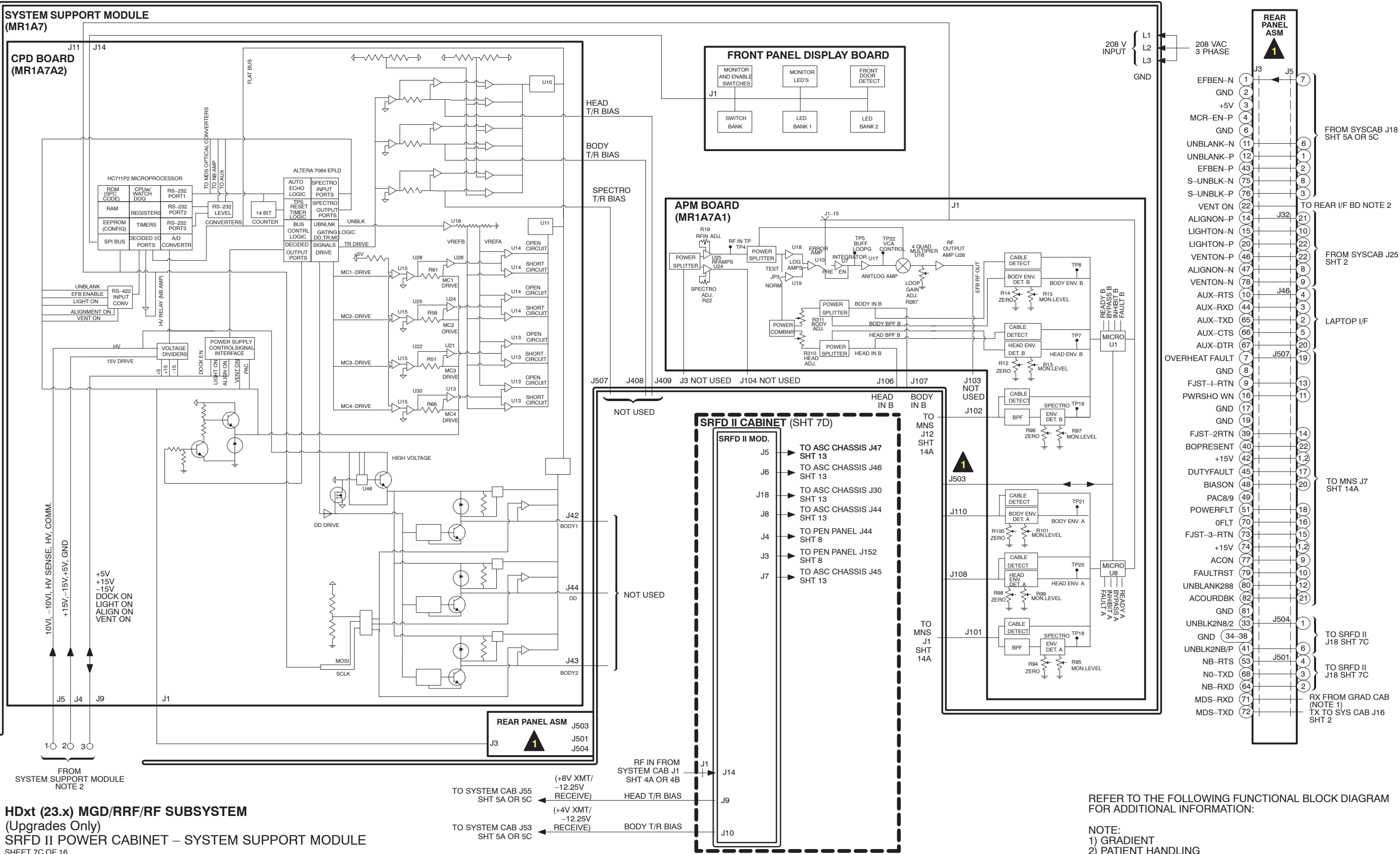


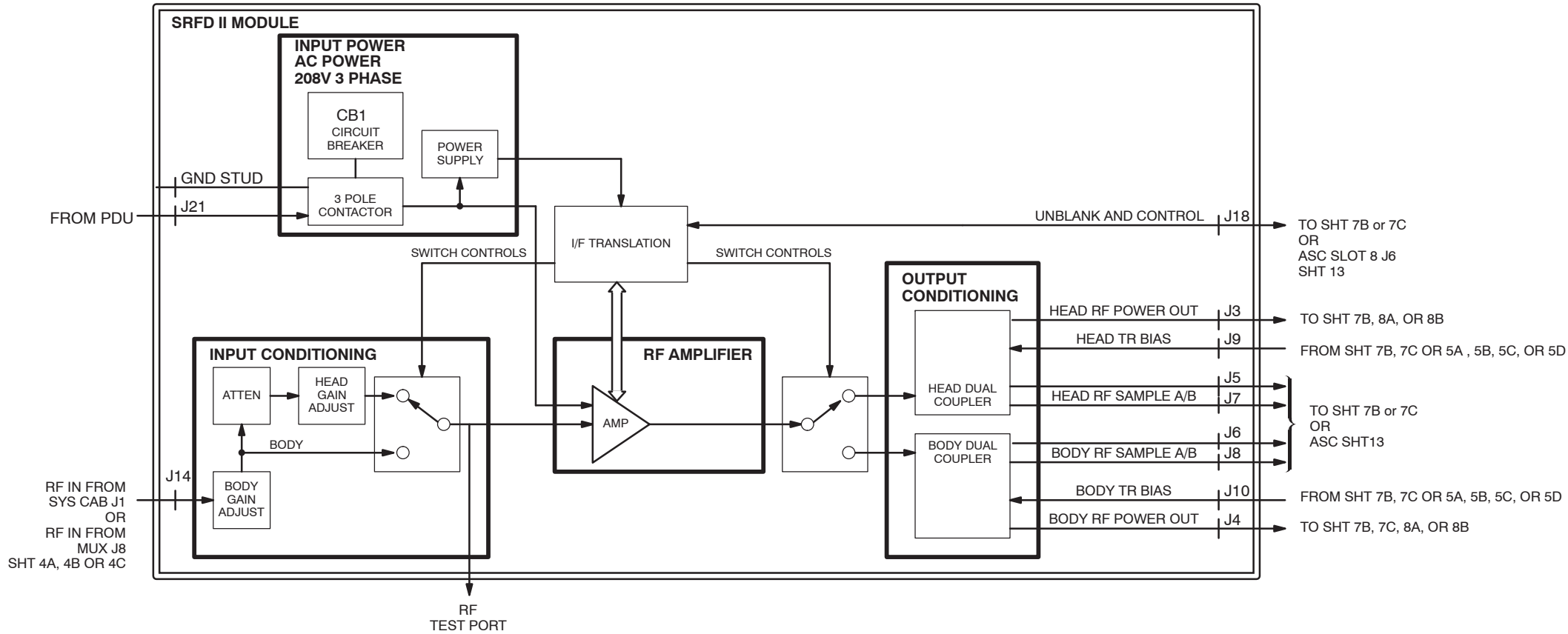


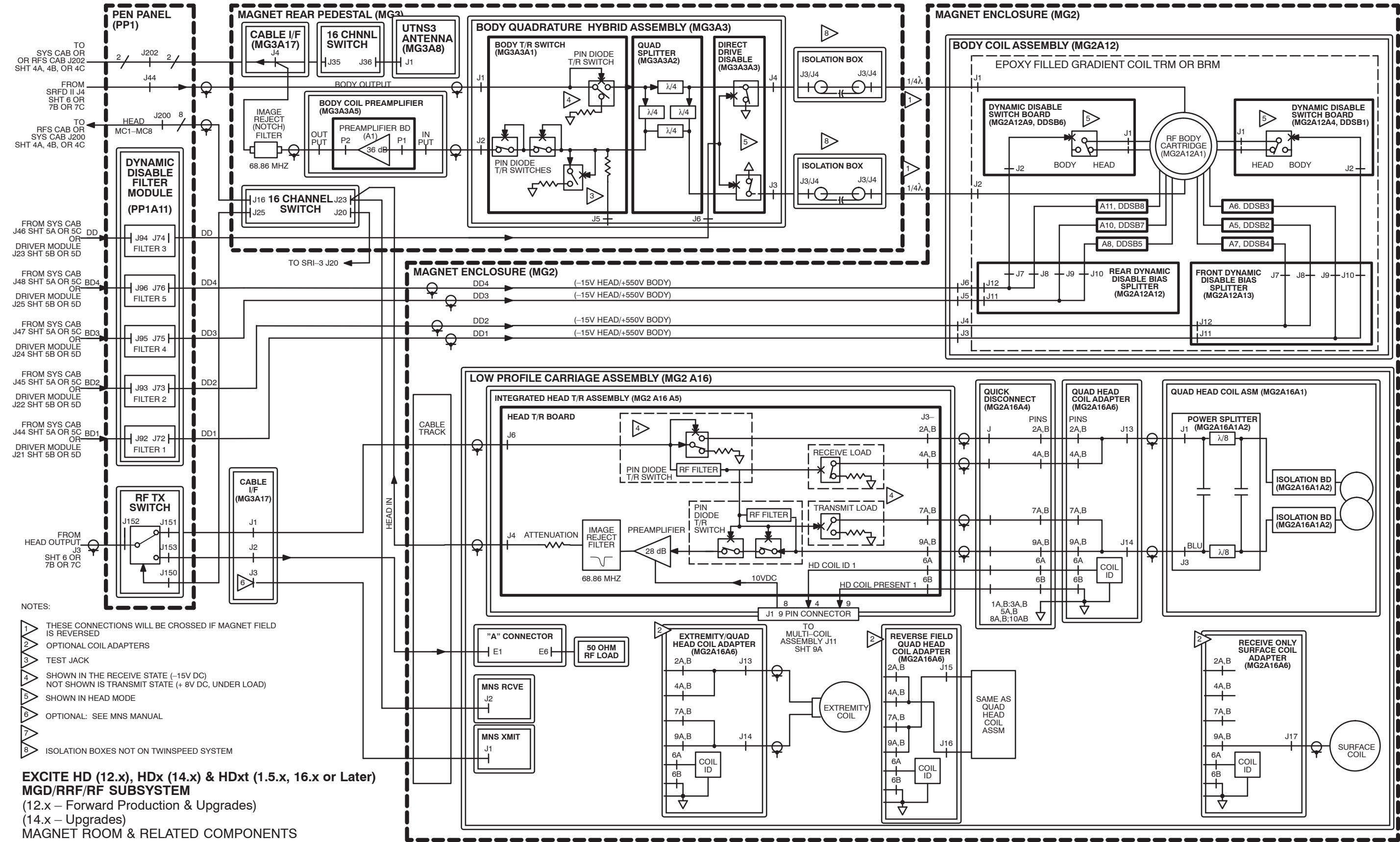


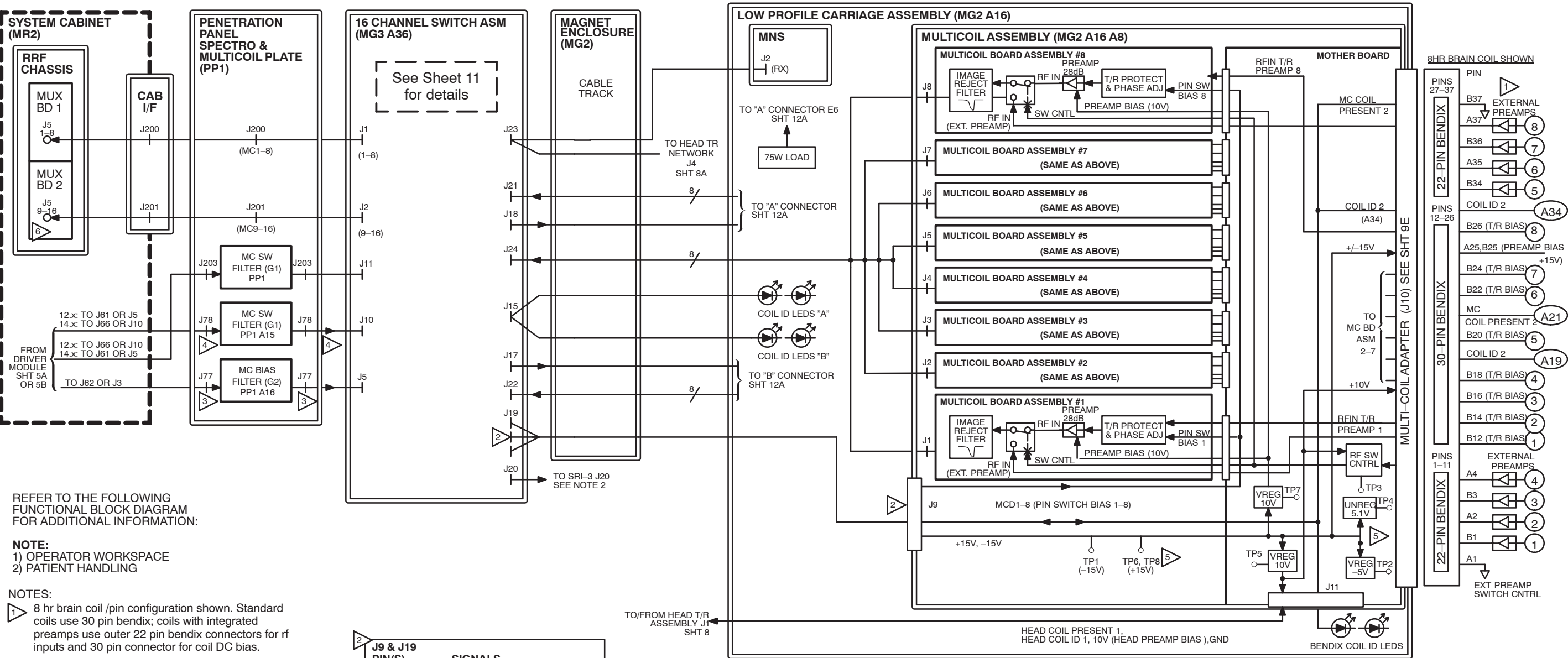
EXCITE HD (12.x), HDx (14.x) & HDxt (1.5.x, 16.x or Later)
MGD/RRF/RF SUBSYSTEM (Upgrades Only)
SRFD CABINET – RFI AND SOLID STATE AMPS
SHEET 7A OF 16







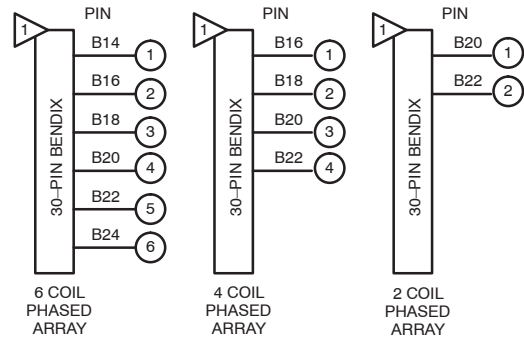




REFER TO THE FOLLOWING
FUNCTIONAL BLOCK DIAGRAM
FOR ADDITIONAL INFORMATION:

NOTE:
1) OPERATOR WORKSPACE
2) PATIENT HANDLING

NOTES:
1) 8 hr brain coil /pin configuration shown. Standard coils use 30 pin bendix; coils with integrated preamps use outer 22 pin bendix connectors for rf inputs and 30 pin connector for coil DC bias.



EXCITE HD (12.x) or HDx (14.x Upgrade)
MGD/RRF/RF SUBSYSTEM
MULTI-COIL MODULES
SHEET 9A OF 16

J9 & J19	PIN(S)	SIGNALS
1,14		-15V
2,15		+15V
3, 4, 23-25	...	GND
5		MCD8
6		MCD4
7		MCD7
8		MCD3
9		MCD6
10		MCD2
11		MCD5
12		MCD1
13		CTL_PREAMP (NOTE 1)
16		RED_LED
17		YEL_LED
18		GRN_LED
19		COIL PRESENT #1
20		COIL PRESENT #2
21		COIL ID #1
22		COIL ID #2

NOTE 1: PIN 13 DEFAULT LOW.
LOW: PREAMP POWER ON.
HIGH: PREAMP POWER OFF.

J62 OR J3 & J77	PIN	SIGNAL NAME
1		XPT_DATA_POS
2		XPT_STROBE_POS
3		XPT_OE_N_POS
4		AGND
5		AGND
6		+15V (NOTE 1)
7		+15V (NOTE 1)
8		+15V (NOTE 1)
9		XPT_DATA_NEG
10		XPT_STROBE_NEG
11		XPT_OE_N_NEG
12		AGND
13		AGND
14		-15V (MC COILS)
15		-15V (MC COILS)

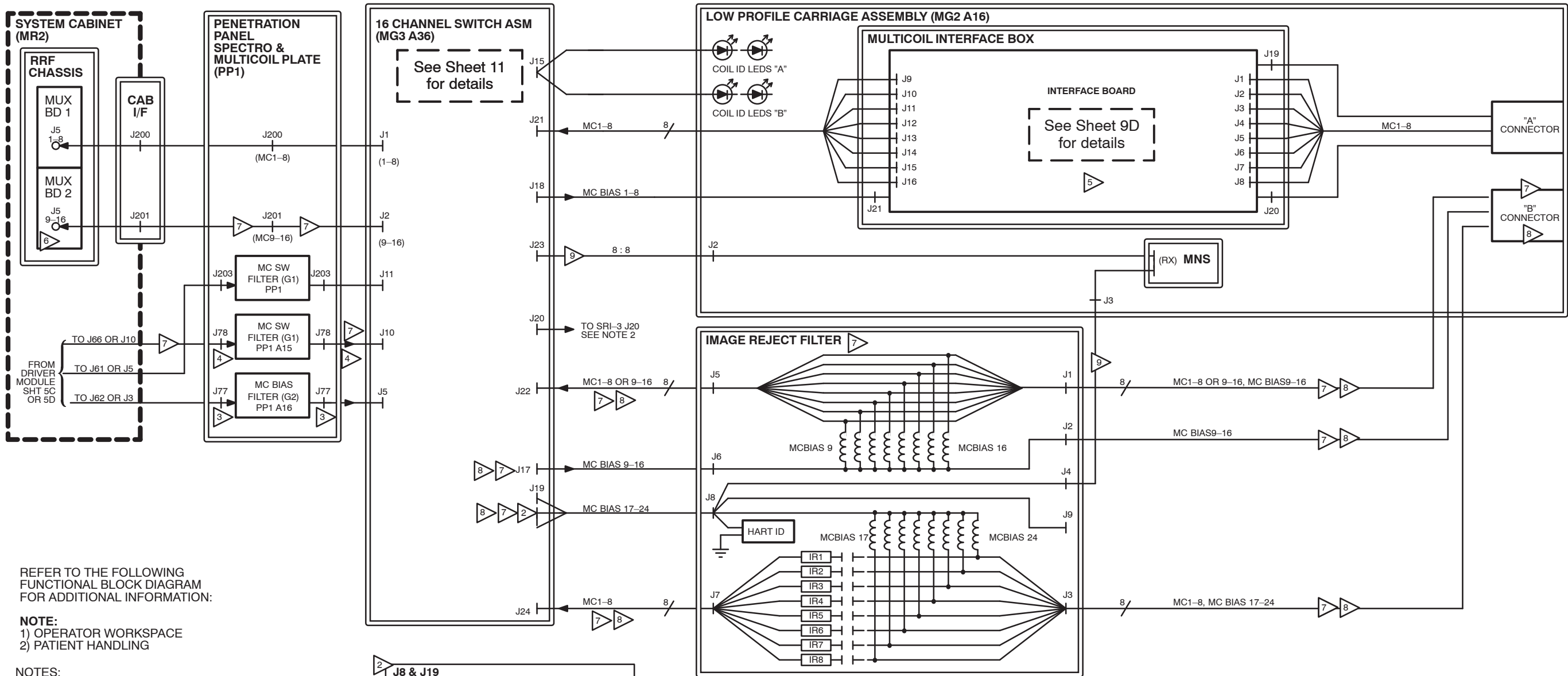
NOTE 1: ANALOG POWER FOR
PREAMPS (MC & HEAD),
CROSSPOINT & MC COILS.

J10 & J78	PIN(S)	SIGNAL
1-8		MCD1-8 (NOTE 1)
9		MC CABLE PRESENT_O IN (NOTE 2)
10,11		NO CONNECT
12-19		MCD9-16 (NOTE 3)
20-27		GND
29		MC CABLE PRESENT_1 IN (NOTE 2)
30-37		GND

NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V).
NOTE 2: THESE TWO PINS CONNECTED TOGETHER
ON J10 FOR DIAGNOSTIC LOOPBACK.
NOTE 3: NOT USED/CONNECTED AT J10 INSIDE
CROSSPOINT SWITCH ASSEMBLY.

LOCATED BEHIND MULTICOIL TEST BD ASSY	POINT	FUNCTION
7		TP1 -15V
6		TP2 -5V
6		TP3 .. RF SW CNTRL SIG
2, 3		TP4 5.1V
1, 2		TP5 +10V_B
1, 2		TP6 +15V
7, 8		TP7 +10V_A
7, 8		TP8 +15V

6) Only 16 Channel Systems have a
second MUX board. J201 not used
in 8 and 4 Channel systems.



REFER TO THE FOLLOWING
FUNCTIONAL BLOCK DIAGRAM
FOR ADDITIONAL INFORMATION:

NOTE:
1) OPERATOR WORKSPACE
2) PATIENT HANDLING

NOTES:

1

2	J8 & J19
PIN(S)	SIGNALS
1,14	 -15V
2,15	 +15V
3, 4, 23-25	 GND
5	 MCD8
6	 MCD4
7	 MCD7
8	 MCD3
9	 MCD6
10	 MCD2
11	 MCD5
12	 MCD1
13	 CTL_PREAMP (NOTE 1)
16	 RED_LED
17	 YEL_LED
18	 GRN_LED
19	 COIL PRESENT #1
20	 COIL PRESENT #2
21	 COIL ID #1
22	 COIL ID #2

NOTE 1: PIN 13 DEFAULT LOW.
LOW: PREAMP POWER ON.
HIGH: PREAMP POWER OFF.

3	J62 OR J3 & J77
PIN	SIGNAL NAME
1	 XPT_DATA_POS
2	 XPT_STROBE_POS
3	 XPT_OE_N_POS
4	 AGND
5	 AGND
6	 +15V (NOTE 1)
7	 +15V (NOTE 1)
8	 +15V (NOTE 1)
9	 XPT_DATA_NEG
10	 XPT_STROBE_NEG
11	 XPT_OE_N_NEG
12	 AGND
13	 AGND
14	 -15V (MC COILS)
15	 -15V (MC COILS)

NOTE 1: ANALOG POWER FOR
PREAMPS (MC & HEAD),
CROSSPOINT & MC COILS.

4	J10 & J66
PIN(S)	SIGNAL
1-8	 MCD1-8 (NOTE 1)
9	 MC CABLE PRESENT_O IN (NOTE 2)
10,11	 NO CONNECT
12-19	 MCD9-16 (NOTE 3)
20-27	 GND
29	 MC CABLE PRESENT_1 IN (NOTE 2)
30-37	 GND

NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V).
NOTE 2: THESE TWO PINS CONNECTED TOGETHER
ON J10 FOR DIAGNOSTIC LOOPBACK.
NOTE 3: NOT USED/CONNECTED AT J10 INSIDE
CROSSPOINT SWITCH ASSEMBLY.

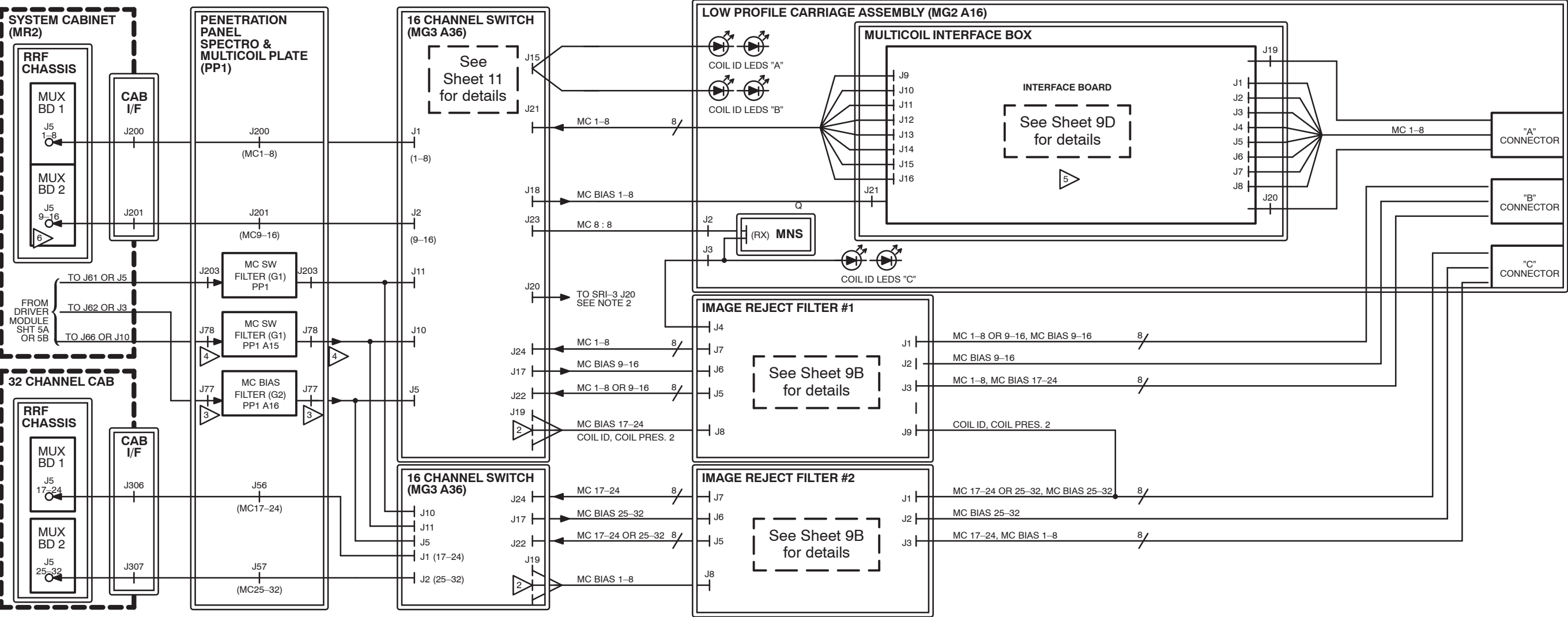
5	TESTPOINT . FUNCTION
TP1	 SYSTEM PREAMP ENABLE (+15V OR 0V)
TP2	 PREAMP SELECT 1
TP3	 PREAMP SELECT 2
TP4	 8.5V_A
TP5	 +10V_B CNTRL
TP6	 +15V
TP7	 +5.1V
TP8	 +15V
TP9	 +15V
TP10	 8.5V_B
TP11	 8.5V_C
TP12	 -15V
TP13	 -8.5V

6 Only 16 Channel Systems have a second MUX board.

7 Required on 16 or 32 Channel systems.

8 B Connector and MC Bias lines are not included in newer
8 Channel systems.

9 Optional: See MNS manual



REFER TO THE FOLLOWING
FUNCTIONAL BLOCK DIAGRAM
FOR ADDITIONAL INFORMATION:

NOTE:
1) OPERATOR WORKSPACE
2) PATIENT HANDLING

NOTES:
1

**HDx (14.x) & HDxt (15.x) MGD/RRF/RF
SUBSYSTEM**
32 CHANNEL MULTI-COIL
SHEET 9C OF 16

J8 & J19	PIN(S) SIGNALS
1,14	 -15V
2,15	 +15V
3, 4, 23-25	... GND
5	 MCD8
6	 MCD4
7	 MCD7
8	 MCD3
9	 MCD6
10	 MCD2
11	 MCD5
12	 MCD1
13	 CTL_PREAMP (NOTE 1)
16	 RED_LED
17	 YEL_LED
18	 GRN_LED
19	 COIL PRESENT #1
20	 COIL PRESENT #2
21	 COIL ID #1
22	 COIL ID #2

NOTE 1: PIN 13 DEFAULT LOW.
LOW: PREAMP POWER ON.
HIGH: PREAMP POWER OFF.

J62 OR J3 & J77	PIN SIGNAL NAME
1	 XPT_DATA_POS
2	 XPT_STROBE_POS
3	 XPT_OE_N_POS
4	 AGND
5	 AGND
6	 +15V (NOTE 1)
7	 +15V (NOTE 1)
8	 +15V (NOTE 1)
9	 XPT_DATA_NEG
10	 XPT_STROBE_NEG
11	 XPT_OE_N_NEG
12	 AGND
13	 AGND
14	 -15V (MC COILS)
15	 -15V (MC COILS)

NOTE 1: ANALOG POWER FOR
PREAMPS (MC & HEAD),
CROSSPOINT & MC COILS.

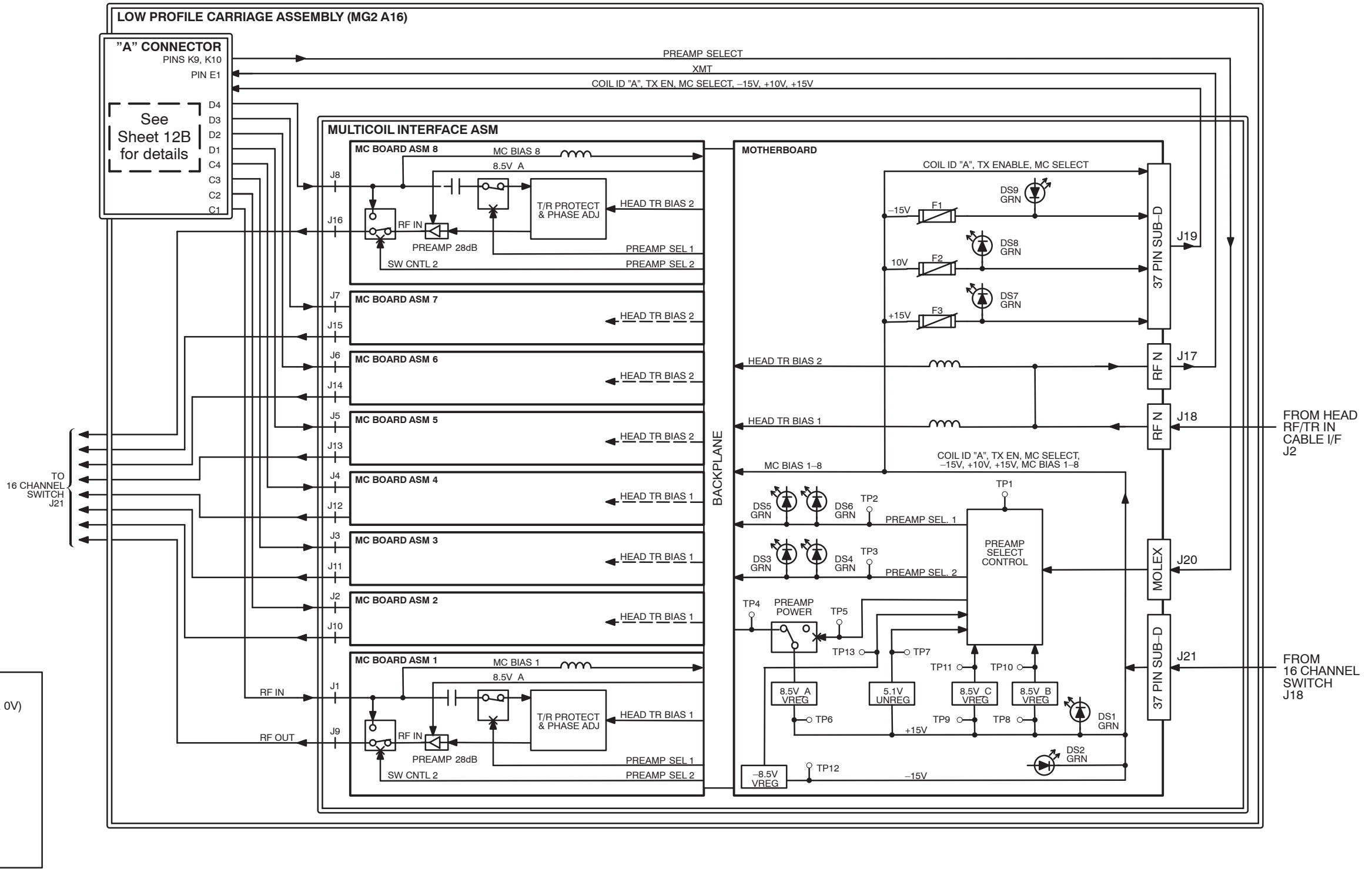
J10 & J66	PIN(S) SIGNAL
1-8	 MCD1-8 (NOTE 1)
9	 MC CABLE PRESENT_O IN (NOTE 2)
10,11	 NO CONNECT
12-19	 MCD9-16 (NOTE 3)
20-27	 GND
29	 MC CABLE PRESENT_1 IN (NOTE 2)
30-37	 GND

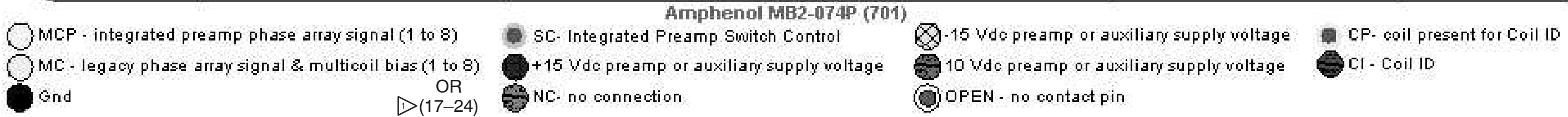
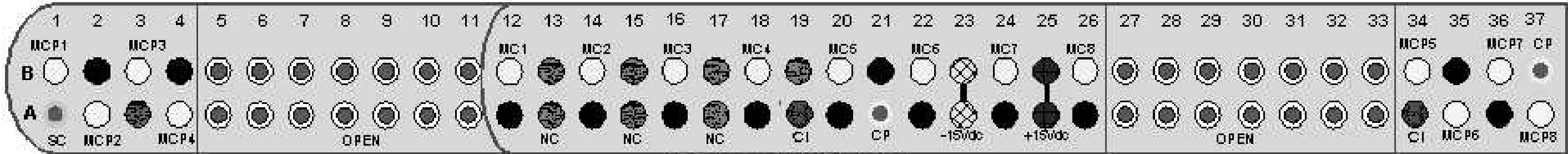
NOTE 1: MULTICOIL BIAS (+3V,+5V,+7V,-5V).
NOTE 2: THESE TWO PINS CONNECTED TOGETHER
ON J10 FOR DIAGNOSTIC LOOPBACK.
NOTE 3: NOT USED/CONNECTED AT J10 INSIDE
CROSSPOINT SWITCH ASSEMBLY.

5

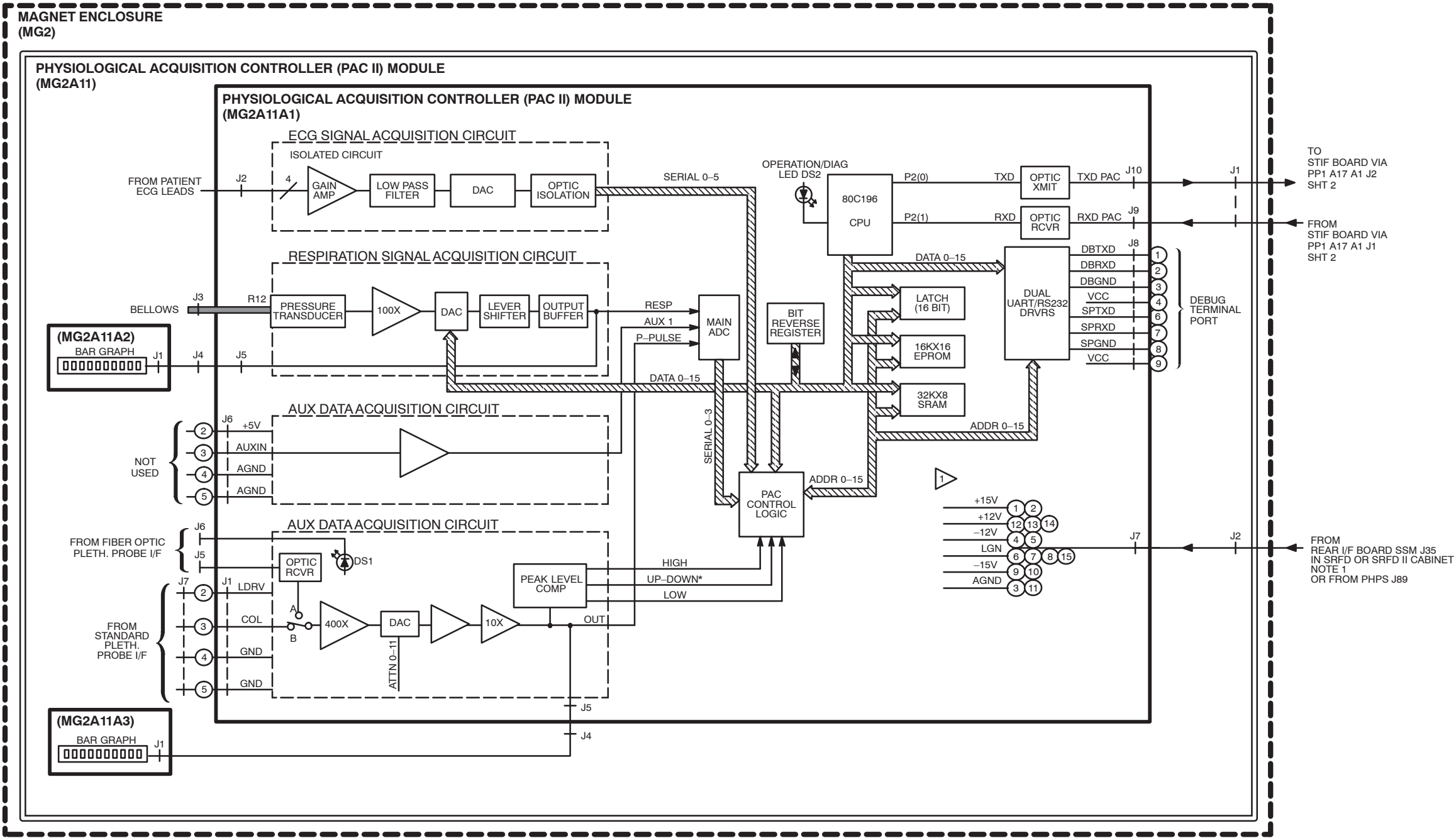
TESTPOINT . FUNCTION	
TP1	SYSTEM PREAMP ENABLE (+15V OR 0V)
TP2	PREAMP SELECT 1
TP3	PREAMP SELECT 2
TP4	8.5V_A
TP5	+10V_B CNTRL
TP6	+15V
TP7	+5.1V
TP8	+15V
TP9	+15V
TP10	8.5V_B
TP11	8.5V_C
TP12	-15V
TP13	-8.5V

6 Only 16 Channel Systems have a
second MUX board. J201 not used
in 8 and 4 Channel systems.





NOTES:
▶ MC BIAS is 17-24 on 14.x systems.



NOTES:

1 MULTIPLE WIRES/PINS WITH SAME SIGNAL ARE NOT SHOWN

REFER TO THE FOLLOWING FUNCTIONAL BLOCK DIAGRAM FOR ADDITIONAL INFORMATION:

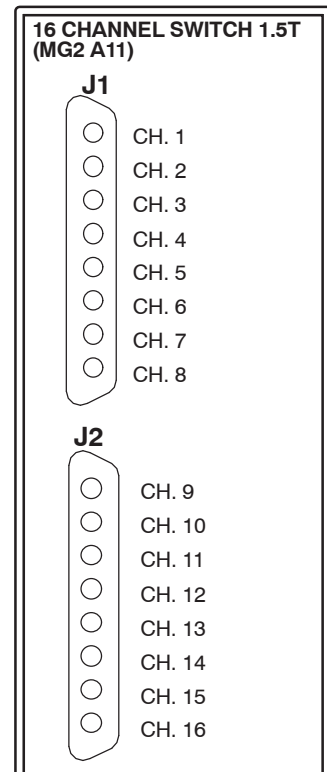
NOTE:

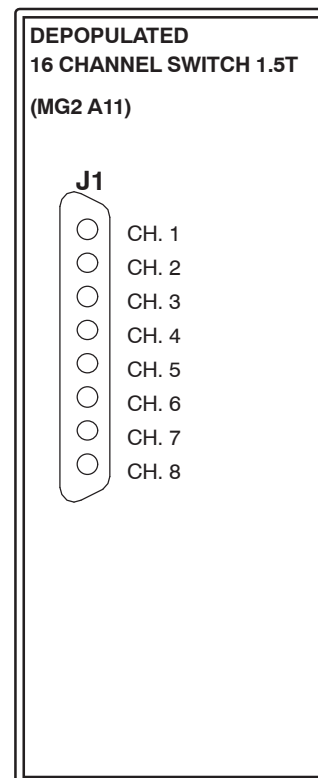
1) PATIENT HANDLING

HDx (14.x) & HDxt (15.x, 16.x or Later) MGD/RRF/RF SUBSYSTEM
PHYSIOLOGICAL ACQUISITION CONTROLLER II

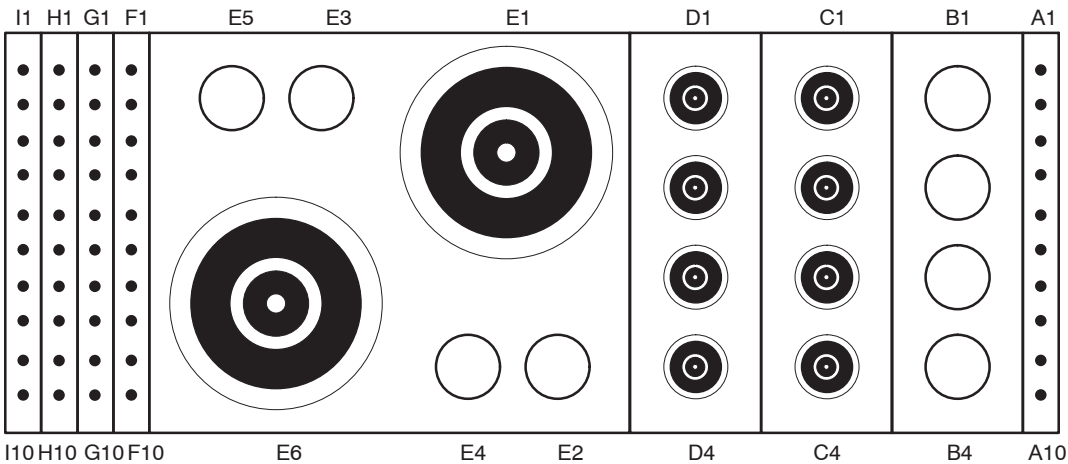


- COIL ID A GREEN
- COIL ID A RED
- COIL ID B GREEN
- COIL ID B RED
- COIL ID LEGACY GREEN
- COIL ID LEGACY RED
- COIL PRESENT A
- COIL PRESENT B
- COIL PRESENT 1
- COIL PRESENT 2
- TX ENABLE A
- TX ENABLE B
- DIAG OUT
- DIAG IN
- -15V POWER
- +15V POWER
- +24V POWER
- J21 ON CH. 1-4
- J22 ON CH. 1-4
- J23 ON CH. 1-4
- J24 ON CH. 1-4
- J21 ON CH. 5-8
- J22 ON CH. 5-8
- J23 ON CH. 5-8
- J24 ON CH. 5-8
- J22 ON CH. 9-12
- J22 ON CH. 13-16



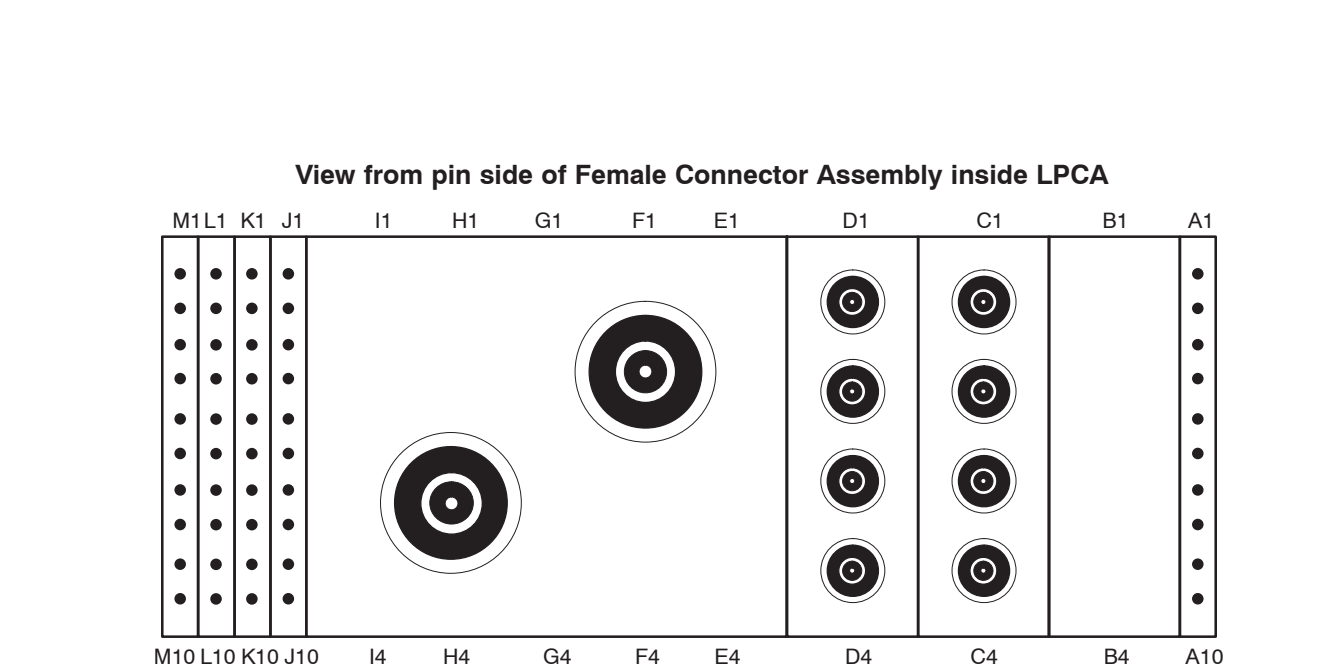


View from pin side of Female Connector Assembly inside LPCA



Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a	Key blocked in receptacle E	
B2	n/a	Key blocked in receptacle F	
B3	n/a	Key blocked in receptacle G	
B4	n/a	Key blocked in receptacle H	
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

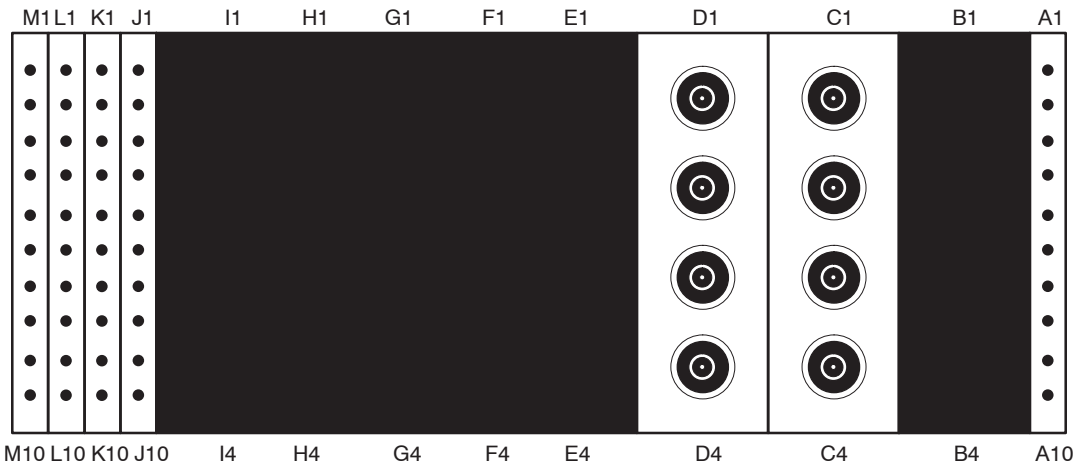
Contact Function Allocation			
Pin	Twisted with	Description	Comments
E1	n/a	Transmit signal up to 2kw (1.5T) or 4kw (3.0T) PEP.	“A” Connector only. Not present in “B” Connector
E2	n/a	Key blocked in receptacle C	
E3	n/a	Key blocked in receptacle A	Blocked for “A”
E4	n/a	Key blocked in receptacle B	Blocked for “B”
E5	n/a	Key blocked in receptacle D	
E6	n/a	Reflected power to 50 ohm load 2.5kw PEP, 50w RMS.	“A” Connector only. Not present in “B” Connector
F1	G1	MC bias 1	MC T/R Bias Line 17 for “A”, 25 for “B”
F2	G2	MC bias 2	MC T/R Bias Line 18 for “A”, 26 for “B”
F3	G3	MC bias 3	MC T/R Bias Line 19 for “A”, 27 for “B”
F4	G4	MC bias 4	MC T/R Bias Line 20 for “A”, 28 for “B”
F5	G5	MC bias 5	MC T/R Bias Line 21 for “A”, 29 for “B”
F6	G6	MC bias 6	MC T/R Bias Line 22 for “A”, 30 for “B”
F7	G7	MC bias 7	MC T/R Bias Line 23 for “A”, 31 for “B”
F8	G8	MC bias 8	MC T/R Bias Line 24 for “A”, 32 for “B”
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for “A”, 25 for “B”
G2	F2	MC bias return 2	MC T/R Bias Line 18 for “A”, 26 for “B”
G3	F3	MC bias return 3	MC T/R Bias Line 19 for “A”, 27 for “B”
G4	F4	MC bias return 4	MC T/R Bias Line 20 for “A”, 28 for “B”
G5	F5	MC bias return 5	MC T/R Bias Line 21 for “A”, 29 for “B”
G6	F6	MC bias return 6	MC T/R Bias Line 22 for “A”, 30 for “B”
G7	F7	MC bias return 7	MC T/R Bias Line 23 for “A”, 31 for “B”
G8	F8	MC bias return 8	MC T/R Bias Line 24 for “A”, 32 for “B”
G9	n/a	Reserved	
G10	n/a	Reserved	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	-15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	



Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a	Key blocked in receptacle E	
B2	n/a	Key blocked in receptacle F	
B3	n/a	Key blocked in receptacle G	
B4	n/a	Key blocked in receptacle H	
C1	n/a	MC-A1	Preamp out 1, no bias, DC blocked
C2	n/a	MC-A2	Preamp out 2, no bias, DC blocked
C3	n/a	MC-A3	Preamp out 3, no bias, DC blocked
C4	n/a	MC-A4	Preamp out 4, no bias, DC blocked
D1	n/a	MC-A5	Preamp out 5, no bias, DC blocked
D2	n/a	MC-A6	Preamp out 6, no bias, DC blocked
D3	n/a	MC-A7	Preamp out 7, no bias, DC blocked
D4	n/a	MC-A8	Preamp out 8, no bias, DC blocked

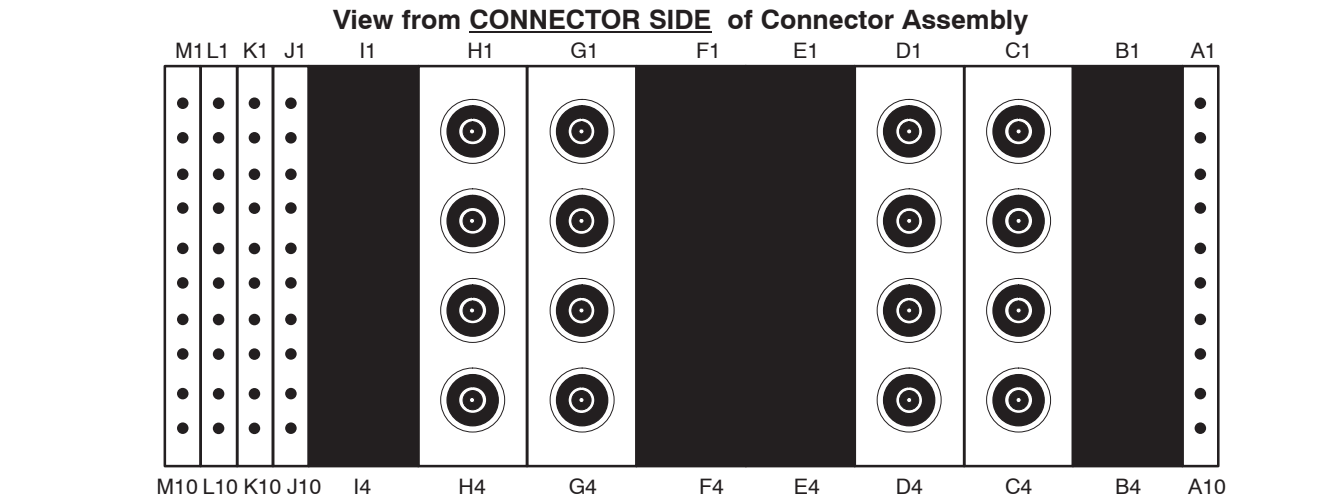
Contact Function Allocation			
Pin	Twisted with	Description	Comments
F2	n/a	Transmit Signal up to 8kw PEP	
H3	n/a	Reflected power to 50 ohm load 2.5kw PEP, 50w RMS	“A” connector only
J1	K1	MC bias 1	
J2	K2	MC bias 2	
J3	K3	MC bias 3	
J4	K4	MC bias 4	
J5	K5	MC bias 5	
J6	K6	MC bias 6	
J7	K7	MC bias 7	
J8	K8	MC bias 8	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1	
K2	J2	MC bias return 2	
K3	J3	MC bias return 3	
K4	J4	MC bias return 4	
K5	J5	MC bias return 5	
K6	J6	MC bias return 6	
K7	J7	MC bias return 7	
K8	J8	MC bias return 8	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	-15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	

View from CONNECTOR SIDE of Connector Assembly



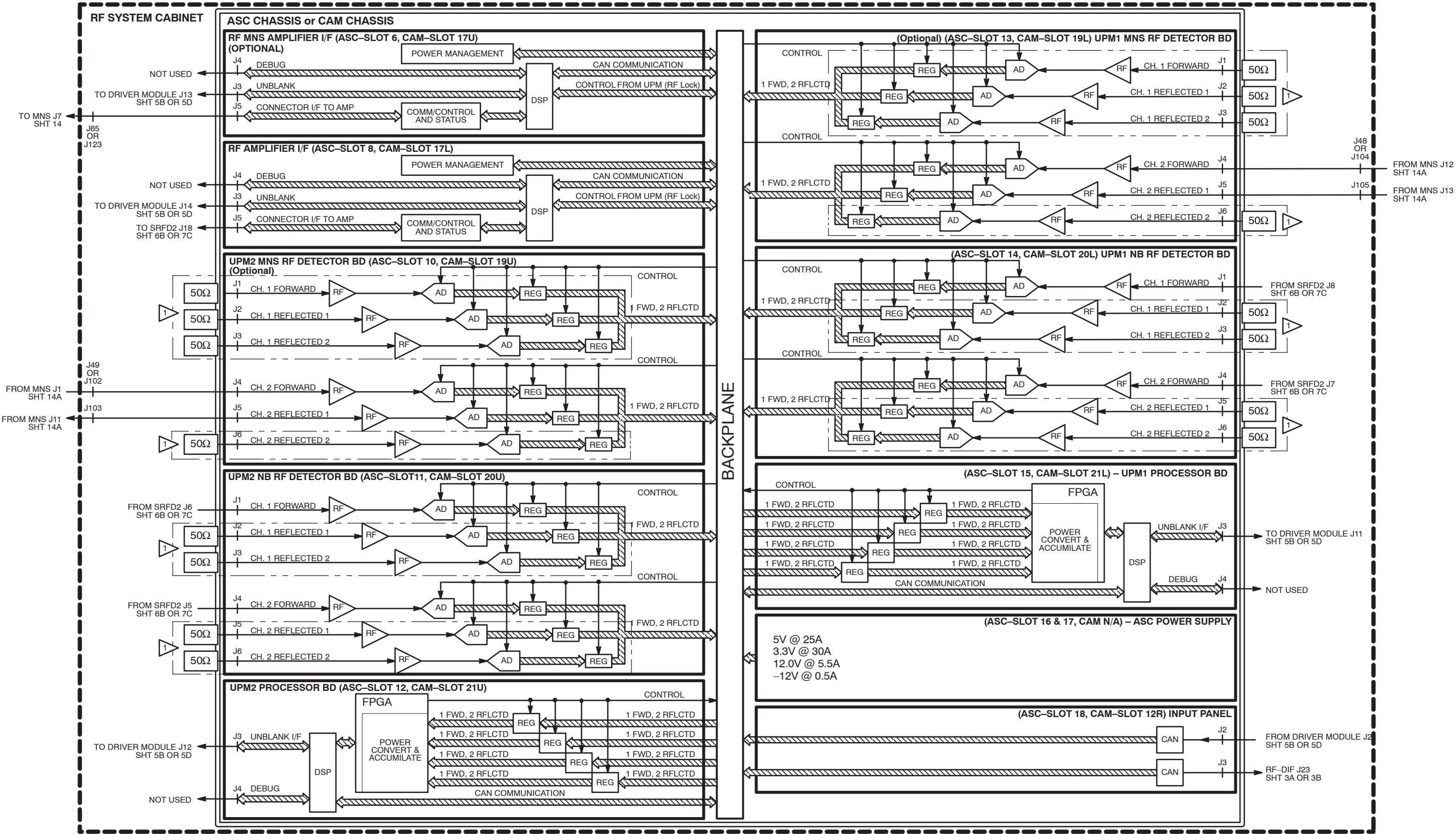
Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a		
B2	n/a		
B3	n/a		
B4	n/a		
C1	n/a	MC-B1	Preamp out 1, MC bias 1 (9)
C2	n/a	MC-B2	Preamp out 2, MC bias 2 (10)
C3	n/a	MC-B3	Preamp out 3, MC bias 3 (11)
C4	n/a	MC-B4	Preamp out 4, MC bias 4 (12)
D1	n/a	MC-B5	Preamp out 5, MC bias 5 (13)
D2	n/a	MC-B6	Preamp out 6, MC bias 6 (14)
D3	n/a	MC-B7	Preamp out 7, MC bias 7 (15)
D4	n/a	MC-B8	Preamp out 8, MC bias 8 (16)

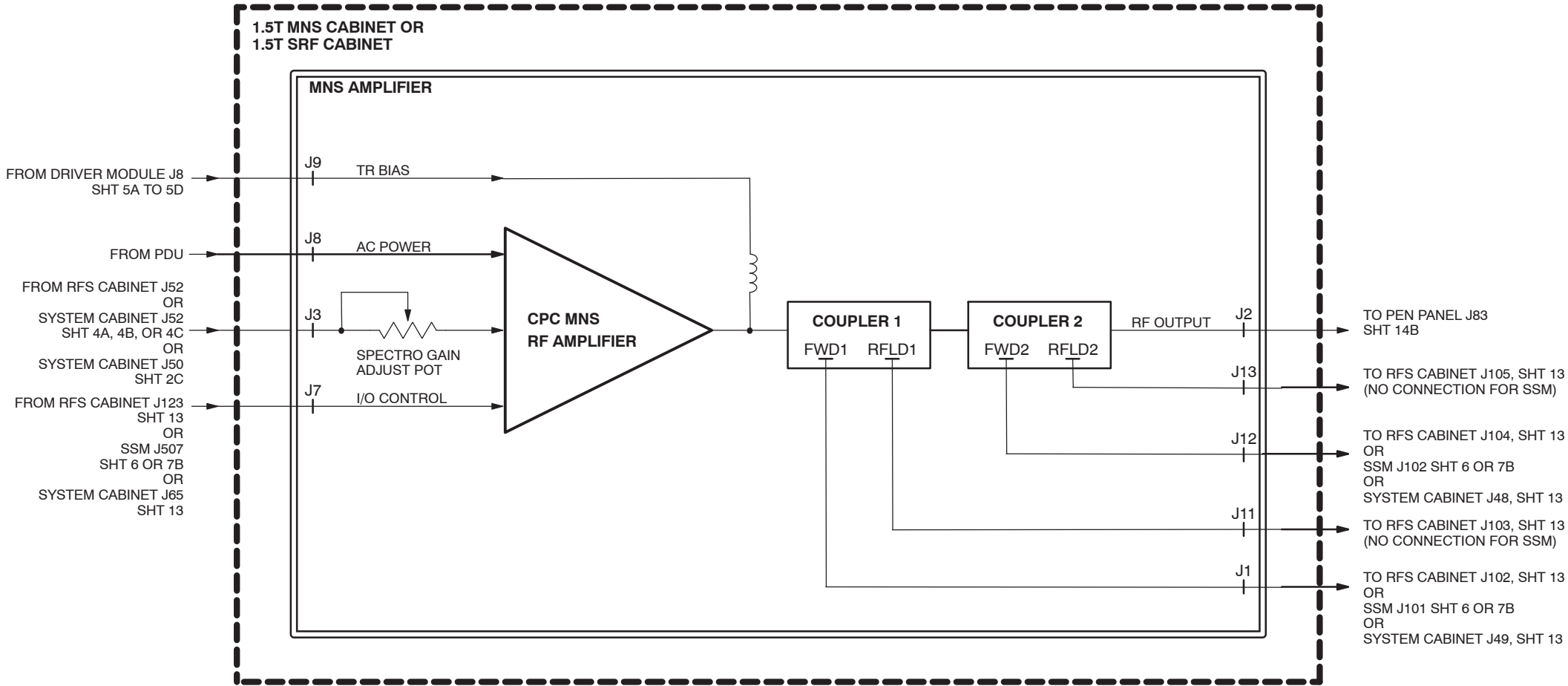
Contact Function Allocation			
Pin	Twisted with	Description	Comments
F2	n/a		
J1	K1	MC bias 1 (9)	
J2	K2	MC bias 2 (10)	
J3	K3	MC bias 3 (11)	
J4	K4	MC bias 4 (12)	
J5	K5	MC bias 5 (13)	
J6	K6	MC bias 6 (14)	
J7	K7	MC bias 7 (15)	
J8	K8	MC bias 8 (16)	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1 (9)	
K2	J2	MC bias return 2 (10)	
K3	J3	MC bias return 3 (11)	
K4	J4	MC bias return 4 (12)	
K5	J5	MC bias return 5 (13)	
K6	J6	MC bias return 6 (14)	
K7	J7	MC bias return 7 (15)	
K8	J8	MC bias return 8 (16)	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	-15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	

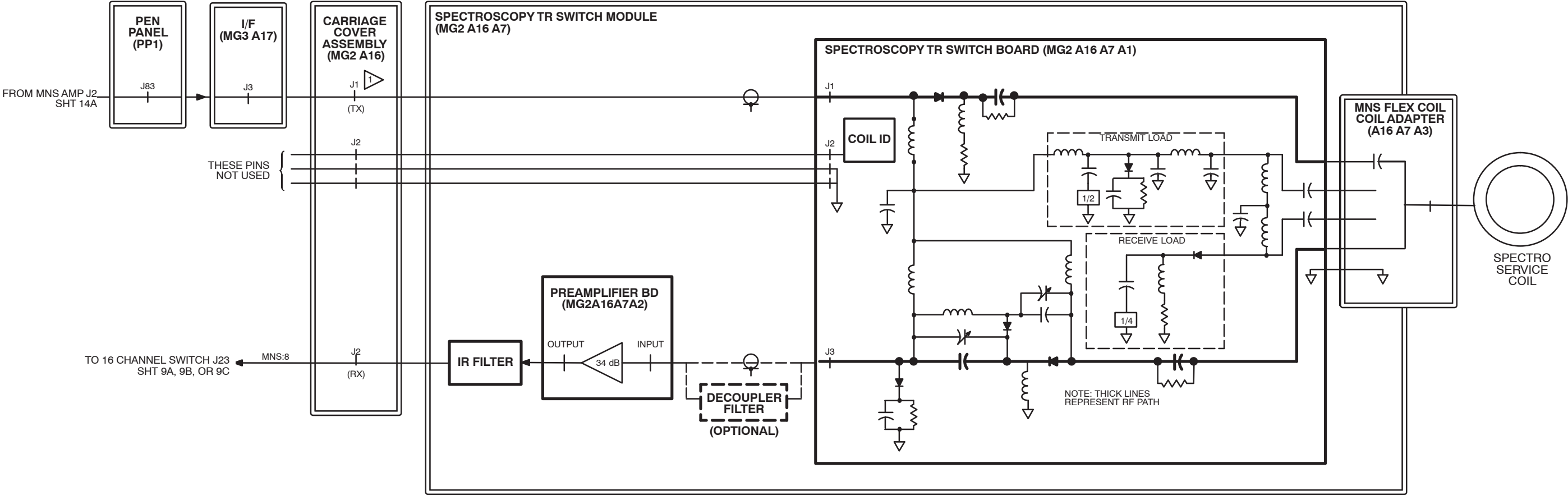


Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
C1	n/a	MC-B1 or MC-C1, MC bias 1B (9), 1C (25)	Preamp out 1, MC bias 1B (9), 1C (25)
C2	n/a	MC-B2 or MC-C2, MC bias 2B (10), 2C (26)	Preamp out 2, MC bias 2B (10), 2C (26)
C3	n/a	MC-B3 or MC-C3, MC bias 3B (11), 3C (27)	Preamp out 3, MC bias 3B (11), 3C (27)
C4	n/a	MC-B4 or MC-C4, MC bias 4B (12), 4C (28)	Preamp out 4, MC bias 4B (12), 4C (28)
D1	n/a	MC-B5 or MC-C5, MC bias 5B (13), 5C (29)	Preamp out 5, MC bias 5B (13), 5C (29)
D2	n/a	MC-B6 or MC-C6, MC bias 6B (14), 6C (30)	Preamp out 6, MC bias 6B (14), 6C (30)
D3	n/a	MC-B7 or MC-C7, MC bias 7B (15), 7C (31)	Preamp out 7, MC bias 7B (15), 7C (31)
D4	n/a	MC-B8 or MC-C8, MC bias 8B (16), 8C (32)	Preamp out 8, MC bias 8B (16), 8C (32)
G1	n/a	MC-B9 or MC-C9, MC bias 9B (25), 9C (1)	Preamp out 1, MC bias 9B (25), 9C (1)
G2	n/a	MC-B10 or MC-C10, MC bias 10B (26), 10C (2)	Preamp out 2, MC bias 10B (26), 10C (2)
G3	n/a	MC-B11 or MC-C11, MC bias 11B (27), 11C (3)	Preamp out 3, MC bias 11B (27), 11C (3)
G4	n/a	MC-B12 or MC-C12, MC bias 12B (28), 12C (4)	Preamp out 4, MC bias 12B (28), 12C (4)
H1	n/a	MC-B13 or MC-C13, MC bias 13B (29), 13C (5)	Preamp out 5, MC bias 13B (29), 13C (5)
H2	n/a	MC-B14 or MC-C14, MC bias 14B (30), 14C (6)	Preamp out 6, MC bias 14B (30), 14C (6)
H3	n/a	MC-B15 or MC-C15, MC bias 15B (31), 15C (7)	Preamp out 7, MC bias 15B (31), 15C (7)
H4	n/a	MC-B16 or MC-C16, MC bias 16B (32), 16C (8)	Preamp out 8, MC bias 16B (32), 16C (8)

Contact Function Allocation			
Pin	Twisted with	Description	Comments
J1	K1	MC bias 1 (9)–B / MC bias 1 (25)–C	
J2	K2	MC bias 2 (10)–B / MC bias 2 (26)–C	
J3	K3	MC bias 3 (11)–B / MC bias 3 (27)–C	
J4	K4	MC bias 4 (12)–B / MC bias 4 (28)–C	
J5	K5	MC bias 5 (13)–B / MC bias 5 (29)–C	
J6	K6	MC bias 6 (14)–B / MC bias 6 (30)–C	
J7	K7	MC bias 7 (15)–B / MC bias 7 (31)–C	
J8	K8	MC bias 8 (16)–B / MC bias 8 (32)–C	
J9	J10	body transmit enable	
J10	J9	body transmit enable return	
K1	J1	MC bias return 1 (25)–B / MC bias return 1 (1)–C	
K2	J2	MC bias return 2 (26)–B / MC bias return 2 (2)–C	
K3	J3	MC bias return 3 (27)–B / MC bias return 3 (3)–C	
K4	J4	MC bias return 4 (28)–B / MC bias return 4 (4)–C	
K5	J5	MC bias return 5 (29)–B / MC bias return 5 (5)–C	
K6	J6	MC bias return 6 (30)–B / MC bias return 6 (6)–C	
K7	J7	MC bias return 7 (31)–B / MC bias return 7 (7)–C	
K8	J8	MC bias return 8 (32)–B / MC bias return 8 (8)–C	
K9	K10	System preamp enable return	
K10	K9	System preamp enable	
L1	n/a	Reserved	
L2	n/a	Reserved	
L3	n/a	Reserved	
L4	n/a	Reserved	
L5	n/a	Reserved	
L6	n/a	Reserved	
L7	n/a	Reserved	
L8	n/a	Reserved	
L9	n/a	Reserved	
L10	n/a	Reserved	
M1	n/a	Reserved	
M2	M5	+15Vdc (auxiliary & preamp power)	
M3	M6	+10Vdc (auxiliary & preamp power)	
M4	n/a	–15Vdc (auxiliary power)	
M5	M2	Return for aux power	
M6	M3	Return for aux power	
M7	M8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
M8	M7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
M9	M10	coil present'	
M10	M9	coil present return'	

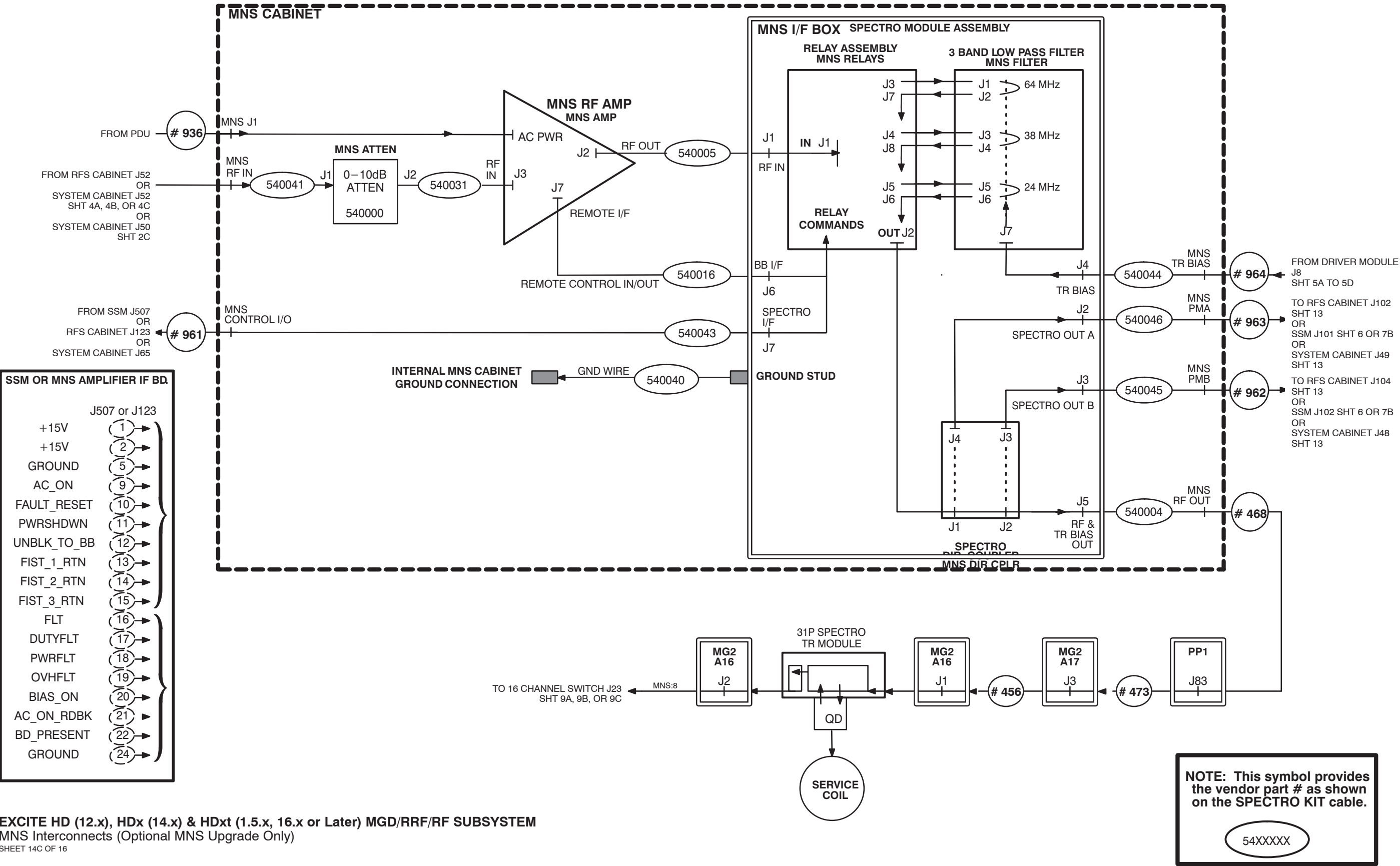


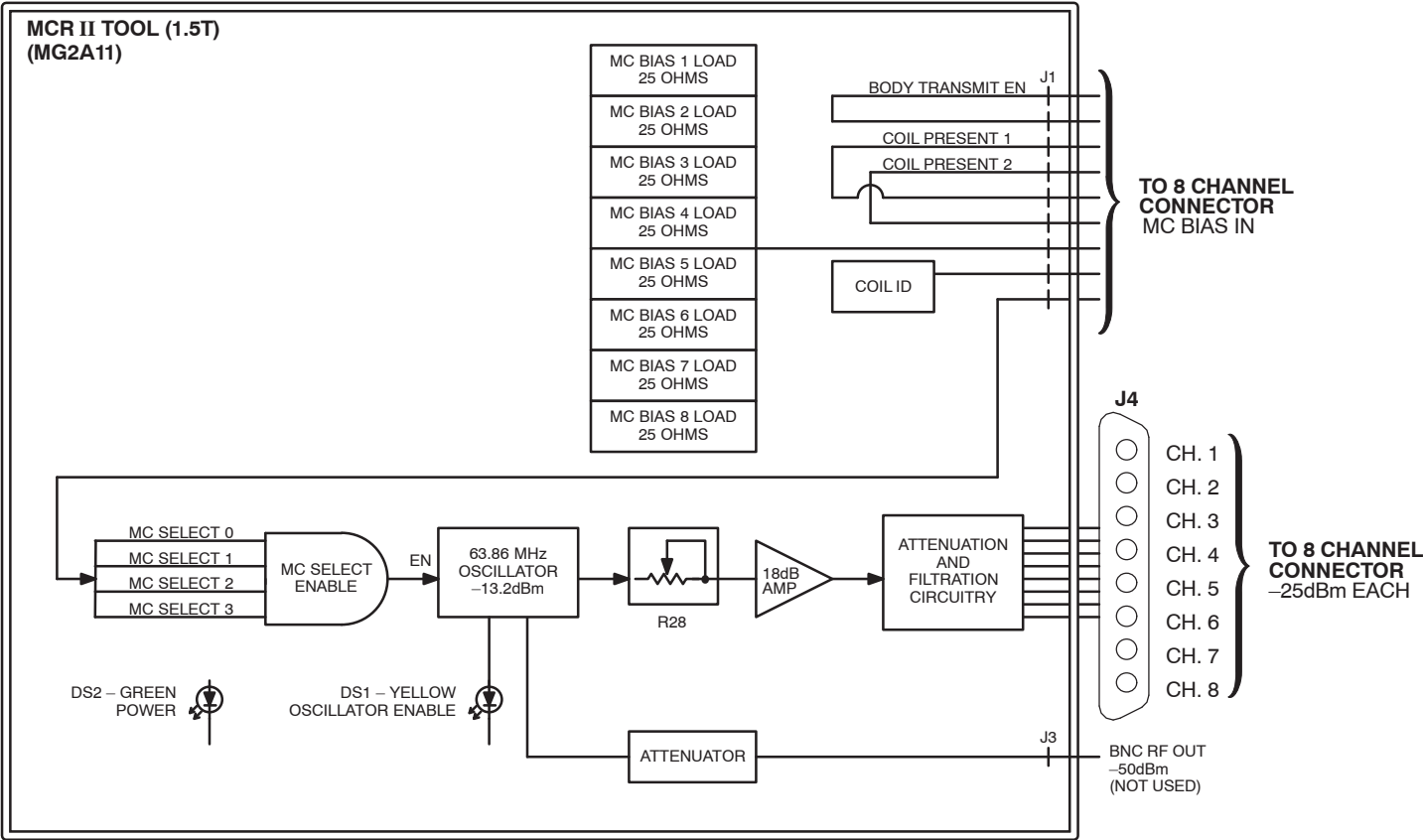




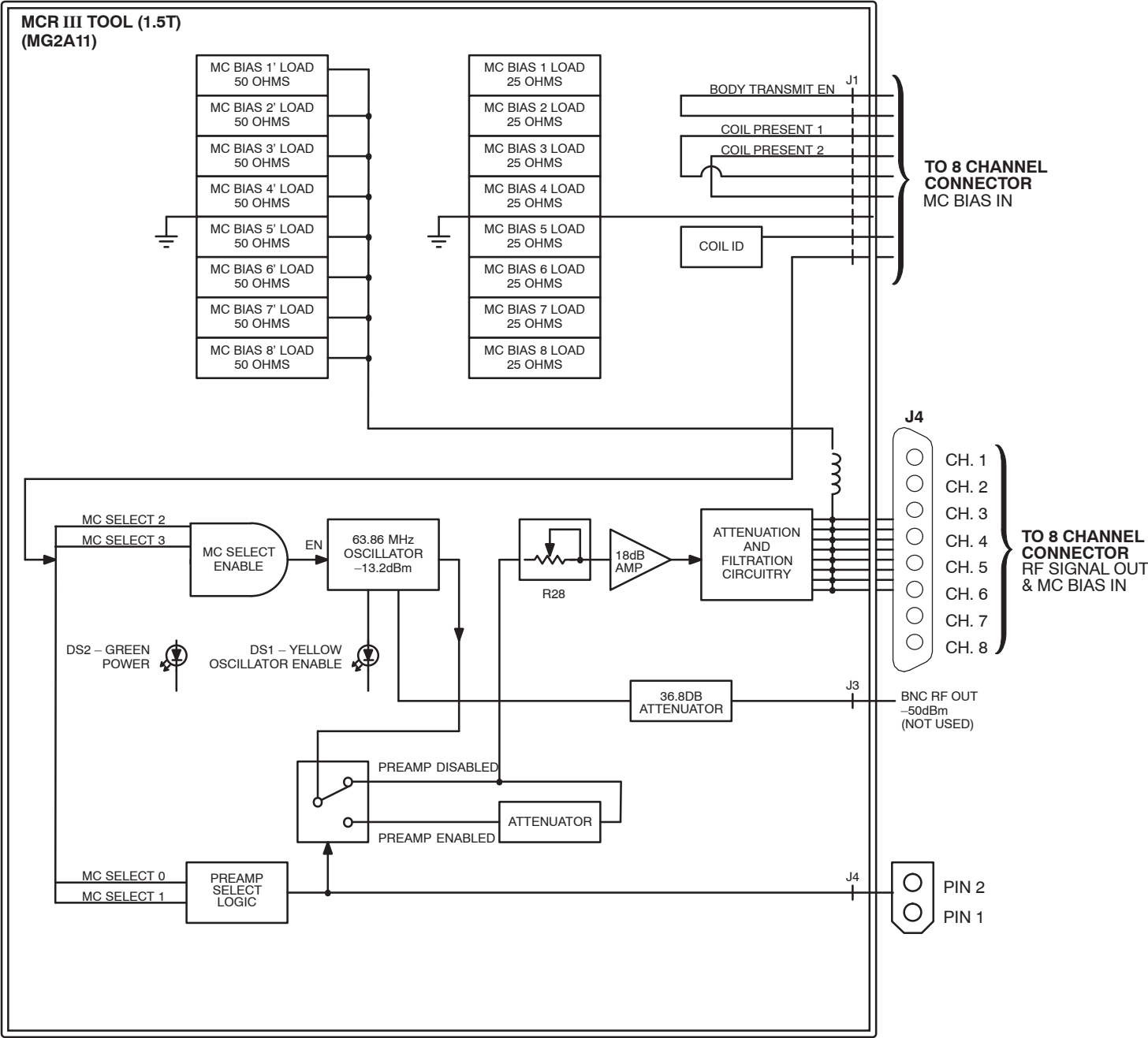
NOTES:

1 ▷ SHV Thick Coax Pasternack PE-4194 connector.



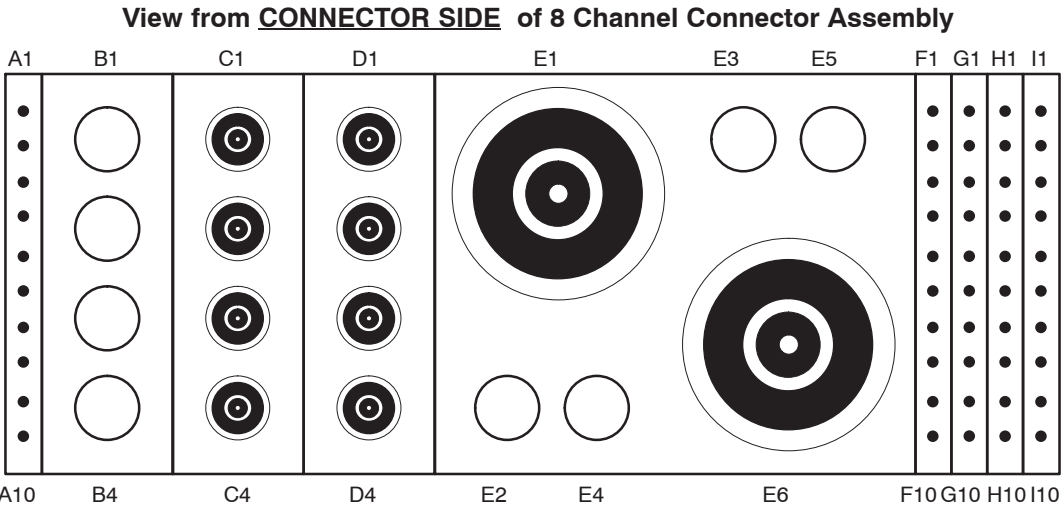


J1	
PINS	SIGNAL
1	MC Bias 1
2	MC Bias 2
3	MC Bias 3
4	MC Bias 4
5	MC Bias 5
6	MC Bias 6
7	MC Bias 7
8	MC Bias 8
9	MC Select 0
10	MC Select 1
11	MC Select 2
12	MC Select 3
13	Body Transmit Enable
14	Coil ID
15	Coil Present 15
16	Coil Present 2
17	+15VDC (Auxiliary Power)
18	+10VDC (Auxiliary Power)
19	-15VDC (Auxiliary Power)
20	MC Bias Return 1
21	MC Bias Return 2
22	MC Bias Return 3
23	MC Bias Return 4
24	MC Bias Return 5
25	MC Bias Return 6
26	MC Bias Return 7
27	MC Bias Return 8
28	MC Select 0 Return
29	MC Select 1 Return
30	MC Select 2 Return
31	MC Select 3 Return
32	Body Transmit Enable Return
33	Coil ID Return
34	Coil Present Return 1
35	Coil Present Return 2
36	Auxiliary Power Return
37	Auxiliary Power Return



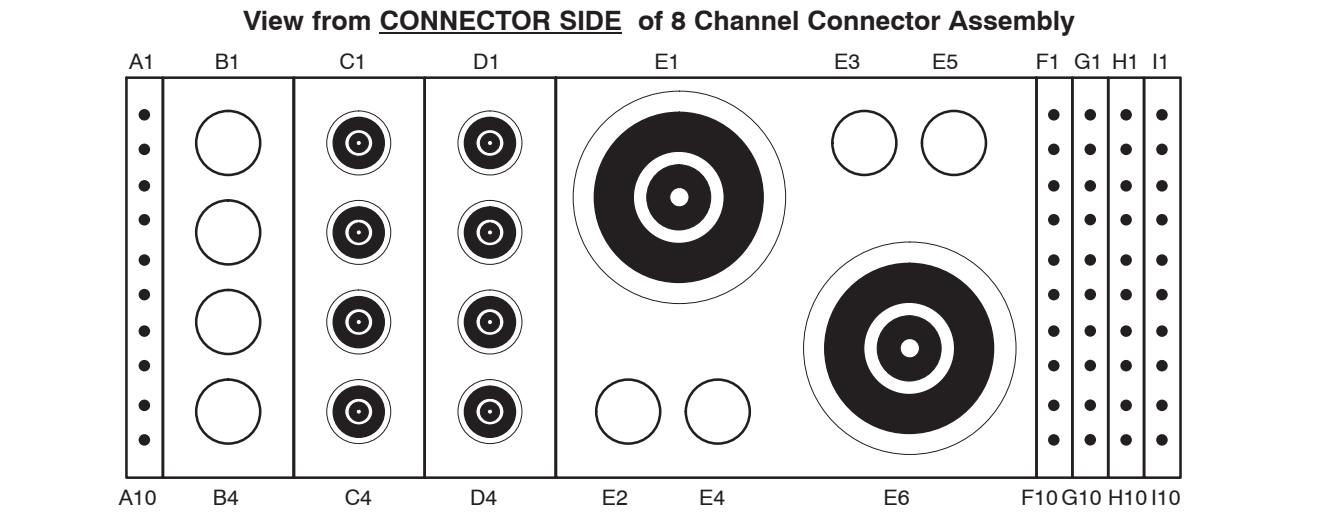
J1	
PINS	SIGNAL
1	MC Bias 1
2	MC Bias 2
3	MC Bias 3
4	MC Bias 4
5	MC Bias 5
6	MC Bias 6
7	MC Bias 7
8	MC Bias 8
9	MC Select 0
10	MC Select 1
11	MC Select 2
12	MC Select 3
13	Body Transmit Enable
14	Coil ID
15	Coil Present 15
16	Coil Present 2
17	+15VDC (Auxiliary Power)
18	+10VDC (Auxiliary Power)
19	-15VDC (Auxiliary Power)
20	MC Bias Return 1
21	MC Bias Return 2
22	MC Bias Return 3
23	MC Bias Return 4
24	MC Bias Return 5
25	MC Bias Return 6
26	MC Bias Return 7
27	MC Bias Return 8
28	MC Select 0 Return
29	MC Select 1 Return
30	MC Select 2 Return
31	MC Select 3 Return
32	Body Transmit Enable Return
33	Coil ID Return
34	Coil Present Return 1
35	Coil Present Return 2
36	Auxiliary Power Return
37	Auxiliary Power Return

J4	
PINS	SIGNAL
1	Preamp Enable
2	Preamp Enable RTN



Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a		Not used
B2	n/a		Not used
B3	n/a		Not used
B4	n/a		Not used
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

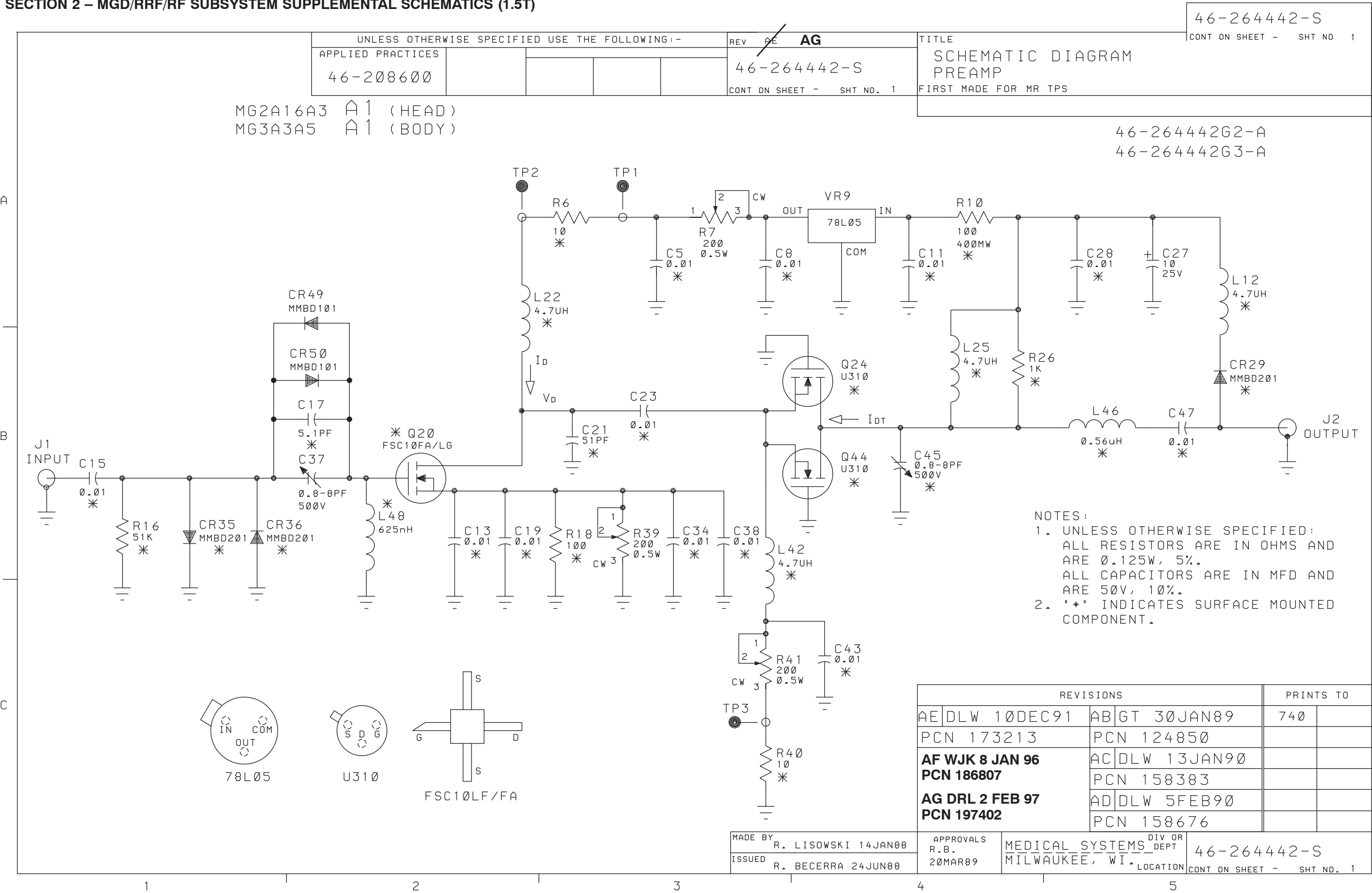
Contact Function Allocation			
Pin	Twisted with	Description	Comments
E1	n/a	Open	Not used
E2	n/a	Open	Not used
E3	n/a	Open	Not used
E4	n/a	Open	Not used
E5	n/a	Open	Not used
E6	n/a	Open	Not used
F1	G1	MC bias 1	MC T/R Bias Line 17 for “A”, 25 for “B”
F2	G2	MC bias 2	MC T/R Bias Line 18 for “A”, 26 for “B”
F3	G3	MC bias 3	MC T/R Bias Line 19 for “A”, 27 for “B”
F4	G4	MC bias 4	MC T/R Bias Line 20 for “A”, 28 for “B”
F5	G5	MC bias 5	MC T/R Bias Line 21 for “A”, 29 for “B”
F6	G6	MC bias 6	MC T/R Bias Line 22 for “A”, 30 for “B”
F7	G7	MC bias 7	MC T/R Bias Line 23 for “A”, 31 for “B”
F8	G8	MC bias 8	MC T/R Bias Line 24 for “A”, 32 for “B”
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for “A”, 25 for “B”
G2	F2	MC bias return 2	MC T/R Bias Line 18 for “A”, 26 for “B”
G3	F3	MC bias return 3	MC T/R Bias Line 19 for “A”, 27 for “B”
G4	F4	MC bias return 4	MC T/R Bias Line 20 for “A”, 28 for “B”
G5	F5	MC bias return 5	MC T/R Bias Line 21 for “A”, 29 for “B”
G6	F6	MC bias return 6	MC T/R Bias Line 22 for “A”, 30 for “B”
G7	F7	MC bias return 7	MC T/R Bias Line 23 for “A”, 31 for “B”
G8	F8	MC bias return 8	MC T/R Bias Line 24 for “A”, 32 for “B”
G9	n/a	Reserved	
G10	n/a	Reserved	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	-15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	



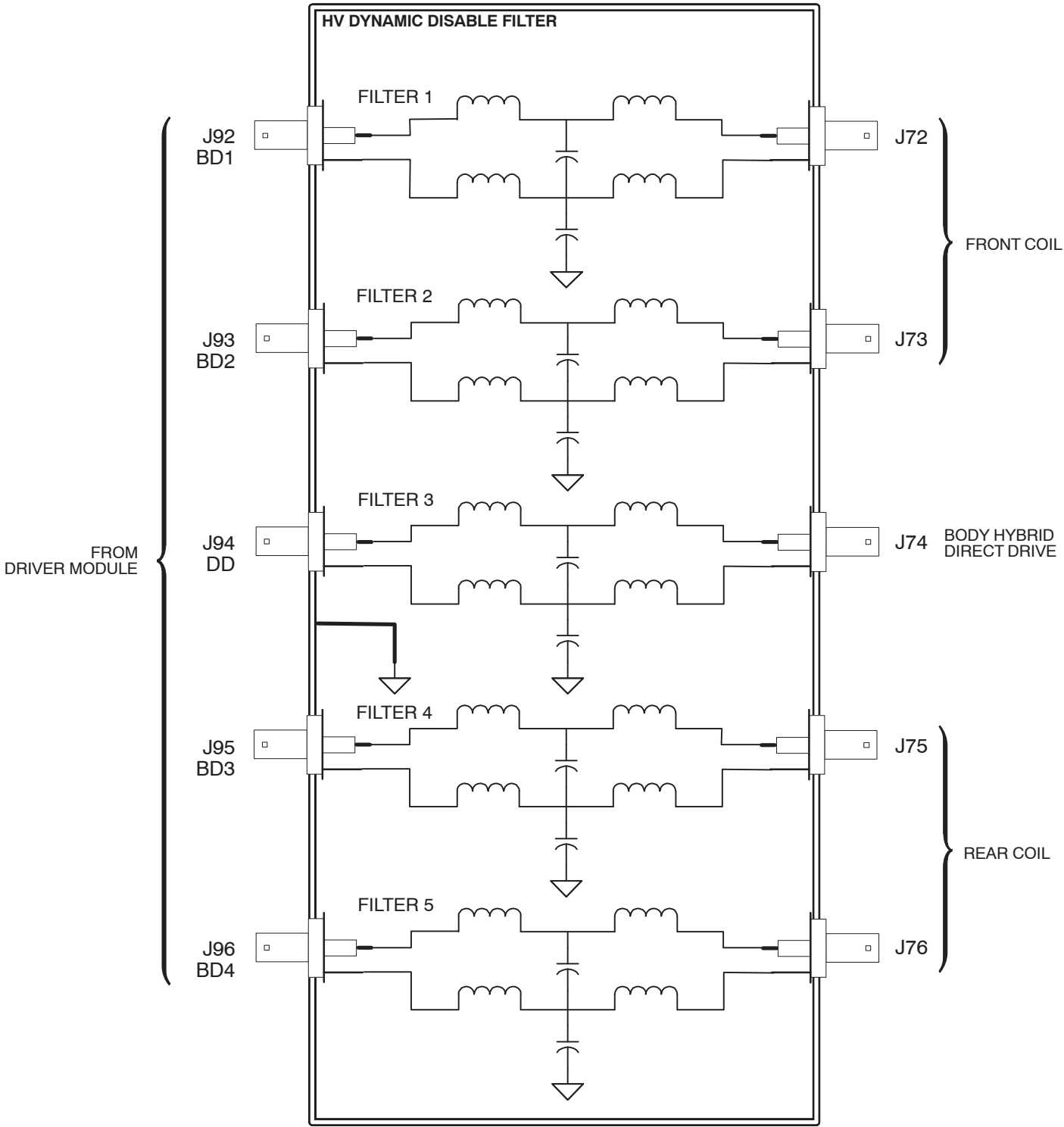
Contact Function Allocation			
Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a		Not used
B2	n/a		Not used
B3	n/a		Not used
B4	n/a		Not used
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

Contact Function Allocation			
Pin	Twisted with	Description	Comments
E1	n/a	Open	Not used
E2	n/a	Open	Not used
E3	n/a	Open	Not used
E4	n/a	Open	Not used
E5	n/a	Open	Not used
E6	n/a	Open	Not used
F1	G1	MC bias 1	MC T/R Bias Line 17 for “A”, 25 for “B”
F2	G2	MC bias 2	MC T/R Bias Line 18 for “A”, 26 for “B”
F3	G3	MC bias 3	MC T/R Bias Line 19 for “A”, 27 for “B”
F4	G4	MC bias 4	MC T/R Bias Line 20 for “A”, 28 for “B”
F5	G5	MC bias 5	MC T/R Bias Line 21 for “A”, 29 for “B”
F6	G6	MC bias 6	MC T/R Bias Line 22 for “A”, 30 for “B”
F7	G7	MC bias 7	MC T/R Bias Line 23 for “A”, 31 for “B”
F8	G8	MC bias 8	MC T/R Bias Line 24 for “A”, 32 for “B”
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for “A”, 25 for “B”
G2	F2	MC bias return 2	MC T/R Bias Line 18 for “A”, 26 for “B”
G3	F3	MC bias return 3	MC T/R Bias Line 19 for “A”, 27 for “B”
G4	F4	MC bias return 4	MC T/R Bias Line 20 for “A”, 28 for “B”
G5	F5	MC bias return 5	MC T/R Bias Line 21 for “A”, 29 for “B”
G6	F6	MC bias return 6	MC T/R Bias Line 22 for “A”, 30 for “B”
G7	F7	MC bias return 7	MC T/R Bias Line 23 for “A”, 31 for “B”
G8	F8	MC bias return 8	MC T/R Bias Line 24 for “A”, 32 for “B”
G9	G10	Preamp Enable	
G10	G9	Preamp Enable Return	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	-15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	

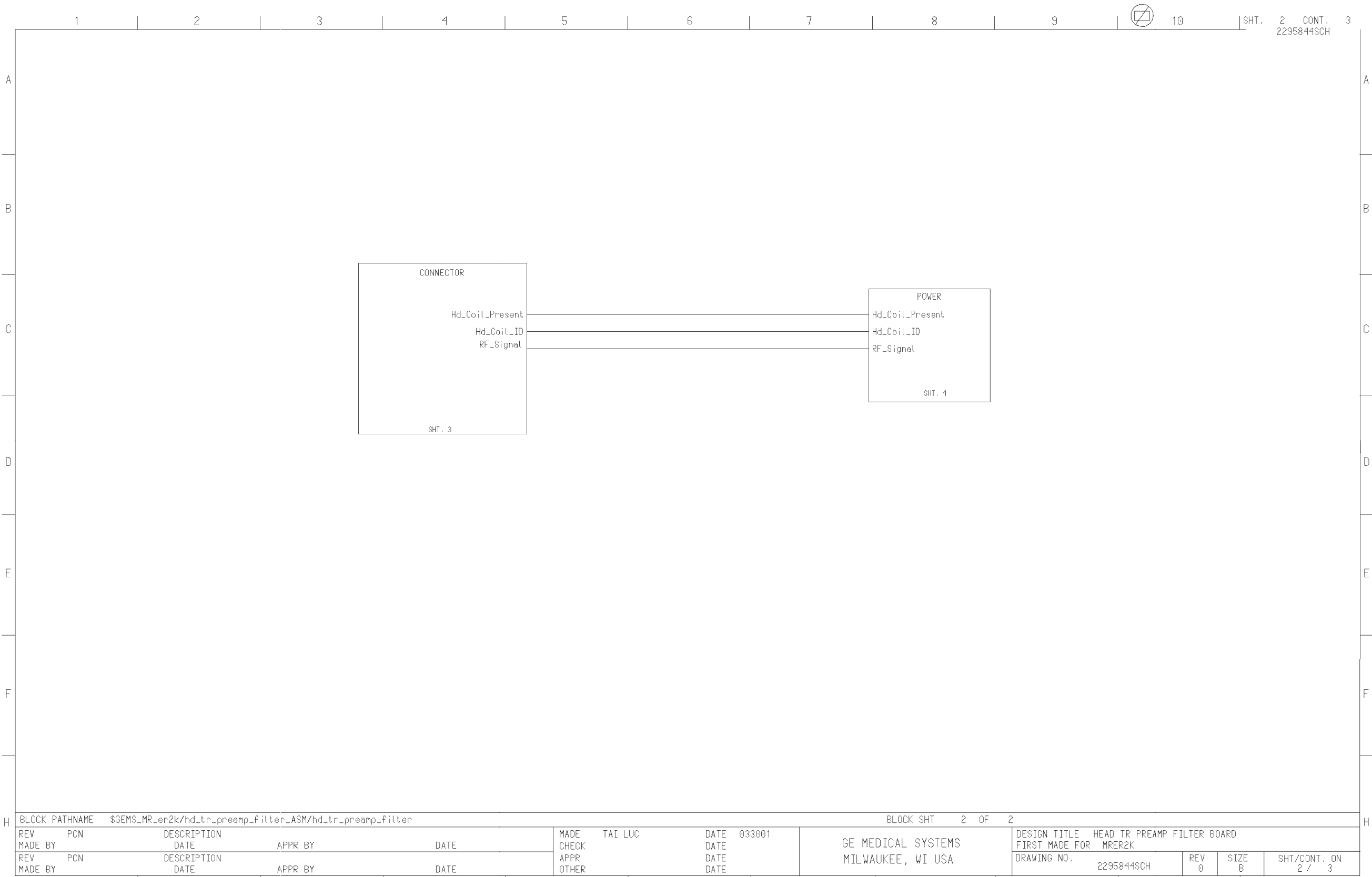
SECTION 2 – MGD/RRF/RF SUBSYSTEM SUPPLEMENTAL SCHEMATICS (1.5T)



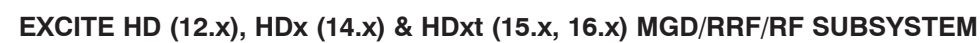
EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) MGD/RRF/RF SUBSYSTEM

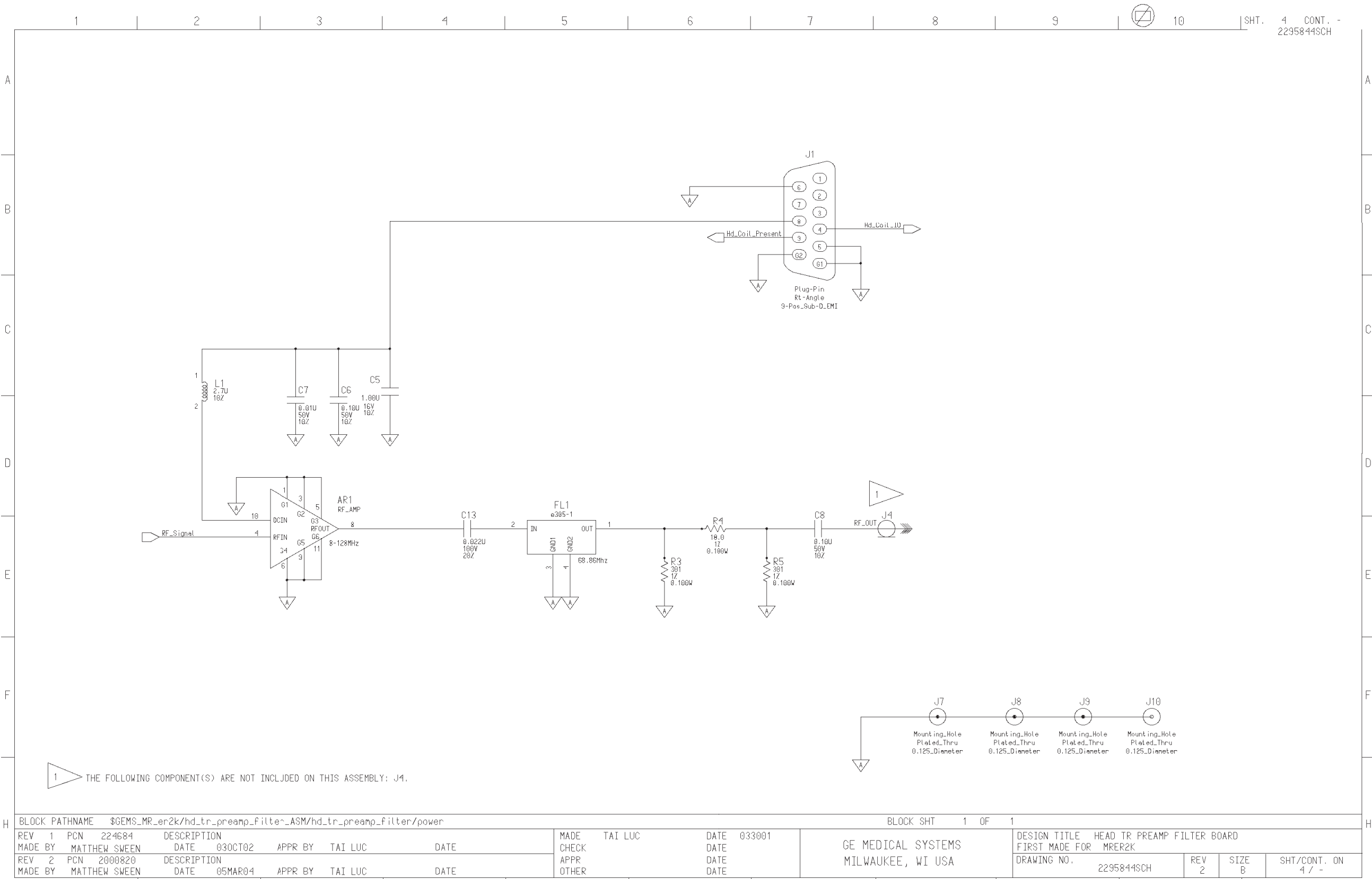


NOTE: Connector shields are electrically isolated from enclosure housing.

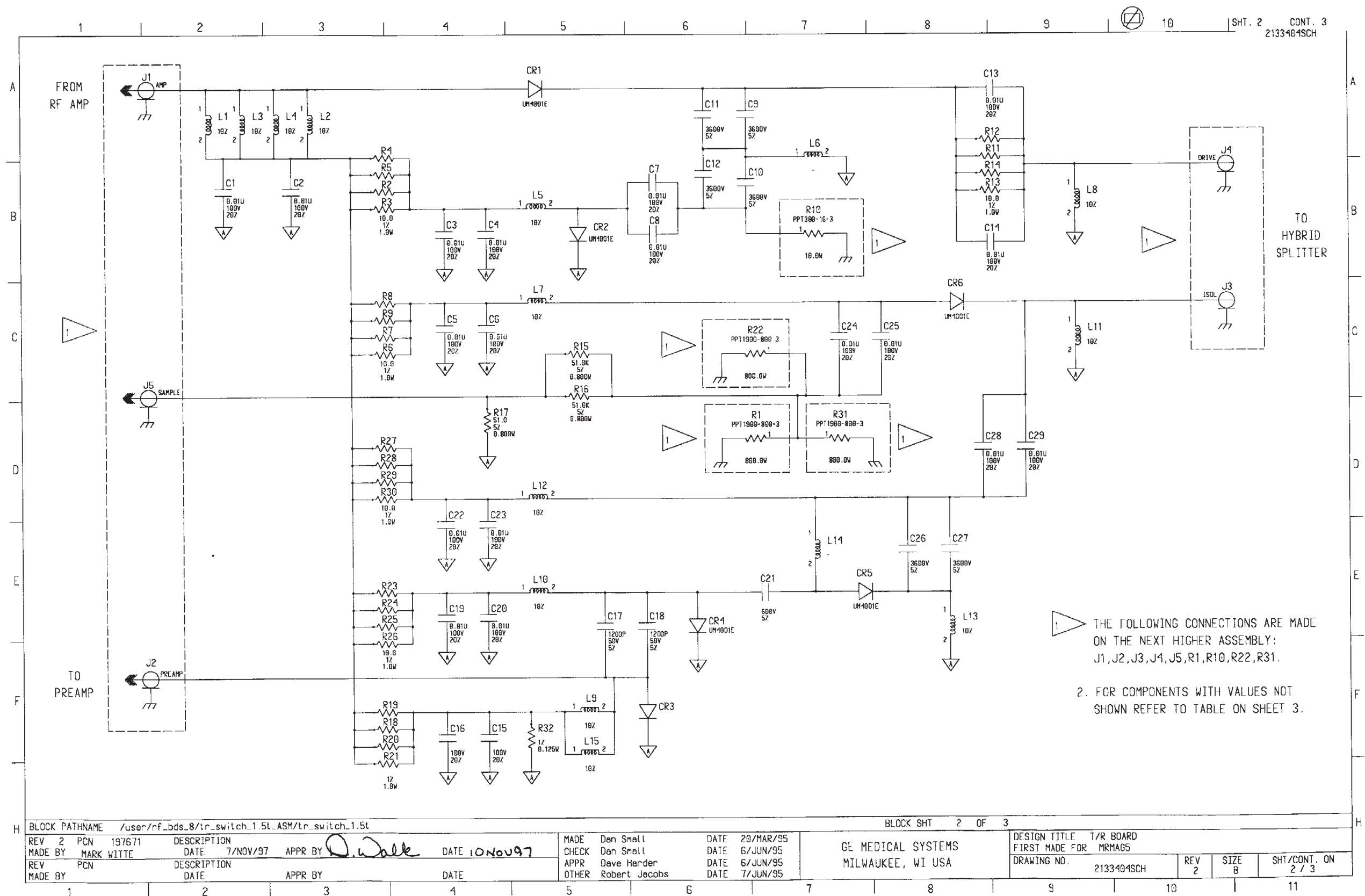


EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) MGD/RRF/RF SUBSYSTEM

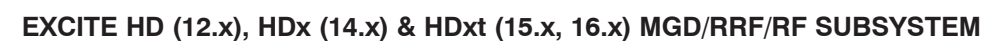


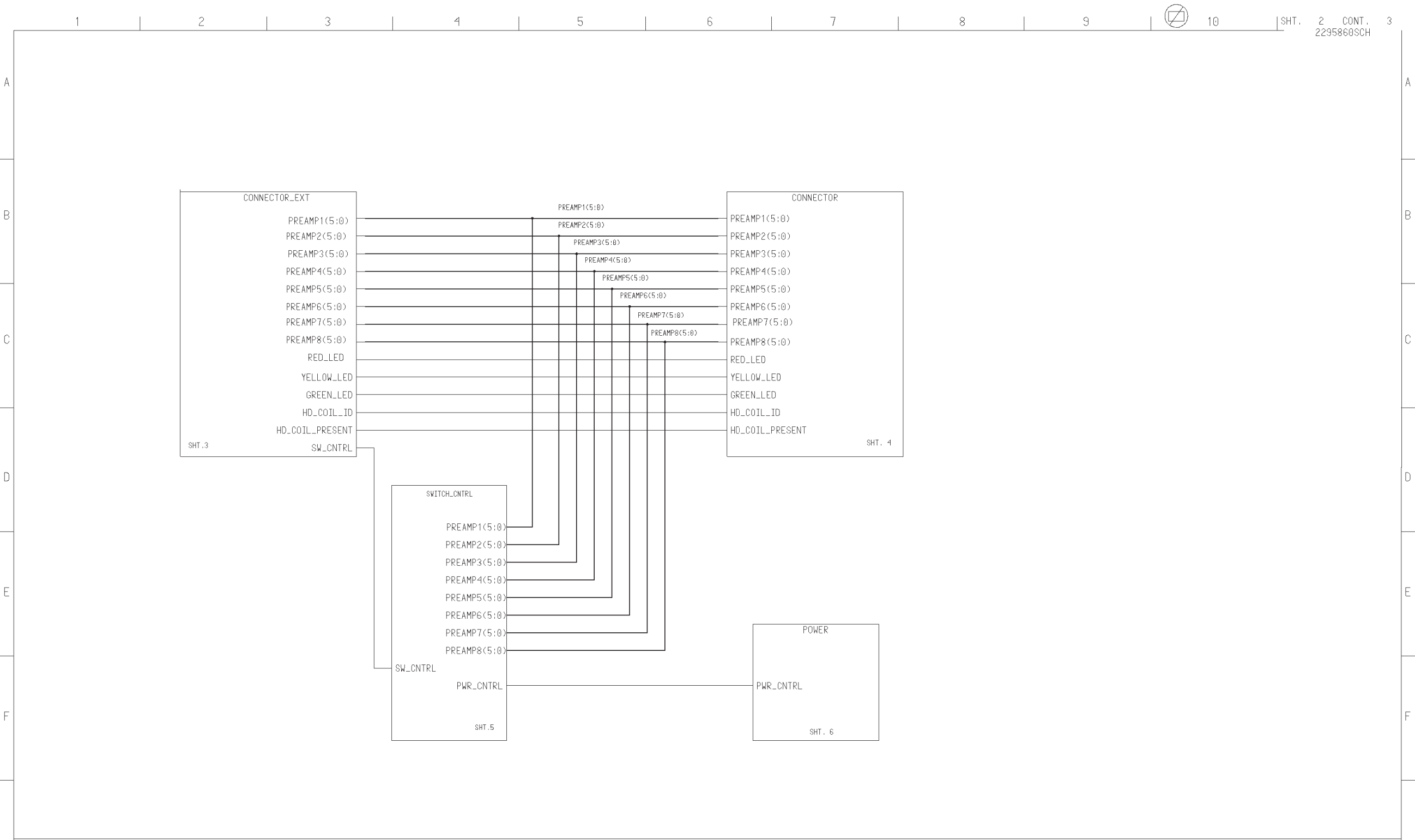


EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) MGD/RRF/RF SUBSYSTEM

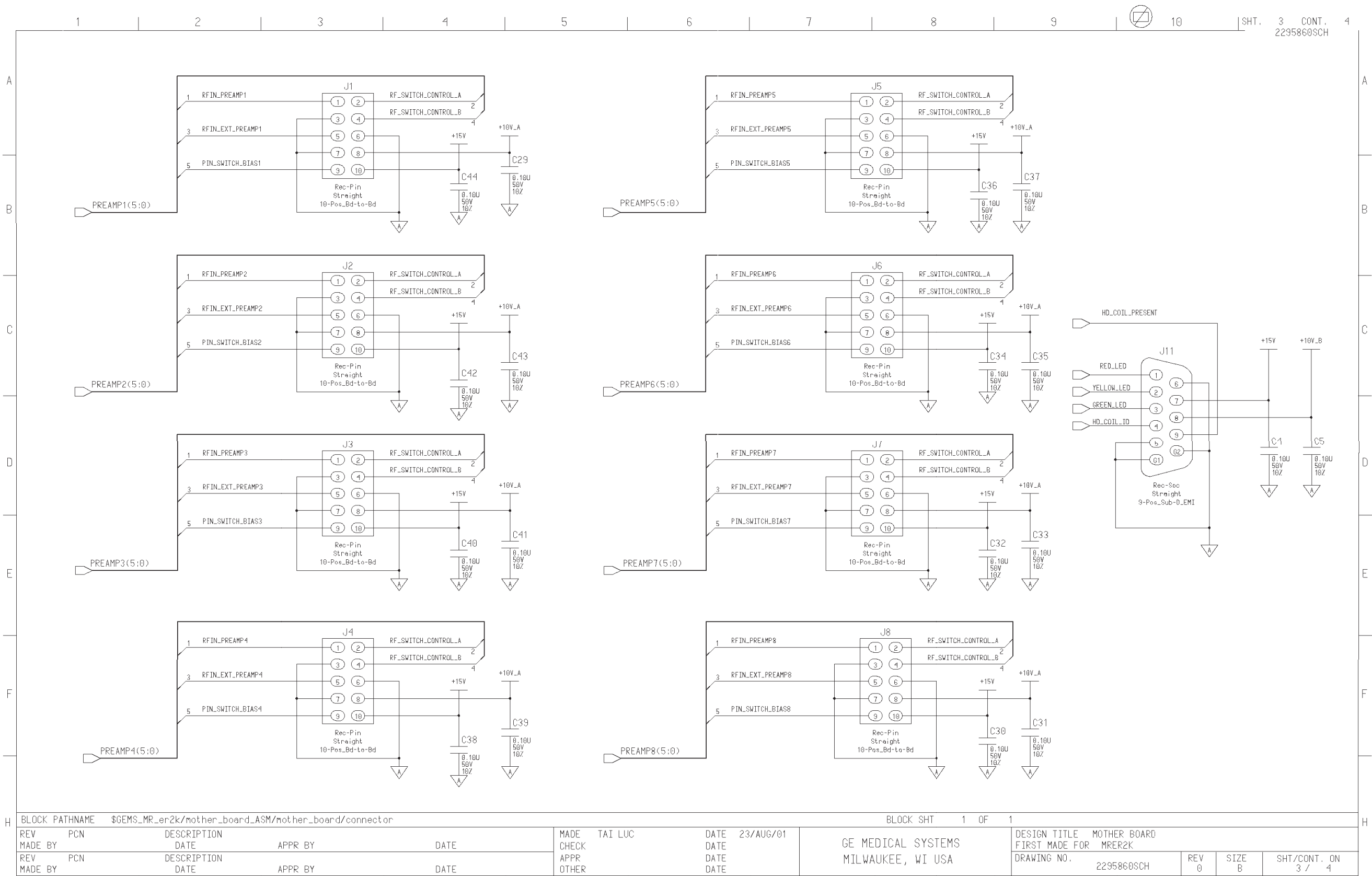


EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) MGD/RRF/RF SUBSYSTEM

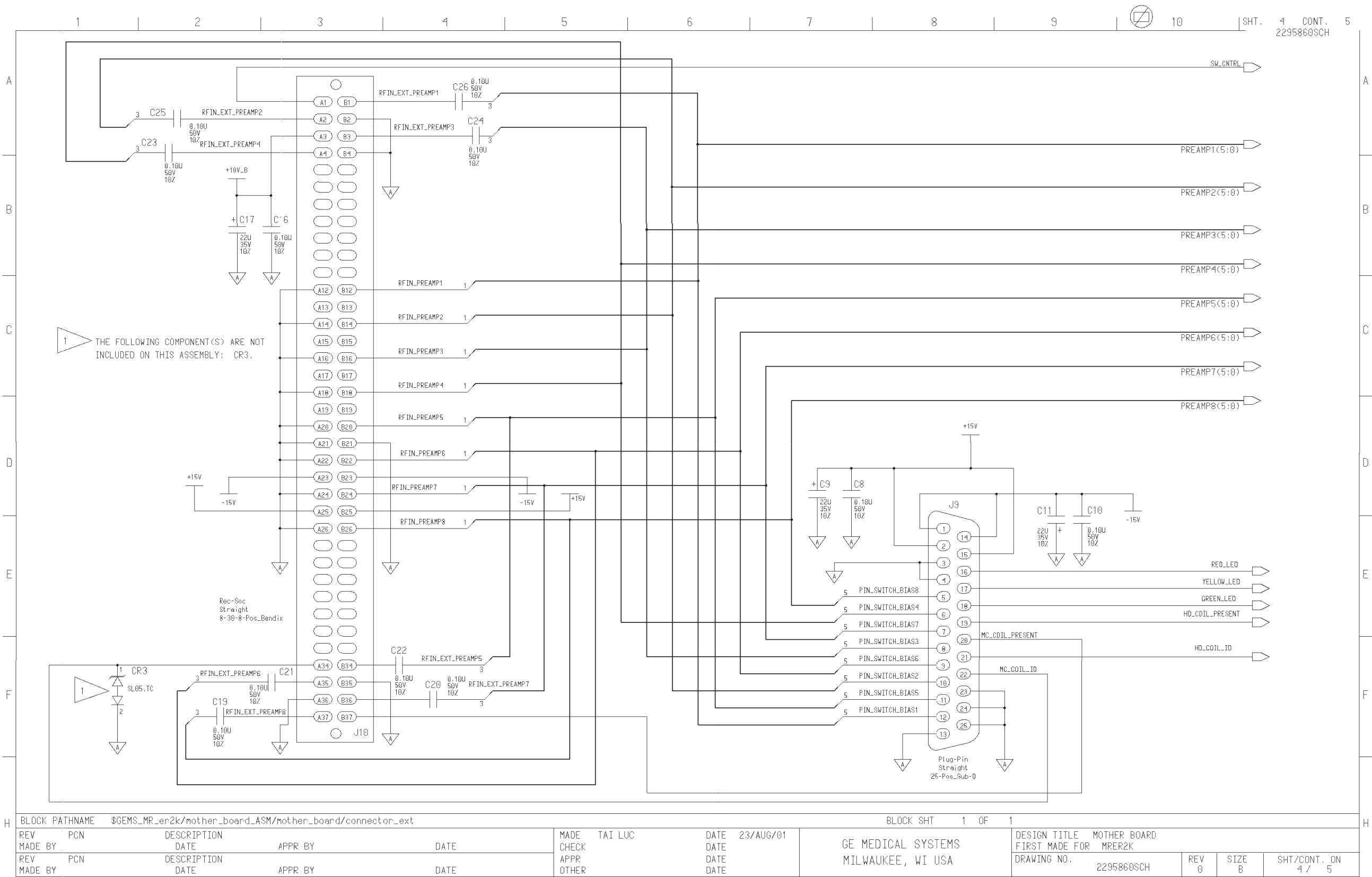


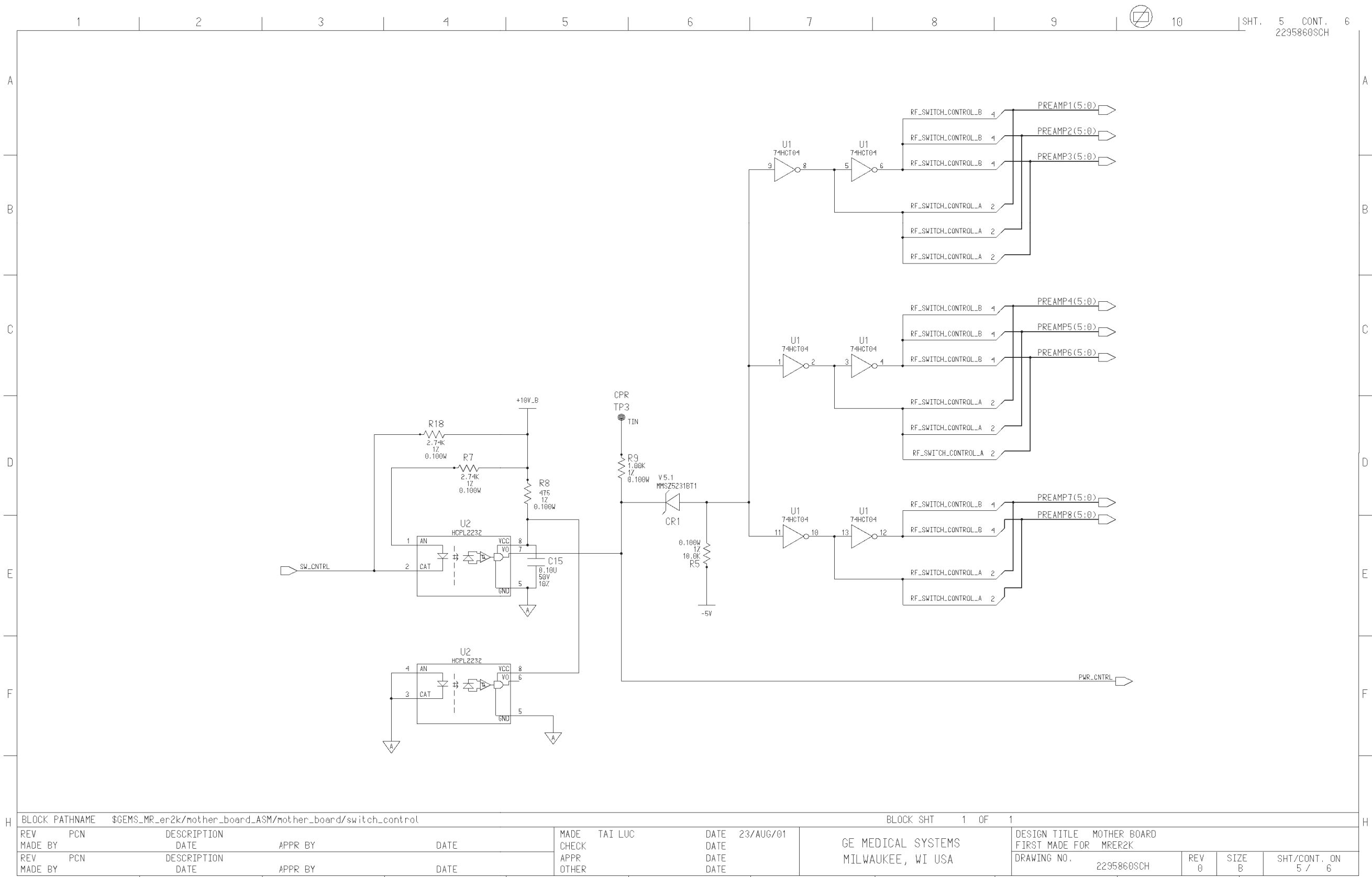


BLOCK PATHNAME \$GEMS_MR_er2k/mother_board_ASM/mother_board				BLOCK SHT 2 OF 2			
REV	PCN	DESCRIPTION	DATE	MADE	TAI LUC	DATE 23/AUG/01	DESIGN TITLE MOTHER BOARD
MADE BY		DATE	APPR BY	CHECK		DATE	FIRST MADE FOR MRER2K
REV	PCN	DESCRIPTION	DATE	APPR		DATE	DRAWING NO. 2295860SCH
MADE BY		DATE	APPR BY	OTHER		DATE	REV 0
						GE MEDICAL SYSTEMS MILWAUKEE, WI USA	SIZE B
							SHT/CONT. ON 2 / 3

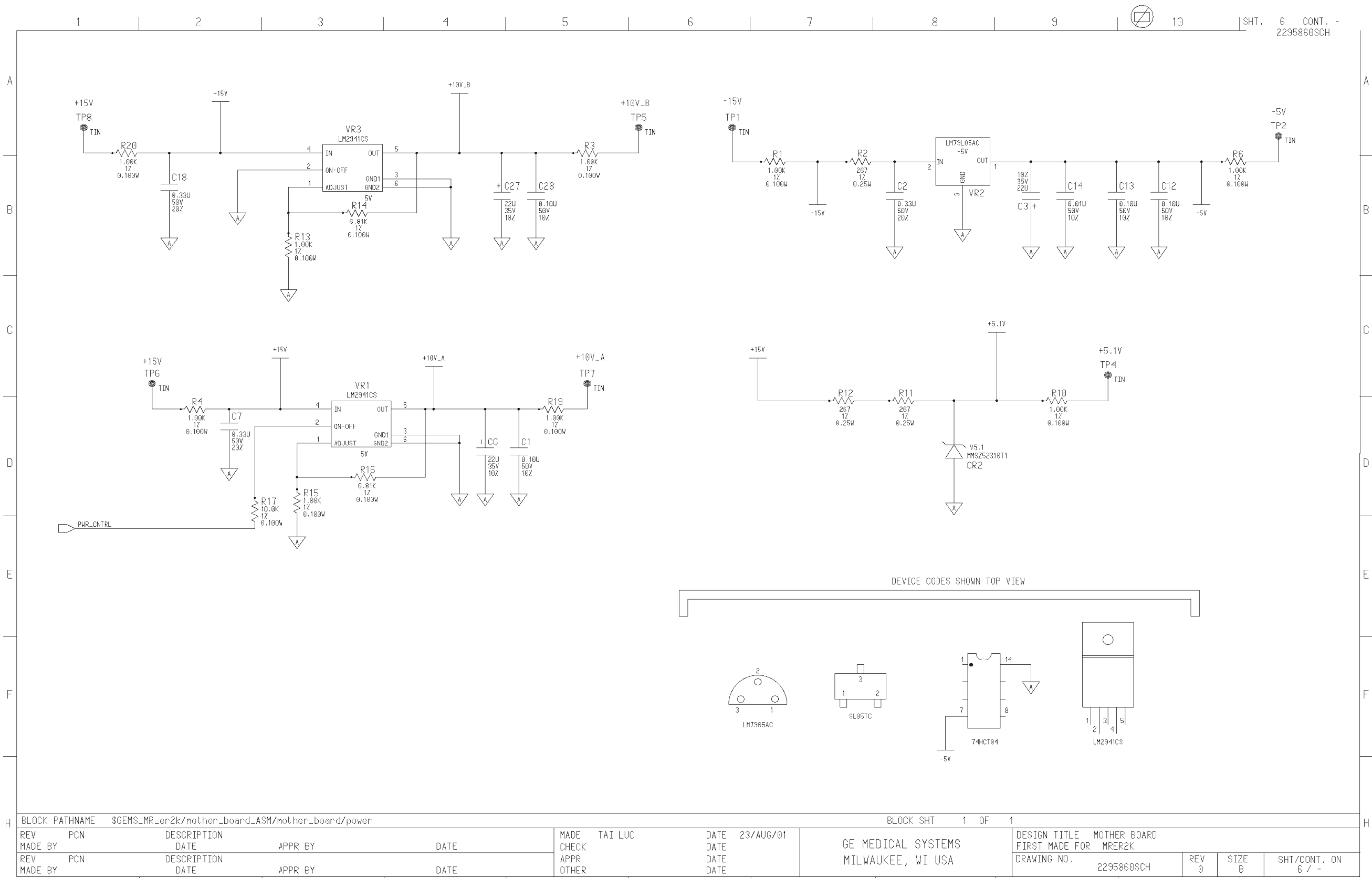


EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) MGD/RRF/RF SUBSYSTEM





EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) MGD/RRF/RF SUBSYSTEM



EXCITE HD (12.x), HDx (14.x) & HDxt (15.x, 16.x) MGD/RRF/RF SUBSYSTEM

COOLING SYSTEM

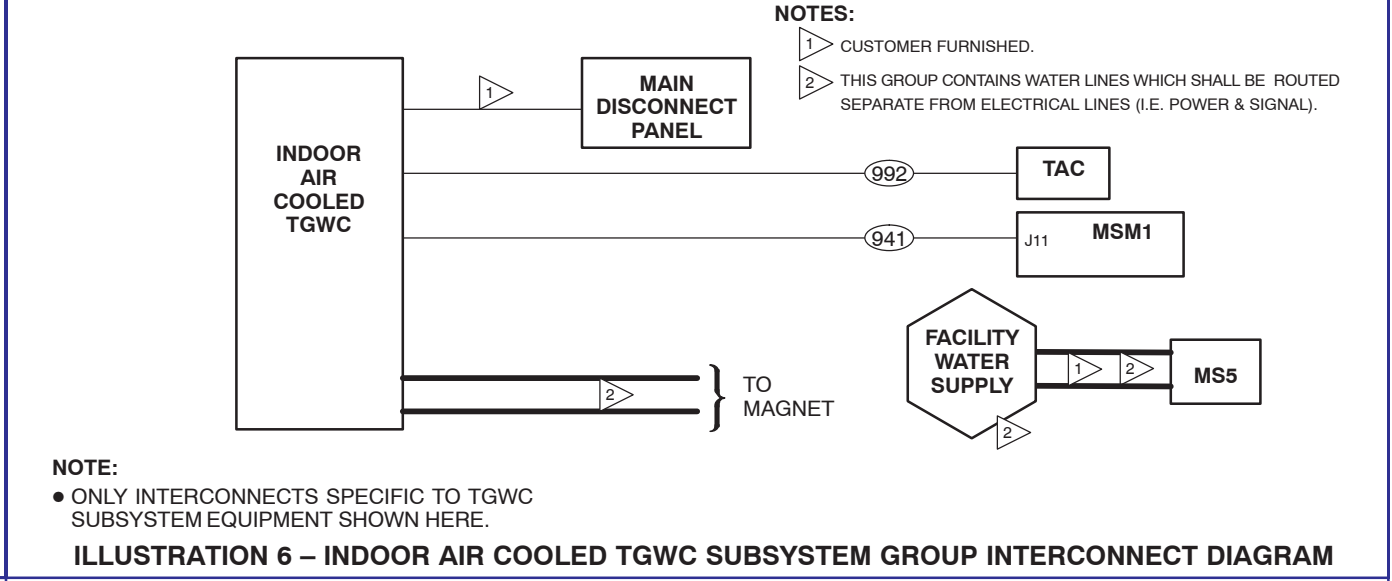
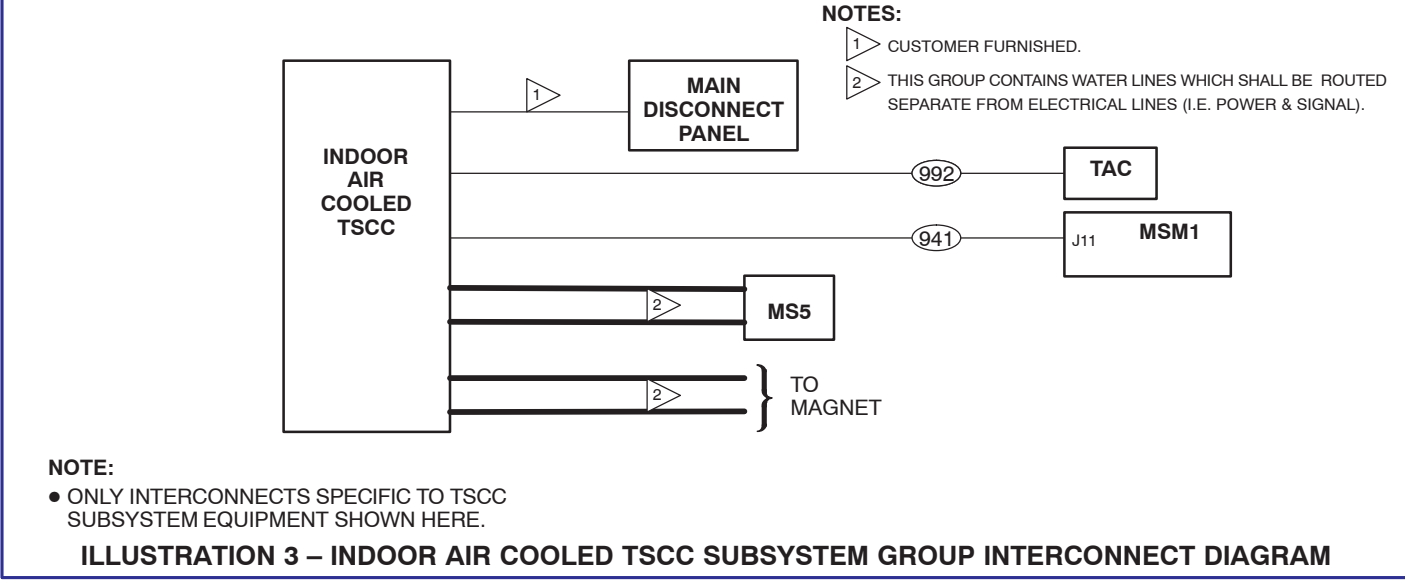
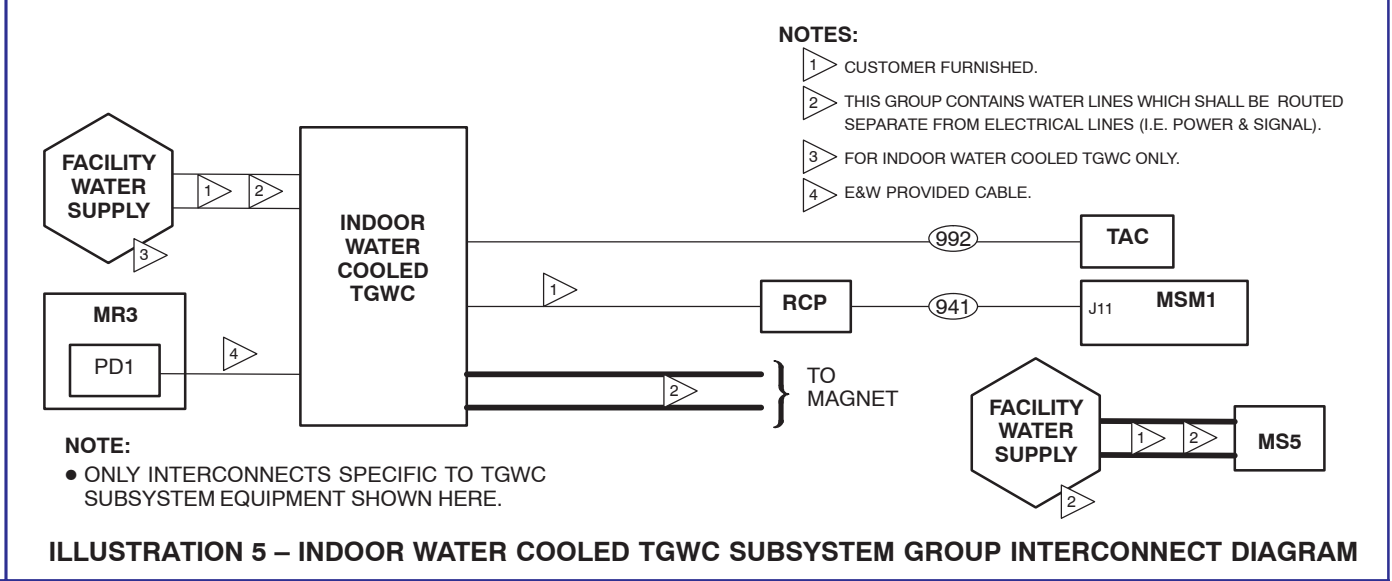
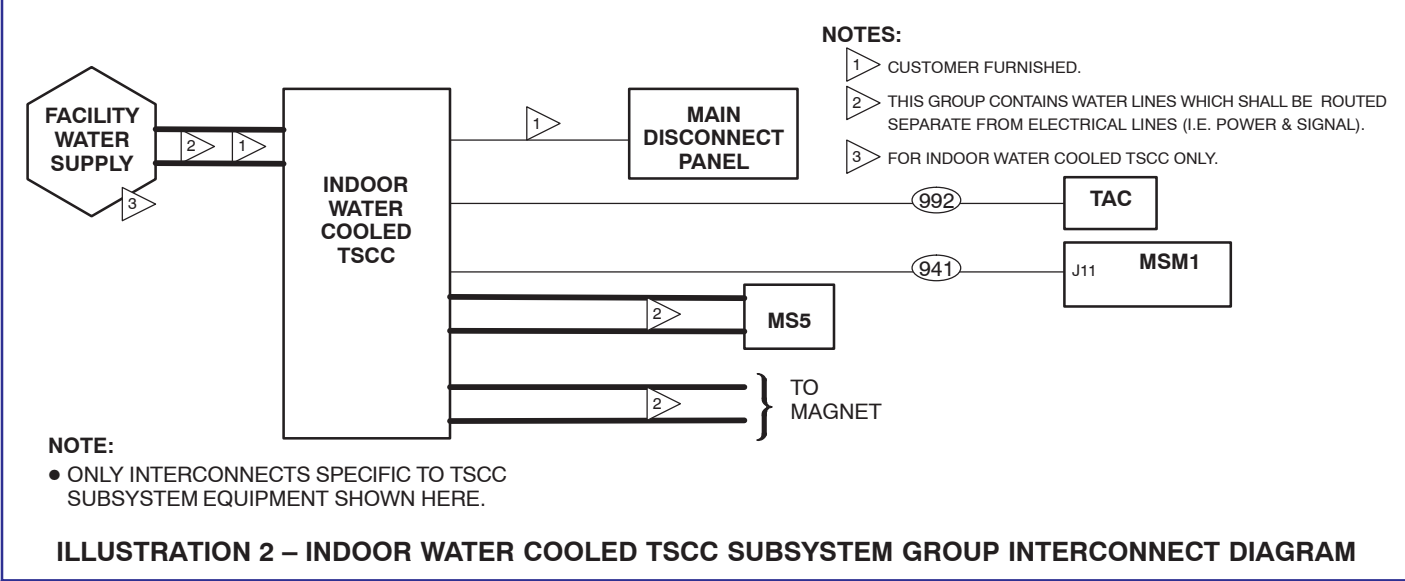
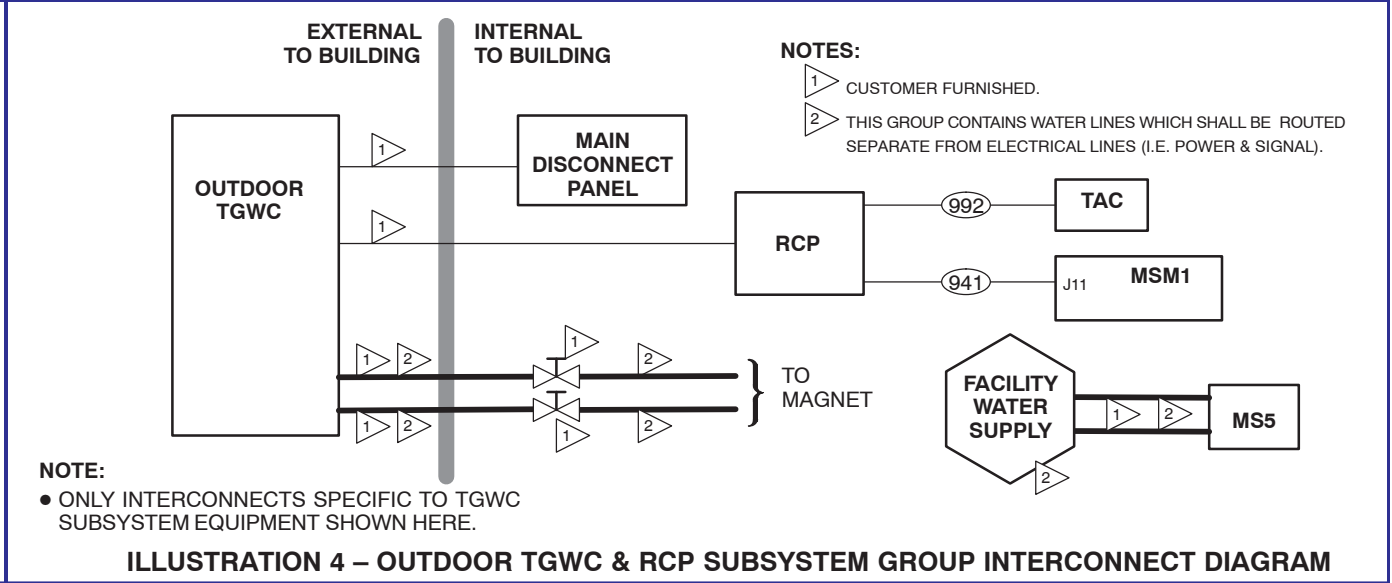
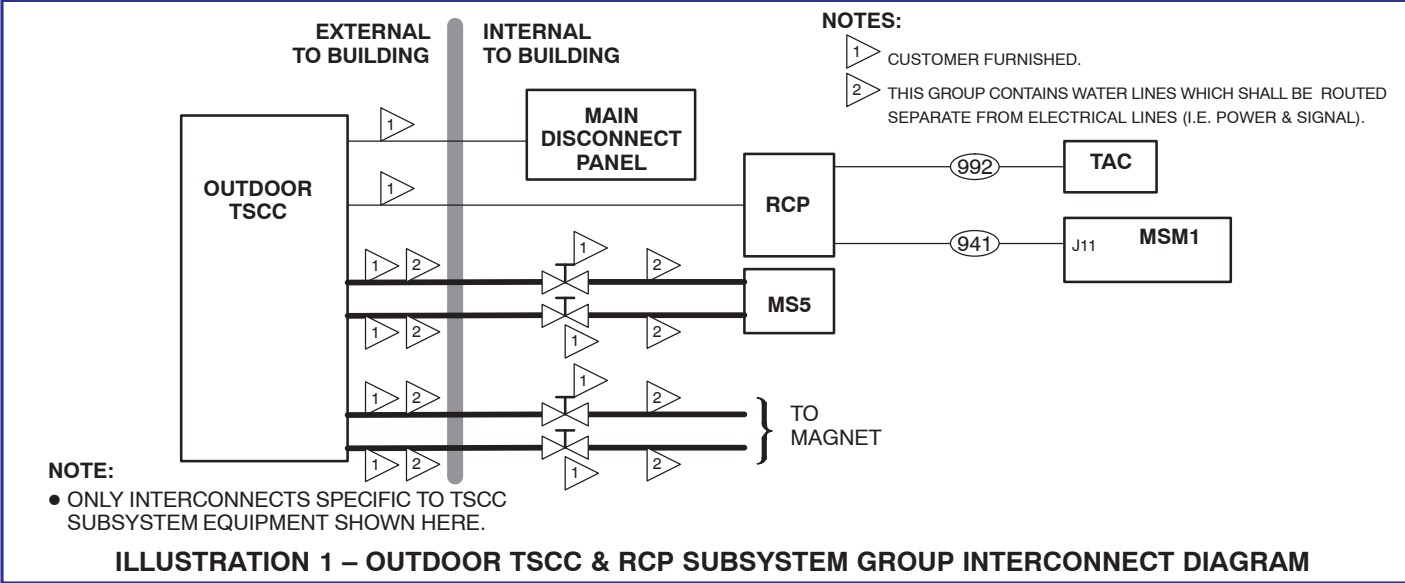
TABLE OF CONTENTS

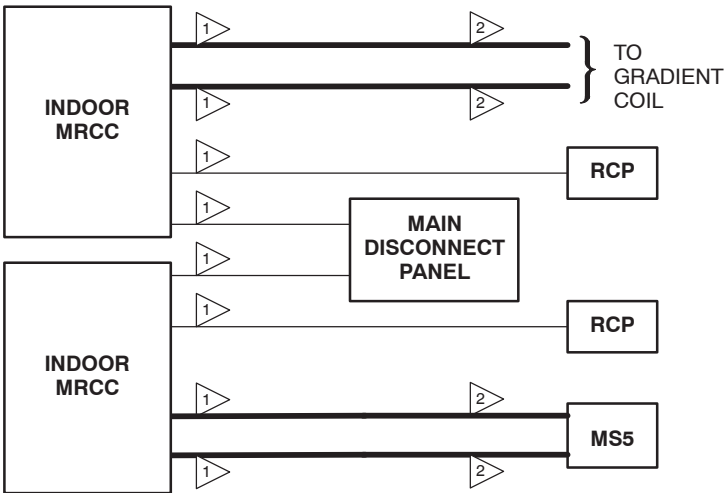
Water Chillers Subsystem Group Interconnect Diagrams

<u>ILLUSTRATION</u>	<u>PAGE 1-1</u>
1	OUTDOOR TSCC & RCP
2	INDOOR WATER COOLED TSCC
3	INDOOR AIR COOLED TSCC
4	OUTDOOR TGWC & RCP
5	INDOOR WATER COOLED TGWC
6	INDOOR AIR COOLED TGWC
7	DUAL INDOOR MRCC
8	INDOOR MRCC
9	DUAL OUTDOOR MRCC
10	OUTDOOR MRCC
11	INDOOR GWHX

Note:

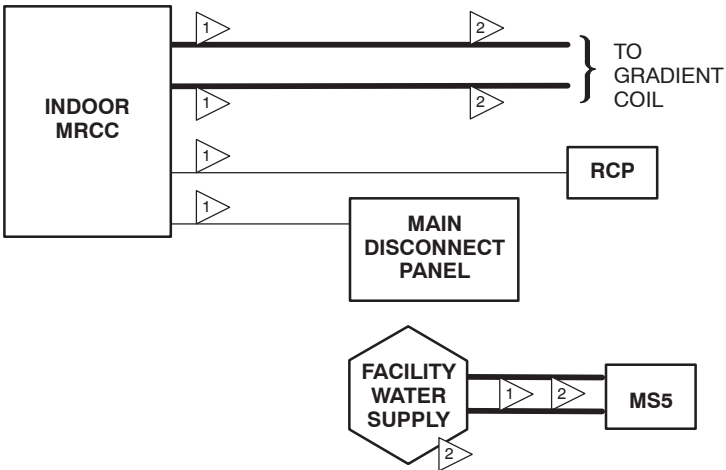
For all Chillers and Cooling System variations (SEE Vendor Manuals on MR Service Methods CDROM)





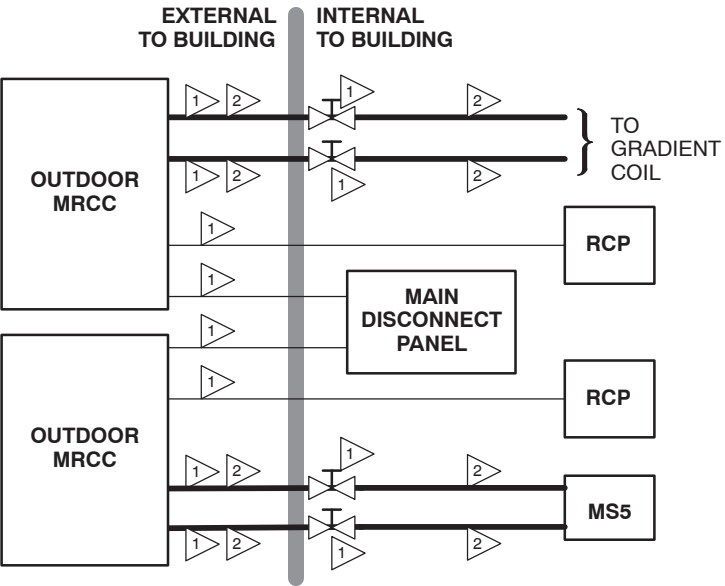
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

ILLUSTRATION 7 – DUAL INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



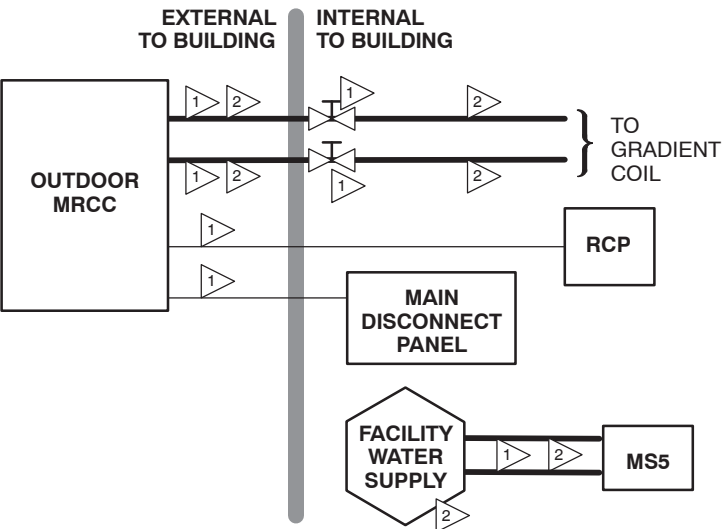
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

ILLUSTRATION 8 – INDOOR MRCC SUBSYSTEM GROUP INTERCONNECT DIAGRAM



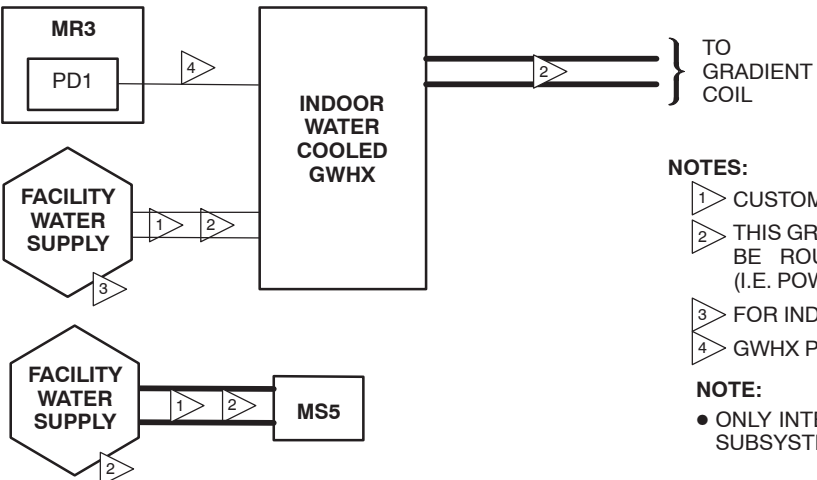
- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

ILLUSTRATION 9 –
DUAL OUTDOOR MRCC SUBSYSTEM
GROUP INTERCONNECT DIAGRAM



- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO MRCC SUBSYSTEM EQUIPMENT SHOWN HERE.

ILLUSTRATION 10 –
OUTDOOR MRCC SUBSYSTEM
GROUP INTERCONNECT DIAGRAM



- NOTES:**
- 1 CUSTOMER FURNISHED.
 - 2 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).
 - 3 FOR INDOOR WATER COOLED GWHX ONLY.
 - 4 GWHX PROVIDED CABLE.
- NOTE:**
- ONLY INTERCONNECTS SPECIFIC TO GWHX SUBSYSTEM EQUIPMENT SHOWN HERE.

ILLUSTRATION 11 – INDOOR GWHX SUBSYSTEM GROUP INTERCONNECT DIAGRAM